



US 20250282714A1

(19) **United States**

(12) **Patent Application Publication**
Hargrove et al.

(10) **Pub. No.: US 2025/0282714 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **COMPOSITIONS FOR AND METHODS OF MODULATING RNA STABILITY**

(71) Applicant: **Duke University**, Durham, NC (US)

(72) Inventors: **Amanda Hargrove**, Durham, NC (US);
Martina Zafferani, Durham, NC (US)

(21) Appl. No.: **18/849,279**

(22) PCT Filed: **Mar. 23, 2023**

(86) PCT No.: **PCT/US2023/064879**
§ 371 (c)(1),
(2) Date: **Sep. 20, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/322,902, filed on Mar. 23, 2022.

Publication Classification

(51) **Int. Cl.**
C07C 257/18 (2006.01)
A61K 31/655 (2006.01)
A61P 35/00 (2006.01)

C07C 317/32 (2006.01)
C07D 211/34 (2006.01)
C07D 211/58 (2006.01)
C07D 213/58 (2006.01)
C07D 213/78 (2006.01)
C07D 295/135 (2006.01)
C07D 295/155 (2006.01)
C07D 307/54 (2006.01)
C07D 401/14 (2006.01)

(52) **U.S. Cl.**
CPC **C07C 257/18** (2013.01); **A61K 31/655** (2013.01); **A61P 35/00** (2018.01); **C07C 317/32** (2013.01); **C07D 211/34** (2013.01); **C07D 211/58** (2013.01); **C07D 213/58** (2013.01); **C07D 213/78** (2013.01); **C07D 295/135** (2013.01); **C07D 295/155** (2013.01); **C07D 307/54** (2013.01); **C07D 401/14** (2013.01); **C07C 2601/02** (2017.05)

(57) **ABSTRACT**

Disclosed herein are DMZ-derived compositions for and methods of modulating the expression and/or function of disease associated and/or diseasing causing secondary RNA structures and tertiary RNA structures.

Specification includes a Sequence Listing.

FIG. 1C

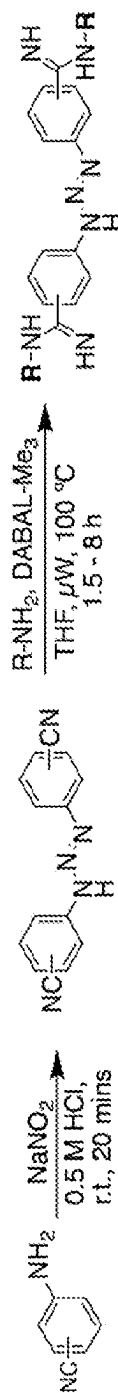
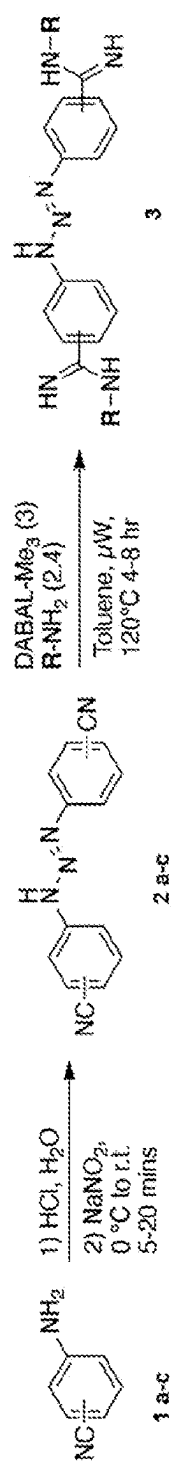


FIG. 2A

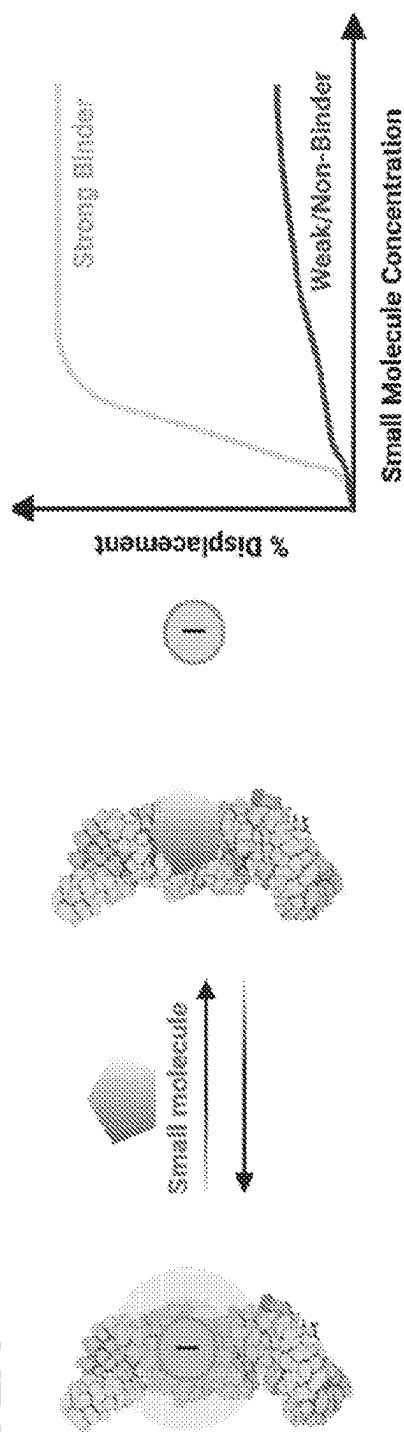


FIG. 2B

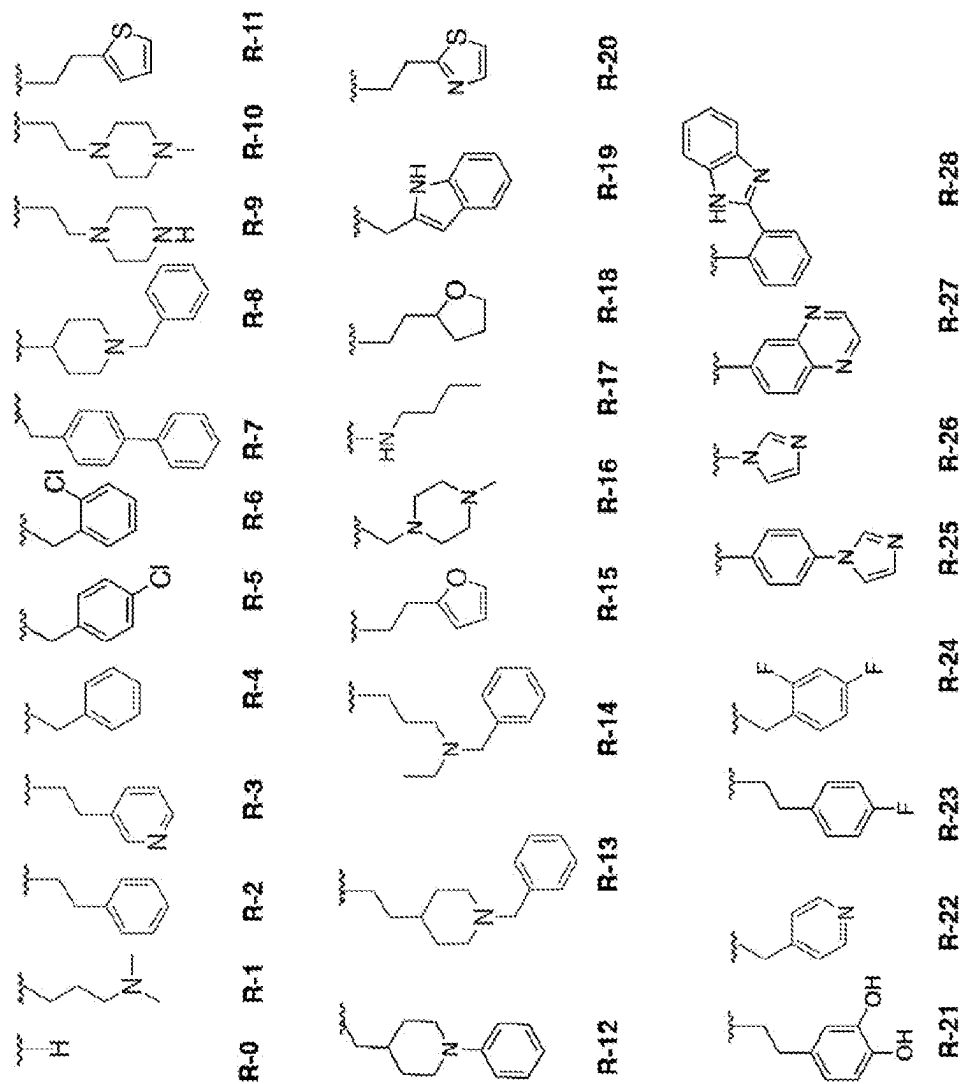


FIG. 2C

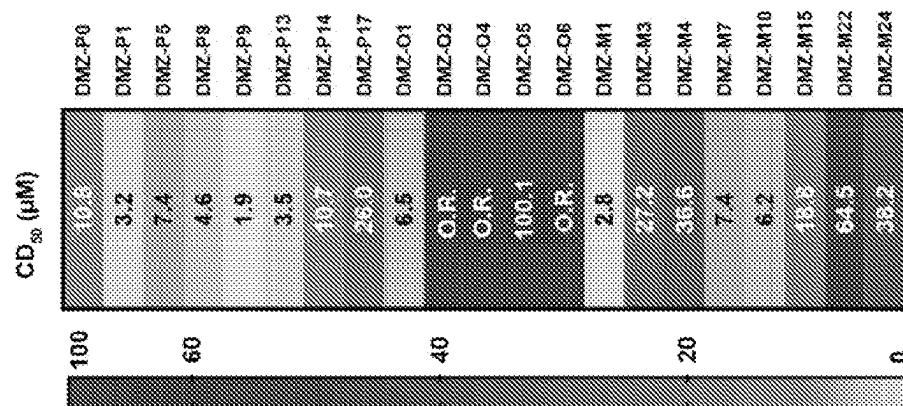


FIG. 3A

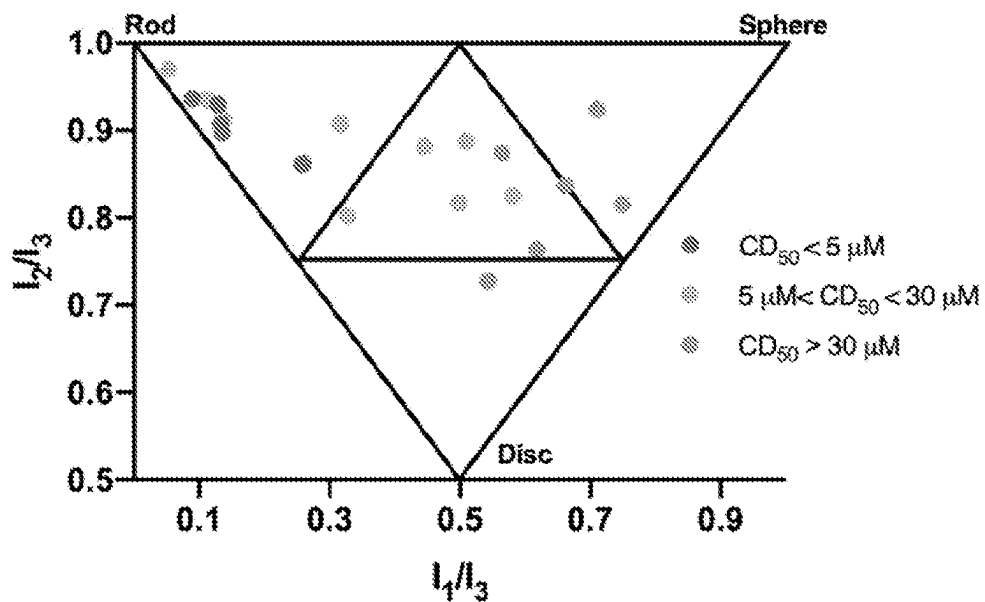


FIG. 3B

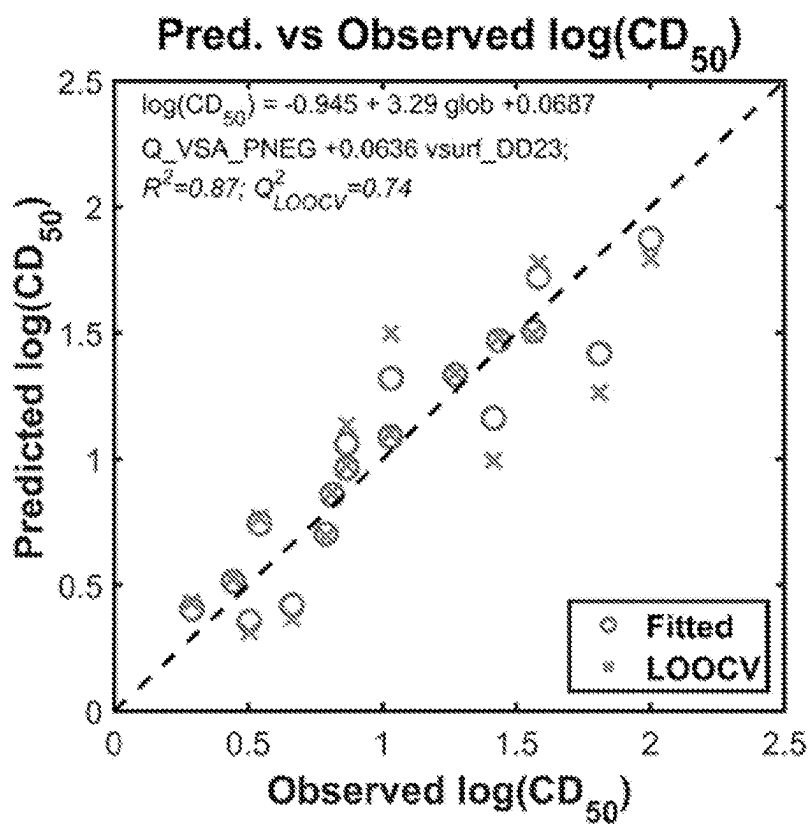


FIG. 4A

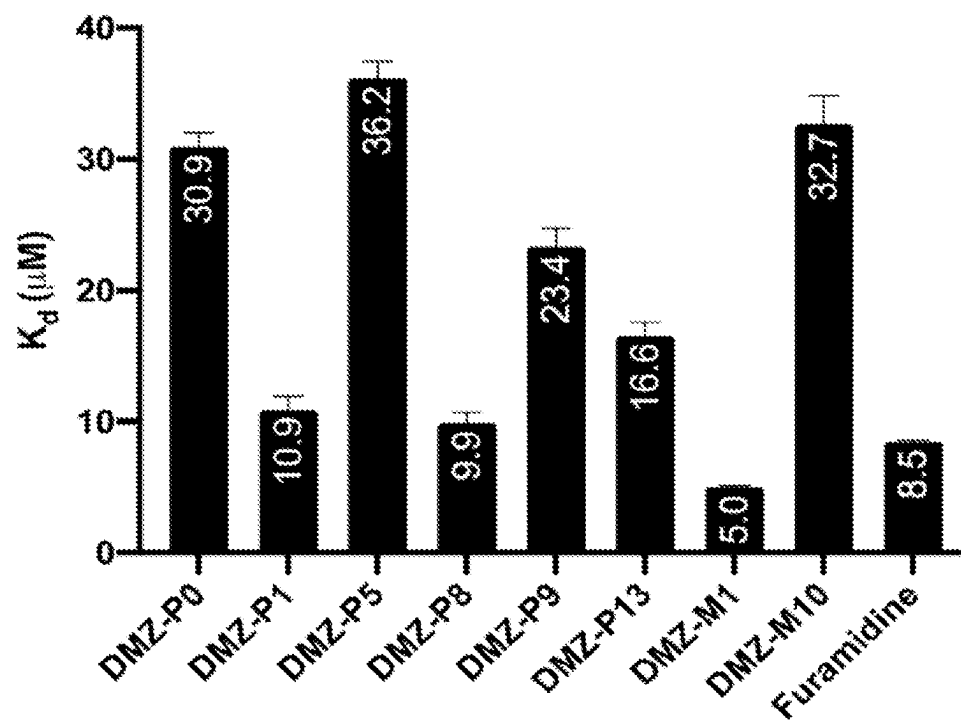


FIG. 4B

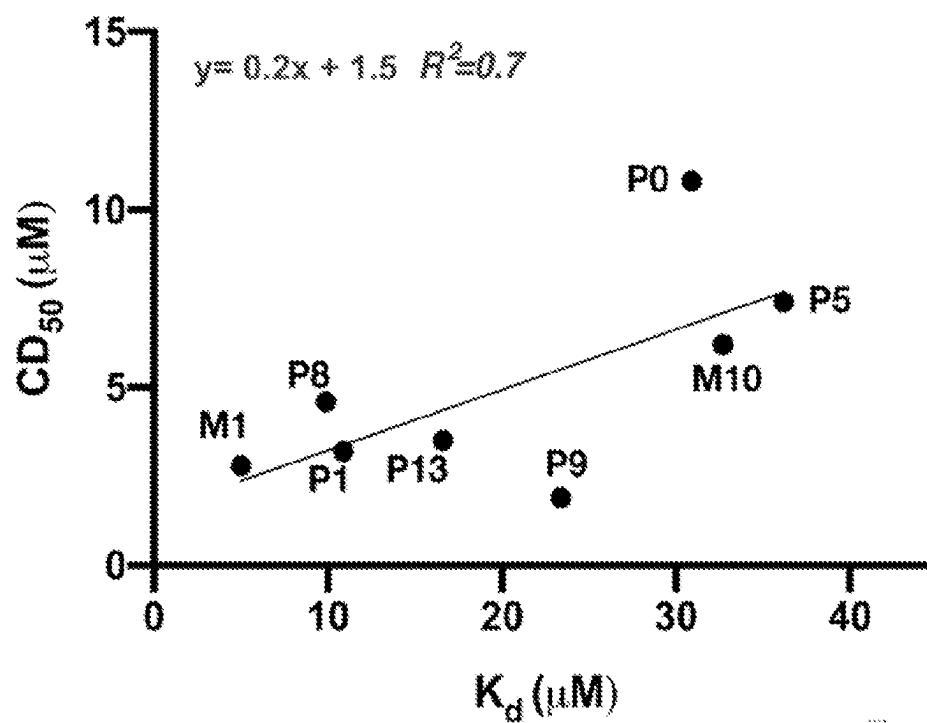


FIG. 5A

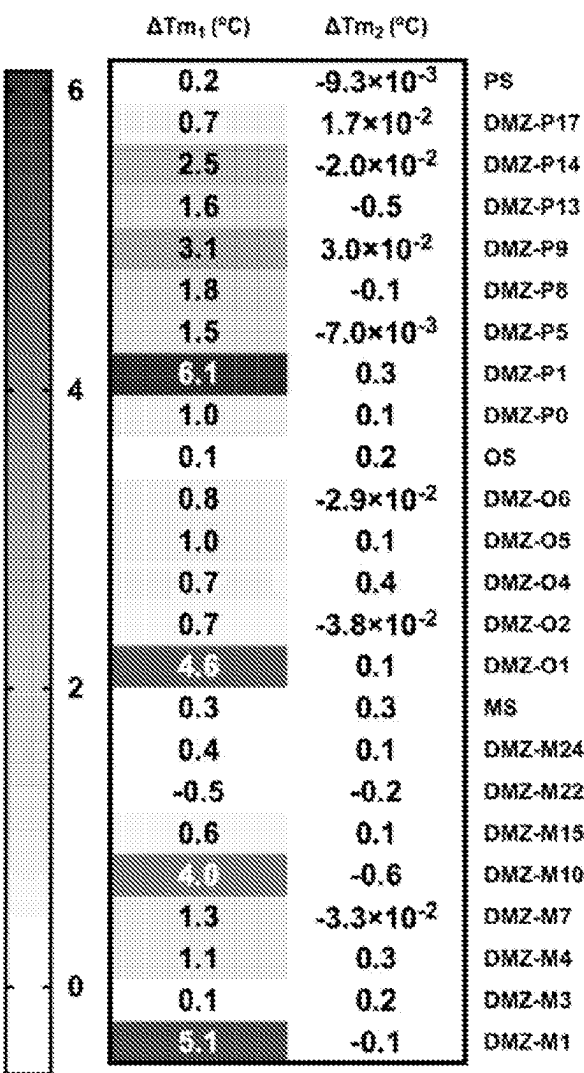


FIG. 5B

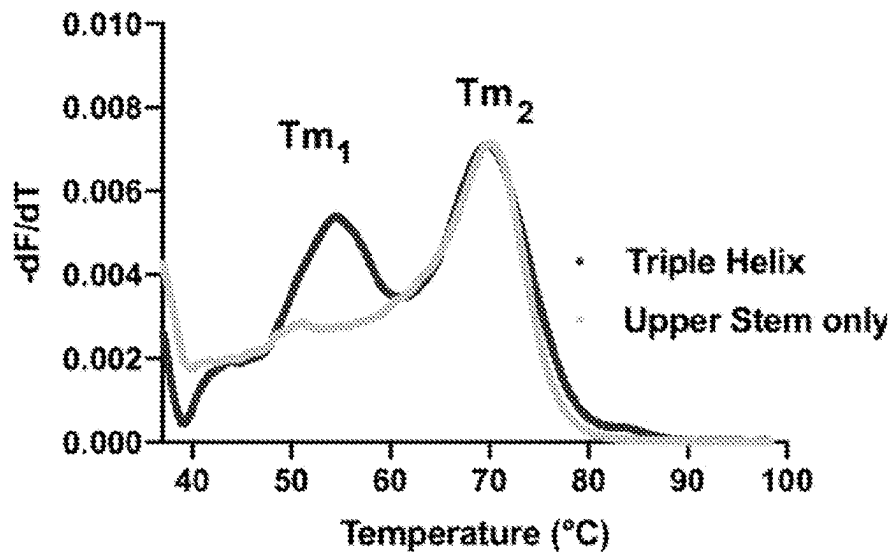


FIG. 5C

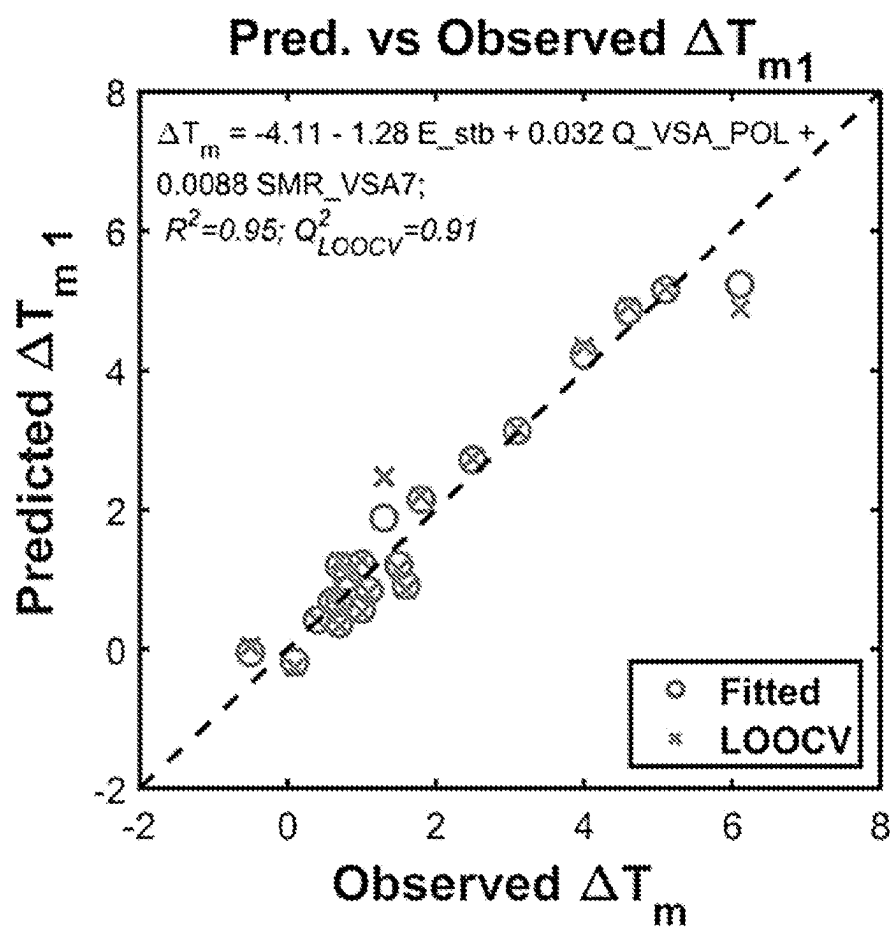


FIG. 6A

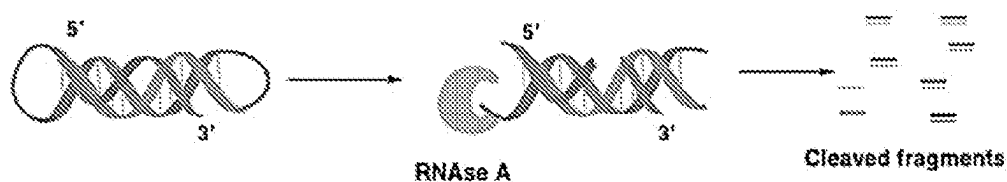


FIG. 6B

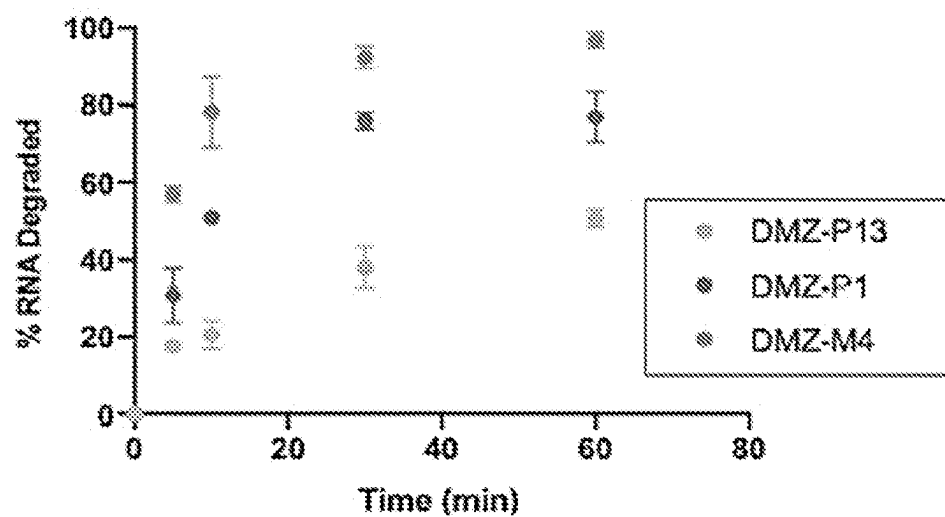


FIG. 6C

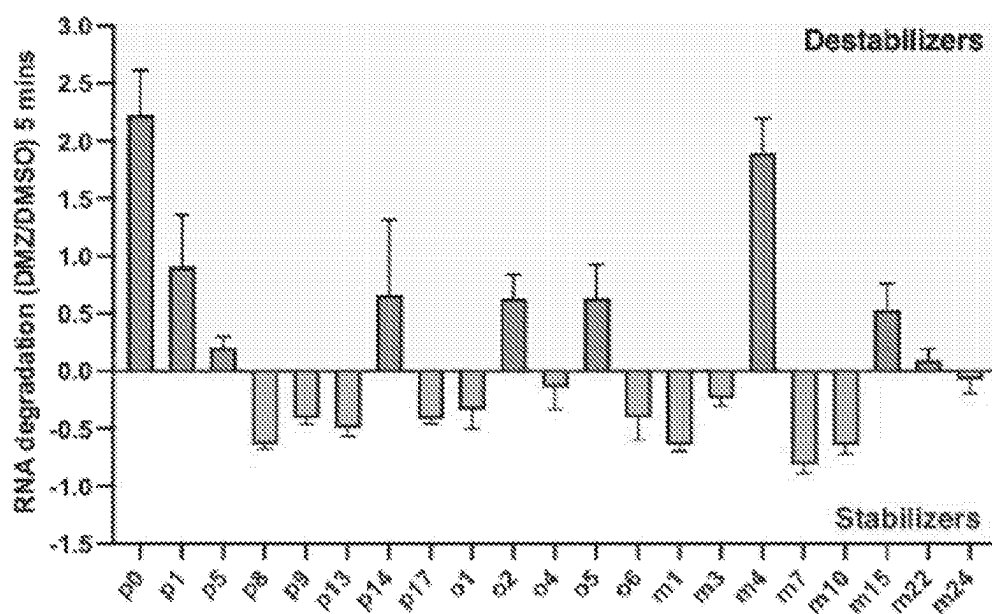


FIG. 6D

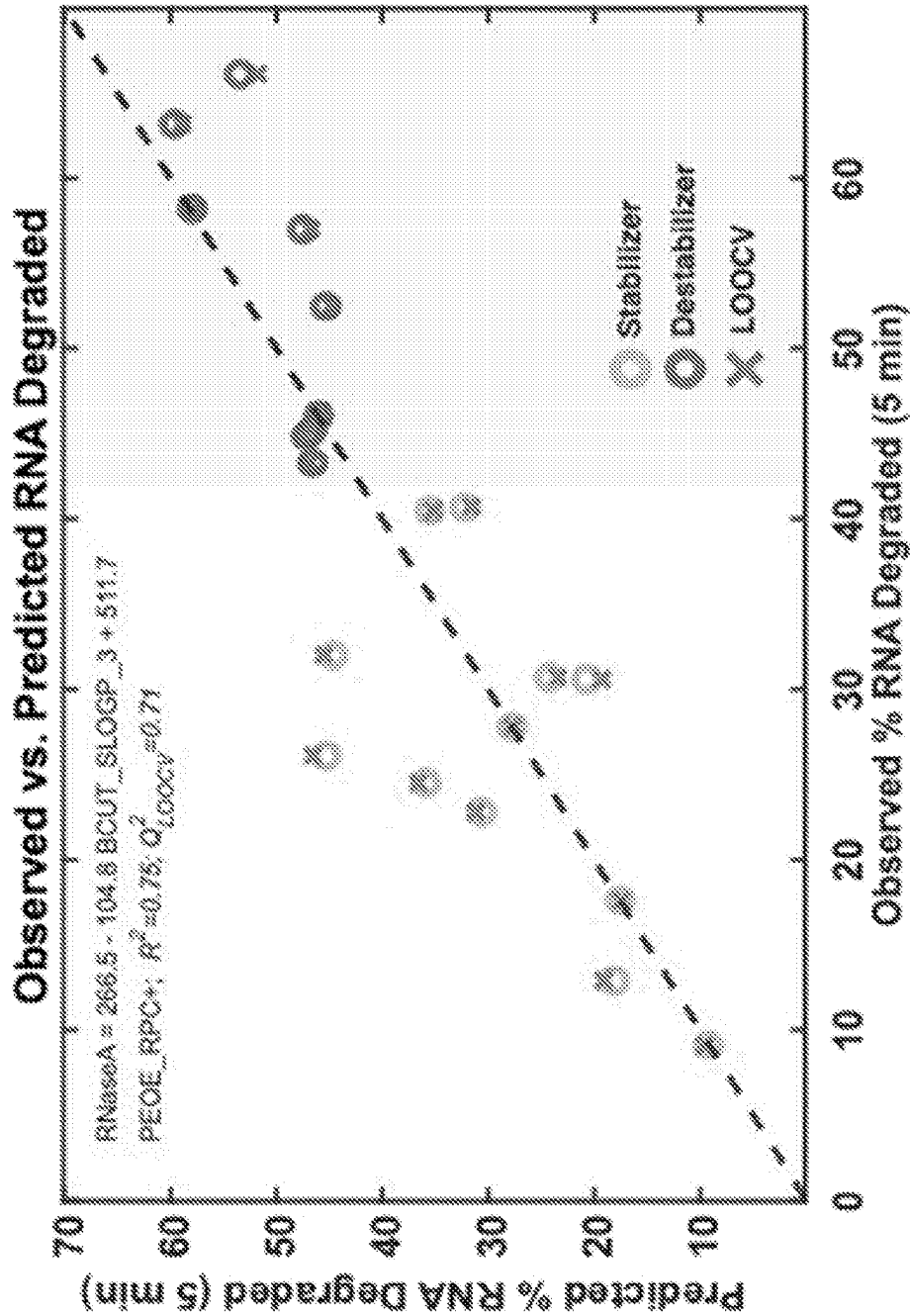


FIG. 7

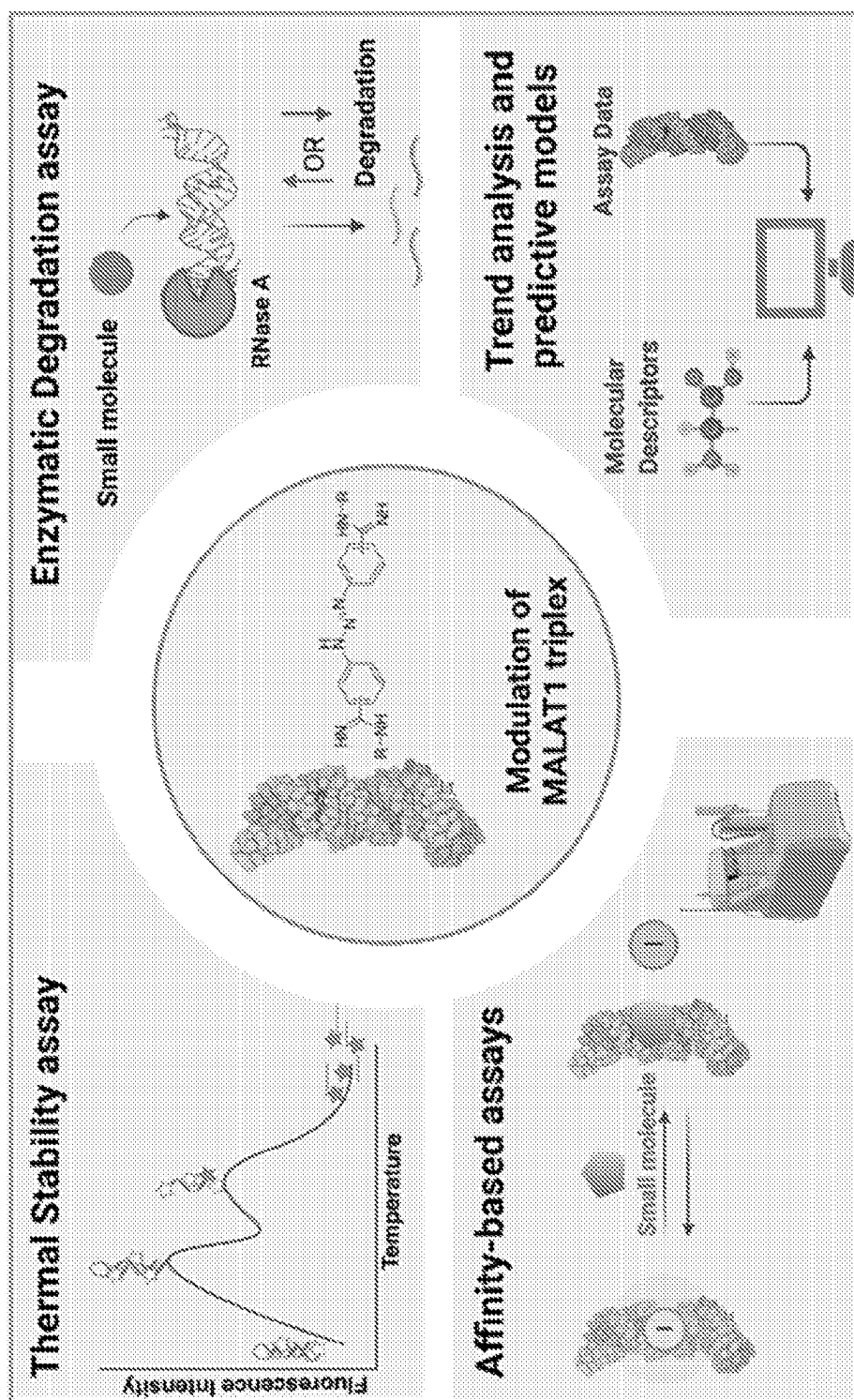


FIG. 8A

Library Design & Selection

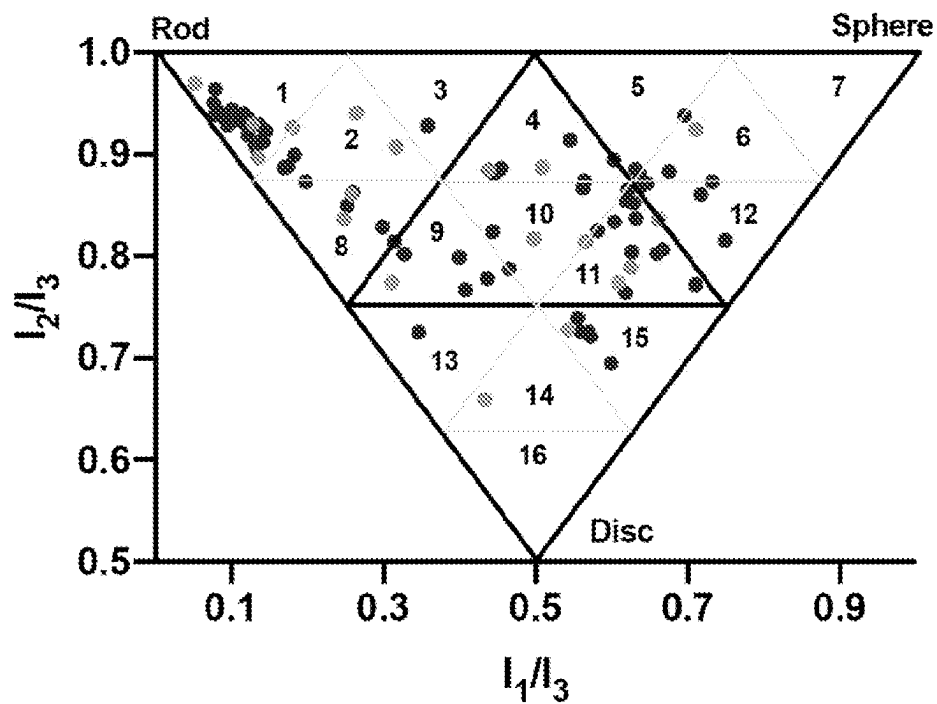


FIG. 8B

Euclidean Distance Replacement

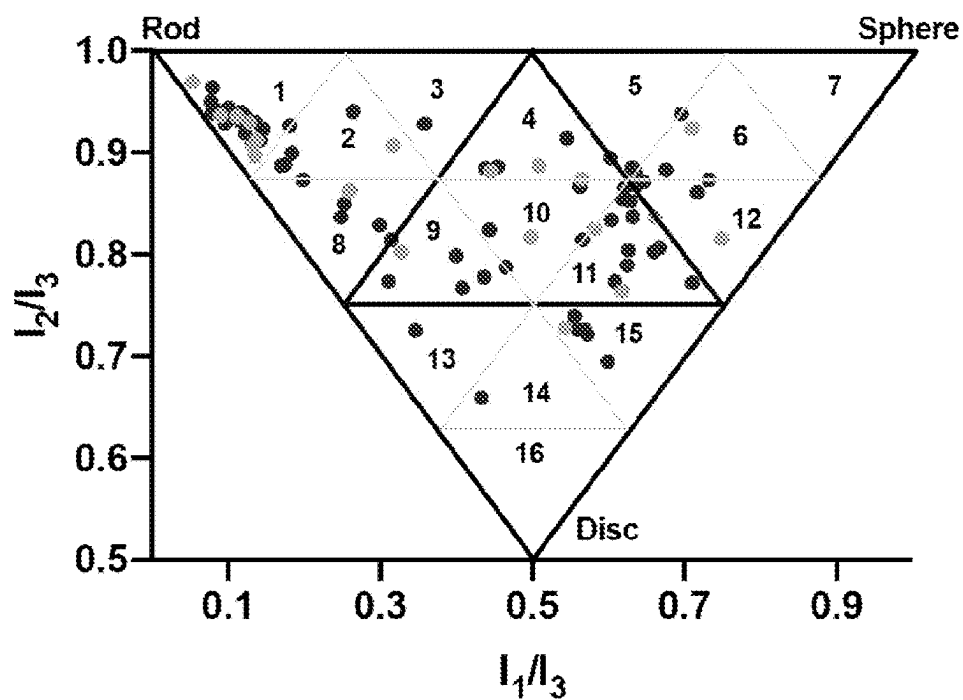


FIG. 10

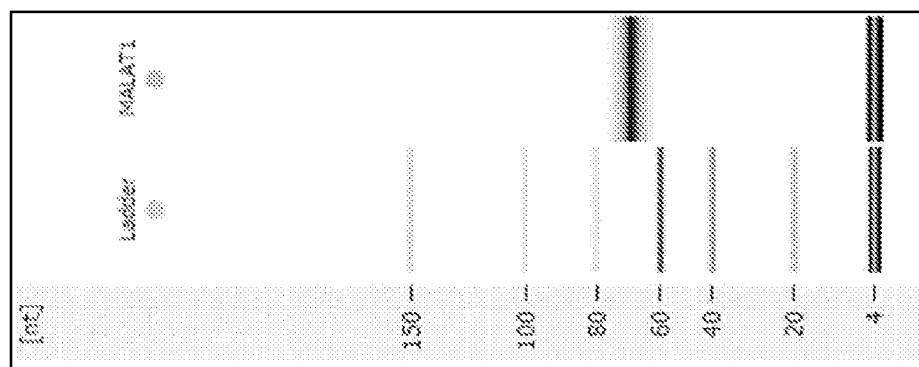


FIG. 9

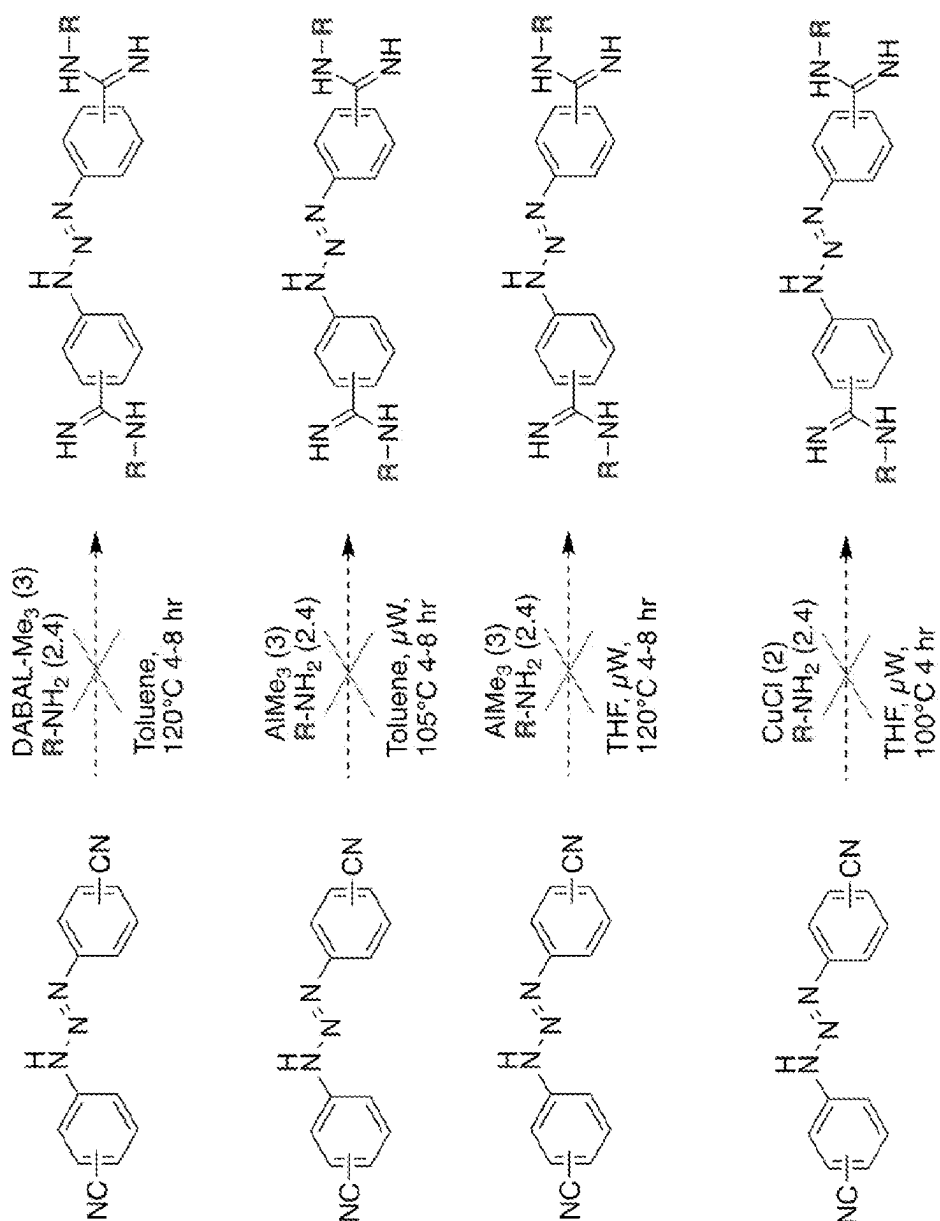


FIG. 11

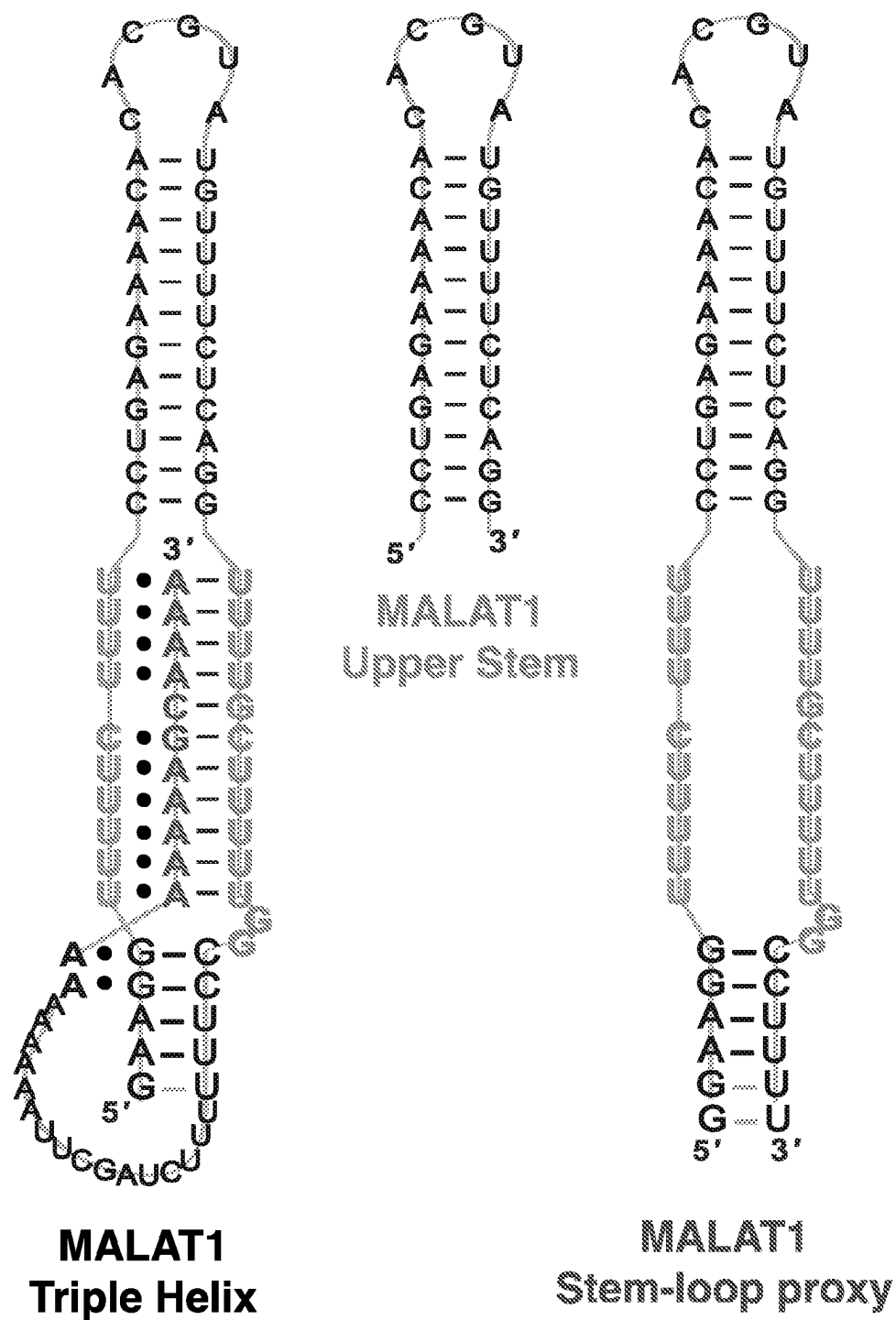


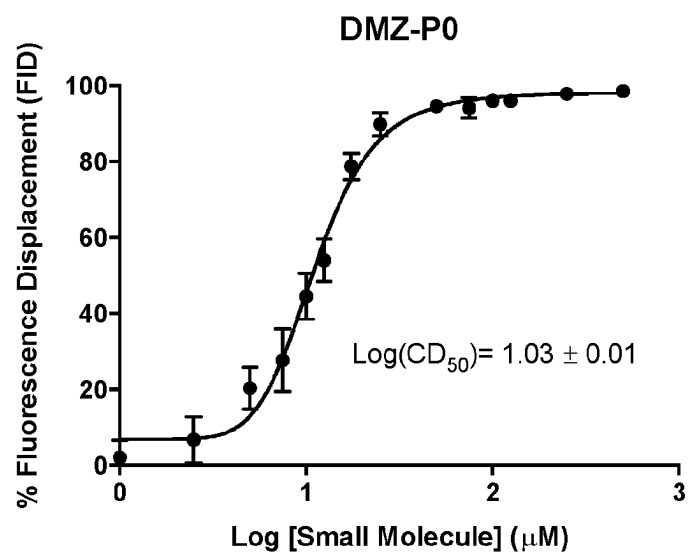
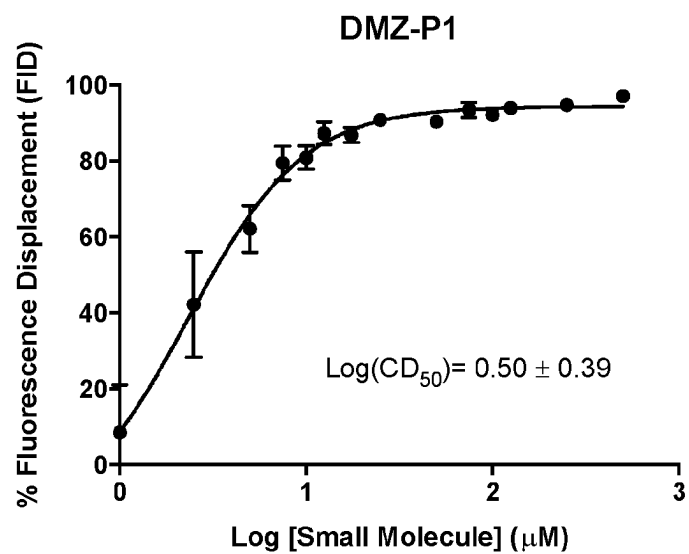
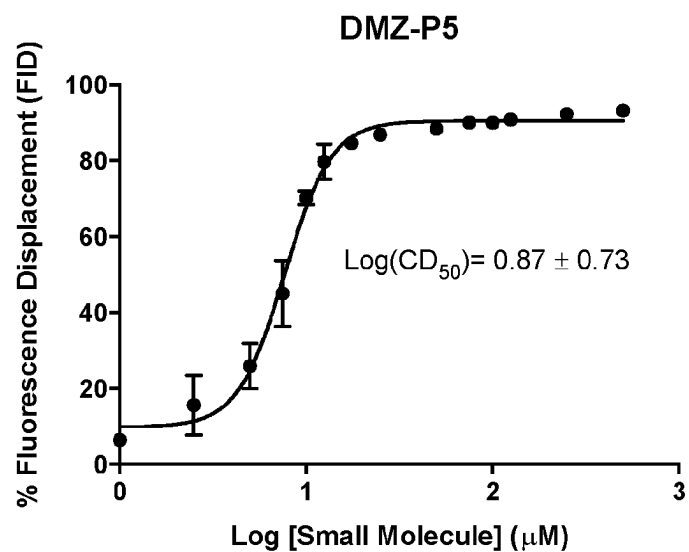
FIG. 12A**FIG. 12B****FIG. 12C**

FIG. 12D

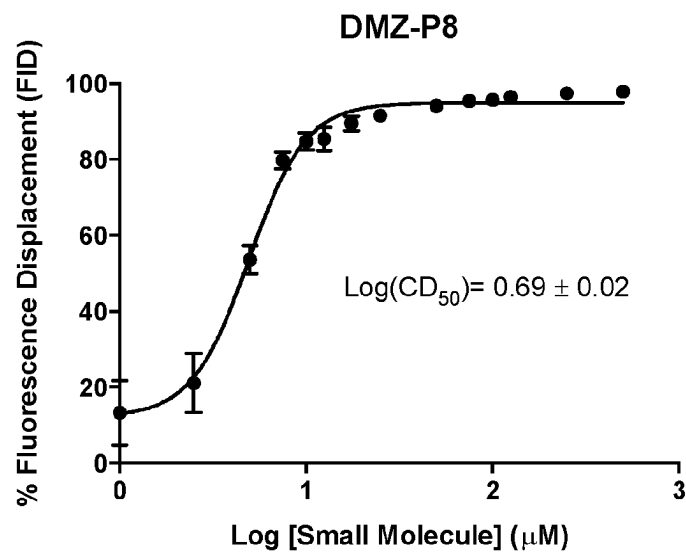


FIG. 12E

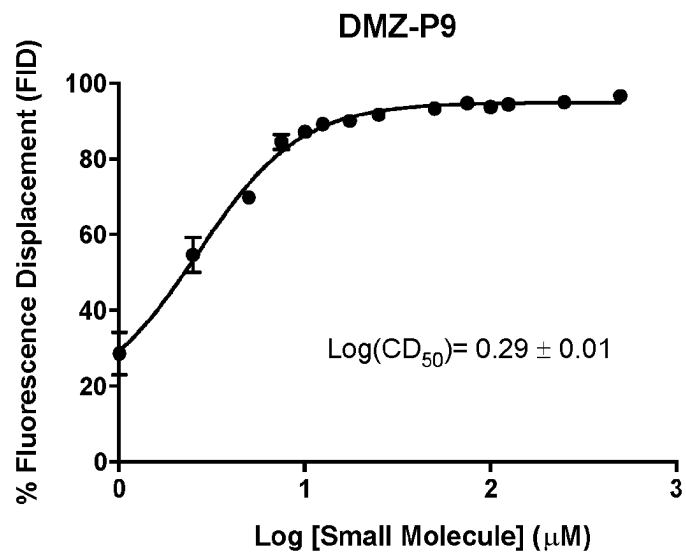


FIG. 12F

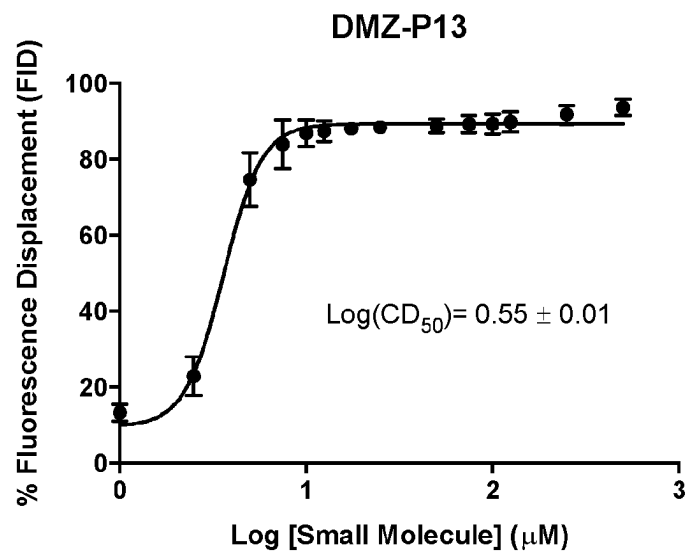


FIG. 12G

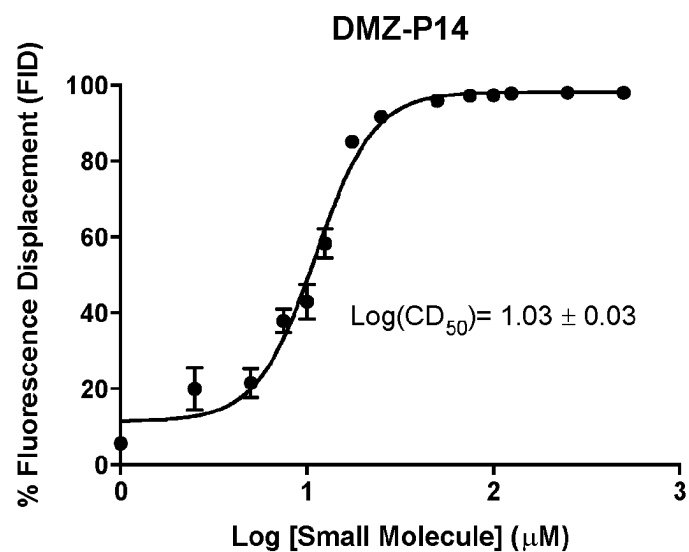


FIG. 12H

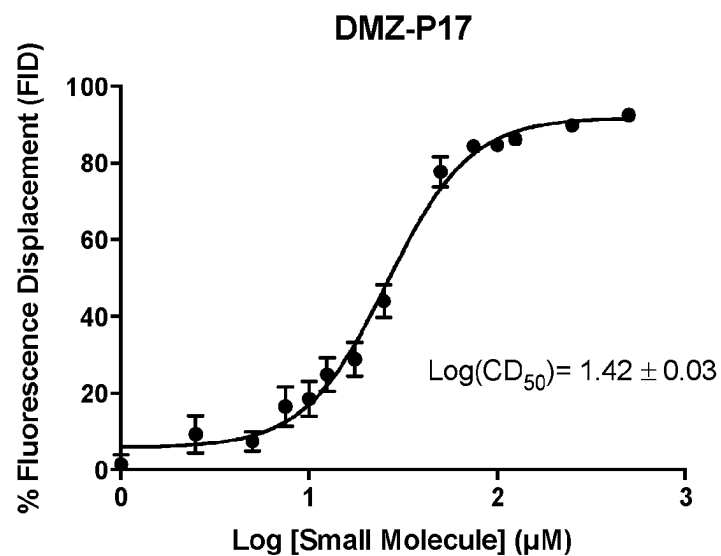


FIG. 12I

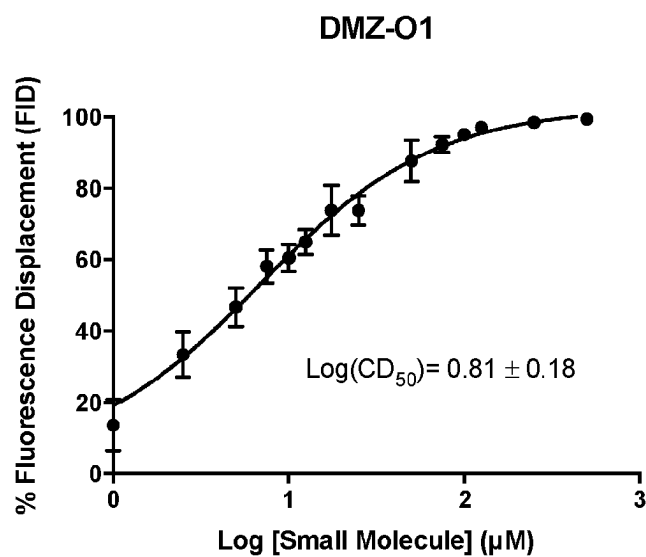


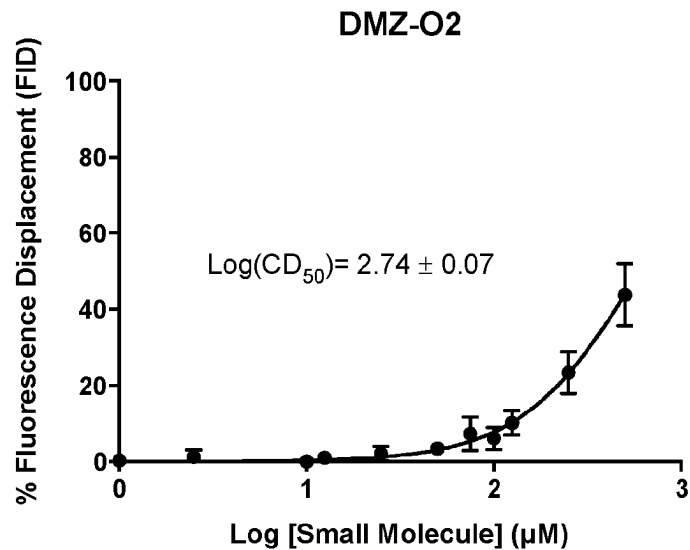
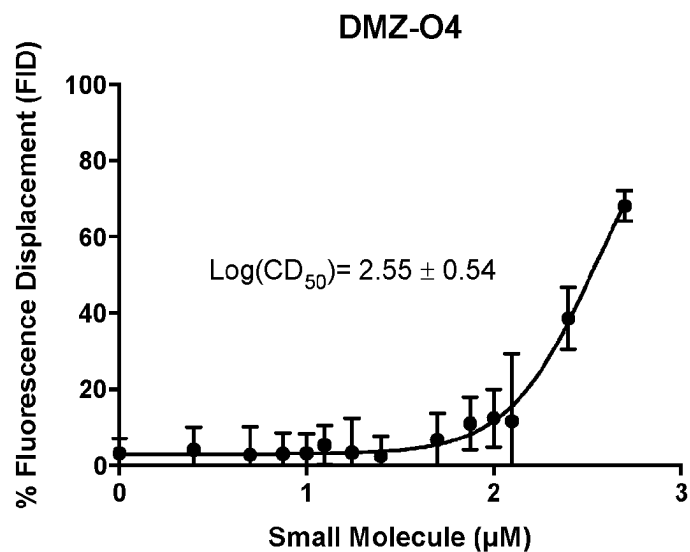
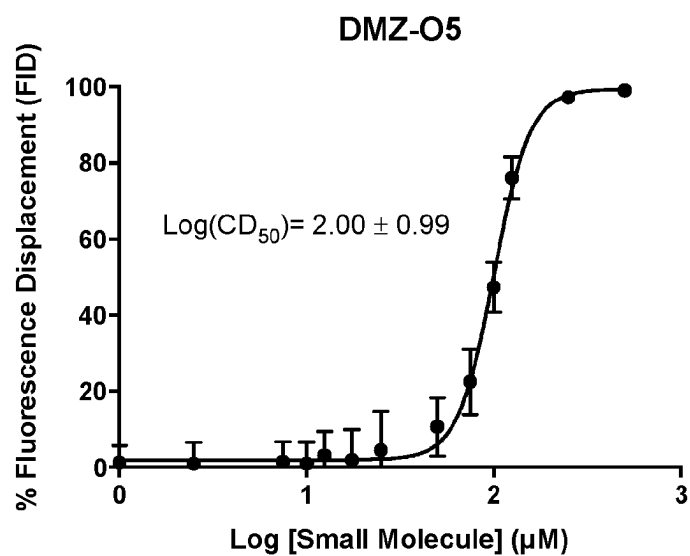
FIG. 12J**FIG. 12K****FIG. 12L**

FIG. 12M

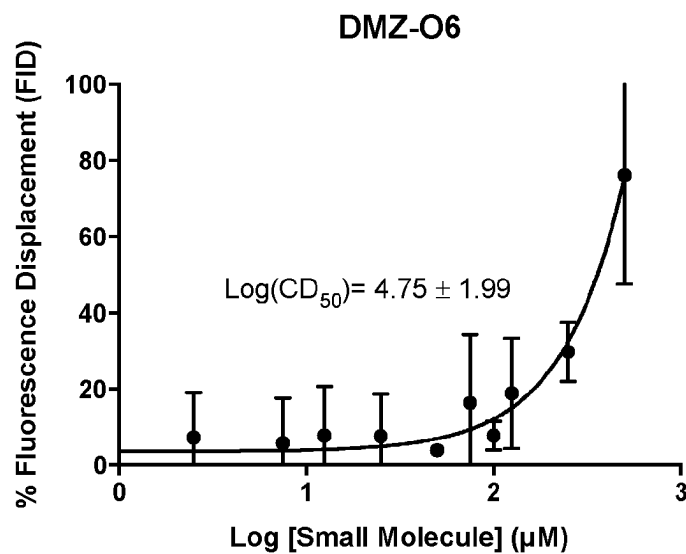


FIG. 12N

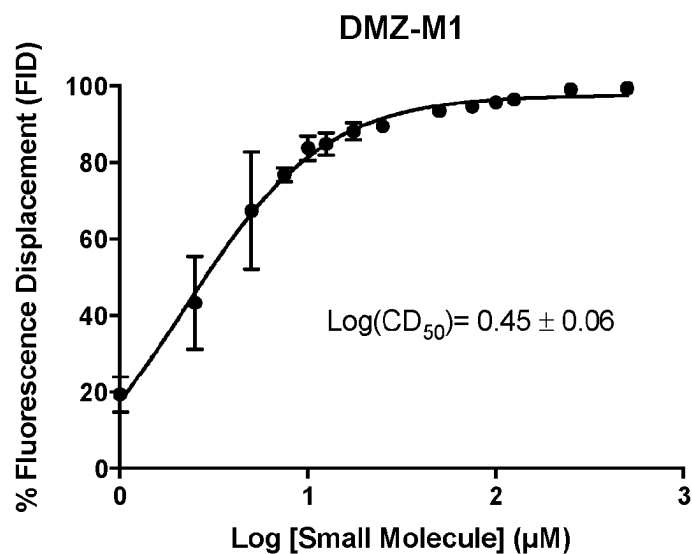


FIG. 12O

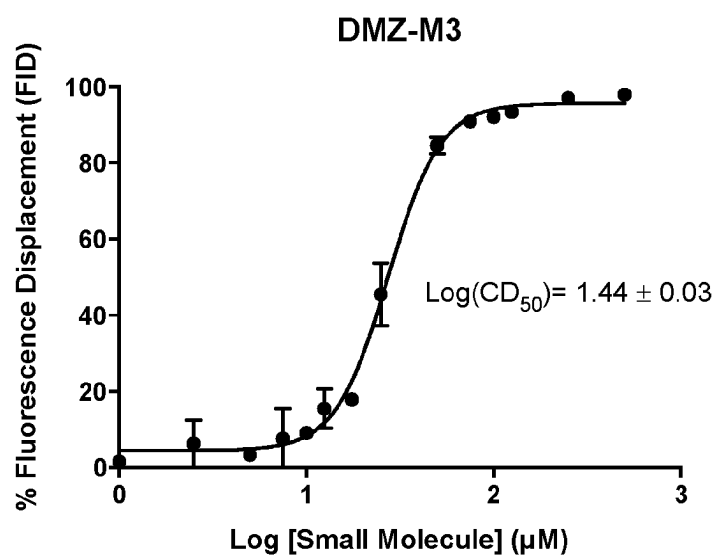


FIG. 12P

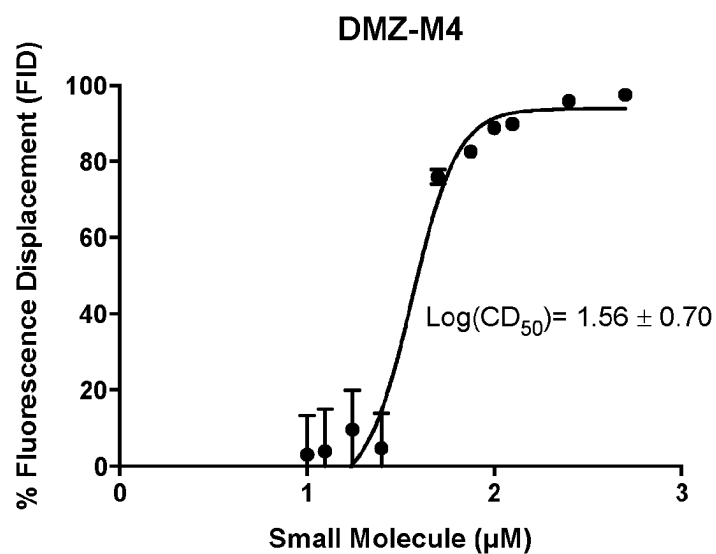


FIG. 12Q

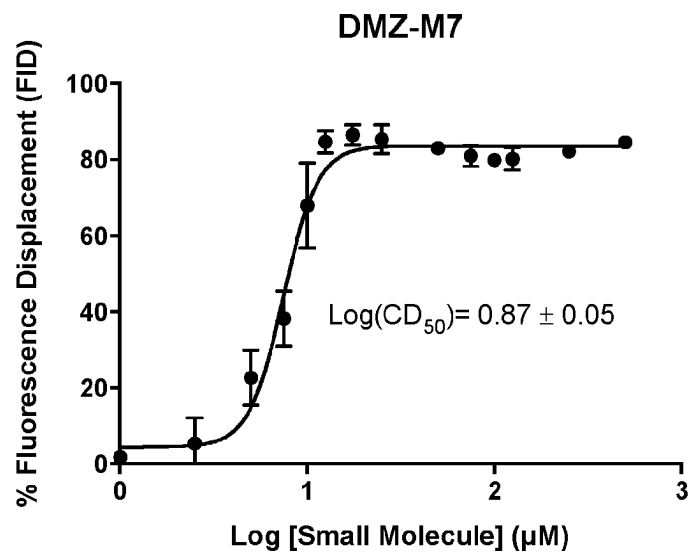


FIG. 12R

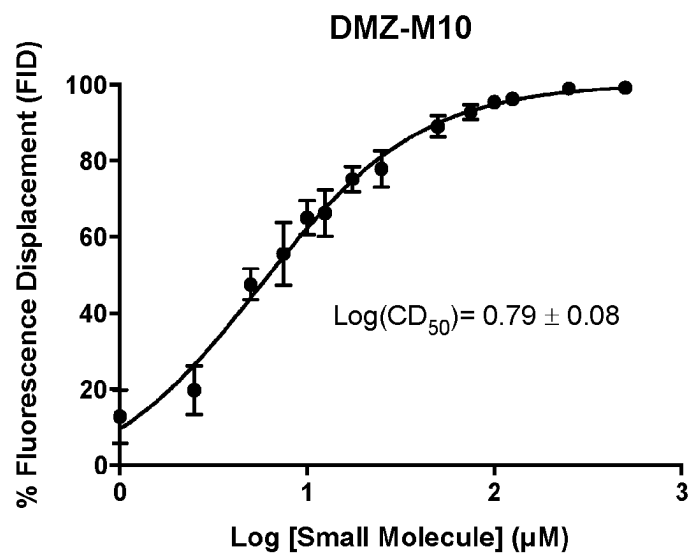


FIG. 12S

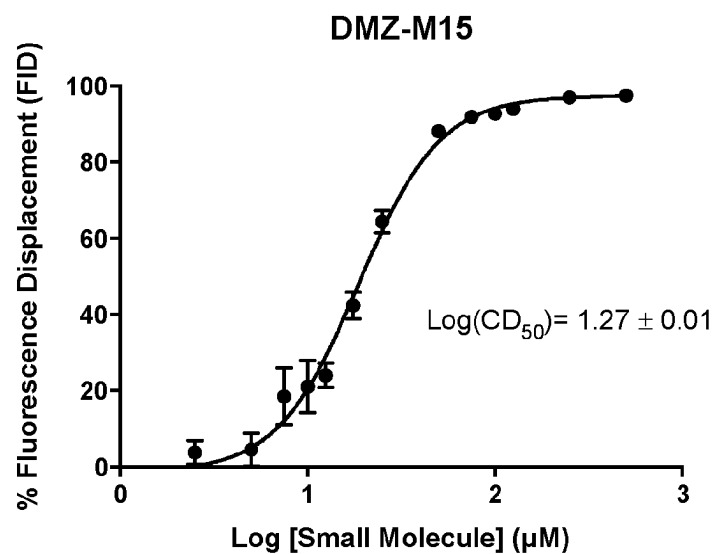


FIG. 12T

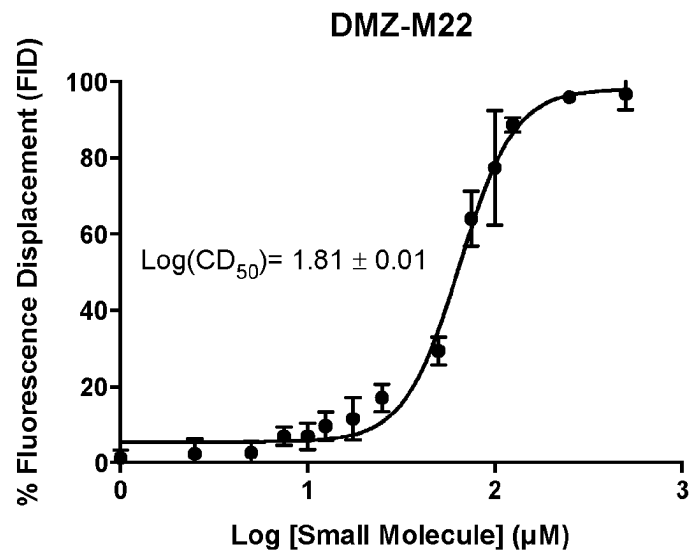


FIG. 12U

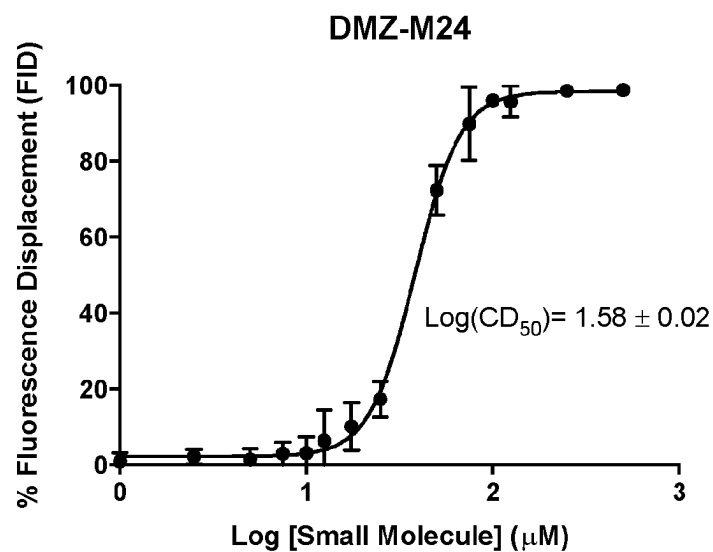


FIG. 13A

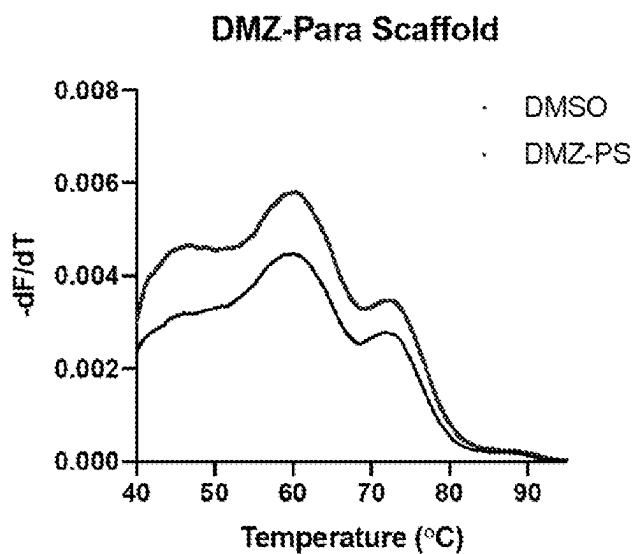


FIG. 13B

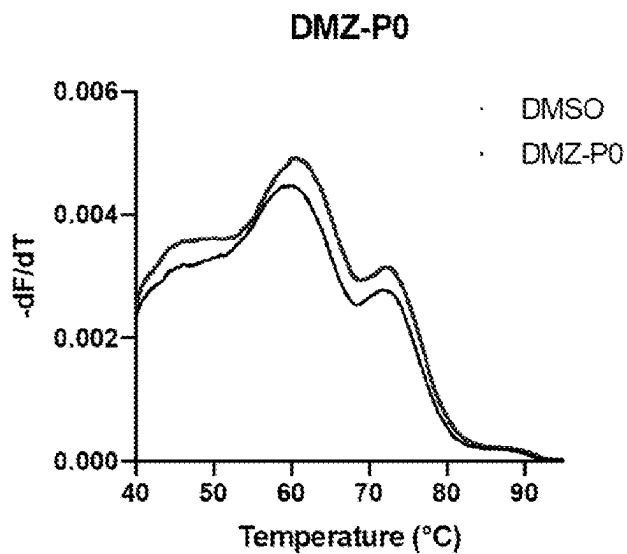


FIG. 13C

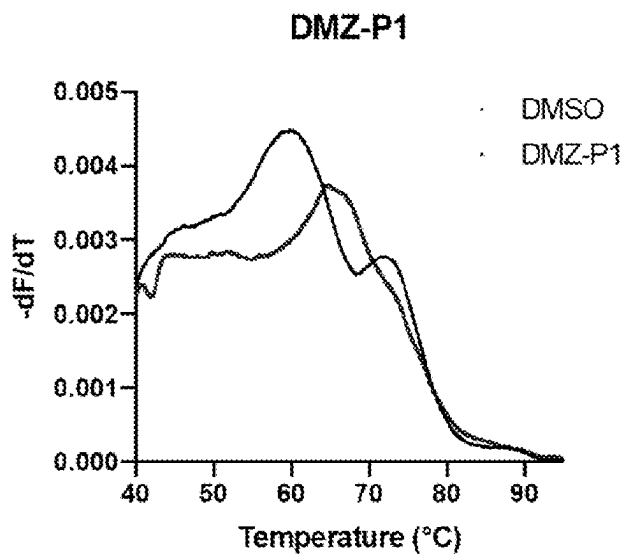


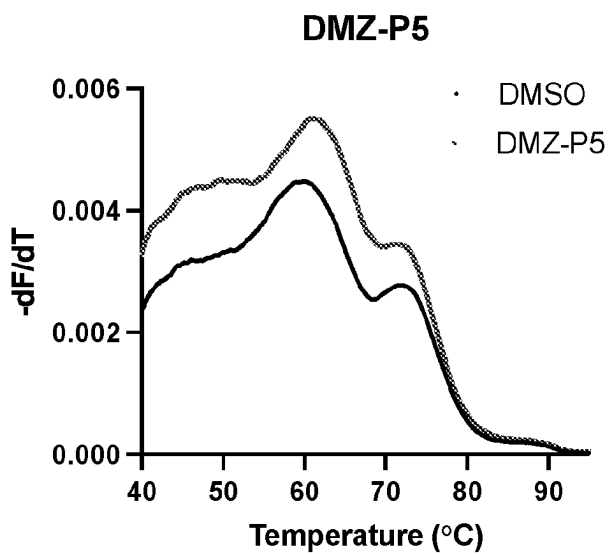
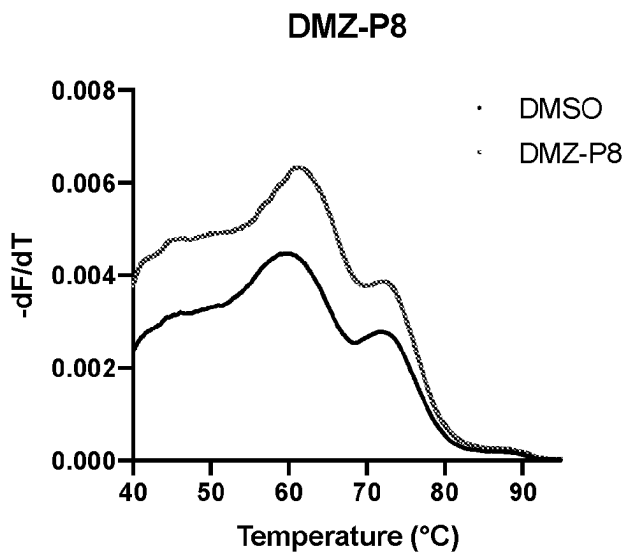
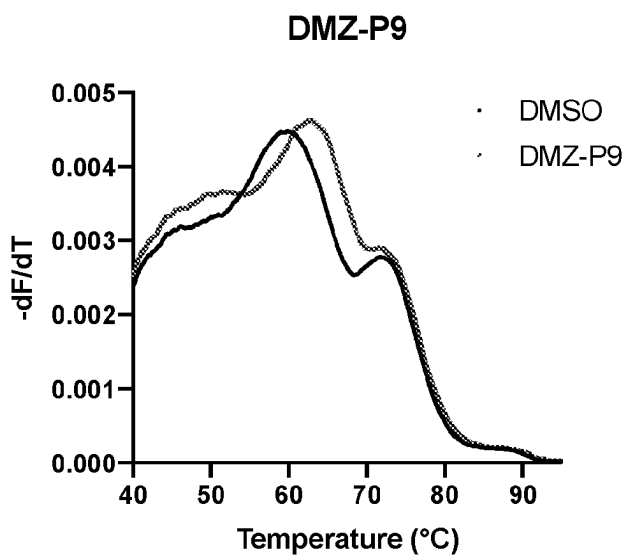
FIG. 13D**FIG. 13E****FIG. 13F**

FIG. 13G

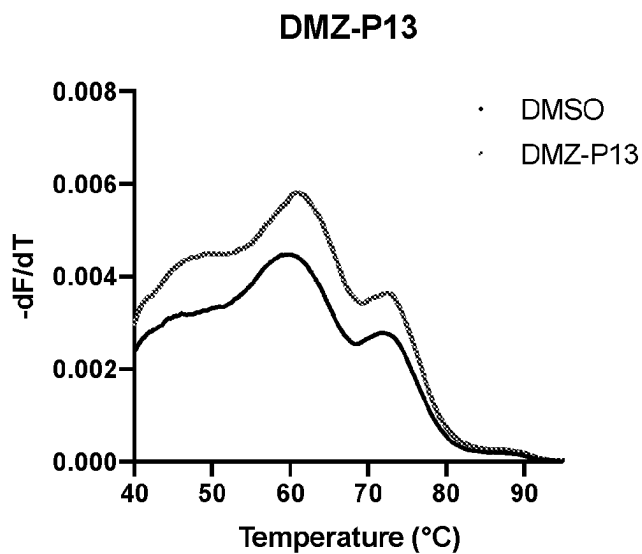


FIG. 13H

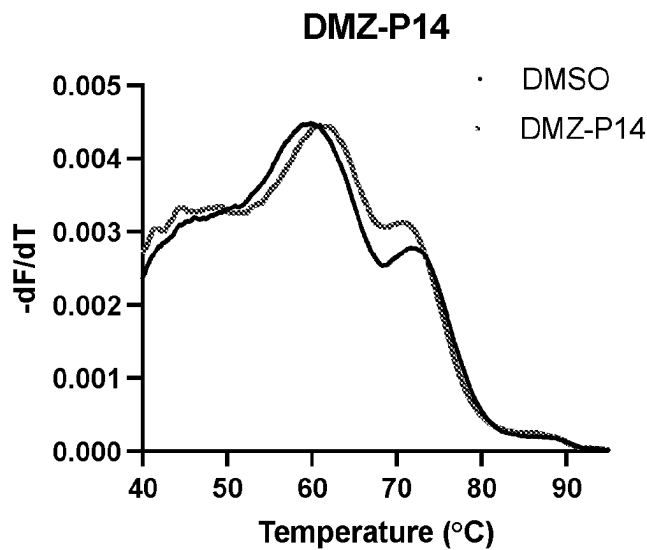


FIG. 13I

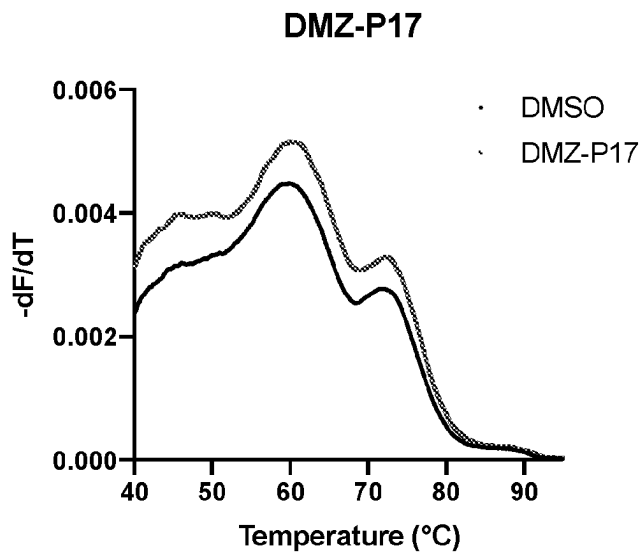


FIG. 13J

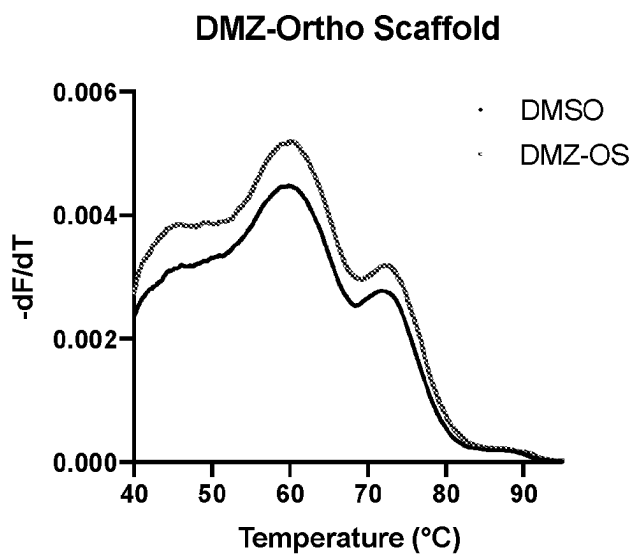


FIG. 13K

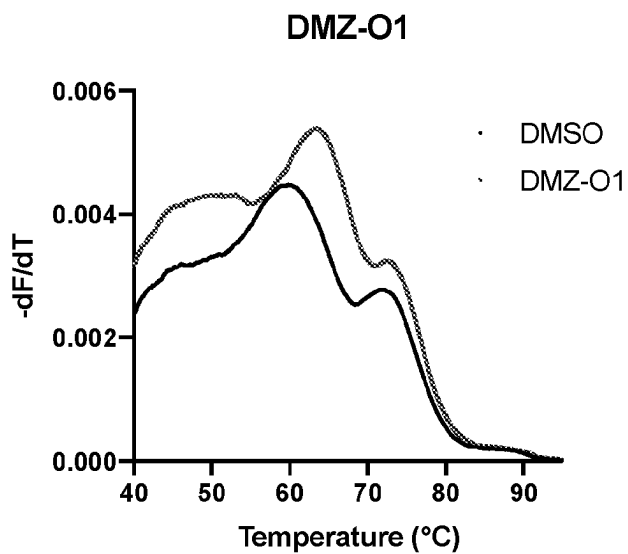


FIG. 13L

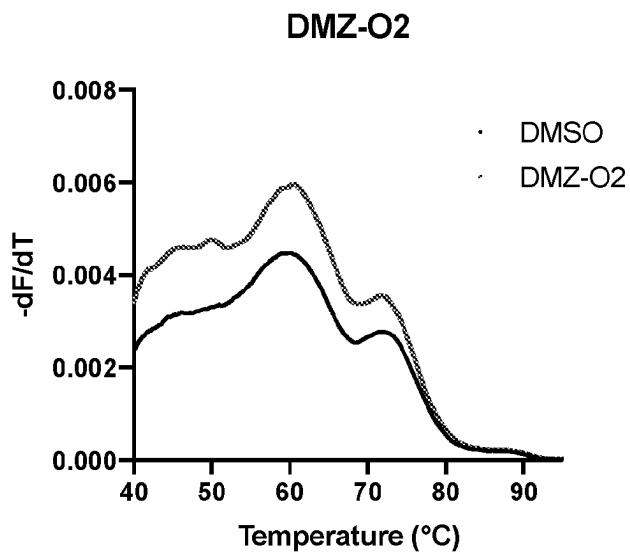


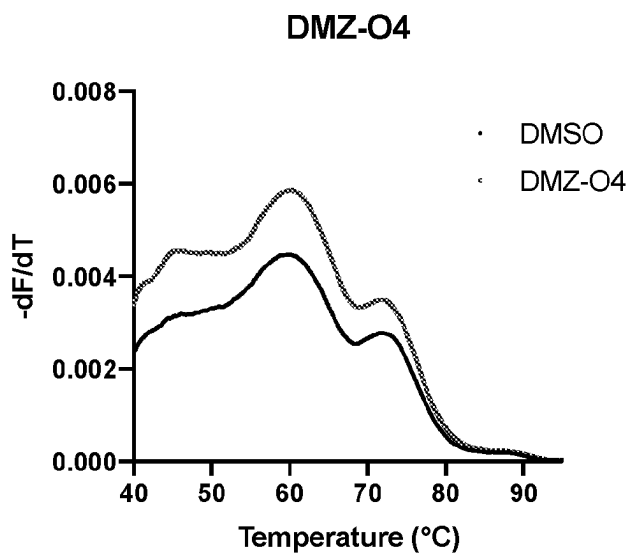
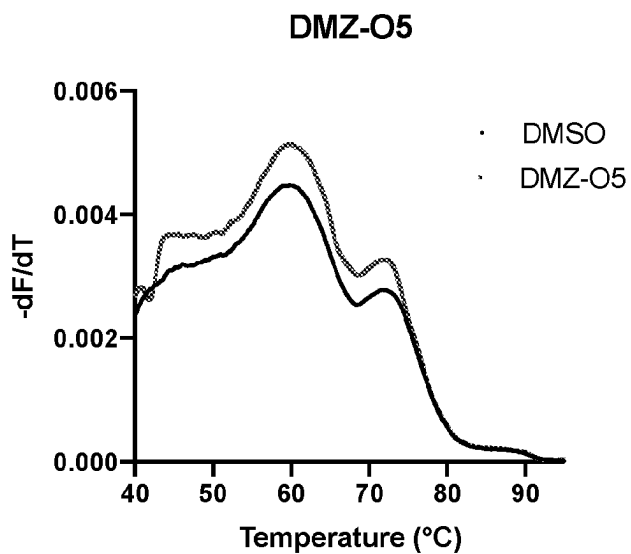
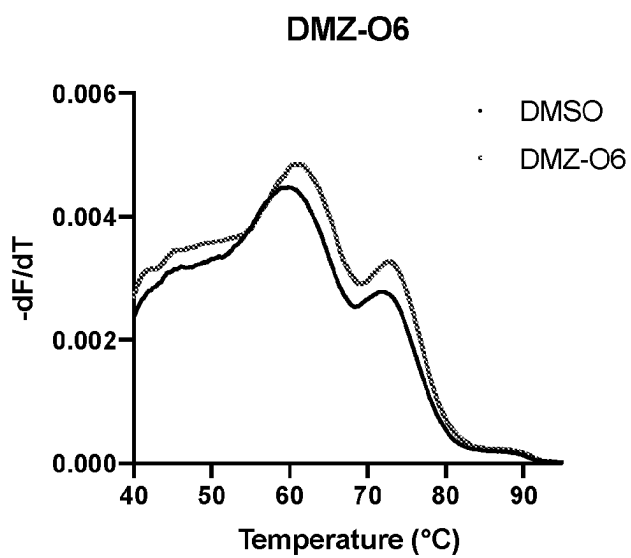
FIG. 13M**FIG. 13N****FIG. 13O**

FIG. 13P

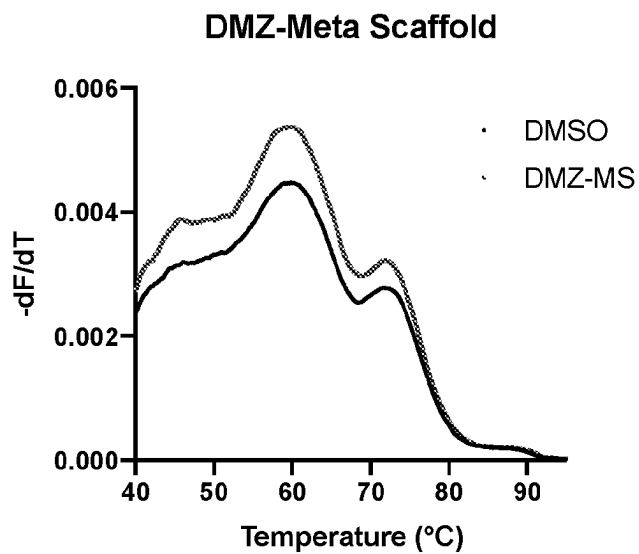


FIG. 13Q

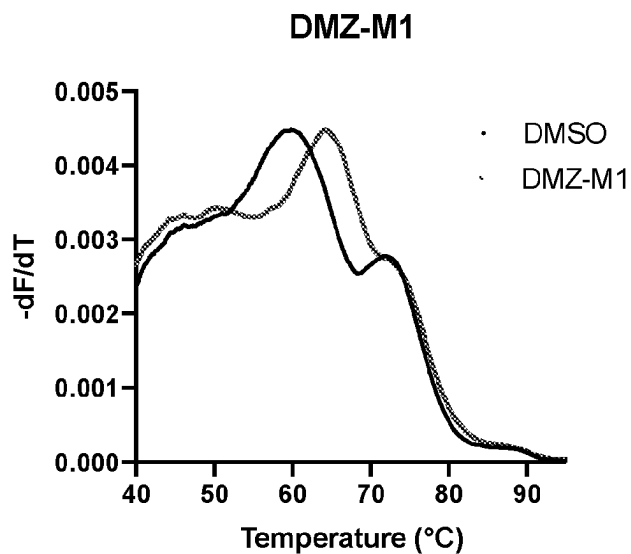


FIG. 13R

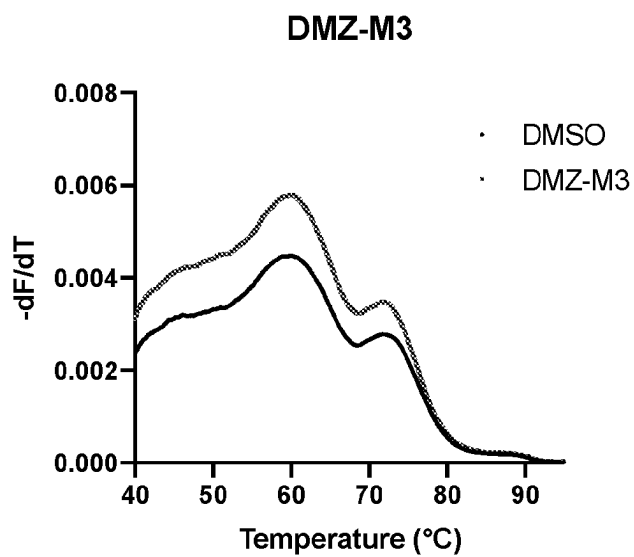


FIG. 13S

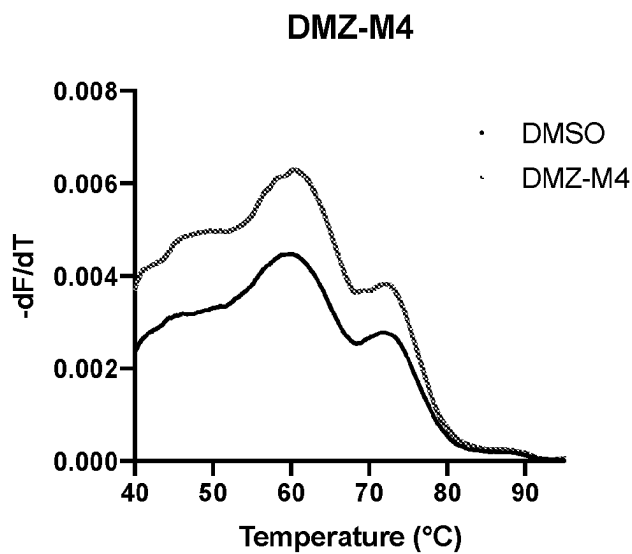


FIG. 13T

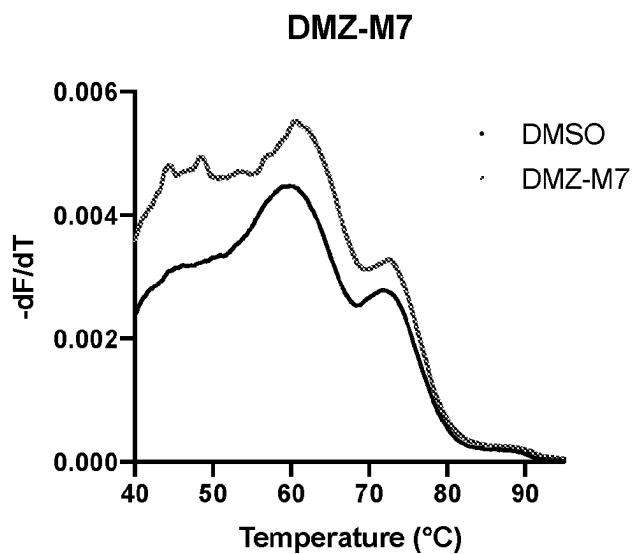


FIG. 13U

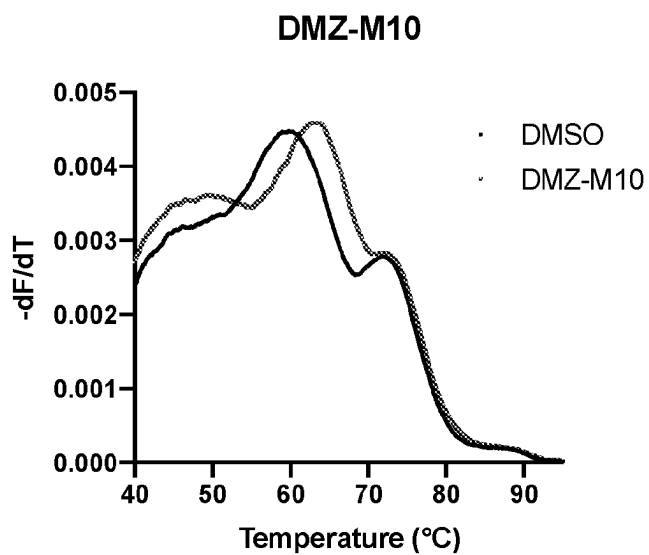


FIG. 13V

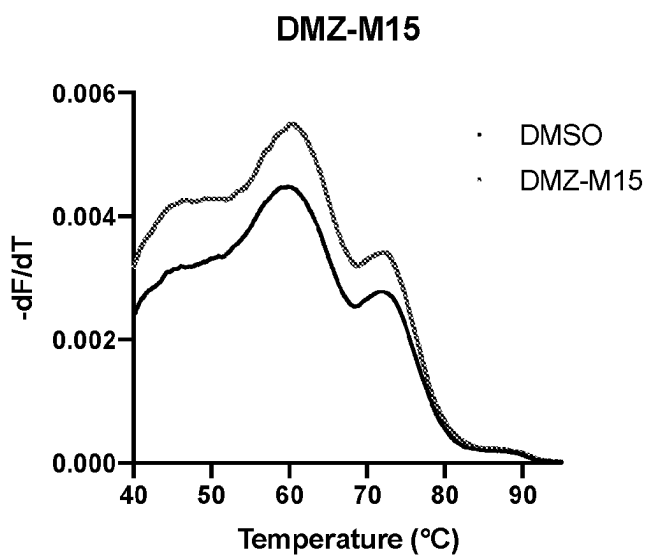


FIG. 13W

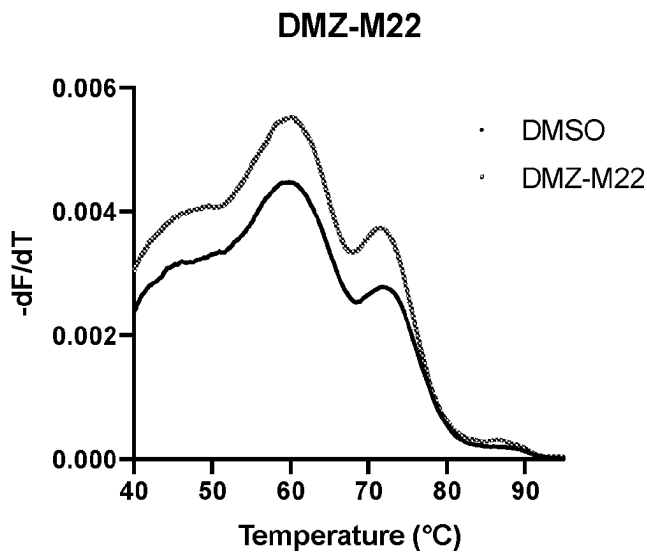


FIG. 13X

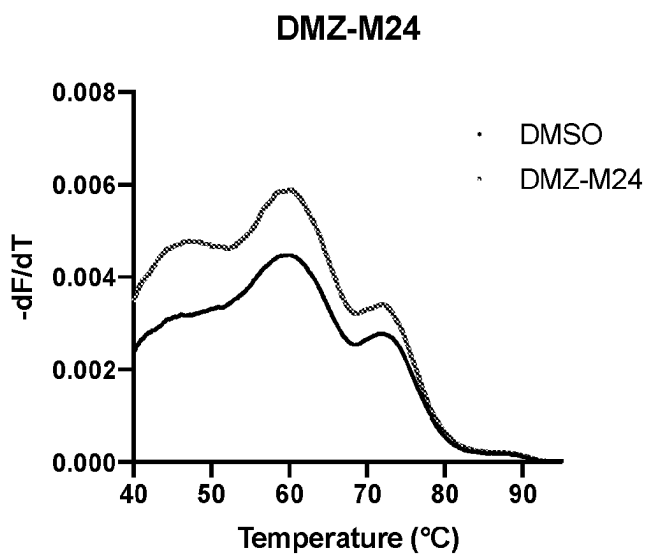


FIG. 13Y

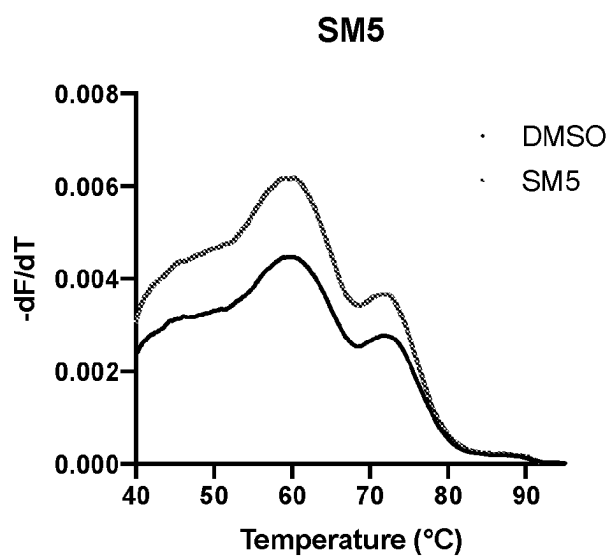


FIG. 13Z

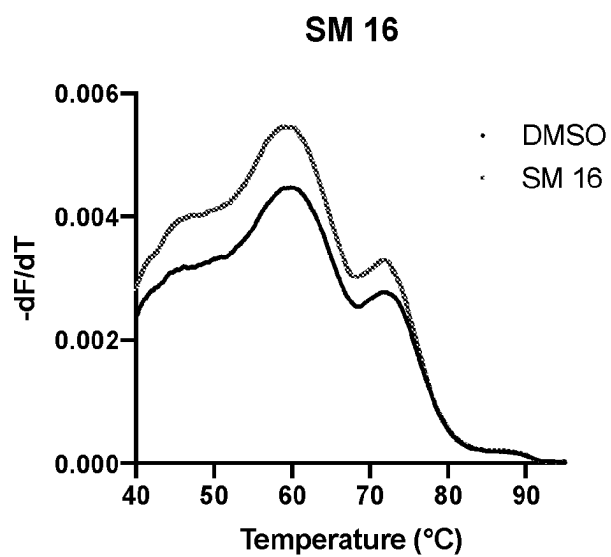


FIG. 14

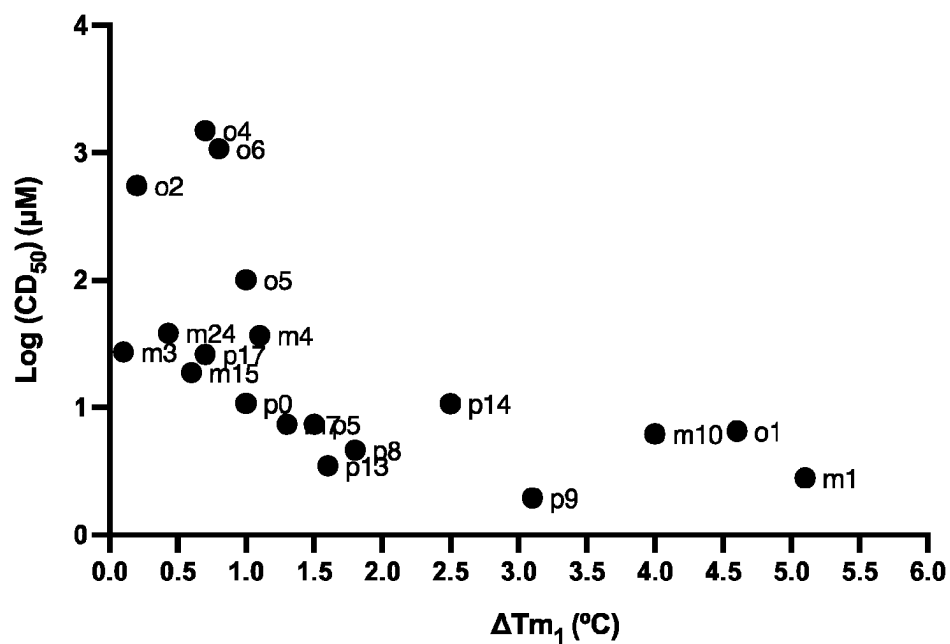


FIG. 15

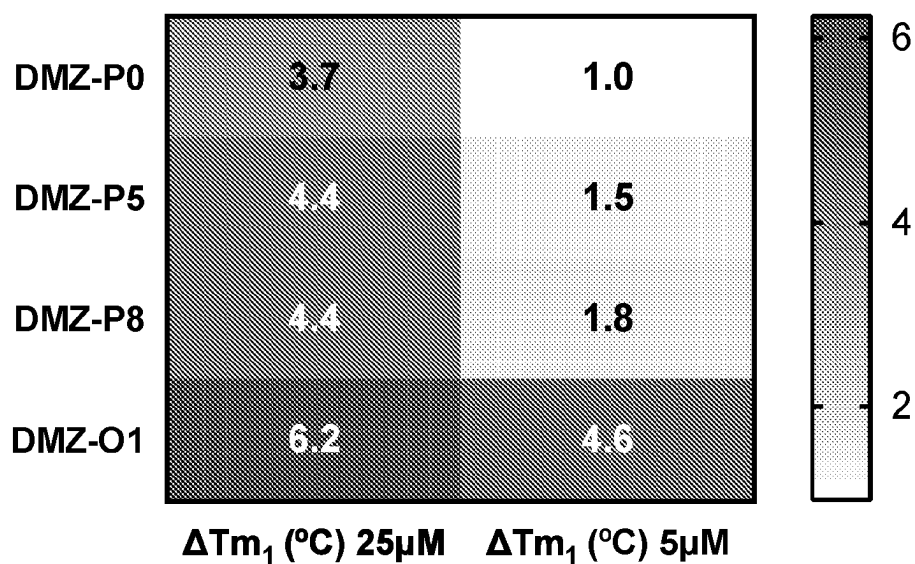


FIG. 16A

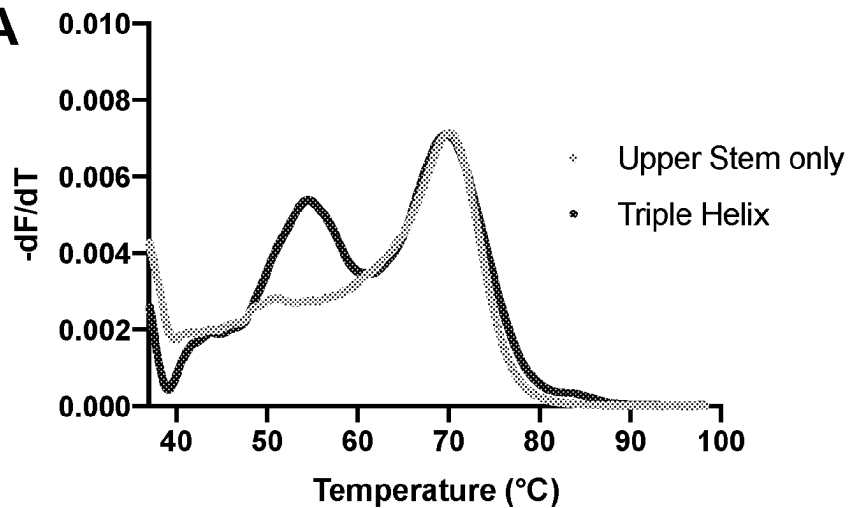


FIG. 16B

Stem DMZ-P1

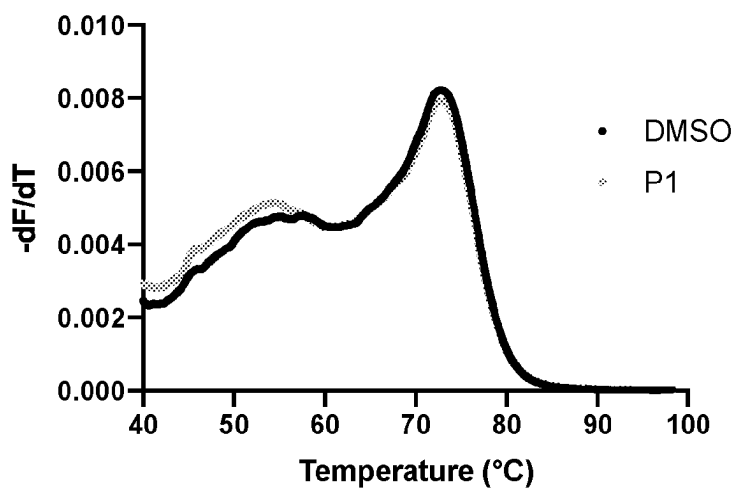


FIG. 16C

Stem DMZ-P8

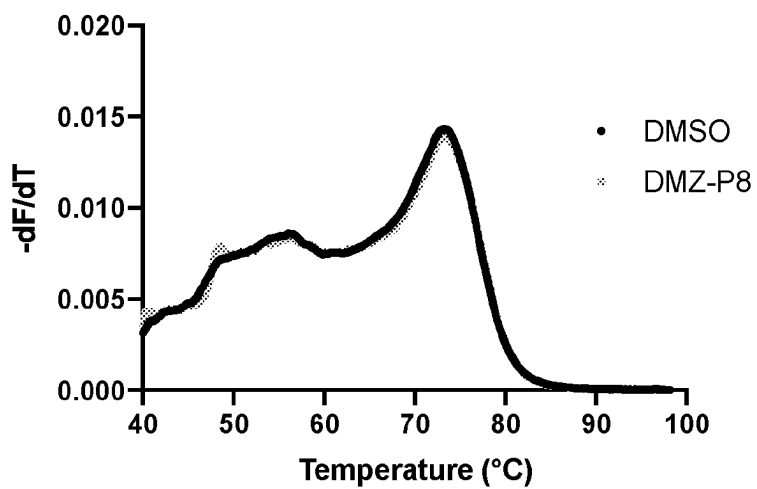


FIG. 16D

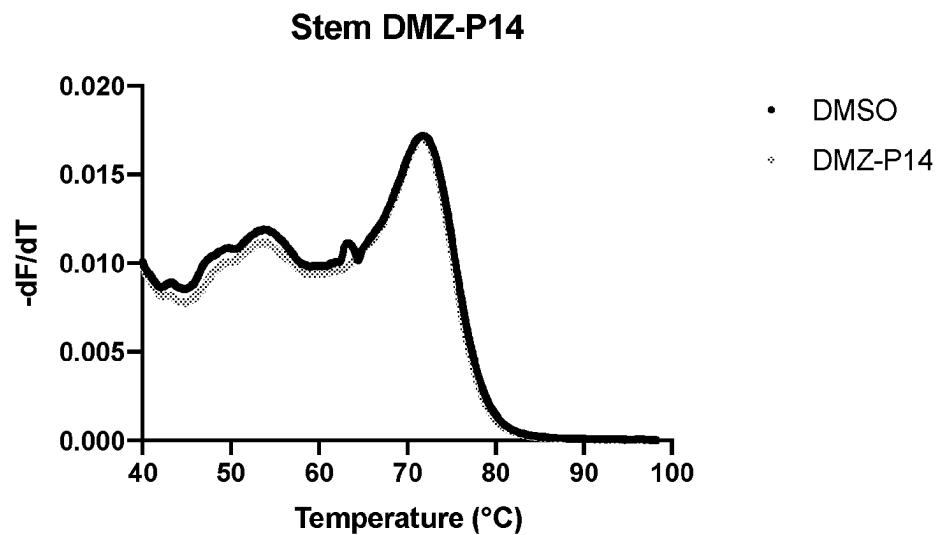


FIG. 16E

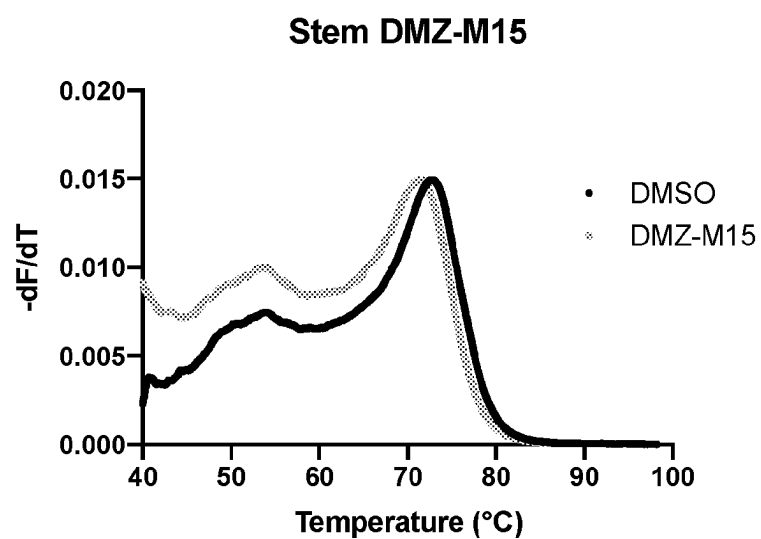


FIG. 17A

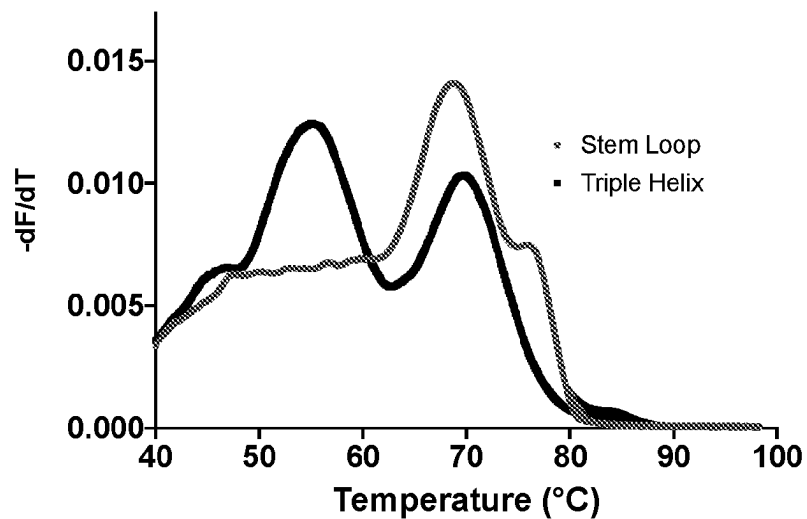


FIG. 17B

Stem Loop DMZ-P1

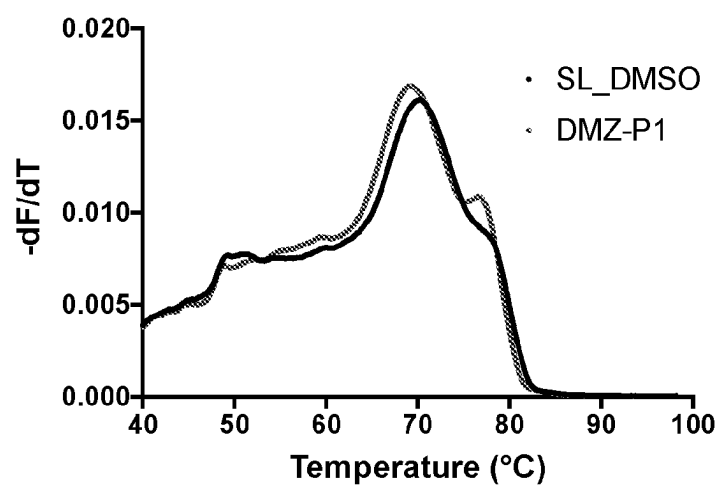


FIG. 17C

Stem Loop DMZ-P8

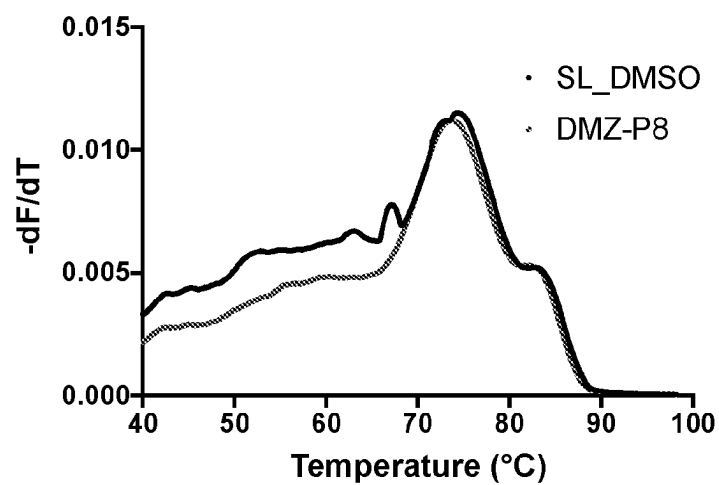


FIG. 17D

Stem Loop DMZ-P14

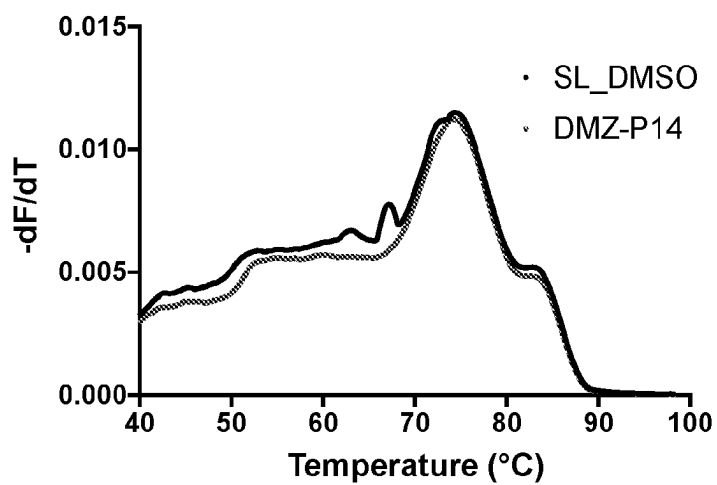
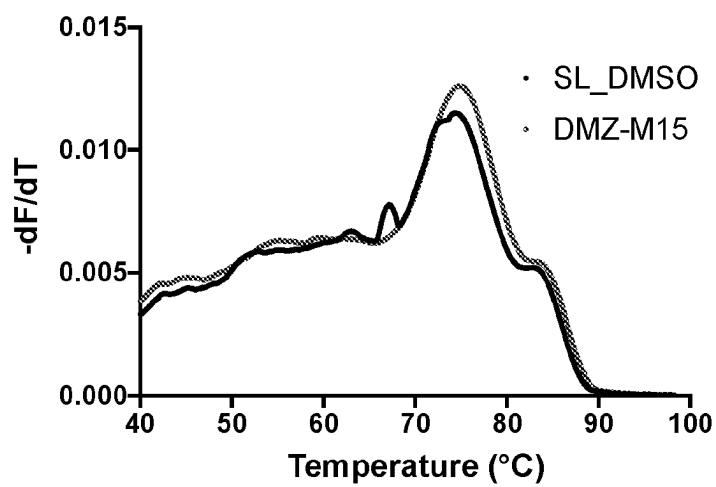


FIG. 17E

Stem Loop DMZ-M15



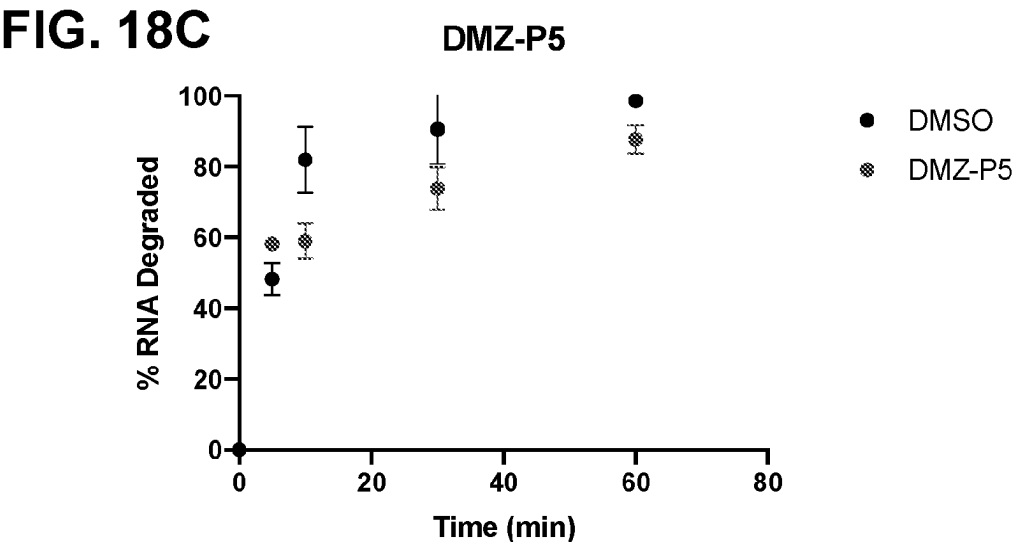
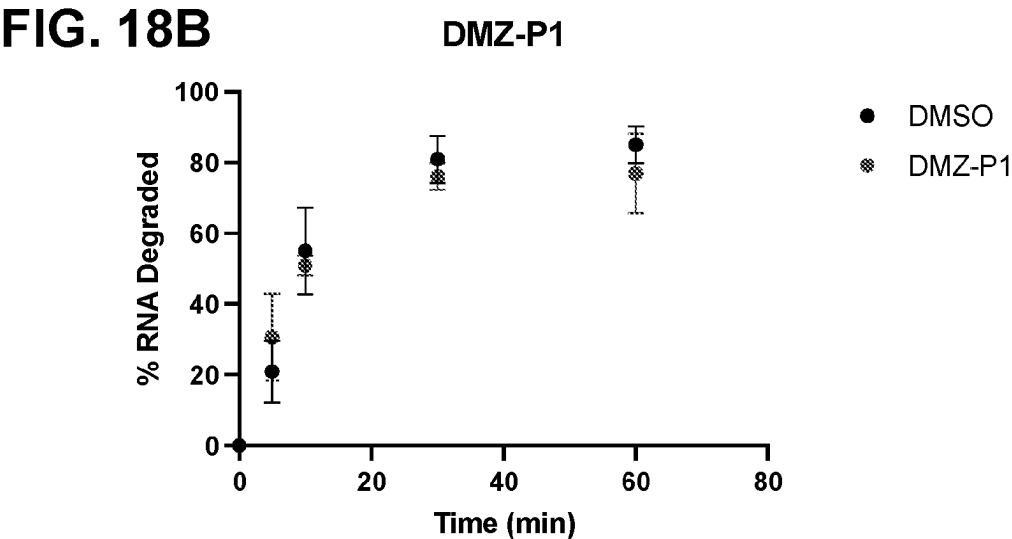
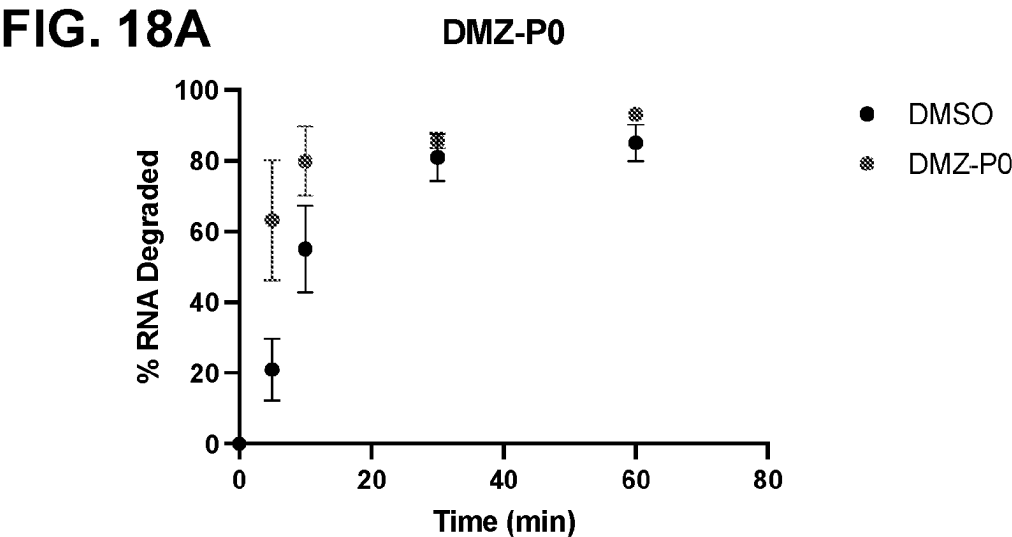


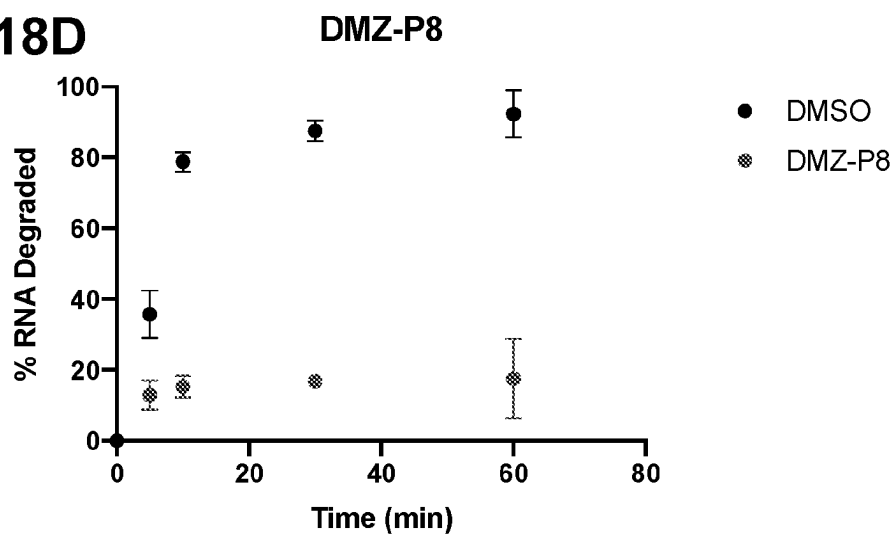
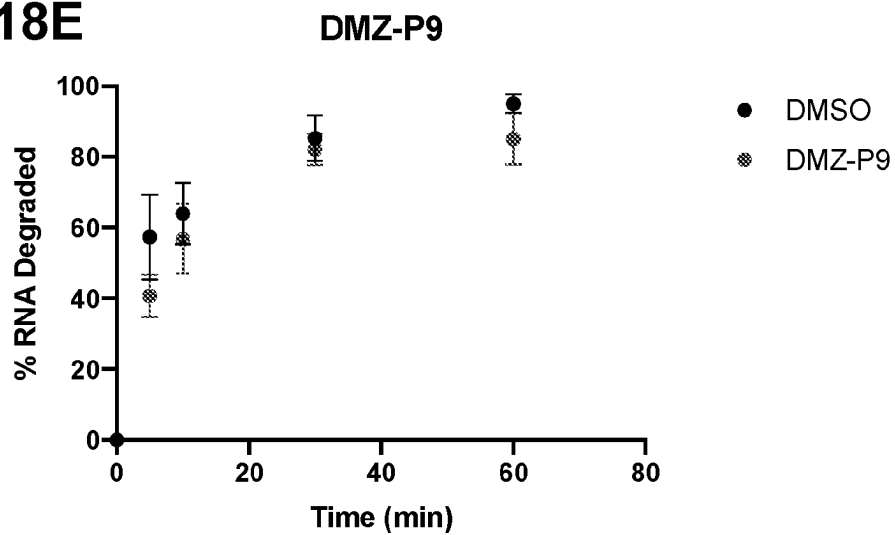
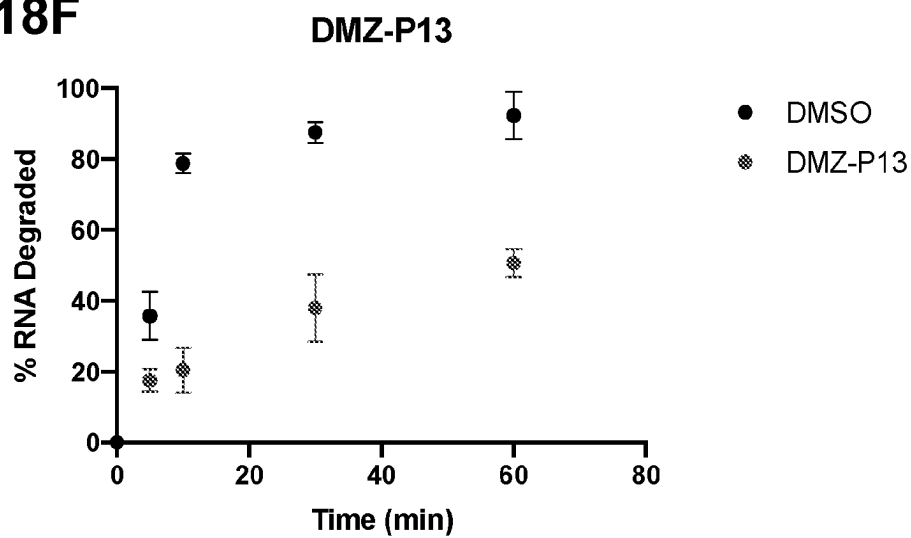
FIG. 18D**FIG. 18E****FIG. 18F**

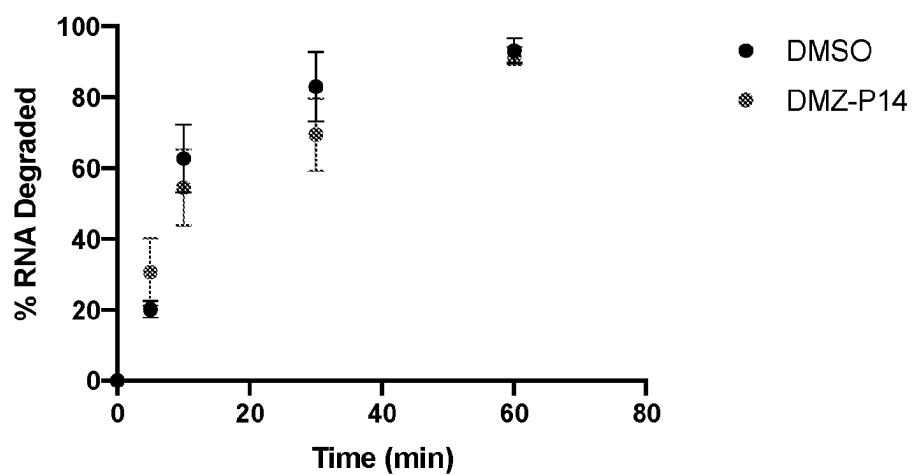
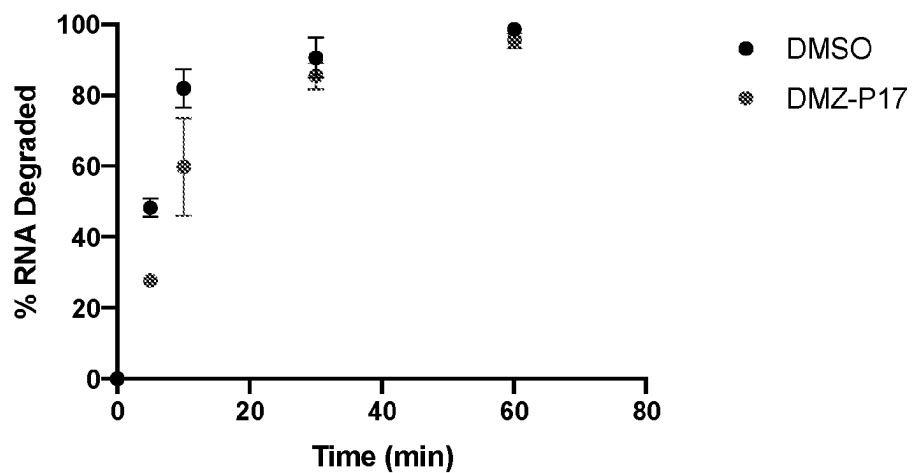
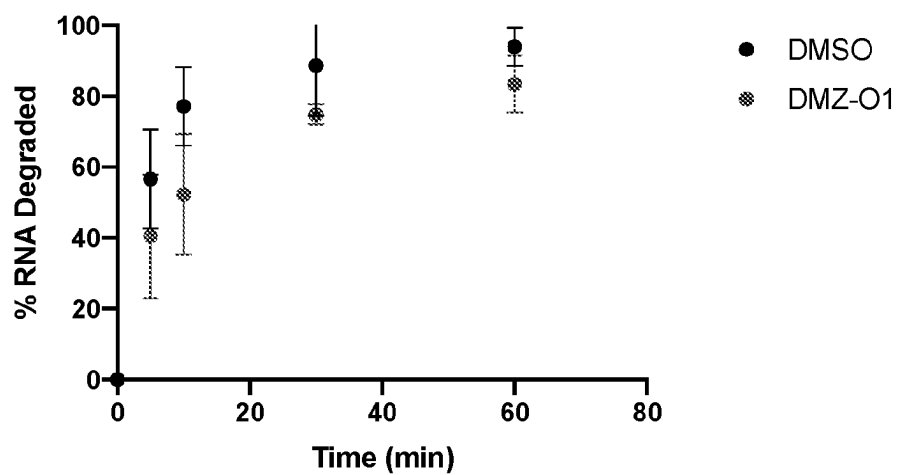
FIG. 18G**DMZ-P14****FIG. 18H****DMZ-P17****FIG. 18I****DMZ-O1**

FIG. 18J

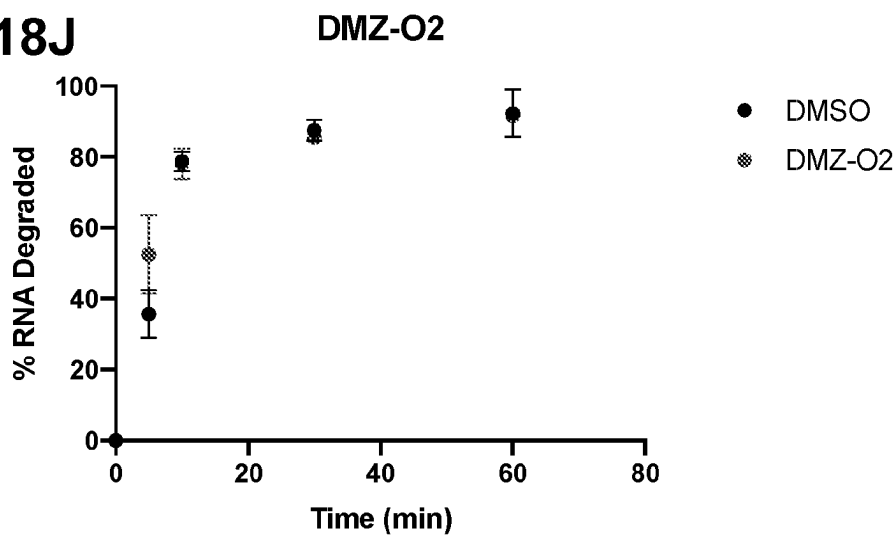


FIG. 18K

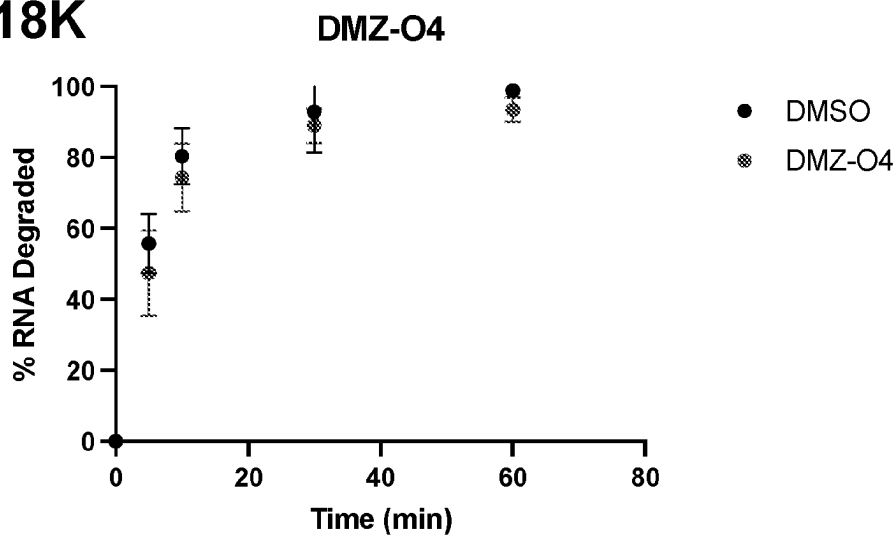


FIG. 18L

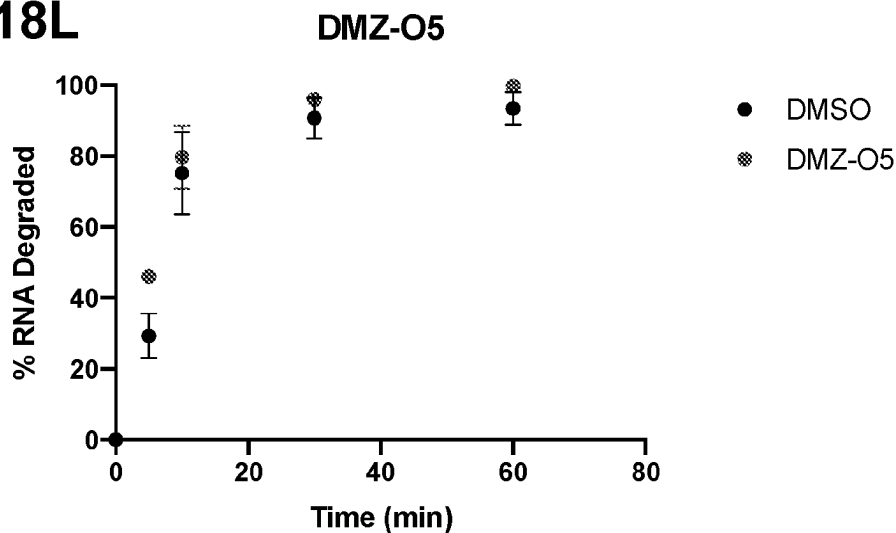


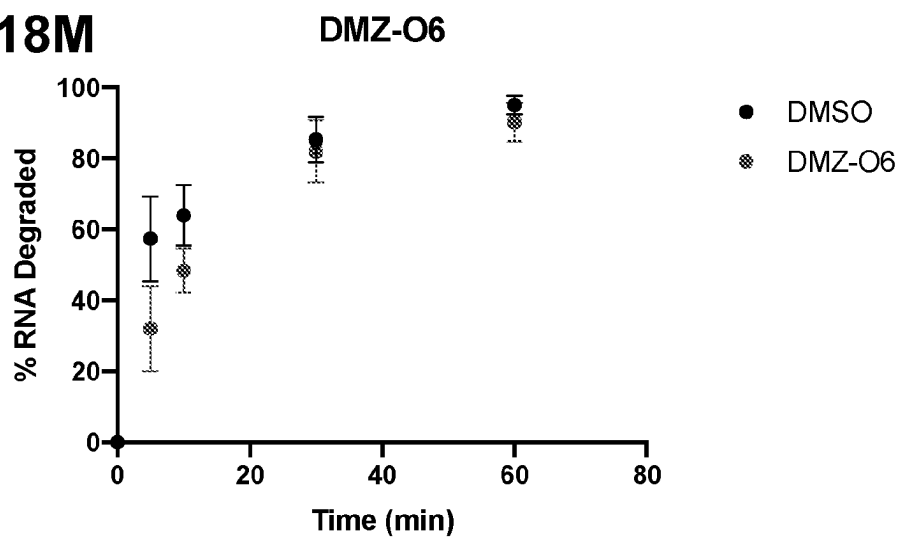
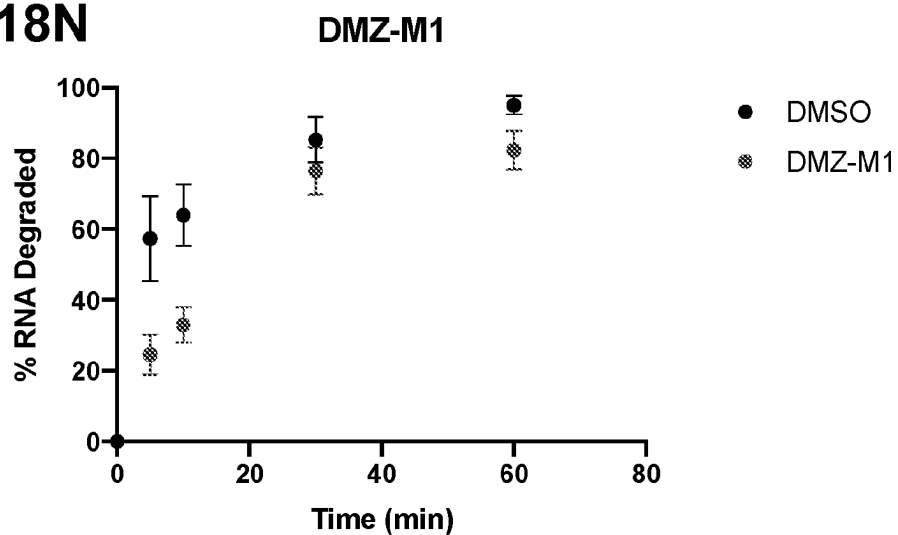
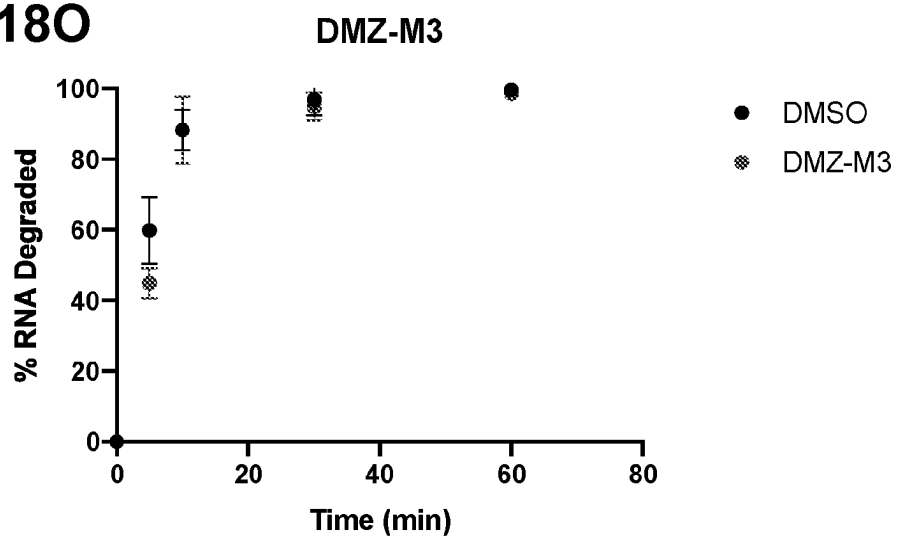
FIG. 18M**FIG. 18N****FIG. 18O**

FIG. 18P

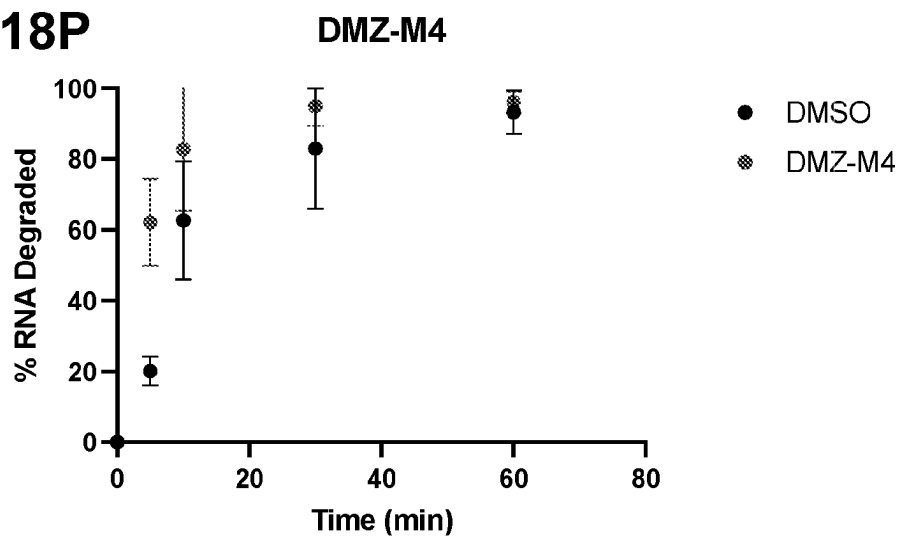


FIG. 18Q

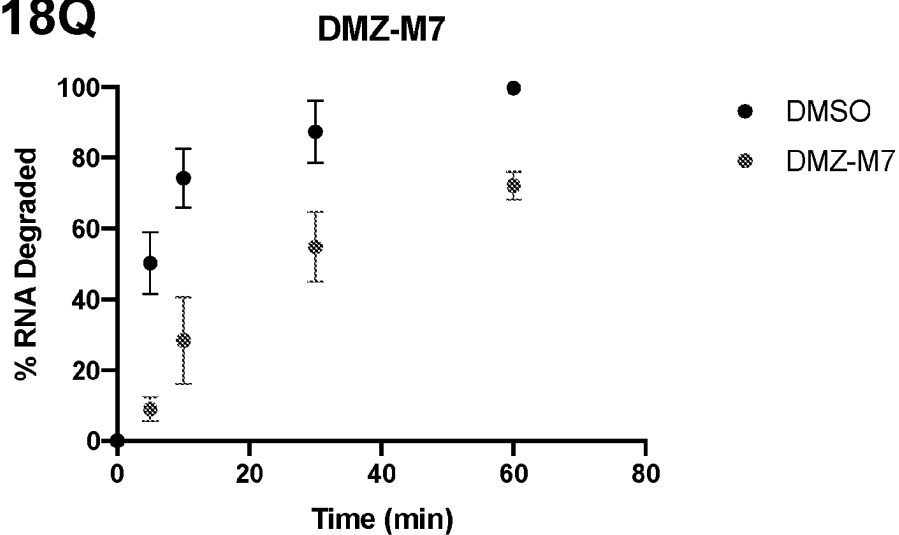


FIG. 18R

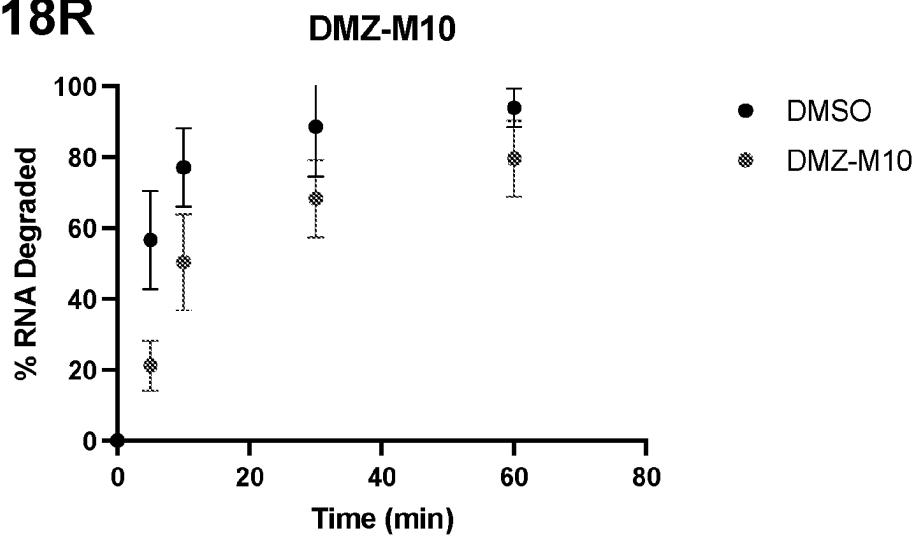


FIG. 18S

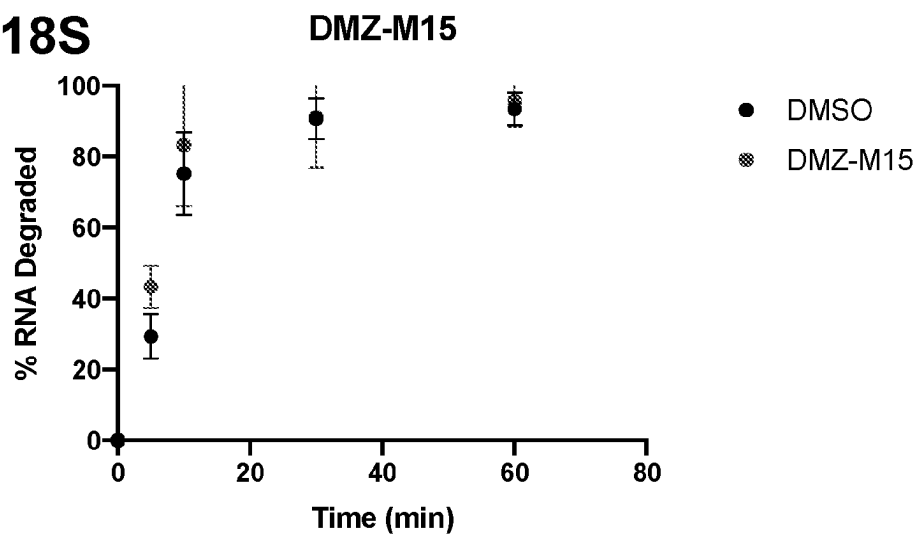


FIG. 18T

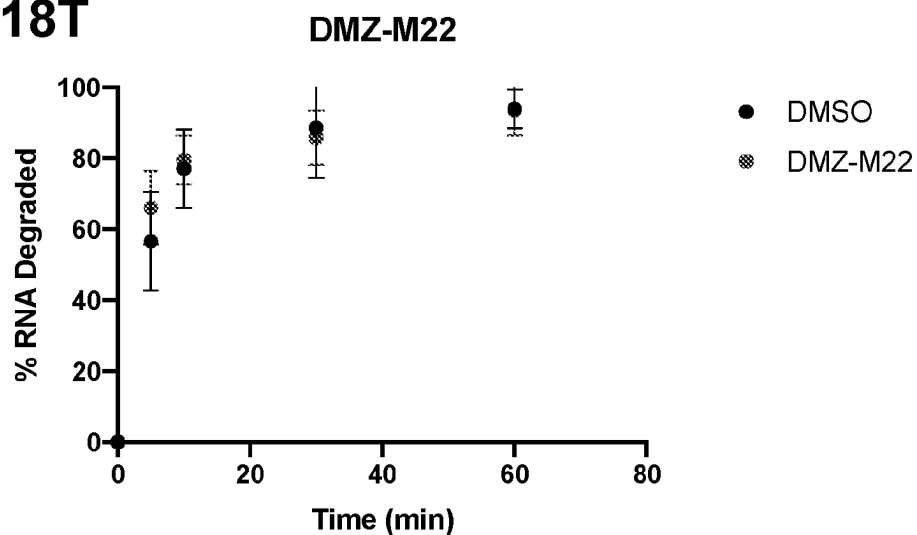
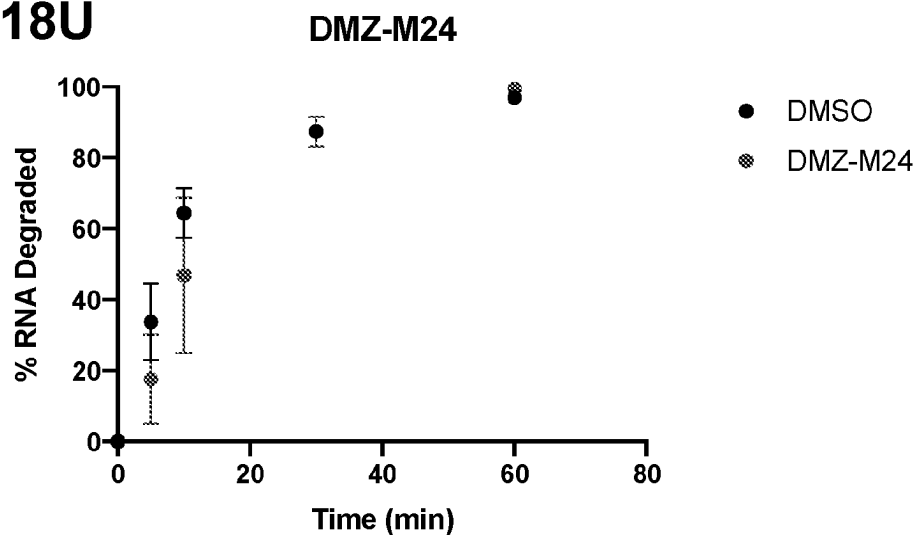


FIG. 18U



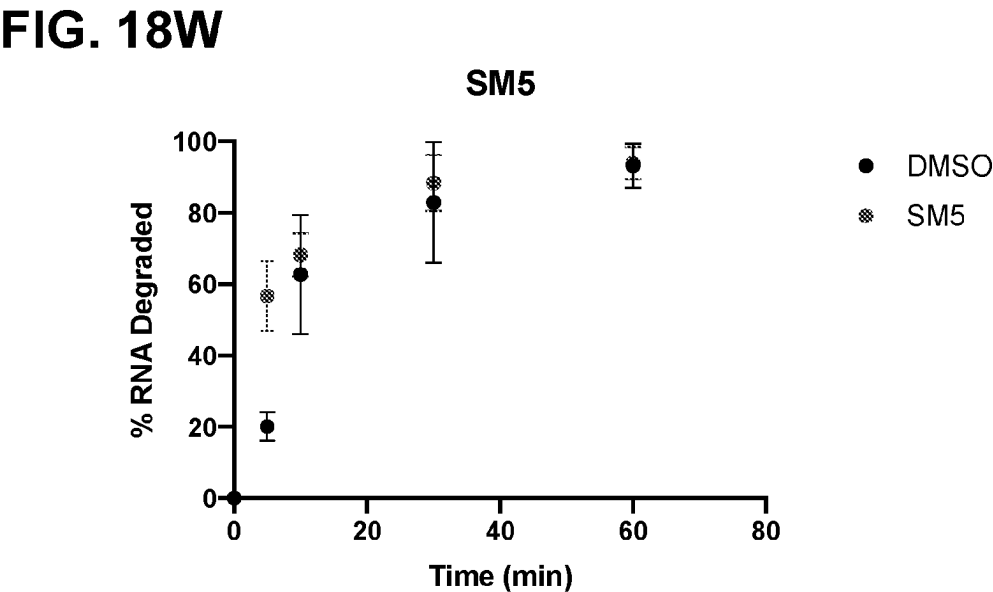
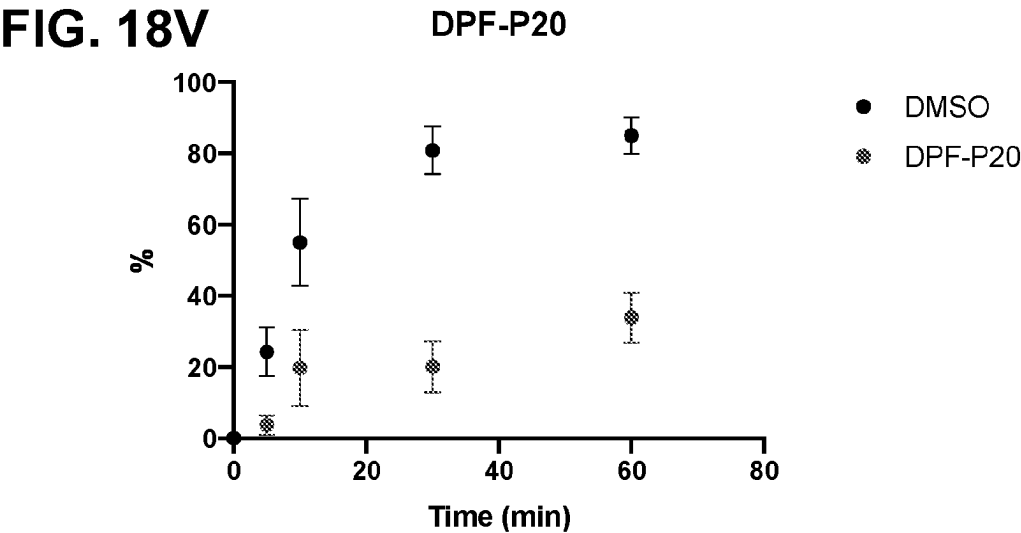


FIG. 19A

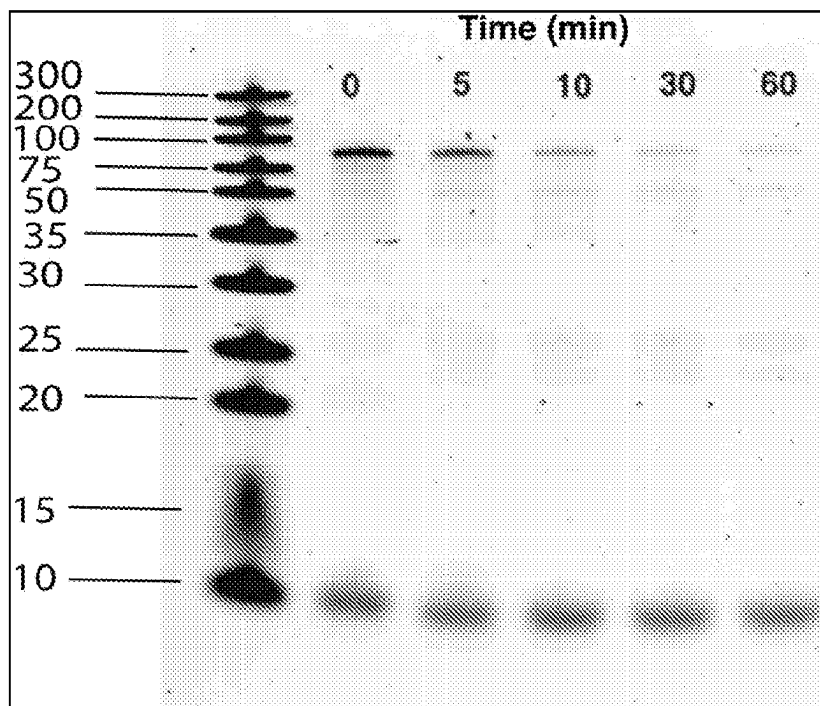


FIG. 19B

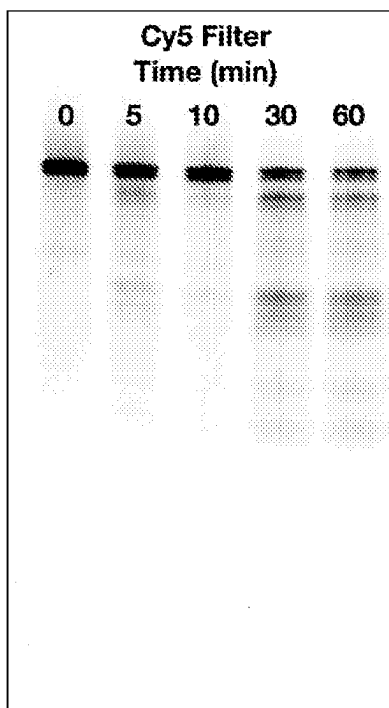


FIG. 20A

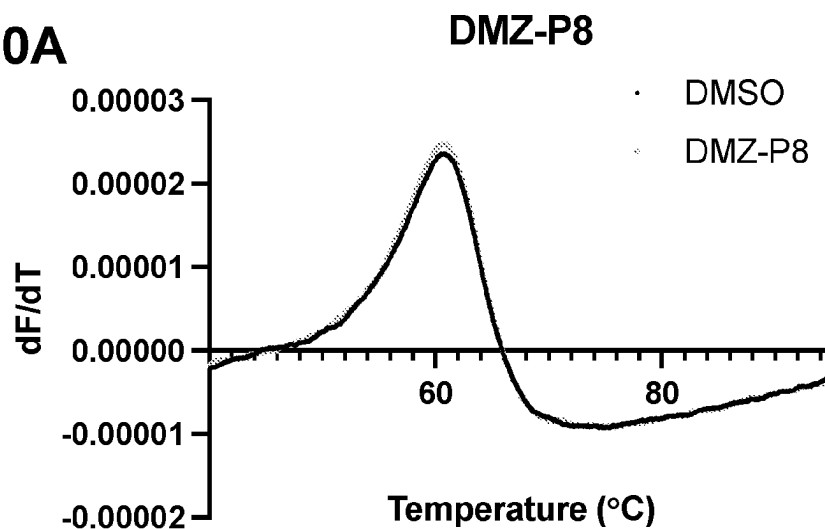


FIG. 20B

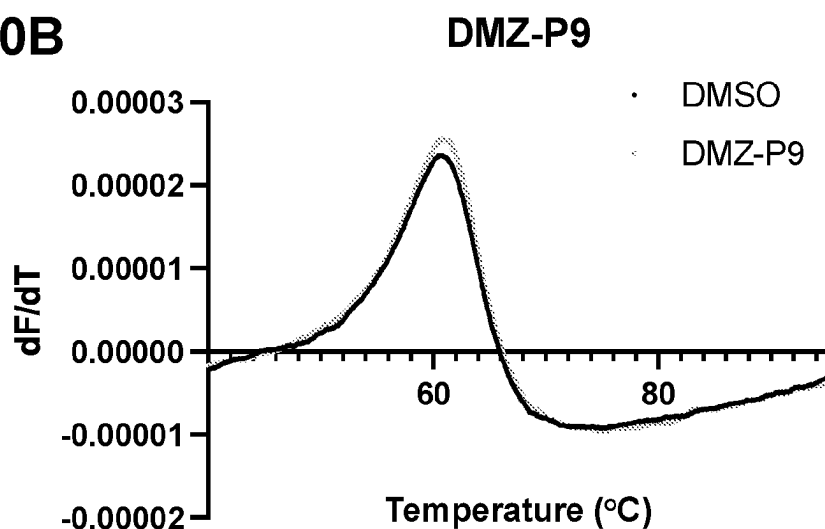


FIG. 20C

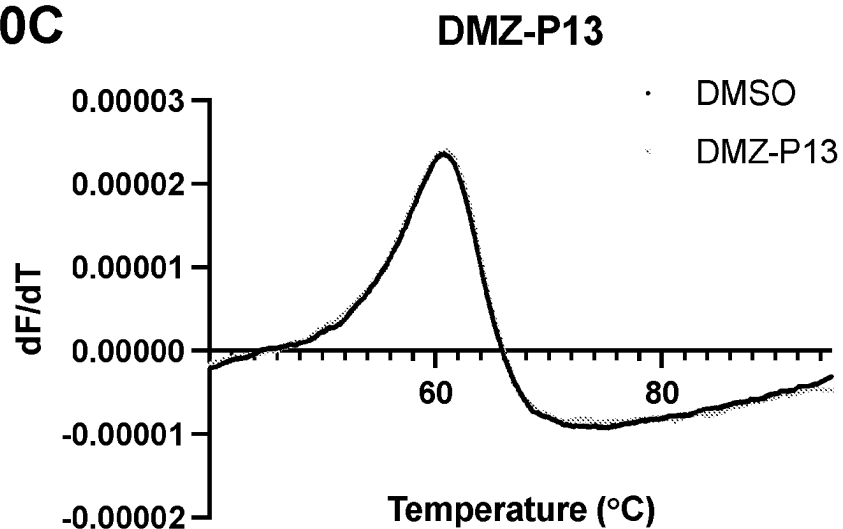


FIG. 20D

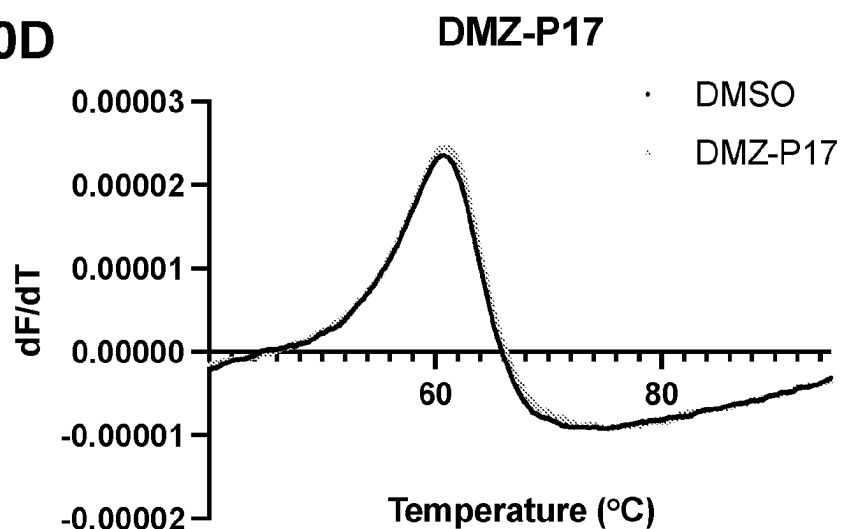


FIG. 20E

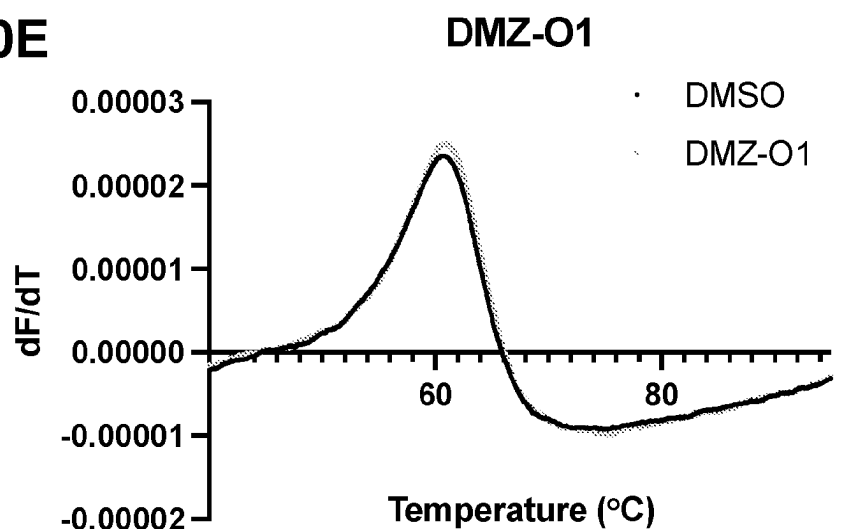


FIG. 20F

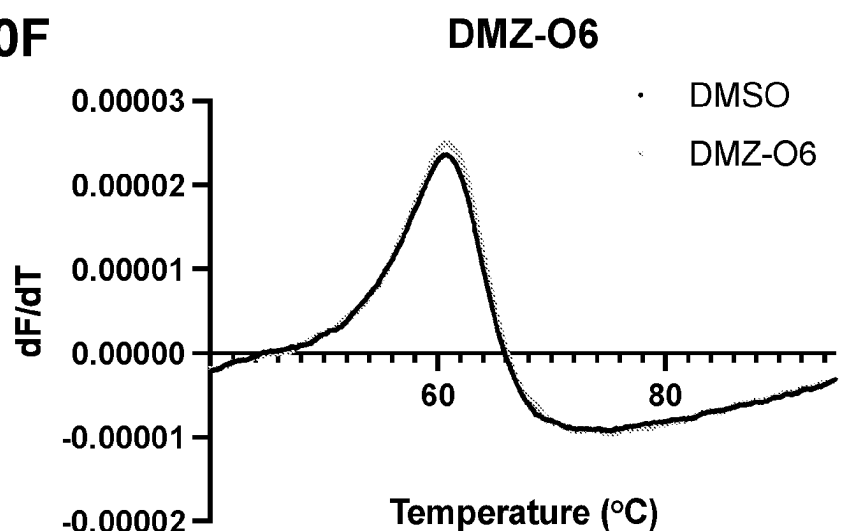


FIG. 20G

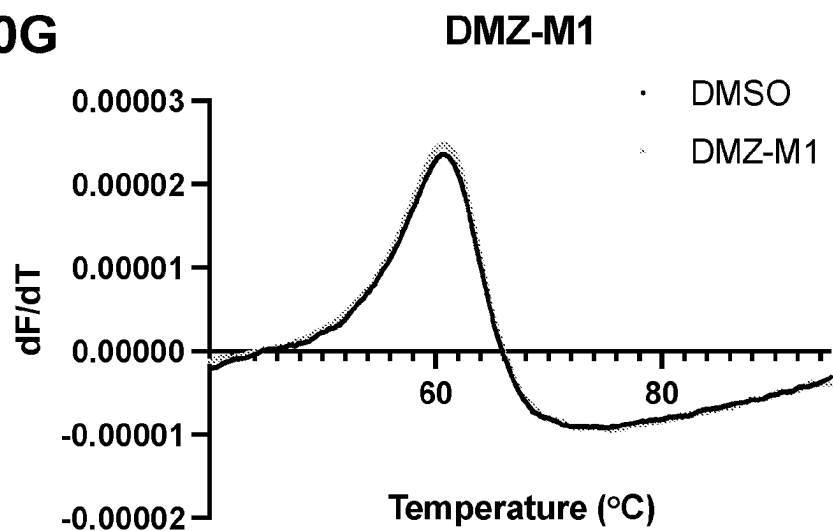


FIG. 20H

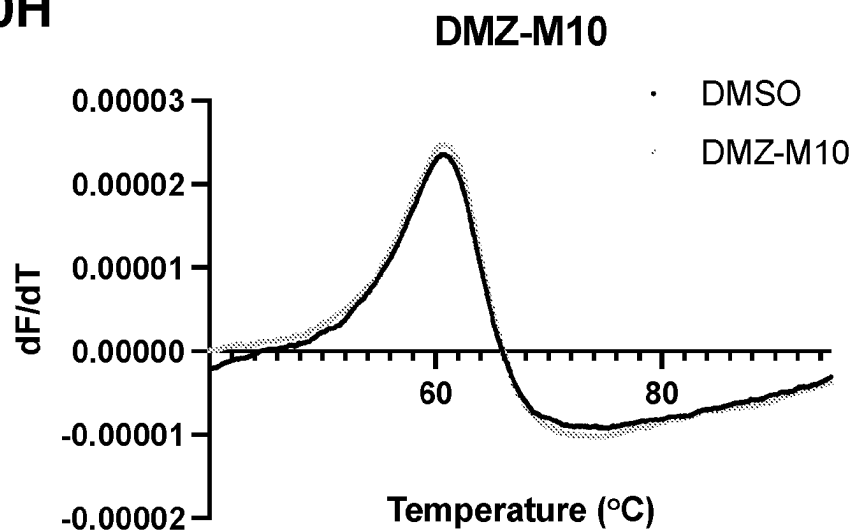


FIG. 21A

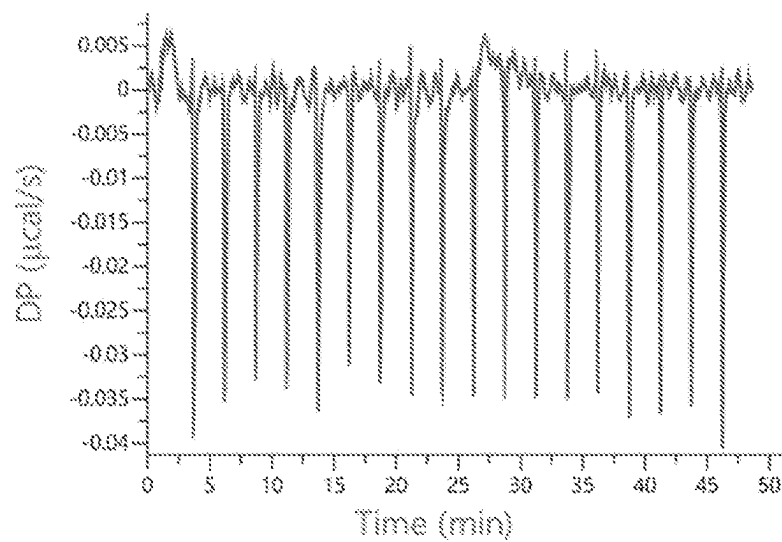


FIG. 21B

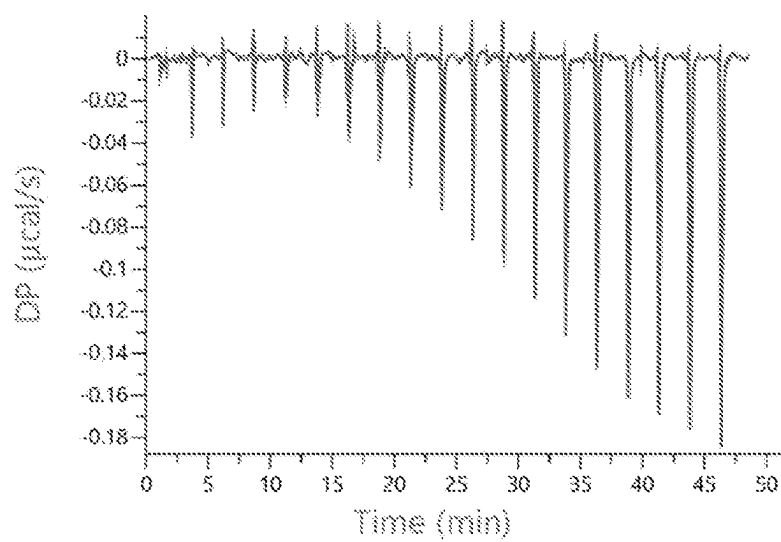


FIG. 21C

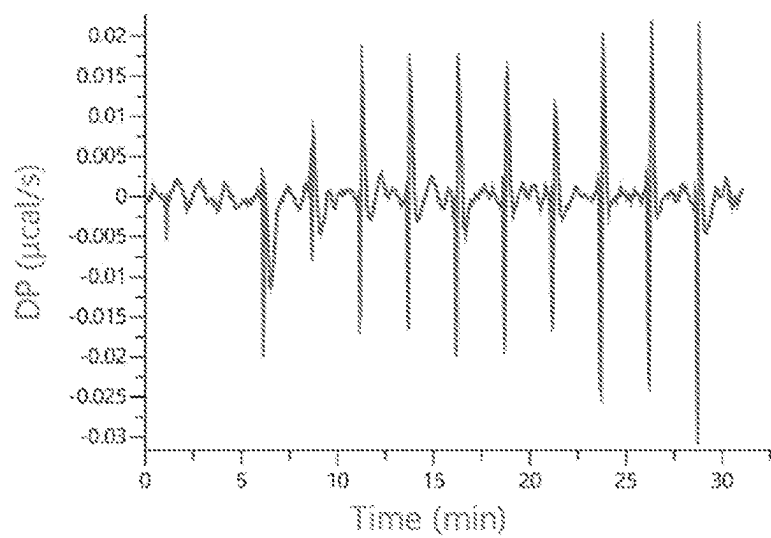


FIG. 22A

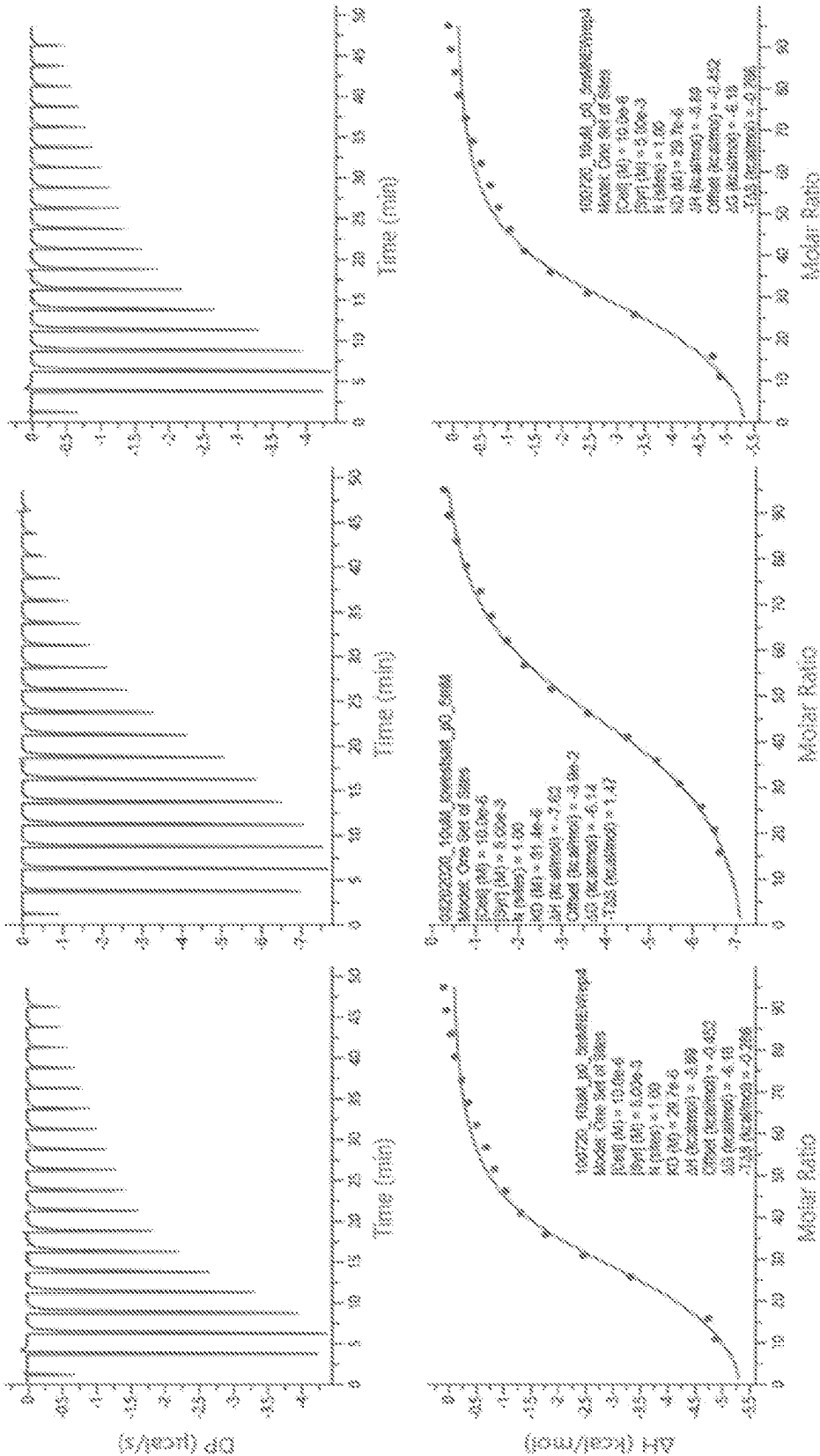


FIG. 22B

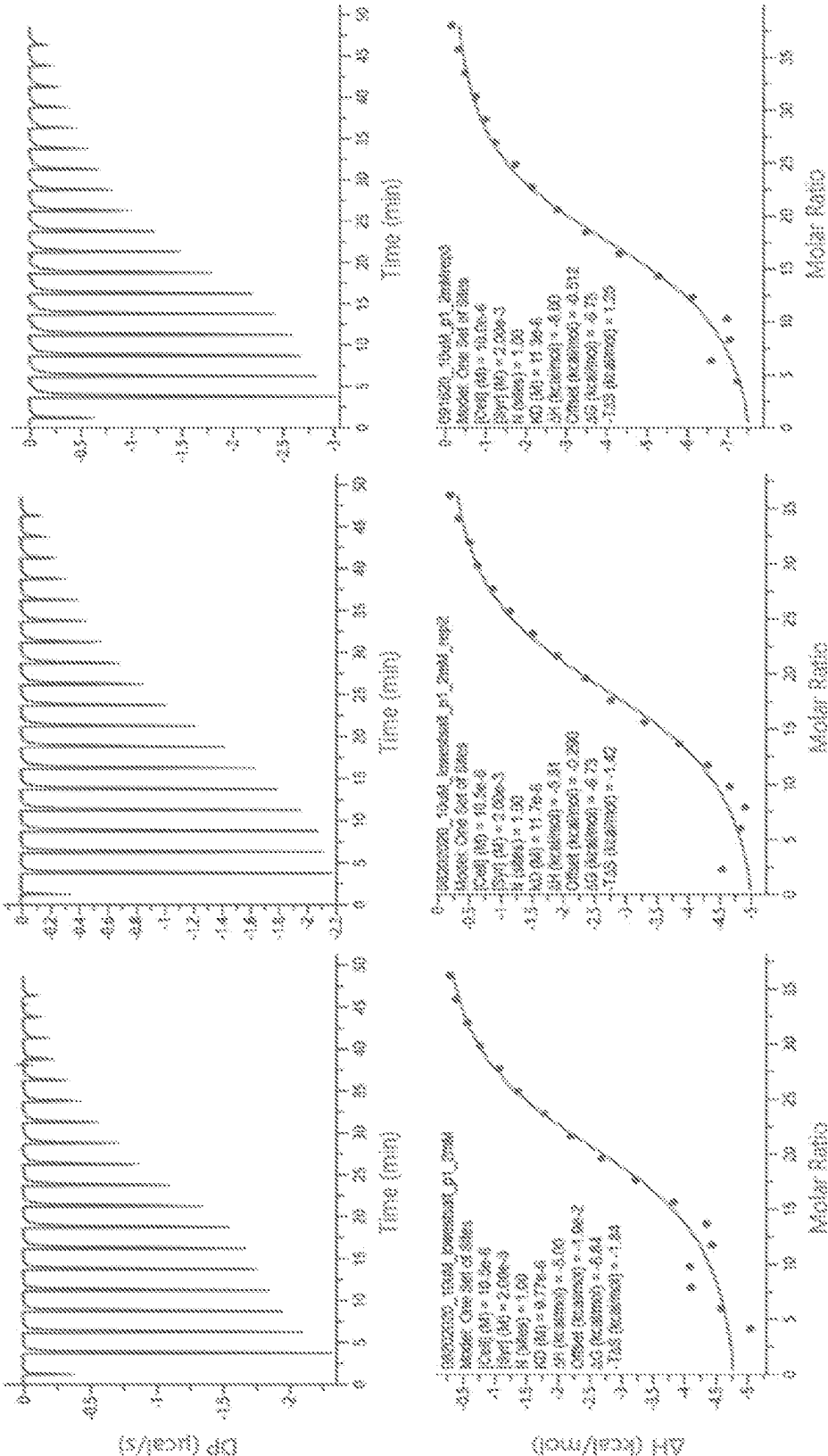


FIG. 22C

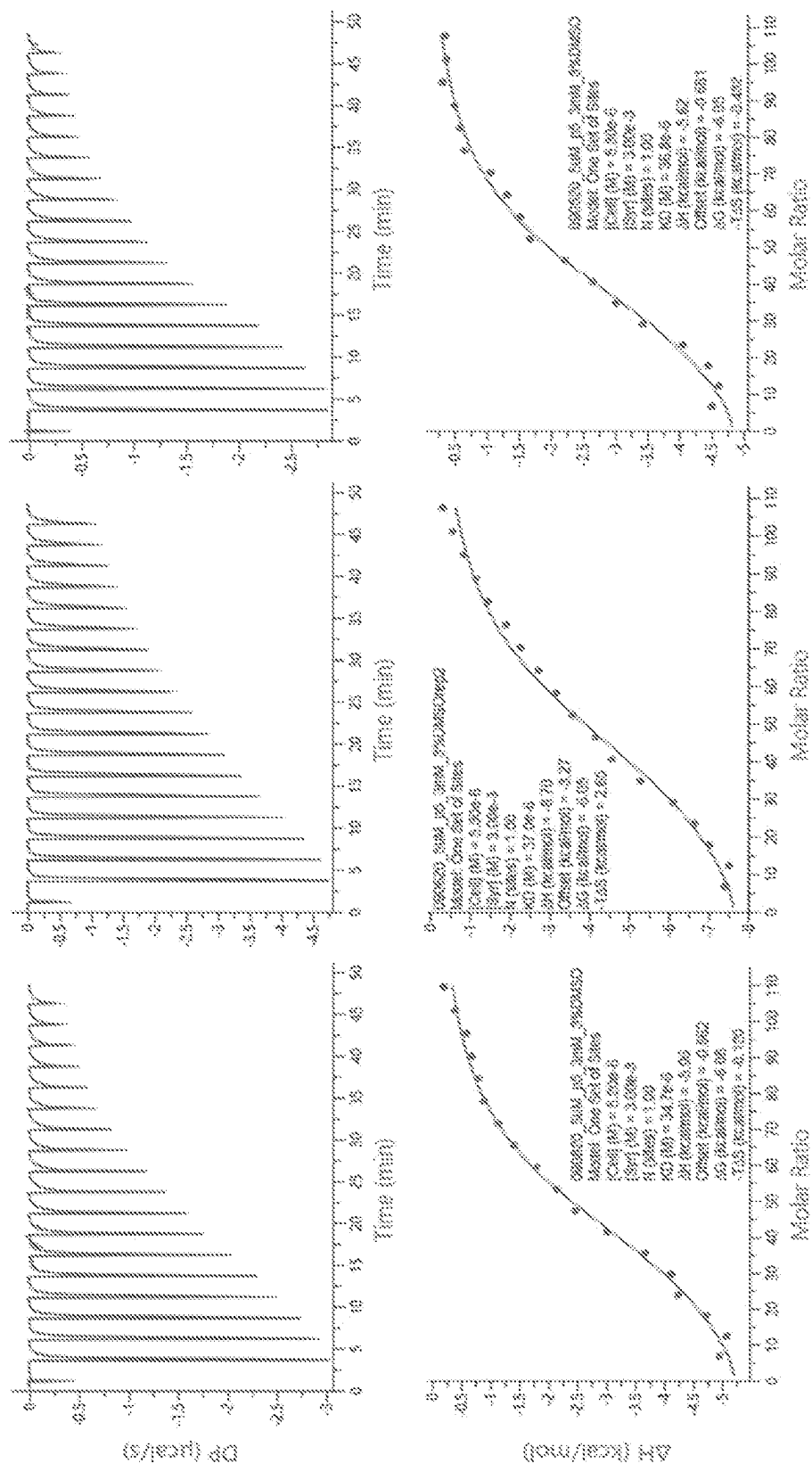


FIG. 22D

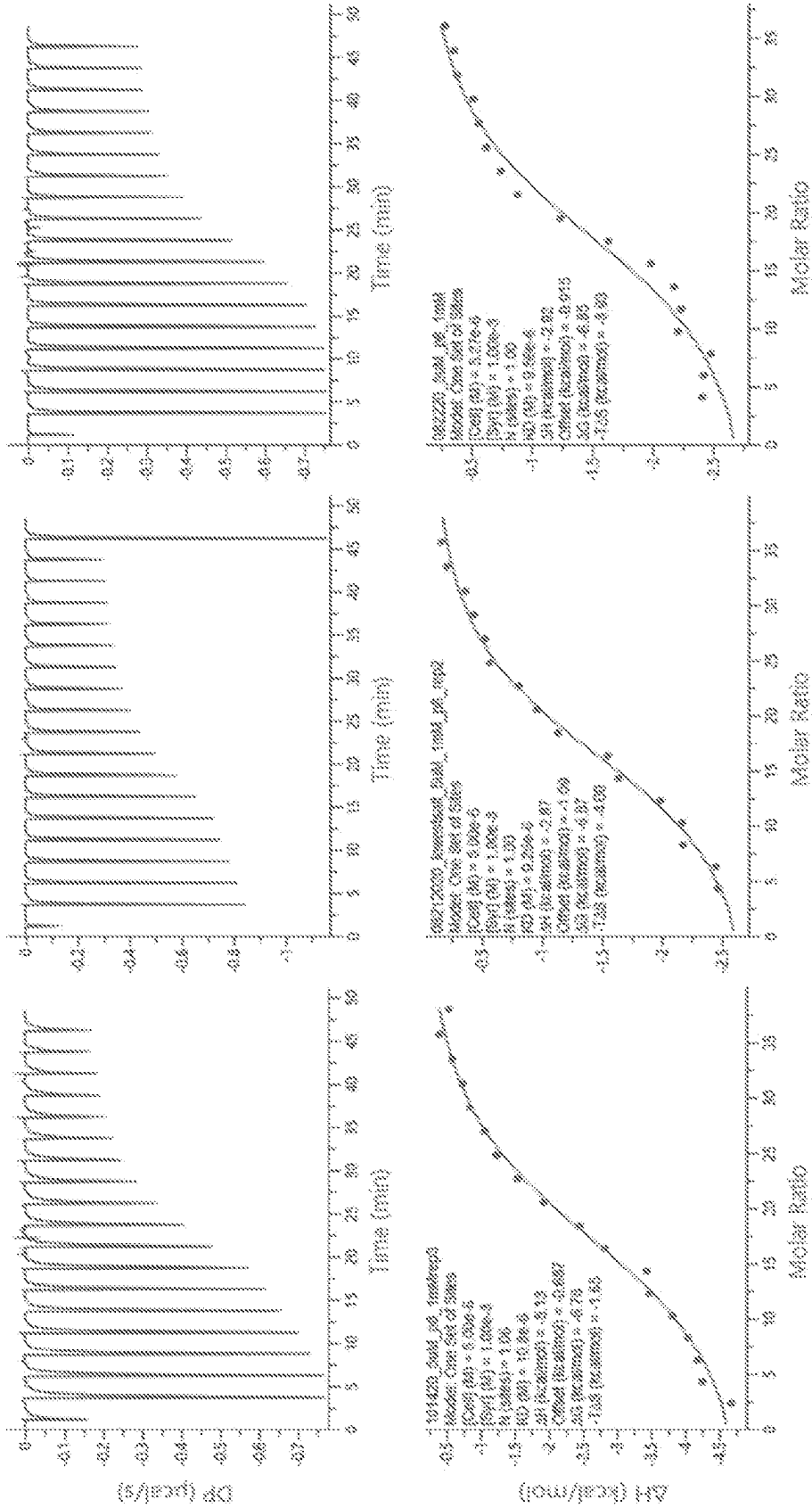


FIG. 22E

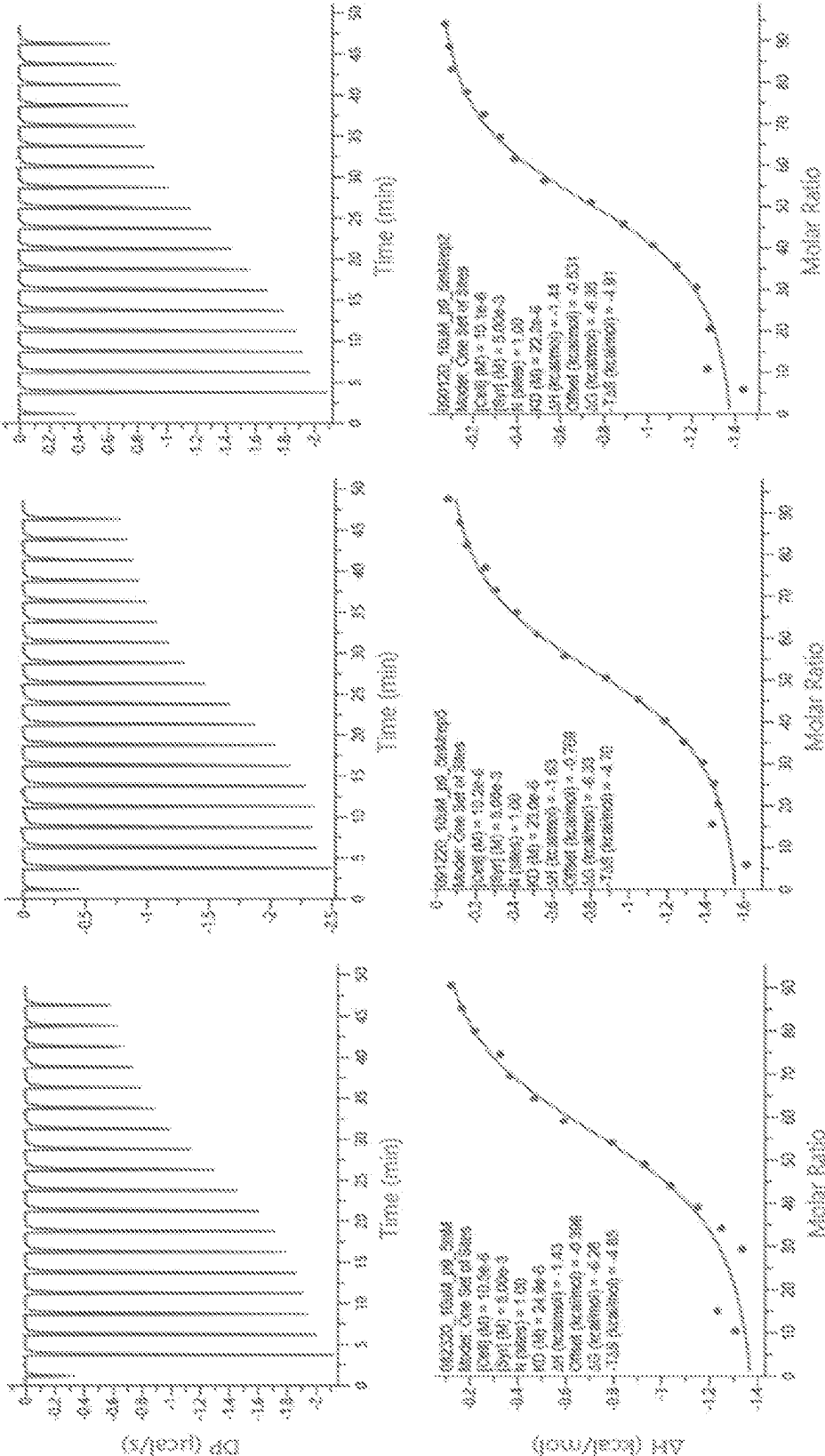


FIG. 22F

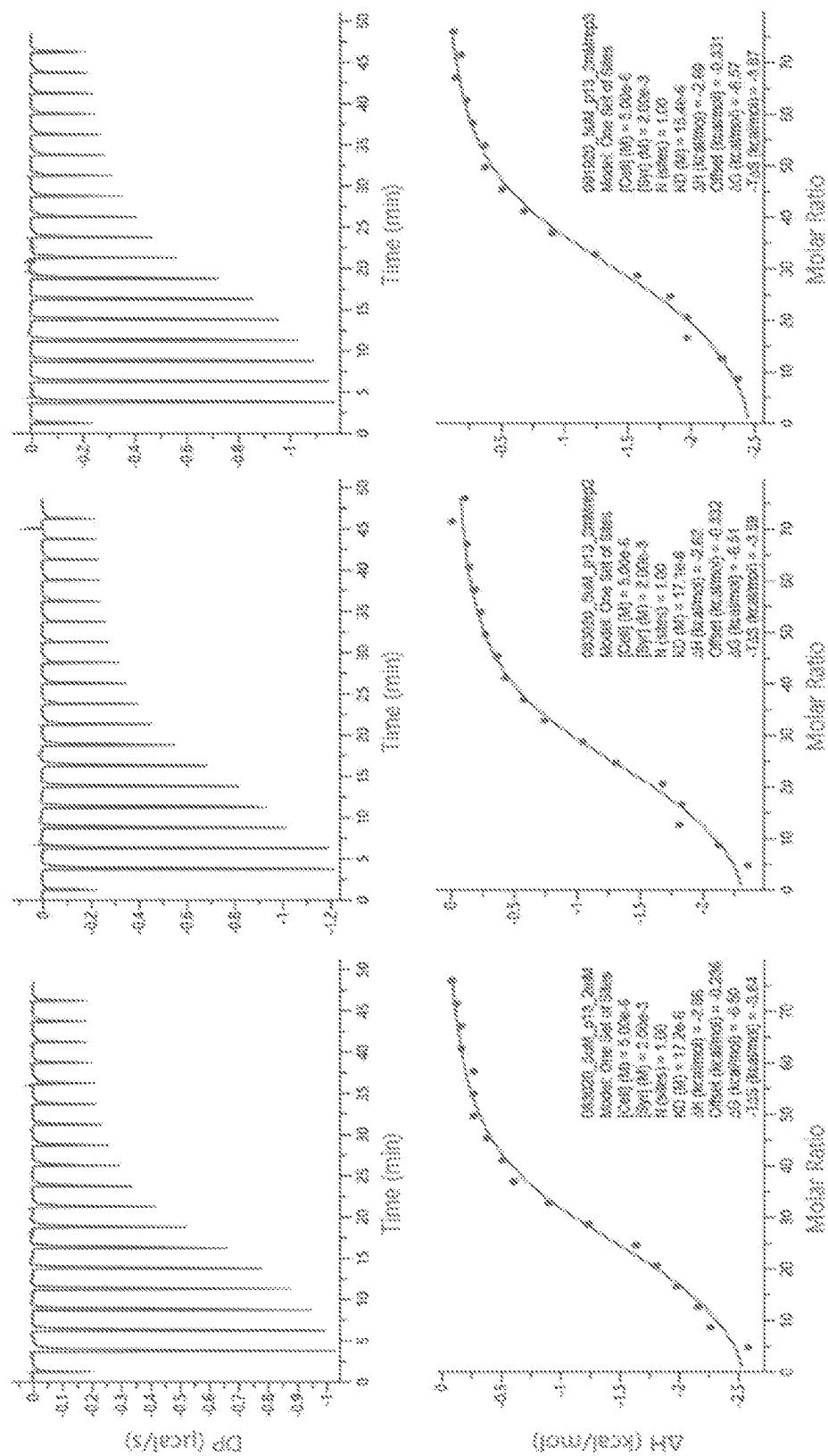


FIG. 22G

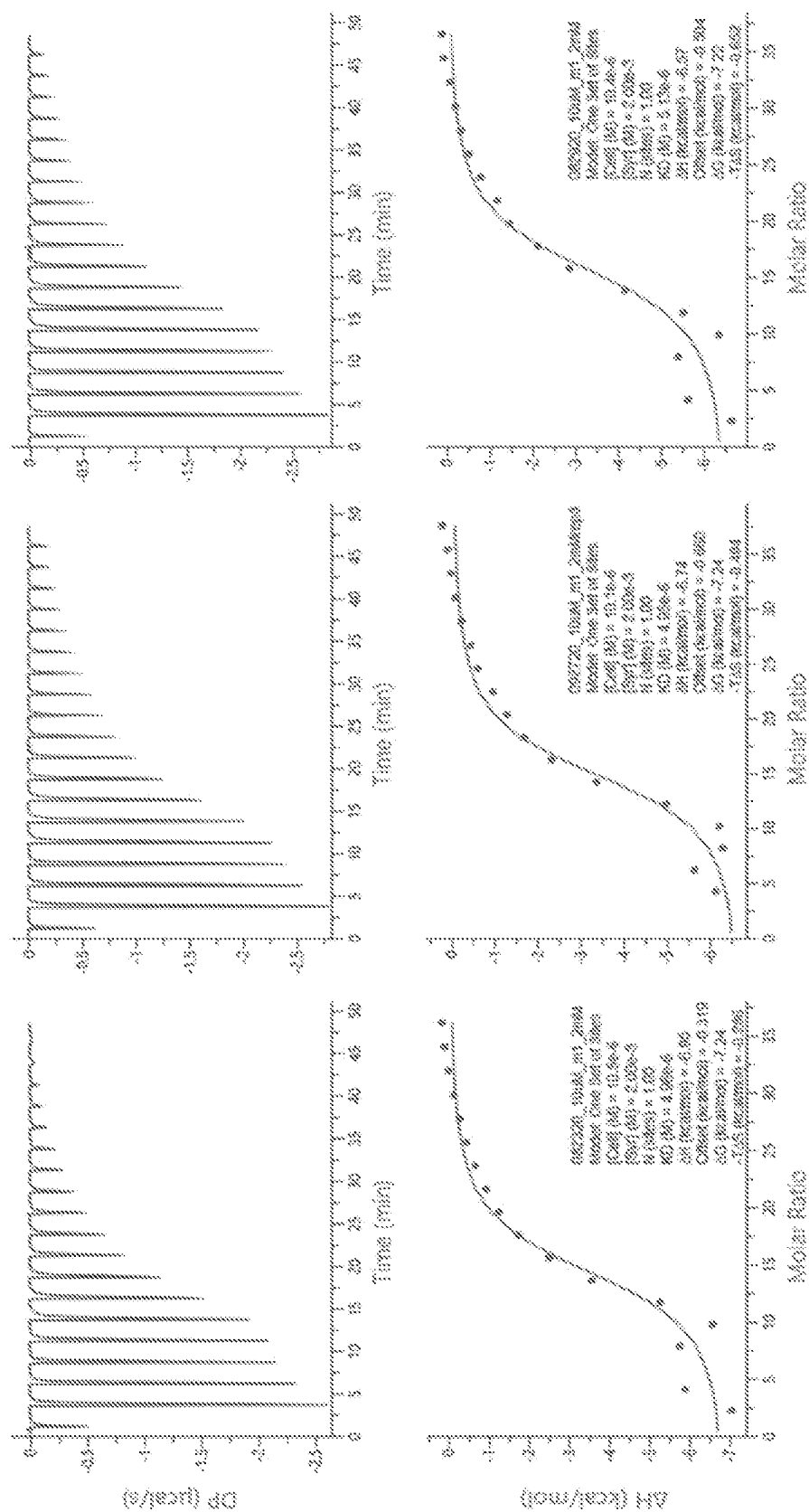


FIG. 22H

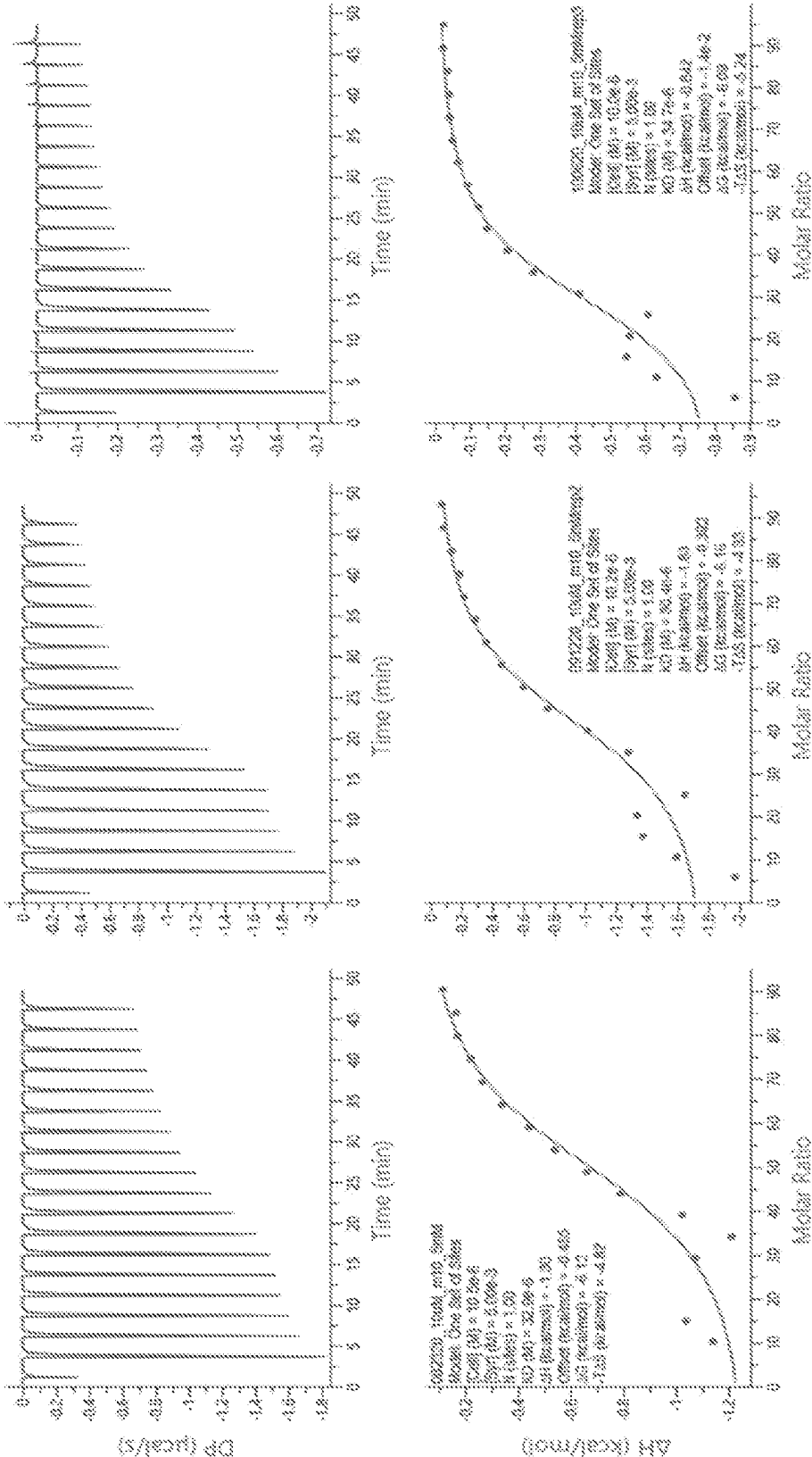


FIG. 22I

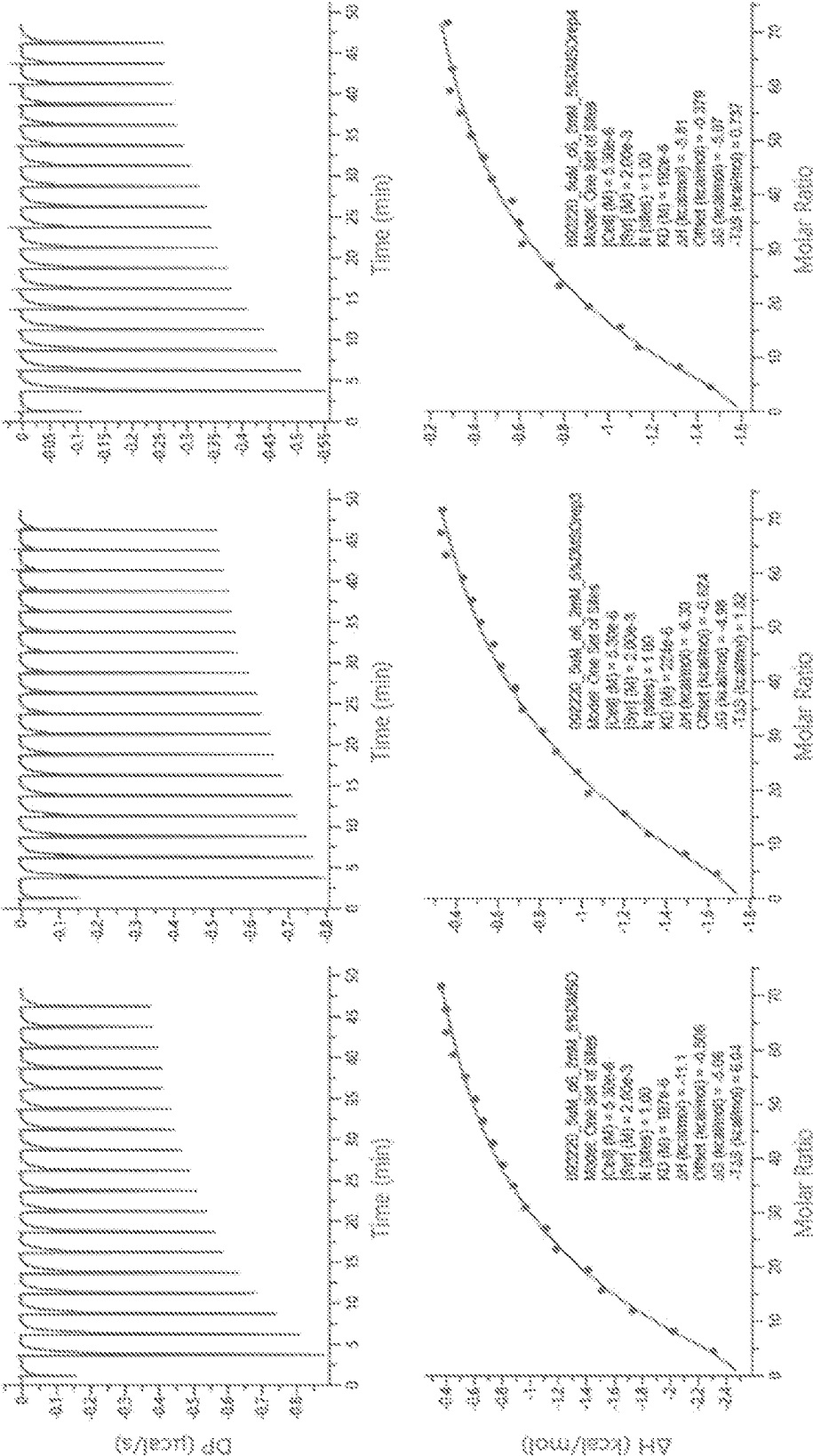


FIG. 22J

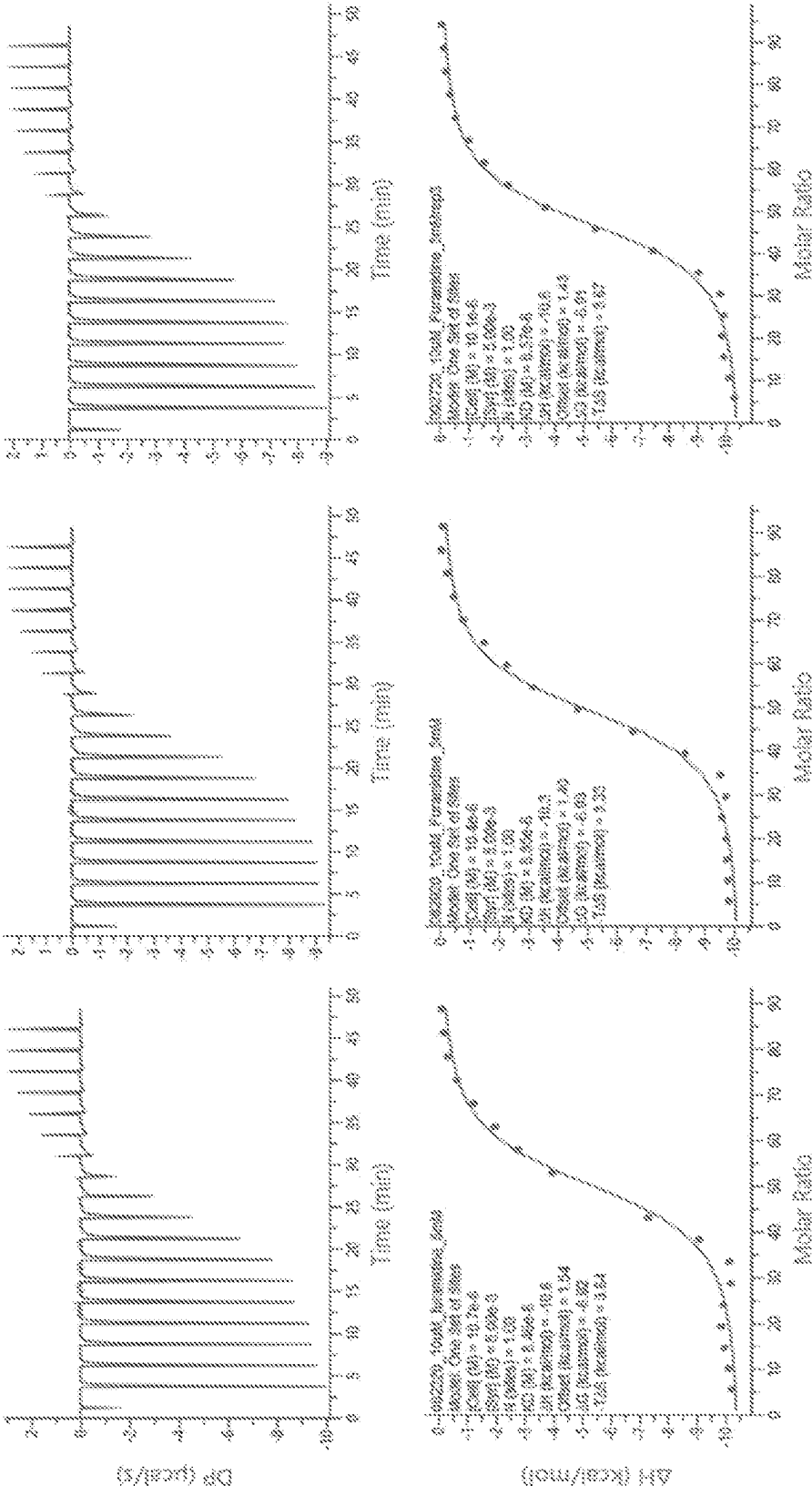


FIG. 23A

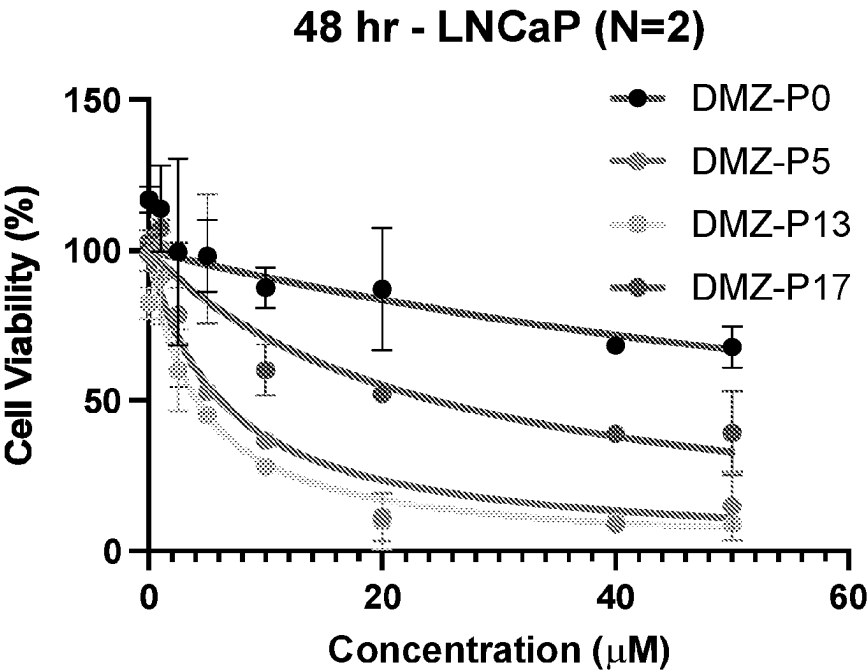
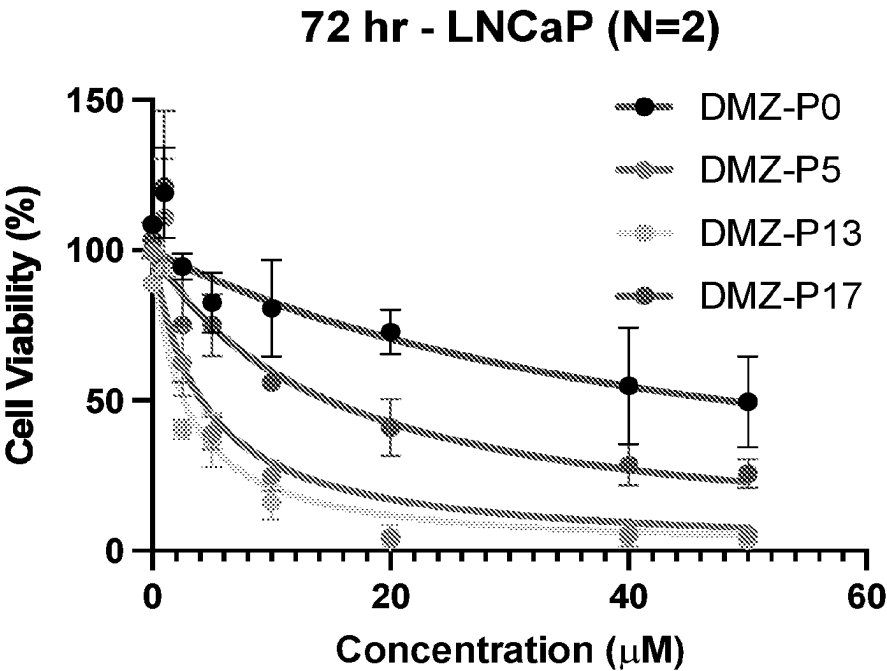


FIG. 23B



COMPOSITIONS FOR AND METHODS OF MODULATING RNA STABILITY

I. CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/322,902 filed 23 Mar. 2022, which is incorporated by reference herein in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] This invention was made with government support under R35GM124785 awarded by the National Institute of General Medical Sciences. The government has certain rights in the invention.

II. REFERENCE TO THE SEQUENCE LISTING

[0003] The Sequence Listing submitted 23 Mar. 2023 as an XML file named "23_2070_WO_Sequence_Listing", created on 23 Mar. 2023 and having a size of 109 KB is hereby incorporated by reference pursuant to 37 C.F.R. § 1.52 (e) (5).

III. BACKGROUND

[0004] The rapidly increasing characterization of RNA tertiary structures has revealed their pervasiveness and active roles in human diseases. Small molecule-mediated modulation of RNA tertiary structures constitutes an attractive avenue for the development of tools for both therapeutically targeting and/or uncovering the pathways associated with these RNA motifs. This potential has been highlighted by targeting of the triple helix present at the 3'-end of the non-coding RNA MALAT1, a transcript implicated in several human diseases. This triplex has been reported to decrease the transcript susceptibility to degradation and promote its cellular accumulation. While small molecules have been shown to bind and impact the stability of the MALAT1 triple helix, the small molecule properties that lead to these structural modulations are not well understood.

[0005] Thus, there remains an unmet need to test the applicability of previously identified RNA-targeted chemical space to more complex RNAs and to develop new methods to better understand these molecular recognition events. Moreover, as it relates to human pathology, there remains an unmet need to develop techniques and/or strategies to modulate the expression and/or function of disease associated and/or diseasing causing RNA secondary and tertiary structures.

IV. BRIEF DESCRIPTION OF THE FIGURES

[0006] FIG. 1A-FIG. 1C show the design and synthesis of the diminazene focused library.

[0007] FIG. 1A shows the representation of MALAT1 triple helix base pairing (left) and crystal structure (right). The triple helix is formed by the recruitment of the genomically encoded poly(A) tail (purple) to a uridine-rich stem loop (green) (adapted from Brown A, et al. (2014) Nat Struct Mol Biol. 21(7):633-640).

[0008] FIG. 1B is the envelope diagram of principal moments of inertia calculated for the 90-member theoretical

diminazene library. The 21 small molecules selected using the Kennard Stone algorithm are outlined in red.

[0009] FIG. 1C shows the synthetic scheme designed for synthesis of biscyano substituted scaffolds and coupling of commercially available amines as subunits (R).

[0010] FIG. 2A-FIG. 2C show that an indicator displacement assay (IDA) allowed evaluation of the affinity of the entire library against the MALAT1 triple helix.

[0011] FIG. 2A is a schematic of fluorescence-displacement assays. RiboGreen™ has measurable fluorescence signal when bound to RNA. Upon small molecule binding to the RNA, the fluorophore is displaced, and the fluorescence signal is quenched (left). Hypothetical curves of strong binders (green) and weak or non-binders (blue) where % displacement is plotted as a function of small molecule concentration (right).

[0012] FIG. 2B show the subunits chosen to create the theoretical 90-member library. Subunits selected for synthesis are in red.

[0013] FIG. 2C shows the relative affinity expressed as CD_{50} (small molecule concentration needed to achieve 50% of competitive displacement) obtained via small molecule titrations using the optimized IDA assay with Ribogreen (0.5 mM) and the MALAT1 triplex (0.17 mM) in 20 mM HEPES-KOH, 52 mM KCl, 0.1 mM $MgCl_2$ at pH=7.4. Molecules labeled as out of range (O.R.) indicate small molecules with CD_{50} outside of the range of small molecule concentrations (>500 mM) that lead to minimal dye displacement.

[0014] FIG. 3A-FIG. 3B show the cheminformatic trend analysis and QSAR model generation reveal affinity-based trends.

[0015] FIG. 3A is an envelope diagram of the synthesized diminazene small molecules. Small molecules that exhibited highest affinity for the triplex clustered in the rod-like sub-triangle (red). As the affinity of the small molecules decreases, the density of the ligands in the rod sub-triangle decreases (orange), with weak and non-binders being the most disc and sphere-like ligands (grey).

[0016] FIG. 3B is a plot of the best model obtained via QSAR for the CD_{50} calculated for all 21 small molecules.

[0017] FIG. 4A-FIG. 4B show the validation of indicator displacement assay (IDA) utilizing isothermal titration calorimetry (ITC).

[0018] FIG. 4A shows affinity as measured via isothermal titration calorimetry in three independent replicates for small molecules soluble with less than or equal to 3% DMSO in 20 mM Tris-HCl, 25 mM NaCl, 3 mM $MgCl_2$ buffer at pH=7.4.

[0019] FIG. 4B shows the correlation between affinity measured via indicator displacement assay (IDA) and ITC.

[0020] FIG. 5A-FIG. 5C show the evaluation of the effect of the focused library on the thermal stability of the MALAT1 triplex via differential scanning fluorimetry (DSF).

[0021] FIG. 5A shows the difference (4) in melting temperature between the MALAT1 triplex (5 mM) with small molecules (5 mM) compared to the DMSO vehicle. Difference in the first but not the second peak of the melting curves supports binding in the triplex tract and not the upper stem of the MALAT1 construct.

[0022] FIG. 5B shows the overlap of representative melting curves of the MALAT1 triple helix (blue) and the

isolated upper stem (orange) are in line with the second peak of the triplex bi-phasic melting curve representing the upper stem.

[0023] FIG. 5C is a plot of the best model obtained via QSAR for the ΔTm_1 calculated for all 21 small molecules.

[0024] FIG. 6A-FIG. 6D show the design and optimization of RNase A assay allows evaluation of the effect of the small molecule library on the triple helix enzymatic degradation profile.

[0025] FIG. 6A is a schematic of the RNase A assay; small molecules that stabilize base-paired regions reduce the amount of ssRNA tracts that the enzyme can cleave, resulting in a decrease of enzymatic degradation over time.

[0026] FIG. 6B shows that five time points were collected for each small molecule (2 mM) and triplex (0.2 mM) reaction, allowing the plotting of a degradation curve. The 5-minute time-point was chosen to compare small molecules assuming it is in the linear phase of decay of both stabilizing and destabilizing small molecules. Amount of RNA degraded over time was assessed by denaturing gel electrophoresis, Diamond staining, and Image J quantification.

[0027] FIG. 6C shows bar-graphs representing the fold-change of RNA degradation at the 5-minute time point for all the small molecules. RNA degradation is normalized to the same time point of DMSO as a control. All small molecules were evaluated in three independent experiments. A 5% DMSO control was included for every set of small molecules loaded on the gel.

[0028] FIG. 6D shows plot of the predictive model yielded by QSAR analysis. Stabilizers (orange) and destabilizers (blue) present separately in the graph.

[0029] FIG. 7 shows a suite of powerful orthogonal biophysical tools for the evaluation of small molecule:RNA triplex interactions that generate predictive models and will allow small molecule interrogation of the growing body of disease-associated RNA triple helices.

[0030] FIG. 8A shows an envelope diagram representing 3D shape diversity of the 90-member theoretical library of diminazene analogues (blue). A sub-set of 20 chosen for synthetic investigation using the Kennard Stone algorithm (pink). FIG. 8B shows unreactive and unstable small molecule candidates were then substituted with small molecules closest in the envelope diagram and afforded a total of 20 small molecule analogues and the Berenil® scaffold (purple).

[0031] FIG. 9 shows the coupling conditions attempted for R-19, R-25, R-27, and R-28 that did not yield any product formation.

[0032] FIG. 10 shows the MALAT1 triple helix construct run on Small RNA chip on Agilent bioanalyzer. Construct size is within 25% confidence value of sizing for the gel chip.

[0033] FIG. 11 shows a cartoon representation of constructs used for DSF selectivity experiments.

[0034] FIG. 12A-FIG. 12U show the small molecule affinity (CD_{50}) determined via RiboGreen™ IDA for the library as screened in three independent replicates in a 16-point titration.

[0035] FIG. 13A-FIG. 13Z show the melting profiles of the MALAT1 triple helix with equimolar small molecule. The entire diminazene (DMZ) library was screened in three independent replicates. Two small molecules previously published and identified as destabilizers showed little to no effect on the thermal stability of the triplex, thereby indi-

cating that DSF might not be a suitable assay to identify destabilizing small molecules.

[0036] FIG. 14 shows the correlation between the difference in melting temperature of the first peak of the bi-phasic melting curve of the triplex (ΔTm_1) and IDA-derived small molecule affinity ($\text{Log}(CD_{50})$). The small molecules with the highest affinity for the MALAT1 triple helix are also the ones that cause the largest shifts in thermal stability. Spearman's correlation -0.89 ($p\text{-value} < 0.0001$).

[0037] FIG. 15 shows the dose-dependent increase in thermal stability of the MALAT1 triple helix as measured by DSF. Shown is a representative set of small molecules identified as the best stabilizers.

[0038] FIG. 16A-FIG. 16F show the melting profiles of the MALAT1 upper stem and triple helix (top) and of the upper stem with equimolar small molecule (bottom). A sub-set of DMZ small molecules were screened in three independent replicates. Small molecules were chosen for testing based on their stabilization of ΔTm_1 but not ΔTm_2 of the MALAT1 triple helix. No small molecules caused a significant shift in the melting profile of the upper stem, indicating selective interactions with the triplex construct.

[0039] FIG. 17A-FIG. 17E show the melting profiles of the MALAT1 stem loop proxy and triple helix (top) and of the stem loop proxy with equimolar small molecule (bottom). A sub-set of DMZ small molecules were screened in three independent replicates. Small molecules were chosen for testing based on their stabilization of ΔTm_1 but not ΔTm_2 of the MALAT1 triple helix. No small molecules caused a significant shift in the melting profile of the stem loop proxy, indicating selective interactions with the triplex construct.

[0040] FIG. 18A-FIG. 18W show the RNase A enzymatic degradation assay of the entire diminazene (DMZ) library and two controls, a stabilizer (DPFp20) and a destabilizer (SM5). The two controls exhibit the same effects observed via RNase R enzymatic degradation study, thereby validating the fitness of the new, faster RNase A assay. A total of five time-points were collected for each small molecules in three independent replicates and the time point of 5 minutes was chosen to classify small molecules as stabilizers or destabilizers as it is in the linear phase of decay.

[0041] FIG. 19A shows representative 15% TBE denaturing gel of DMSO RNase A enzymatic degradation assay with ladder. At the zero time-point no degradation of the triplex is observed. As time progresses the triplex is degraded is shorter fragments. A short dsDNA loading control (10 bp band) was added to each loaded sample to normalize the intensity of the bands to account for loading and pipetting errors (left).

[0042] FIG. 19B shows imaging Cy5 reveals 5' fragments, leading to identification of potential RNase A cleavage sites (right).

[0043] FIG. 20A-FIG. 20H show the melting profiles of RNase A enzyme with (orange) and without (black) small molecule measured via differential scanning fluorimetry (DSF) in three independent replicates.

[0044] FIG. 21A-FIG. 21C show the controls performed with MALAT1 triple helix and small molecules for ITC. FIG. 21A shows 10 μM MALAT1 triple helix control in buffer (20 mM Tris-HCl, 25 mM NaCl, 3 mM MgCl_2 pH=7.4). FIG. 21B shows 10 μM MALAT1 triple helix control in buffer (20 mM Tris-HCl, 25 mM NaCl, 3 mM

MgCl₂ pH=7.4) and 3% DMSO. FIG. 21C shows representative 5 mM small molecule control in the syringe and buffer in the sample cell.

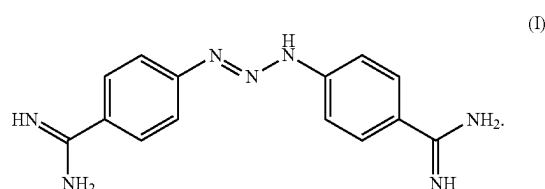
[0045] FIG. 22A-FIG. 22J show the MALAT1 ITC titrations across 3 independent replicates for various small molecules.

[0046] FIG. 23A-FIG. 23B show MTT experiments with LNCaP cells at 48 hrs (FIG. 23A) and 72 hours (FIG. 23B).

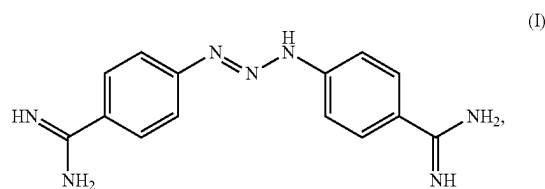
V. BRIEF SUMMARY

[0047] Disclosed herein is a compound comprising a small molecule having a diminazene scaffold. Disclosed herein is a compound comprising a small molecule having a diminazene scaffold and a diamidine moiety.

[0048] Disclosed herein is a compound of formula



[0049] Disclosed herein is a compound of formula



having one or more of ortho, para, and/or meta substitutions.

[0050] Disclosed herein is a compound identified as DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or a pharmaceutically acceptable salt, a hydrate, a prodrug, an ester, or a derivative thereof.

[0051] Disclosed herein is a composition comprising a disclosed compound, and one or more carriers and/or excipients.

[0052] Disclosed herein is a composition comprising a disclosed compound comprising a small molecule having a diminazene scaffold, and one or more carriers and/or excipients.

[0053] Disclosed herein is a composition comprising a disclosed compound comprising a small molecule having a diminazene scaffold and a diamidine moiety, and one or more carriers and/or excipients.

[0054] Disclosed herein is a composition comprising DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, a pharmaceutically acceptable salt, a hydrate, a prodrug, an ester, or a derivative thereof, or any combination thereof.

[0055] Disclosed herein is a pharmaceutical formulation comprising a disclosed compound and one or more pharmaceutically acceptable carriers. Disclosed herein is a pharmaceutical formulation comprising a disclosed composition and one or more pharmaceutically acceptable carriers. Disclosed herein is a pharmaceutical formulation comprising one or more disclosed compounds and one or more pharmaceutically acceptable carriers. Disclosed herein is a pharmaceutical formulation comprising one or more disclosed compositions and one or more pharmaceutically acceptable carriers.

[0056] Disclosed herein is a library comprising one or more disclosed small molecules having a diminazene scaffold. Disclosed herein is a library comprising one or more disclosed shape diverse small molecules having a diminazene scaffold. Disclosed herein is a library comprising one or more disclosed small molecules having a diminazene scaffold and a diamidine moiety.

[0057] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations.

[0058] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, and treating a disease, condition, or disorder in the subject. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, and treating a disease, condition, or

disorder in the subject. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, and treating a disease, condition, or disorder in the subject.

[0059] Disclosed herein is a method of modulating one or more secondary RNA structures and/or tertiary RNA structures, the method comprising contacting one or more RNA structures with one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof.

[0060] Disclosed herein is a method of assessing a library of small molecules that target RNA, the method comprising using a multi-dimensional approach comprising designing and optimizing affinity-based and structural stability-based assays.

[0061] Disclosed herein is a method of assessing a library of shape diverse small molecule analogs that modulate a targeted RNA or a targeted RNA structure, the method comprising choosing a candidate small molecule; generating a theoretical library of shape diverse small molecule analogs based on the candidate small molecule; evaluating the three-dimensional shape of one or more shape diverse small molecule analogs of the theoretical library; selecting one or more shape diverse small molecules for synthesis; synthesizing one or more shape diverse small molecule analogs in the theoretic library; and characterizing the one or more synthesized shape diverse small molecule analogs.

VI. DETAILED DESCRIPTION

[0062] The present disclosure describes formulations, compounded compositions, kits, capsules, containers, and/or methods thereof. It is to be understood that the inventive aspects of which are not limited to specific synthetic methods unless otherwise specified, or to particular reagents unless otherwise specified, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, example methods and materials are now described.

[0063] All publications mentioned herein are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited. The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention.

A. Relevant Definitions

[0064] Before the present compounds, compositions, articles, systems, devices, and/or methods are disclosed and described, it is to be understood that they are not limited to specific synthetic methods unless otherwise specified, or to particular reagents unless otherwise specified, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting. Although any methods and materials similar or equivalent to those

described herein can be used in the practice or testing of the present invention, example methods and materials are now described.

[0065] This disclosure describes inventive concepts with reference to specific examples. However, the intent is to cover all modifications, equivalents, and alternatives of the inventive concepts that are consistent with this disclosure.

[0066] As used in the specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

[0067] The phrase “consisting essentially of” limits the scope of a claim to the recited components in a composition or the recited steps in a method as well as those that do not materially affect the basic and novel characteristic or characteristics of the claimed composition or claimed method. The phrase “consisting of” excludes any component, step, or element that is not recited in the claim. The phrase “comprising” is synonymous with “including”, “containing”, or “characterized by”, and is inclusive or open-ended. “Comprising” does not exclude additional, unrecited components or steps.

[0068] As used herein, when referring to any numerical value, the term “about” means a value falling within a range that is $\pm 10\%$ of the stated value.

[0069] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, a further aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms a further aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint and independently of the other endpoint. It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed as “about” that particular value in addition to the value itself. For example, if the value “10” is disclosed, then “about 10” is also disclosed. It is also understood that each unit between two particular units are also disclosed. For example, if 10 and 15 are disclosed, then 11, 12, 13, and 14 are also disclosed.

[0070] References in the specification and concluding claims to parts by weight of a particular element or component in a composition denotes the weight relationship between the element or component and any other elements or components in the composition or article for which a part by weight is expressed. Thus, in a compound containing 2 parts by weight component X and 5 parts by weight component Y, X and Y are present at a weight ratio of 2:5, and are present in such ratio regardless of whether additional components are contained in the compound.

[0071] As used herein, the terms “optional” or “optionally” means that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not. In an aspect, a disclosed method can optionally comprise one or more additional steps, such as, for example, repeating an administering step or altering an administering step.

[0072] As used herein, the term “subject” refers to the target of administration, e.g., a human being. The term “subject” also includes domesticated animals (e.g., cats, dogs, etc.), livestock (e.g., cattle, horses, pigs, sheep, goats, etc.), and laboratory animals (e.g., mouse, rabbit, rat, guinea

pig, fruit fly, etc.). Thus, the subject of the herein disclosed methods can be a vertebrate, such as a mammal, a fish, a bird, a reptile, or an amphibian. Alternatively, the subject of the herein disclosed methods can be a human, non-human primate, horse, pig, rabbit, dog, sheep, goat, cow, cat, guinea pig, or rodent. The term does not denote a particular age or sex, and thus, adult and child subjects, as well as fetuses, whether male or female, are intended to be covered. In an aspect, a subject can be a human patient. In an aspect, a subject can have a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0073] The term “biological sample” as used herein includes, but is not limited to, a sample containing tissues, cells, and/or biological fluids isolated from a subject. Examples of biological samples include, but are not limited to, tissues, cells, biopsies, blood, lymph, serum, plasma, urine, saliva, mucus, and tears. In an aspect, the biological sample can be a biopsy (such as a tumor biopsy). A biological sample can be obtained directly from a subject (e.g., by blood or tissue sampling) or from a third party (e.g., received from an intermediary, such as a healthcare provider or lab technician).

[0074] As used herein, the term “diagnosed” means having been subjected to an examination by a person of skill, for example, a physician, and found to have a condition that can be diagnosed or treated by one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof, or by one or more of the disclosed methods. For example, “diagnosed with a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof” means having been subjected to an examination by a person of skill, for example, a physician, and found to have a condition that can be treated by one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof, or by one or more of the disclosed methods. In an aspect, an examination can be physical, can involve various tests (e.g., blood tests, genotyping, biopsies, etc.) and assays (e.g., enzymatic assay), or a combination thereof.

[0075] A “patient” can refer to a subject that has been diagnosed with or is suspected of having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, a patient can refer to a subject that has been diagnosed with or is suspected of having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression

of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof and is seeking treatment or receiving treatment.

[0076] As used herein, the phrase “identified to be in need of treatment for a disorder,” or the like, refers to selection of a subject based upon need for treatment of the disorder. For example, a subject can be identified as having a need for treatment of a disorder (e.g., such as a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof) based upon an earlier diagnosis by a person of skill and thereafter subjected to treatment for the disorder (e.g., a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). In an aspect, the identification can be performed by a person different from the person making the diagnosis. In an aspect, the administration can be performed by one who performed the diagnosis.

[0077] As used herein, “inhibit,” “inhibiting,” and “inhibition” mean to diminish or decrease an activity, level, response, expression, condition, severity, disease, or other biological parameter. This can include, but is not limited to, the complete ablation of the activity, level, response, expression, condition, severity, disease, or other biological parameter. This can also include, for example, a 10% inhibition or reduction in the activity, level, response, condition, severity, disease, or other biological parameter as compared to the native or control level (e.g., a subject not having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). Thus, in an aspect, the inhibition or reduction can be a 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, or any amount of reduction in between as compared to native or control levels. In an aspect, the inhibition or reduction can be 10-20%, 20-30%, 30-40%, 40-50%, 50-60%, 60-70%, 70-80%, 80-90%, or 90-100% as compared to native or control levels. In an aspect, the inhibition or reduction can be 0-25%, 25-50%, 50-75%, or 75-100% as compared to native or control levels. In an aspect, a native or control level can be a pre-disease or pre-disorder level.

[0078] The words “treat” or “treating” or “treatment” include palliative treatment, that is, treatment designed for the relief of symptoms rather than the curing of the disease, pathological condition, or disorder; preventative treatment, that is, treatment directed to minimizing or partially or completely inhibiting the development of the associated disease, pathological condition, or disorder; and supportive treatment, that is, treatment employed to supplement another specific therapy directed toward the improvement of the associated disease, pathological condition, or disorder (such as a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). In an aspect, the terms cover any treatment of a subject, including a mammal (e.g., a human), and includes: (i) preventing the undesired physiological change, disease, pathological condition, or disorder from occurring in a subject that can be predisposed to the disease but has not yet

been diagnosed as having it; (ii) inhibiting the physiological change, disease, pathological condition, or disorder, i.e., arresting its development; or (iii) relieving the physiological change, disease, pathological condition, or disorder, i.e., causing regression of the disease. For example, in an aspect, treating a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof can reduce the severity of an established disease in a subject by 1%-100% as compared to a control (such as, for example, an individual not having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). In an aspect, treating can refer to a 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 100% reduction in the severity of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. For example, treating a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof can reduce one or more symptoms of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof in a subject by 1%-100% as compared to a control (such as, for example, an individual not having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). In an aspect, treating can refer to 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100% reduction of one or more symptoms of an established disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. It is understood that treatment does not necessarily refer to a cure or complete ablation or eradication of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. However, in an aspect, treatment can refer to a cure or complete ablation or eradication of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0079] As used herein, a “biomarker” refers to a defined characteristic that is measured as an indicator of normal biological processes, pathogenic processes, or response to an exposure of intervention. In an aspect, a biomarker can be diagnostic (i.e., detects or classifies a pathological condition), prognostic (i.e., predicts the probability of disease occurrence or progression), pharmacodynamic/responsive (i.e., identifies a change in response to a therapeutic inter-

vention), predictive (i.e., predicts how an individual or subject might respond to a particular intervention or event). In an aspect, a biomarker can be diagnostic, prognostic, pharmacodynamic/responsive, and/or predictive at the same time. In an aspect, a biomarker can be diagnostic, prognostic, pharmacodynamic/responsive, and/or predictive at different times (e.g., first a biomarker can be diagnostic and then later, the same biomarker can be prognostic, pharmacodynamic/responsive, and/or predictive). A biomarker can be an objective measure that can be linked to a clinical outcome assessment. A biomarker can be used by the skilled person to make a clinical decision based on its context of use.

[0080] As used herein, “peptide,” “polypeptide,” and “protein” are used interchangeably, and refer to a compound comprised of amino acid residues covalently linked by peptide bonds. A protein must contain at least two amino acids and there is no limitation on the maximum number of amino acids that can comprise a protein’s sequence. The term “peptide” can refer to a short chain of amino acids including, for example, natural peptides, recombinant peptides, synthetic peptides, or any combination thereof. Proteins and peptides can include, for example, biologically active fragments, substantially homologous polypeptides, oligopeptides, homodimers, heterodimers, variants of polypeptides, modified polypeptides, derivatives, analogs, and fusion proteins, among others.

[0081] “Nucleic acid” or “oligonucleotide” or “polynucleotide” as used herein means at least two nucleotides covalently linked together. The depiction of a single strand can also define the sequence of the complementary strand. Thus, a nucleic acid can encompass the complementary strand of a depicted single strand. Many variants of a nucleic acid can be used for the same purpose as a given nucleic acid. Thus, a nucleic acid can encompass substantially identical nucleic acids and complements thereof. A single strand can provide a probe that can hybridize to a target sequence under stringent hybridization conditions. Thus, a nucleic acid can encompass a probe that hybridizes under stringent hybridization conditions. A nucleic acid can be single-stranded, or double-stranded, or can contain portions of both double-stranded and single-stranded sequence. The nucleic acid can be DNA, both genomic and cDNA, RNA, or a hybrid, where the nucleic acid can contain combinations of deoxyribo- and ribo-nucleotides, and combinations of bases including uracil, adenine, thymine, cytosine, guanine, inosine, xanthine hypoxanthine, isocytosine and isoguanine. Nucleic acids can be obtained by chemical synthesis methods or by recombinant methods. Also as used herein, the terms “nucleic acid,” “nucleic acid molecule,” “nucleic acid construct,” “nucleotide sequence”, and “polynucleotide” can refer to RNA or DNA that is linear or branched, single or double stranded, or a hybrid thereof. The term can encompass RNA/DNA hybrids. When dsRNA is produced synthetically, less common bases, such as inosine, 5-methylcytosine, 6-methyladenine, hypoxanthine and others can also be used for antisense, dsRNA, and ribozyme pairing. For example, polynucleotides that contain C-5 propyne analogues of uridine and cytidine have been shown to bind RNA with high affinity and to be potent antisense inhibitors of gene expression. Other modifications, such as modification to the phosphodiester backbone, or the 2'-hydroxy in the ribose sugar group of the RNA can also be made. A “synthetic” nucleic acid or polynucleotide, as used herein, refers to a nucleic

acid or polynucleotide that is not found in nature but is constructed by the hand of man and therefore is not a product of nature.

[0082] A “polynucleotide” is a sequence of nucleotide bases, and may be RNA, DNA, or DNA-RNA hybrid sequences (including both naturally occurring and non-naturally occurring nucleotides).

[0083] A “fragment” or “portion” of a nucleotide sequence can be understood to mean a nucleotide sequence of reduced length relative (e.g., reduced by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more nucleotides) to a reference nucleic acid or nucleotide sequence and comprising, consisting essentially of, or consisting of a nucleotide sequence of contiguous nucleotides identical or almost identical (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical) to the reference nucleic acid or nucleotide sequence. Such a nucleic acid fragment or portion according to the disclosure can be, where appropriate, included in a larger polynucleotide of which it is a constituent. In an aspect, a fragment or portion of a nucleotide sequence or nucleic acid sequence can comprise the sequence encoding an exon having one or more mutations. In an aspect, a fragment or portion of a nucleotide sequence or nucleic acid sequence can comprise a target of interest.

[0084] A “fragment” or “portion” of an amino acid sequence can be understood to mean an amino acid sequence of reduced length relative (e.g., reduced by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 250, or more amino acids) to a reference amino acid sequence and comprising, consisting essentially of, or consisting of an amino acid sequence of contiguous amino acids identical or almost identical (e.g., 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% identical) to the reference amino acid sequence. Such an amino acid fragment or portion according to the disclosure can be, where appropriate, included in a larger amino acid sequence of which it is a constituent.

[0085] “Complement” or “complementary” as used herein means a nucleic acid can mean Watson-Crick (e.g., A-T/U and C-G) or Hoogsteen base pairing between nucleotides or nucleotide analogs of nucleic acid molecules. “Complementarity” refers to a property shared between two nucleic acid sequences, such that when they are aligned antiparallel to each other, the nucleotide bases at each position will be complementary.

[0086] As used herein, the term “prevent” or “preventing” or “prevention” refers to precluding, averting, obviating, forestalling, stopping, or hindering something from happening, especially by advance action. It is understood that where reduce, inhibit, or prevent are used herein, unless specifically indicated otherwise, the use of the other two words is also expressly disclosed. In an aspect, preventing progression of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof is intended. The words “prevent” and “preventing” and “prevention” also refer to prophylactic or preventative measures for protecting or precluding a subject (e.g., an individual) not having a disease, condition, or disorder caused by,

related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof or a related complication from progressing to that complication.

[0087] As used herein, the terms “administering” and “administration” refer to any method of providing one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof to a subject. Such methods are well known to those skilled in the art and include, but are not limited to, the following routes: oral administration, transdermal administration, administration by inhalation, nasal administration, topical administration, in utero administration, intrahepatic administration, intravaginal administration, ophthalmic administration, intraaural administration, otic administration, intracerebral administration, rectal administration, sublingual administration, buccal administration, and parenteral administration, including injectable such as intravenous administration, intra-CSF administration, intra-arterial administration, intramuscular administration, and subcutaneous administration. Administration can also include hepatic intra-arterial administration or administration through the hepatic portal vein (HPV). Administration of a disclosed therapeutic agent, a disclosed pharmaceutical composition, or a combination thereof can comprise administration directly into the CNS (e.g., intraparenchymal, intracerebroventricular, intrathecal cisternal, intrathecal (lumbar), deep gray matter delivery, convection-enhanced delivery to deep gray matter) or the PNS. Administration can be continuous or intermittent.

[0088] In an aspect, a “therapeutic agent” can be a “biologically active agent” or “biologic active agent” or “bioactive agent”, which refers to an agent that is capable of providing a local or systemic biological, physiological, or therapeutic effect in the biological system to which it is applied. For example, the bioactive agent can act to control infection or inflammation, enhance cell growth and tissue regeneration, control tumor growth, act as an analgesic, promote anti-cell attachment, and enhance bone growth, among other functions. Other suitable bioactive agents can include anti-viral agents, vaccines, hormones, antibodies (including active antibody fragments sFv, Fv, and Fab fragments), aptamers, peptide mimetics, functional nucleic acids, therapeutic proteins, peptides, or nucleic acids. Other bioactive agents include prodrugs, which are agents that are not biologically active when administered but, upon administration to a subject are converted to bioactive agents through metabolism or some other mechanism. Additionally, any of the compositions of the invention can contain combinations of two or more bioactive agents. It is understood that a biologically active agent can be used in connection with administration to various subjects, for example, to humans (i.e., medical administration) or to animals (i.e., veterinary administration). As used herein, the recitation of a biologically active agent inherently encompasses the pharmaceutically acceptable salts thereof.

[0089] In an aspect, a “therapeutic agent” can be any agent that effects a desired clinical outcome in a subject having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof, suspected of having a disease, condition, or disorder caused by,

related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof, and/or likely to develop or acquire a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, a disclosed therapeutic agent can be an oligonucleotide therapeutic agent. A disclosed oligonucleotide therapeutic agent can comprise a single-stranded or double-stranded DNA, iRNA, shRNA, siRNA, mRNA, non-coding RNA (ncRNA), an antisense molecule, miRNA, a morpholino, a peptide-nucleic acid (PNA), or an analog or conjugate thereof. In an aspect, a disclosed oligonucleotide therapeutic agent can be an ASO or an RNAi. In an aspect, a disclosed oligonucleotide therapeutic agent can comprise one or more modifications at any position applicable.

[0090] By “determining the amount” is meant both an absolute quantification of a particular analyte (e.g., a disclosed secondary RNA structure and/or a disclosed tertiary RNA structure) or a determination of the relative abundance of a particular analyte (e.g., an amount as compared to a control amount). The phrase includes both direct or indirect measurements of abundance (e.g., individual mRNA transcripts may be quantified or the amount of amplification of an mRNA sequence under certain conditions for a certain period may be used a surrogate for individual transcript quantification) or both.

[0091] As used herein, “modifying the method” can comprise modifying or changing one or more features or aspects of one or more steps of a disclosed method. For example, in an aspect, a method can be altered by changing the amount of one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof administered to a subject, or by changing the frequency of administration of one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof to a subject, by changing the duration of time one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof are administered to a subject, or by substituting for one or more of the disclosed components and/or reagents with a similar or equivalent component and/or reagent. The same applies to all the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof.

[0092] In an aspect, a therapeutic agent can be a “drug” or a “vaccine” and means a molecule, group of molecules, complex or substance administered to an organism for diagnostic, therapeutic, preventative medical, or veterinary purposes. This term includes externally and internally administered topical, localized and systemic human and animal pharmaceuticals, treatments, remedies, nutraceuticals, cosmeceuticals, biologicals, devices, diagnostics and contraceptives, including preparations useful in clinical and veterinary screening, prevention, prophylaxis, healing, wellness, detection, imaging, diagnosis, therapy, surgery, monitoring, cosmetics, prosthetics, forensics and the like. This term may also be used in reference to agricultural, workplace, military, industrial and environmental therapeutics or remedies comprising selected molecules or selected nucleic acid sequences capable of recognizing cellular receptors,

membrane receptors, hormone receptors, therapeutic receptors, microbes, viruses or selected targets comprising or capable of contacting plants, animals and/or humans. Examples include but are not limited to a radiosensitizer, the combination of a radiosensitizer and a chemotherapeutic, a steroid, a xanthine, a beta-2-agonist bronchodilator, an anti-inflammatory agent, an analgesic agent, a calcium antagonist, an angiotensin-converting enzyme inhibitors, a beta-blocker, a centrally active alpha-agonist, an alpha-1-antagonist, carbonic anhydrase inhibitors, prostaglandin analogs, a combination of an alpha agonist and a beta blocker, a combination of a carbonic anhydrase inhibitor and a beta blocker, an anticholinergic/antispasmodic agent, a vasopressin analogue, an antiarrhythmic agent, an antiparkinsonian agent, an antiangina/antihypertensive agent, an anticoagulant agent, an antiplatelet agent, a sedative, an anxiolytic agent, a peptidic agent, a biopolymeric agent, an antineoplastic agent, a laxative, an antidiarrheal agent, an antimicrobial agent, an antifungal agent, or a vaccine. In a further aspect, the pharmaceutically active agent can be coumarin, albumin, bromolodine, steroids such as betamethasone, dexamethasone, methylprednisolone, prednisolone, prednisone, triamcinolone, budesonide, hydrocortisone, and pharmaceutically acceptable hydrocortisone derivatives; xanthines such as theophylline and doxophylline; beta-2-agonist bronchodilators such as salbutamol, fenterol, clenbuterol, bambuterol, salmeterol, fenoterol; anti-inflammatory agents, including antiasthmatic anti-inflammatory agents, antiarthritis anti-inflammatory agents, and non-steroidal anti-inflammatory agents, examples of which include but are not limited to sulfides, mesalamine, budesonide, salazopyrin, diclofenac, pharmaceutically acceptable diclofenac salts, nimesulide, naproxene, acetaminophen, ibuprofen, ketoprofen and piroxicam; analgesic agents such as salicylates; calcium channel blockers such as nifedipine, amlodipine, and nicardipine; angiotensin-converting enzyme inhibitors such as captopril, benazepril hydrochloride, fosinopril sodium, trandolapril, ramipril, lisinopril, enalapril, quinapril hydrochloride, and moexipril hydrochloride; beta-blockers (i.e., beta adrenergic blocking agents) such as sotalol hydrochloride, timolol maleate, timol hemihydrate, levobunolol hydrochloride, esmolol hydrochloride, carteolol, propranolol hydrochloride, betaxolol hydrochloride, penbutolol sulfate, metoprolol tartrate, metoprolol succinate, acebutolol hydrochloride, atenolol, pindolol, and bisoprolol fumarate; centrally active alpha-2-agonists (i.e., alpha adrenergic receptor agonist) such as clonidine, bromidone tartrate, and apraclonidine hydrochloride; alpha-1-antagonists such as doxazosin and prazosin; anticholinergic/antispasmodic agents such as dicyclomine hydrochloride, scopolamine hydrobromide, glycopyrrolate, clidinium bromide, flavoxate, and oxybutynin; vasopressin analogues such as vasopressin and desmopressin; prostaglandin analogs such as latanoprost, travoprost, and bimatoprost; cholinergics (i.e., acetylcholine receptor agonists) such as pilocarpine hydrochloride and carbachol; glutamate receptor agonists such as the N-methyl D-aspartate receptor agonist memantine; anti-Vascular endothelial growth factor (VEGF) aptamers such as pegaptanib; anti-VEGF antibodies (including but not limited to anti-VEGF-A antibodies) such as ranibizumab and bevacizumab; carbonic anhydrase inhibitors such as methazolamide, brinzolamide, dorzolamide hydrochloride, and acetazolamide; antiarrhythmic agents such as quinidine, lidocaine, tocainide hydrochloride,

mexiletine hydrochloride, digoxin, verapamil hydrochloride, propafenone hydrochloride, flecainide acetate, procainamide hydrochloride, moricizine hydrochloride, and diisopyramide phosphate; antiparkinsonian agents, such as dopamine, L-Dopa/Carbidopa, selegiline, dihydroergocryptine, pergolide, lisuride, apomorphine, and bromocryptine; antiangina agents and antihypertensive agents such as isosorbide mononitrate, isosorbide dinitrate, propranolol, atenolol and verapamil; anticoagulant and antiplatelet agents such as coumadin, warfarin, acetylsalicylic acid, and ticlopidine; sedatives such as benzodiazepines and barbiturates; anxiolytic agents such as lorazepam, bromazepam, and diazepam; peptidic and biopolymeric agents such as calcitonin, leuprolide and other LHRH agonists, hirudin, cyclosporin, insulin, somatostatin, protirelin, interferon, desmopressin, somatotropin, thymopentin, pidotimod, erythropoietin, interleukins, melatonin, granulocyte/macrophage-CSF, and heparin; antineoplastic agents such as etoposide, etoposide phosphate, cyclophosphamide, methotrexate, 5-fluorouracil, vincristine, doxorubicin, cisplatin, hydroxyurea, leucovorin calcium, tamoxifen, flutamide, asparaginase, altretramine, mitotane, and procarbazine hydrochloride; laxatives such as senna concentrate, casanthranol, bisacodyl, and sodium picosulphate; antidiarrheal agents such as difenoxine hydrochloride, loperamide hydrochloride, furazolidone, diphenoxylate hydrochloride, and microorganisms; vaccines such as bacterial and viral vaccines; antimicrobial agents such as penicillins, cephalosporins, and macrolides, antifungal agents such as imidazolic and triazolic derivatives; and nucleic acids such as DNA sequences encoding for biological proteins, and antisense oligonucleotides. It is understood that a pharmaceutically active agent can be used in connection with administration to various subjects, for example, to humans (i.e., medical administration) or to animals (i.e., veterinary administration). As used herein, the recitation of a pharmaceutically active agent inherently encompasses the pharmaceutically acceptable salts thereof.

[0093] “Sequence identity” and “sequence similarity” can be determined by alignment of two peptide or two nucleotide sequences using global or local alignment algorithms. Sequences may then be referred to as “substantially identical” or “essentially similar” when they are optimally aligned. For example, sequence similarity or identity can be determined by searching against databases such as FASTA, BLAST, etc., but hits should be retrieved and aligned pairwise to compare sequence identity. Two proteins or two protein domains, or two nucleic acid sequences can have “substantial sequence identity” if the percentage sequence identity is at least 70%, 75%, 80%, 85%, 90%, 95%, 98%, 99% or more, preferably 90%, 95%, 98%, 99% or more. Such sequences are also referred to as “variants” herein, e.g., other variants of glycogen branching enzymes and amylases. Sequences with substantial sequence identity do not necessarily have the same length and may differ in length. For example, sequences that have the same nucleotide sequence but of which one has additional nucleotides on the 3'- and/or 5'-side are 100% identical.

[0094] In an aspect, the skilled person can determine an efficacious dose, an efficacious schedule, and an efficacious route of administration for one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof so as to treat or prevent a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of,

and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, the skilled person can also alter, change, or modify an aspect of an administering step to improve efficacy of one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof. In an aspect, the skilled person can determine an efficacious dose, an efficacious schedule, and an efficacious route of administration for any of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof.

[0095] As used herein, “modifying the method” can comprise modifying or changing one or more features or aspects of one or more steps of a disclosed method. For example, in an aspect, a method can be altered by changing the amount of one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof administered to a subject, or by changing the frequency of administration of one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof, or by changing the duration of time that the one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof are administered to a subject.

[0096] As used herein, “isolated” refers to a nucleic acid molecule or a nucleic acid sequence that has been substantially separated, produced apart from, or purified away from other biological components in the cell or tissue of an organism in which the component occurs, such as other cells, chromosomal and extrachromosomal DNA and RNA, and proteins. Nucleic acids and proteins that have been “isolated” include nucleic acids and proteins purified by standard purification methods. The term also embraces nucleic acids and proteins prepared by recombinant expression in a host cell as well as chemically synthesized nucleic acids and proteins. Isolated proteins or nucleic acids, or cells containing such, in some examples are at least 50% pure, such as at least 75%, at least 80%, at least 90%, at least 95%, at least 98%, or at least 100% pure.

[0097] As used herein, “concurrently” means (1) simultaneously in time, or (2) at different times during the course of a common treatment schedule.

[0098] The term “contacting” as used herein refers to bringing one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof together with a target area or intended target area in such a manner that the one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof exert an effect on the intended target or targeted area either directly or indirectly. A target area or intended target area can be one or more of a subject's organs (e.g., lungs, heart, liver, kidney, brain, etc.). In an aspect, a target area or intended target area can be any cell or any organ affected by one or more disease-associated RNAs or disease relevant RNAs. In an aspect, a target area or intended target area can be the brain or various neuronal populations. In an aspect, a target area or intended target area can be any cell or any organ infected by an over-expression or an under-expression of one or more disease-associated RNAs or disease relevant RNAs.

[0099] As used herein, “determining” can refer to measuring or ascertaining the expression and/or activity of one or more disease-associated RNAs or disease relevant RNAs. Methods and techniques used to determine the expression and/or activity of one or more disease-associated RNAs or disease relevant RNAs are typically known to the medical arts. For example, the art is familiar with the ways to identify and/or diagnose the presence, severity, or both of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, “determining” can also refer to measuring or ascertaining the level of one or more proteins or peptides in a bio sample, or measuring or ascertaining the level of one or more RNAs or miRNAs in a bio sample. Methods and techniques for determining the expression and/or activity level of relevant proteins, peptides, mRNA, DNA, or any combination thereof known to the art and are disclosed herein.

[0100] As used herein, “effective amount” and “amount effective” can refer to an amount that is sufficient to achieve the desired result such as, for example, the treatment and/or prevention of a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. As used herein, the terms “effective amount” and “amount effective” can refer to an amount that is sufficient to achieve the desired an effect on an undesired condition (e.g., a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). For example, a “therapeutically effective amount” refers to an amount that is sufficient to achieve the desired therapeutic result or to have an effect on undesired symptoms, but is generally insufficient to cause adverse side effects. In an aspect, “therapeutically effective amount” means an amount of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof that (i) treats the particular disease, condition, or disorder (e.g., a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof), (ii) attenuates, ameliorates, or eliminates one or more symptoms of the particular disease, condition, or disorder (e.g., a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof), or (iii) delays the onset of one or more symptoms of the particular disease, condition, or disorder described herein (e.g., a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof). The specific therapeutically effective dose level for any particular patient will depend upon a variety of factors including the disorder being treated and the severity of the disorder; the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof employed; the disclosed methods employed; the age, body weight, general health, sex and diet of the patient; the

time of administration; the route of administration; the rate of excretion of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof employed; the duration of the treatment; drugs used in combination or coincidental with the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof employed, and other like factors well known in the medical arts. For example, it is well within the skill of the art to start doses of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof at levels lower than those required to achieve the desired therapeutic effect and to gradually increase the dosage until the desired effect is achieved. If desired, then the effective daily dose can be divided into multiple doses for purposes of administration. Consequently, a single dose of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof can contain such amounts or submultiples thereof to make up the daily dose. The dosage can be adjusted by the individual physician in the event of any contraindications. Dosage can vary, and can be administered in one or more dose administrations daily, for one or several days. Guidance can be found in the literature for appropriate dosages for given classes of pharmaceutical products. In further various aspects, a preparation can be administered in a “prophylactically effective amount”; that is, an amount effective for prevention of a disease or condition, such as, for example, a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0101] As used herein, the term “pharmaceutically acceptable carrier” refers to sterile aqueous or nonaqueous solutions, dispersions, suspensions or emulsions, as well as sterile powders for reconstitution into sterile injectable solutions or dispersions just prior to use. Examples of suitable aqueous and nonaqueous carriers, diluents, solvents, or vehicles include water, ethanol, polyols (such as glycerol, propylene glycol, polyethylene glycol and the like), carboxymethylcellulose and suitable mixtures thereof, vegetable oils (such as olive oil) and injectable organic esters such as ethyl oleate. In an aspect, a pharmaceutical carrier employed can be a solid, liquid, or gas. In an aspect, examples of solid carriers can include lactose, terra alba, sucrose, talc, gelatin, agar, pectin, acacia, magnesium stearate, and stearic acid. In an aspect, examples of liquid carriers can include sugar syrup, peanut oil, olive oil, and water. In an aspect, examples of gaseous carriers can include carbon dioxide and nitrogen. In preparing a disclosed composition for oral dosage form, any convenient pharmaceutical media can be employed. For example, water, glycols, oils, alcohols, flavoring agents, preservatives, coloring agents and the like can be used to form oral liquid preparations such as suspensions, elixirs and solutions; while carriers such as starches, sugars, microcrystalline cellulose, diluents, granulating agents, lubricants, binders, disintegrating agents, and the like can be used to form oral solid preparations such as powders, capsules and tablets. Because of their ease of administration, tablets and capsules are the preferred oral dosage units whereby solid pharmaceutical carriers are employed. Optionally, tablets can be coated by standard aqueous or nonaqueous techniques. Proper fluidity can be maintained, for example, by the use of coating materials

such as lecithin, by the maintenance of the required particle size in the case of dispersions and by the use of surfactants. These compositions can also contain adjuvants such as preservatives, wetting agents, emulsifying agents and dispersing agents. Prevention of the action of microorganisms can be ensured by the inclusion of various antibacterial and antifungal agents such as paraben, chlorobutanol, phenol, sorbic acid and the like. It can also be desirable to include isotonic agents such as sugars, sodium chloride and the like. Prolonged absorption of the injectable pharmaceutical form can be brought about by the inclusion of agents, such as aluminum monostearate and gelatin, which delay absorption. Injectable depot forms are made by forming microencapsule matrices of the drug in biodegradable polymers such as polylactide-polyglycolide, poly(orthoesters) and poly(anhydrides). Depending upon the ratio of drug to polymer and the nature of the particular polymer employed, the rate of drug release can be controlled. Depot injectable formulations are also prepared by entrapping the drug in liposomes or microemulsions that are compatible with body tissues. The injectable formulations can be sterilized, for example, by filtration through a bacterial-retaining filter or by incorporating sterilizing agents in the form of sterile solid compositions which can be dissolved or dispersed in sterile water or other sterile injectable media just prior to use. Suitable inert carriers can include sugars such as lactose. Desirably, at least 95% by weight of the particles of the active ingredient have an effective particle size in the range of 0.01 to 10 micrometers.

[0102] As used herein, the term “excipient” refers to an inert substance which is commonly used as a diluent, vehicle, preservative, binder, or stabilizing agent, and includes, but is not limited to, proteins (e.g., serum albumin, etc.), amino acids (e.g., aspartic acid, glutamic acid, lysine, arginine, glycine, histidine, etc.), fatty acids and phospholipids (e.g., alkyl sulfonates, caprylate, etc.), surfactants (e.g., SDS, polysorbate, nonionic surfactant, etc.), saccharides (e.g., sucrose, maltose, trehalose, etc.) and polyols (e.g., mannitol, sorbitol, etc.). See, also, for reference, Remington's Pharmaceutical Sciences, (1990) Mack Publishing Co., Easton, Pa., which is hereby incorporated by reference in its entirety.

[0103] As used herein, the term “package insert” is used to refer to instructions customarily included in commercial packages of therapeutic products, that contain information about the indications, usage, dosage, administration, contraindications and/or warnings concerning the use of such therapeutic products.

[0104] As used herein, the term “in combination” in the context of the administration of one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof includes the use of more than one therapy (e.g., additional therapeutic agents). Administration “in combination with” one or more additional therapeutic agents includes simultaneous (e.g., concurrent) and consecutive administration in any order. The use of the term “in combination” does not restrict the order in which therapies are administered to a subject. By way of non-limiting example, a first therapy (e.g., one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof) may be administered prior to (e.g., 1 minute, 15 minutes, 30 minutes, 45 minutes, 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 12 hours, 24

hours, 48 hours, 72 hours, 96 hours, 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, 6 weeks, 8 weeks, 8 weeks, 9 weeks, 10 weeks, 11 weeks, or 12 weeks), concurrently, or after (e.g., 1 minute, 15 minutes, 30 minutes, 45 minutes, 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 12 hours, 24 hours, 48 hours, 72 hours, 96 hours, 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, 6 weeks, 7 weeks, 8 weeks, 9 weeks, 10 weeks, 11 weeks, or 12 weeks or longer) the administration of a second therapy (e.g., one or more of the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof or one or more additional therapeutic agents) to a subject having or diagnosed with a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0105] Disclosed are the components to be used to prepare the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof as well the disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof used within the methods disclosed herein. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective combinations and permutation of these compounds cannot be explicitly disclosed, each is specifically contemplated and described herein. For example, if a particular compound is disclosed and discussed and a number of modifications that can be made to a number of molecules including the compounds are discussed, specifically contemplated is each and every combination and permutation of the compound and the modifications that are possible unless specifically indicated to the contrary. Thus, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F and an example of a combination molecule, A-D is disclosed, then even if each is not individually recited each is individually and collectively contemplated meaning combinations, A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are considered disclosed. Likewise, any subset or combination of these is also disclosed. Thus, for example, the sub-group of A-E, B-F, and C-E would be considered disclosed. This concept applies to all aspects of this application including, but not limited to, steps in methods of making and using the compositions of the invention. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspects or combination of aspects of the disclosed methods.

B. RNA Tertiary Structures

[0106] RNA tertiary structure is defined as the arrangement and interaction of secondary structure building blocks in three-dimensional (3D) space. RNA tertiary structure is defined as the three-dimensional arrangement of RNA building blocks, which include helical duplexes, triple-stranded structures, and other components that are held together through connections collectively termed RNA tertiary interactions. Biophysical techniques are being used to elucidate the driving forces for tertiary structure formation and the mechanisms for its stabilization.

[0107] RNA tertiary folding is promoted by maximization of base stacking, much like the hydrophobic effect that drives protein folding. RNA folding also requires electrostatic stabilization, both through charge screening and site binding of metals, and it is enhanced by desolvation of the phosphate backbone. A major determinant for overall tertiary RNA architecture is local conformation in secondary-structure junctions, which are regions from which two or more duplexes project. At junctions and other structures, such as pseudoknots and kissing loops, adjacent helices stack on one another, and these coaxial stacks play a major role in dictating the overall architectural form of an RNA molecule. In addition to RNA junction topology, a second determinant for RNA tertiary structure is the formation of sequence-specific interactions. Networks of triple helices, tetraloop receptor interactions, and other sequence-specific contacts establish the framework for the overall tertiary fold. The third determinant of tertiary structure is the formation of stabilizing stacking and backbone interactions, and many are not sequence specific. For example, ribose zippers allow 2'-hydroxyl groups on different RNA strands to form networks of interdigitated hydrogen bonds, serving to seal strands together and thereby stabilize adjacent substructures. These motifs often require monovalent and divalent cations, which can interact diffusely or through chelation to specific RNA functional groups.

Stacking and Coaxial Helices

[0108] Stacking of the aromatic nucleic acid bases is one of the most important driving forces during formation of RNA structure. In general, an RNA molecule will attempt to maximize base stacking, particularly at helical termini. If two helices are next to each other (e.g., separated by a phosphodiester linkage), their terminal base pairs will stack, and the helices will become colinear, resulting in a "coaxially stacked" substructure. The earliest glimpses of nucleic acid tertiary structure, in the form of tRNA crystal structures, revealed that coaxial stacking of helices determines the overall molecular shape. Each of the four helices in the "cloverleaf" tRNA secondary structure chooses a stacking partner, and these pairs of helices stack end-on-end, forming two long helices that ultimately arrange themselves through tertiary interactions. Numerous subsequent studies have shown that the choice of stacking partners among sets of helices is a critical determinant of RNA structural fate. The coaxial stacking of helices at junctions is thermodynamically favorable, and the free energy gained is sequence dependent, closely following the trends observed for nearest neighbor interactions in formation of RNA secondary structure. From 0.5 to 3.0 kcal/mol of free energy is released upon formation of a coaxial stack, and these energetic differences can influence choice of coaxial stacking partners at a junction.

[0109] Tertiary structures such as kissing loops and pseudoknots are composed of coaxially stacked helices and therefore represent RNA structures that are almost completely dictated by the forces driving coaxial stacking. At RNA junctions, duplexes immediately next to each other tend to coaxially stack, to minimize folding free energy. However, determining which helices are actually adjacent can be difficult because "linker" nucleotides are not always flexible. Rather, linker nucleotides often form noncanonical base pairs that extend the helical terminus, thereby presenting an alternative interface with other helices. For example,

A G, C C, and G U pairs are common at helical termini, where they serve as energetically favorable interfaces for coaxial stacks. Although the role of sequence is central, coaxial stacking choice can be strongly dictated by ionic conditions and by the topological constraints of intervening junctions.

Kissing Loops

[0110] Long-range base pairings between hairpin stem loops are known as "kissing" loops. These interactions are mediated by loop nucleotides that interact through complementary Watson Crick base pairs. Kissing loops have been found in many RNAs, including tRNA, mRNA, and rRNA. Stable loop-loop interactions can occur with as few as two base pairs between the loops. Kissing loop interactions are commonly used by retroviruses to initiate dimerization of genomic RNA. The HIV-1 dimerization initiation site (DIS) is a well-studied example of this interaction. The self-complementary loop forms a stable interaction involving six base pairs that coaxially stack within a continuous helix that contains bulged purines.

Kink Turns

[0111] Structural Features. Kink turns (or K-turns) were identified during analysis of the large ribosomal subunit crystal structure, and they have been found in a diversity of RNAs. The K-turn is a helix loop helix motif that bends the RNA helical axis by $\sim 120^\circ$, resulting in a close juxtaposition of helical minor grooves. One helix is canonical, whereas the other is noncanonical with tandem G-A base pairs. The bend is initiated by a three-nucleotide loop that is typically purine-rich. The severe kink in the loop is facilitated by the G-A pairs, which form cross-strand stacking interactions that twist the backbone in a characteristic manner. The adenines of the tandem G-A pairs stack on each other and participate in A-minor interactions across the junction.

[0112] K-turns are often bound and stabilized by proteins. In isolation, the K-turn is in equilibrium between the kinked form and a more extended conformation. The kinked population is stabilized by high concentrations of metal ions, but it remains in equilibrium with an extended form. K-turns are not believed to provide a thermodynamic driving force for RNA tertiary folding, but rather require cooperation from surrounding proteins or RNA tertiary structure to stabilize them. The "hook turn" is a recurrent motif that was originally characterized during crystallographic investigations of a loop E motif from the 5S rRNA of a purple sulfur bacterium. This RNA, which undergoes a sudden reverse in direction stabilized by a sheared G-A pair and a reverse-Hoogsteen pair, was found to be recurrent by scanning structures of the ribosome using the motif-finding algorithm Primos. The same algorithm was also used to identify the constituent K-turns and S-turns within the ribosome, revealing that the S-turn occurs in two distinct forms.

Pseudoknots

[0113] Pseudoknots are RNA tertiary structures that arise from energetically favorable long-range interactions between loops and hairpins that are stabilized by coaxial stacking of the newly formed RNA helices. Such structures have been found and mapped across many species and RNAs, from human telomerase RNA to viral genomes, with biological functions that are still being elucidated. Notably,

pseudoknots have been found to be essential for viral replication in a number of viruses, including severe acute respiratory syndrome coronavirus-2 (SARS-COV), where their main function is in 1 programmed ribosomal frameshifting. Frameshifting is a desirable target in some viruses as essential proteins encoded by the secondary open reading frame (ORF1b) region are normally out of frame, thus requiring a re-arrangement of the ribosome to proceed with elongation and translation. In coronaviruses, re-arrangement takes place through the presence of a slippery site, a linker region, and the downstream regulatory pseudoknot. While normal ribosome positioning starts at the zero position of the slippery site, the encounter between the ribosome and the pseudoknot tertiary structure causes the ribosome to pause and re-arrange at the 1 position, which puts ORF1b in frame and allows for ribosomal elongation upon pseudoknot unwinding. Mutational studies as well as antisense peptide nucleic acid (PNA) targeting of the pseudoknot revealed its potential as an antiviral target to dramatically decrease replication and virion production.

Triple-Stranded RNA Structures

[0114] Base triples occur frequently in RNA tertiary structure. For example, there are 27 base triples in the 50S ribosomal subunit, and 10 triples in the 30S subunit. The number of possible base triples constrained by at least three hydrogen bonds is 840. In the *Tetrahymena* intron, the active site contains a sandwich of four base triples. Triple-stranded RNA structures contain a Watson Crick base-paired duplex that hydrogen bonds to a third strand. While some triples occur in the minor groove (A-minor motifs, vide supra), base triples can also occur within the narrow major groove of RNA. Base triples that involve a third strand in the major groove often utilize the “Hoogsteen” face of purines. One example of such an interaction is observed in the telomerase pseudoknot, which has a loop that lies in the helical major groove and forms a series of Hoogsteen base triples. Similarly, a series of stacked major groove triples comprises the conserved “catalytic triplex” within the group II intron active site, where it plays a key role in supporting reaction chemistry. A similar triplex network is hypothesized to exist within the active site of the eukaryotic spliceosome.

[0115] Hoogsteen triples are most commonly U-A·U, where the adenine N7 accepts a hydrogen bond from the U imino proton. The base triple C-G·C+ is isosteric with U-A·U but requires protonation of the third strand C in order to form a hydrogen bond with the guanine N7. The SAM-II riboswitch bound to S-adenosylmethionine contains a pseudoknot with an extended major-groove triple helix.

[0116] Another example was recently identified within the Kaposi’s sarcoma-associated herpesvirus, which produces a highly abundant noncoding RNA, the polyadenylated nuclear (PAN) RNA. PAN contains an expression and nuclear retention element (ENE) that prevents degradation of its message. The ENE contains an internal loop is made up of uracils, which clamp onto the poly-A tail of the message, thereby forming a triple stranded U-A·U structure.

Triple Helices

[0117] Base triples occur frequently in RNA and usually result from the insertion of a third single RNA strand into a duplex in the minor groove, via A-minor interactions, or in the major groove through Hoogsteen H-bonds. Base triples

can be found as part of other tertiary and quaternary structures to enhance their stability, such as within pseudoknots or ribosomal subunits. Stand-alone triple helices have recently been mapped within several non-coding RNAs (ncRNAs) and are linked to an increase in the half-life of the transcript by protecting the transcript from degradation. Despite their clear biological relevance, biophysical and structural methods for their characterization in vitro and in vivo are still limited. Triple helices have mainly been structurally investigated through NMR, FRET experiments, ultraviolet (UV)-visible, and X-ray diffraction, which have limitations in the throughput and/or reflection of the conformer population.

G-Quadruplexes

[0118] Initially discovered in DNA decades ago, G4s are folding structures formed within a G-rich strand or between two G-rich strands. Generally, a tetrad of guanines arranges co-planarly via Hoogsteen H-bond interactions further stabilized by interactions with a cation. Computational and experimental methods have helped map G4 structures in the promoter region of many eukaryotic and prokaryotic genes, with the most notable one being telomeric DNA. G4 overall structure depends on strand polarity, orientation of connecting loops, and cation stabilization, thereby making it a diverse topological family. Recent studies have supported the existence of G4s in several ncRNA, though these differ from DNA G4s as they can be transient and highly dynamic. Remarkably, RNA G4s have showed higher stability than their DNA counterparts due to the added H-bonds of the 2′-OH, as well as stronger stacking of the G-quartets. These key differences ultimately distinguish RNA G4s unique topological features, allowing for small molecule discrimination between RNA and DNA.

Tetraloop-Receptor Motifs

[0119] Tetraloop-receptor motifs are among the most common types of long-range RNA tertiary interaction. They have been observed in almost every large RNA crystal structure and have even been employed in the fabrication of RNA nanostructures. In every case, this motif involves a terminal hairpin loop that contains a signature sequence, and known examples of interacting tetraloops are the GNRA and GANC tetraloops. While other conserved tetraloops are known (such as UNCG and CUYG), these serve as stable caps for hairpin termini and do not typically function as tertiary interaction partners.

[0120] There are several types of receptor motifs for tetraloops, and they differ in their level of structural complexity. The most complex type of receptors are highly conserved internal loop motifs, such as the “11-nucleotide motif”, which specifically recognizes GAAA tetraloops, and the IC3 motif, which has a more relaxed specificity for GNRA tetraloops. The interaction between the GAAA tetraloop and the 11-nucleotide receptor is surprisingly stable.

[0121] Molecular interactions between GAAA tetraloops and the 11-nucleotide receptor are extensive and include two major components: (1) The second adenine of the GAAA tetraloop stacks on an A-A platform that is formed within the receptor loop. Note that in some instances, the A-A platform can be A-C. The second adenine of the tetraloop also forms supporting hydrogen bonds to the receptor. (2) The remain-

ing bases of the tetraloop engage in a network of hydrogen bonds with G C base pairs in the adjacent receptor helix.

[0122] A simpler class of receptors are tandem GC pairs, which represent the first type of tetraloop-receptor interaction to be visualized crystallographically. The interaction first appeared as a set of crystal contacts reported during structural studies of the hammerhead ribozyme. Later dubbed “A-minor motifs” (vide supra), the G C pairs within the minor groove of these receptors form hydrogen bonds with base and sugar substituents on tetraloop nucleotides 3 and 4. Similar variations on this subfamily were reported within group I introns, and they have subsequently been observed in many RNA crystal structures. By contrast, the GANC tetraloops interact exclusively through base stacking with their cognate receptors, which are simple extrahelical bulged purines. Occurring only in group IIC introns, these tetraloop-receptor interactions are highly conserved.

RNA Quadruplex Structures

[0123] Repeated stretches of guanine-rich sequences can form quadruple base pairs that stack on each other to form quadruplexes. Telomeric DNA contains such guanine repeats, and it has recently been discovered that telomeres are transcribed into a large noncoding RNA called TERRA (telomeric repeat-containing RNA). TERRA RNA has been implicated in regulation of telomere length, through telomerase inhibition and chromatin remodeling. The crystal structure of a quadruplex derived from human TERRA has been solved. Guanine quadruplexes are stabilized by potassium ions, which are chelated in the center of the quadruplex by the guanine oxygens. Electrospray mass spectrometry reveals that TERRA quadruplexes form stable multimers. An emerging body of evidence suggests that RNA quadruplex structures play an important role in regulating translation of mRNA and splicing of some pre-mRNAs. Such RNA quadruplexes are recognized by the fragile X mental retardation protein (FMRP), which acts as a repressor of translation.

Ribose and the 2'-Hydroxyl Group: Key Components of RNA Tertiary Structure

[0124] The 2'-hydroxyl group of the RNA backbone is a stabilizing component in many tertiary interactions, since it can make two hydrogen bonds, acting as both hydrogen-bond donor and acceptor. The prevalence of 2'-OH contacts was evident from crystal structures of tRNA, and their thermodynamic contribution to RNA tertiary structure was revealed through studies of the Tetrahymena ribozyme. Many subsequent biochemical and structural studies revealed the importance of 2'-hydroxyl groups in ribozymes and other large RNA molecules, where they contribute to many motifs, including the A-minor interaction. Structural studies show that 2'-OH groups usually form networks of interactions, forming arrays with different types of morphologies. The most common type of 2'-OH array is the ribose zipper motif, which brings two backbone strands into close proximity through interdigitated 2'-OH interactions, thereby stabilizing neighboring structural features. Ribose zippers were first reported in early studies of ribozyme structure, where they play a key role in supporting core architecture. Inspection of solved structures shows that groups of ribose zippers tend to surround and buttress more sequence-specific types of tertiary interactions, such as

tetraloop-receptor interactions, kissing loops, and S-turns. In this way, ribose zippers provide the “glue” that maintains stability of entire tertiary substructures. A single 2'-hydroxyl group within a tertiary interaction network may contribute 1-2 kcal/mol of interaction free energy, so the stabilizing influence of a ribose zipper is likely to be considerable. While most ribose zipper motifs would seem to be sequence independent (all nucleotides contain a ribose), some types of ribose zipper motifs are found in specific sequence contexts, and their formation may be directed by the surrounding structural environment.

C. MALAT1

[0125] One example of a biologically relevant triple helix is the one present at the 30-end of the metastasis-associated lung adenocarcinoma transcript 1 (MALAT1). Metastasis Associated Lung Adenocarcinoma Transcript 1 (MALAT1) is also known as Nuclear Enriched Abundant Transcript 2 (NEAT2) and was first identified in a microarray screen of tumors from patients with non-small cell lung cancer. MALAT1 was found to be upregulated in the tumors with a higher propensity to metastasize. The MALAT1 gene is encoded on human chromosome 11q13.1 and mouse chromosome 19qA. It is in a gene dense region with a very high syntenic evolutionary conservation.

[0126] This lncRNA gained attention as a therapeutic target after it was found to be overexpressed in several cancer types, and it has now been implicated in other disease-related pathways. Specifically, the 8.7-kb long immature MALAT1 transcript undergoes a multiple-step processing, including the cleavage of a 30-end 61-nucleotide (nt) tRNA-like structure. The process yields the blunt formation of a 30-end triple helix that has been associated with nuclear accumulation of the transcript and as an essential element for protection against degradation. A reported crystal structure obtained from a truncated version of the wild-type (WT) triple helix revealed 9 U,A-U base triples resulting from the major groove insertion of a genomically encoded A-rich 30-tail, interrupted by an essential C,G-C triple and a C-G doublet (Brown et al., 2014). Recently, two notable examples of small molecule targeting have been reported for the MALAT1 triple helix utilizing a focused library and high-throughput approach, respectively.

[0127] MALAT1 exhibits a remarkable sequence conservation with greater than 50% overall conservation in vertebrates and greater than 80% conservation at the 3' end of the transcript. This is one of the key distinguishing features of MALAT1 as very few lncRNAs show such a high level of evolutionary conservation. Less than 10% of all lncRNAs show exonic as well as structural conservation equivalent to that of protein coding genes. The MALAT1 transcript has been confirmed as a non-coding RNA as it exhibits low protein-coding potential using two independent coding potential calculating algorithms CPC2 and CPAT.

[0128] Human MALAT1 is ~8.7 knt long, whereas the mouse RNA is 6.7 knt long. It is transcribed by RNA polymerase II and its promoter has an accessible open chromatin architecture, which has been shown in several high-throughput studies and DNase sensitivity assays. The steady state expression level of MALAT1 is very high and is comparable to highly transcribed housekeeping genes, such as B-Actin. Further, MALAT1 is ubiquitously expressed across all tissues with an average median expression of about 150 TPM (transcripts per million) with highest

expression in ovaries with a median expression of 287 TPM. The abundance of MALAT1 in cells is likely the consequence of strong promoter activity coupled with increased stability of the transcribed RNA.

[0129] MALAT1 was originally classified as an intron-less transcript with a genomically-encoded poly A tract. However, with a number of deep sequencing efforts, several alternatively spliced isoforms and transcripts with alternative transcription start sites have been identified that are expressed during different physiological states such as cancer. The MALAT1 primary transcript is processed to yield the well characterized nuclear retained MALAT1 transcript, and from its 3' end a tRNA-like small RNA. The biogenesis of the small RNA is mediated by the tRNA processing machinery, RNase P and RNase Z. The 61-nucleotide tRNA-like MALAT1-associated small cytoplasmic RNA (mascRNA) is exported to the cytoplasm. The resultant 3' end of the nuclear MALAT1 transcript post-processing is not polyadenylated, however, it contains a genomically-encoded poly(A)-rich stretch which pairs with an upstream U-rich region and then adopts a unique triple helical conformation. This triple helical structure was first identified in the PAN (polyadenylated nuclear) RNA produced by the human oncogenic Kaposi sarcoma-associated-herpesvirus (KSHV) PAN RNA. The only other human or mouse RNA that exhibits such a structure is the ~20 knt Men-(NEAT1_2) RNA. The triple helical structure has been shown to confer stability and nuclear localization to MALAT1 in the absence of a true poly(A) tail and has been shown to bind several RNA binding proteins (RBPs) including METTL16 which is an m⁶A RNA methyl-transferase.

[0130] Taken together, the MALAT1 locus displays remarkable evolutionarily conserved secondary and tertiary structural features and an unusual 3' end processing mechanism. It is not fully apparent whether the full length MALAT1 RNA with its 3' end triplex structure, the processed tRNA-like RNA, and the natural antisense RNA have a concerted mechanism of action or if each component derived from this interesting locus has a disparate function. Further, high-throughput chemical mapping experiments have highlighted extensive epi-transcriptomic changes in the MALAT1 transcript, for example, m⁶A, pseudouridylation and 5-methyl cytosine. It has been shown that the addition of m⁶A at the A2577 position could destabilize the hairpin stem of MALAT1, making it accessible for RNA-binding proteins such as HNRNPC. Additional detailed molecular studies elucidating the transcriptional and post-transcriptional regulation of the MALAT1 locus will address these issues and allow us to further understand the regulation and function of the MALAT1 locus.

D. SCHLAP1

[0131] SWI/SNF Complex Antagonist Associated with Prostate Cancer 1 (SCHLAP1) is an RNA gene, and is affiliated with the lncRNA class. Diseases associated with SCHLAP1 include prostate disease and prostate cancer. SCHLAP1 is also identified as HGNC 48603, NCBI Entrez Gene 101669767, Ensembl ENSG00000281131, and OMIM 615568. Chromosome locus associated with prostate-1 (SchLAP1) is one of the most important lncRNAs. It is located in the nucleus. SchLAP1 is highly overexpressed in a subset of prostate cancers and is associated with lethal disease. However, little is known about the exact mechanism of SchLAP1 in prostate cancer development.

E. NEAT1

[0132] Nuclear Paraspeckle Assembly Transcript 1 (NEAT1) is an RNA gene and is affiliated with the lncRNA class. NEAT1 RNA, a highly abundant 4 kb ncRNA, is retained in nuclei in approximately 10 to 20 large foci that are completely coincident with paraspeckles, nuclear domains implicated in mRNA nuclear retention. Diseases associated with NEAT1 include Dengue Disease and Gastric Adenocarcinoma. Among its related pathways are Extrafollicular B cell activation by SARS-CoV-2. NEAT1 is also identified as HGNC 30815, NCBI Entrez Gene 283131, Ensembl ENSG00000245532, and OMIM 612769.

F. LMO7

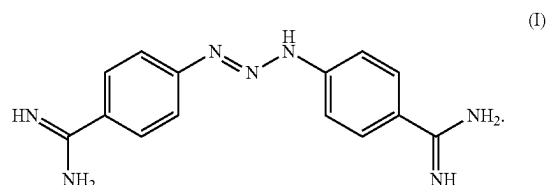
[0133] LMO7 (LIM Domain 7) is a protein coding gene. This gene encodes a protein containing a calponin homology (CH) domain, a PDZ domain, and a LIM domain, and may be involved in protein-protein interactions. Several alternatively spliced transcript variants encoding different isoforms have been found for this gene, however, the full-length nature of some variants is not known. Diseases associated with LMO7 include Townes-Brocks Syndrome and Hyperalphalipoproteinemia 1. Among its related pathways are Class I MHC mediated antigen processing and presentation and Signaling by ALK in cancer. Gene Ontology (GO) annotations related to this gene include ubiquitin-protein transferase activity and actin binding. LMO7 is also identified as HGNC 6646, NCBI Entrez Gene 4008, Ensembl ENSG00000136153, OMIM 604362, and UniProtKB/Swiss-Prot Q8WWI1.

G. Inventive Compounds

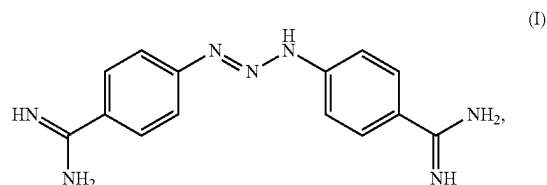
[0134] Disclosed herein are one or more compounds made by a method of assessing a library of shape diverse small molecules. Disclosed herein are one or more compounds made by a disclosed method.

[0135] Disclosed herein is a compound comprising a small molecule having a diminazene (DMZ) scaffold. Disclosed herein is a compound comprising a small molecule having a diminazene scaffold and a diamidine moiety.

[0136] Disclosed herein is a compound of formula



[0137] Disclosed herein is a compound of formula



having one or more of ortho, para, and/or meta substitutions.

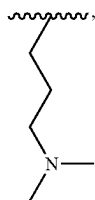
R5 can be

[0138] In an aspect, a disclosed ortho, para, and/or meta substitution can comprise any one of R0-R28 or a combination of any one of R0-R28.

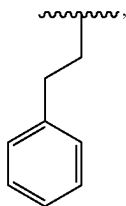
[0139] In an aspect, R0 can be



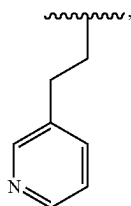
R1 can be



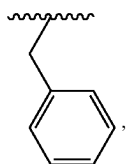
R2 can be



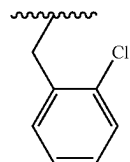
R3 can be



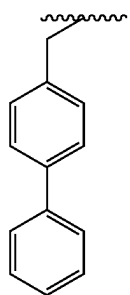
R4 can be



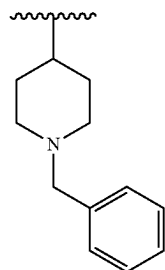
R6 can be



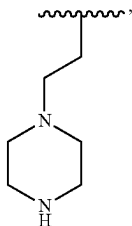
R7 can be



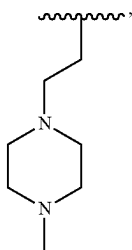
R8 can be



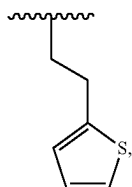
R9 can be



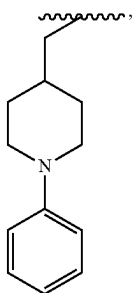
R10 can be



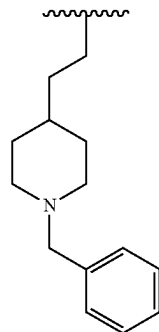
R11 can be



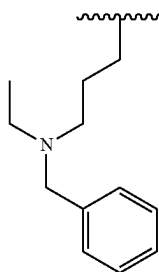
R12 can be



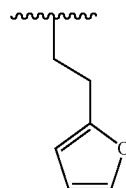
R13 can be



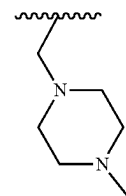
R14 can be



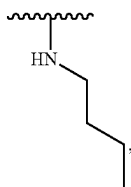
R15 can be



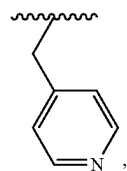
R16 can be



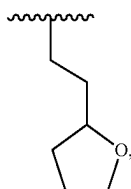
R17 can be



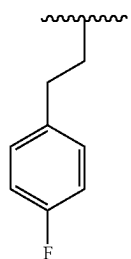
R22 can be



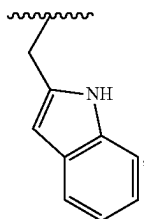
R18 can be



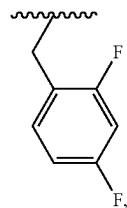
R23 can be



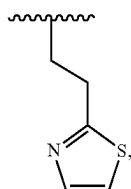
R19 can be



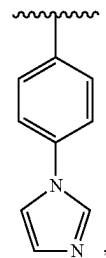
R24 can be



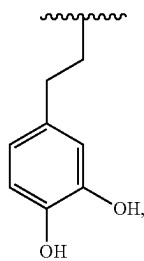
R20 can be



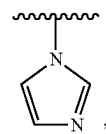
R25 can be



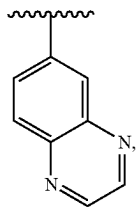
R21 can be



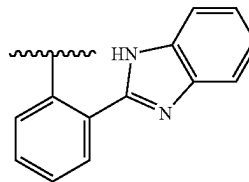
R26 can be



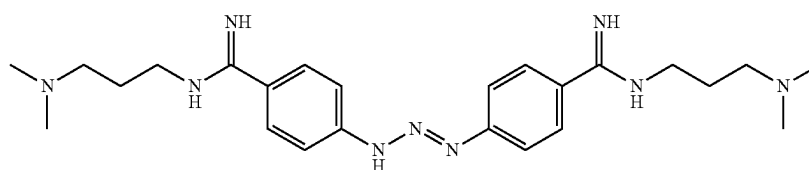
R27 can be



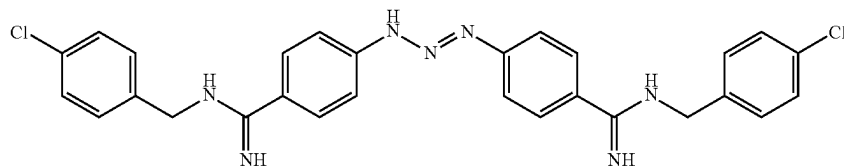
and R28 can be



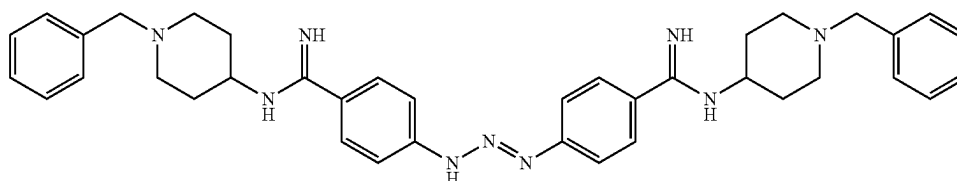
or any combination thereof.

[0140] Disclosed herein is a compound of formula

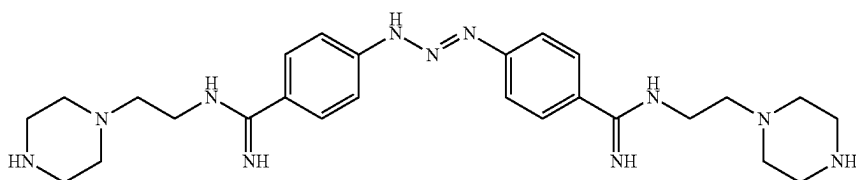
(DMZ-P1)

[0141] Disclosed herein is a compound of formula

(DMZ-P5)

[0142] Disclosed herein is a compound of formula

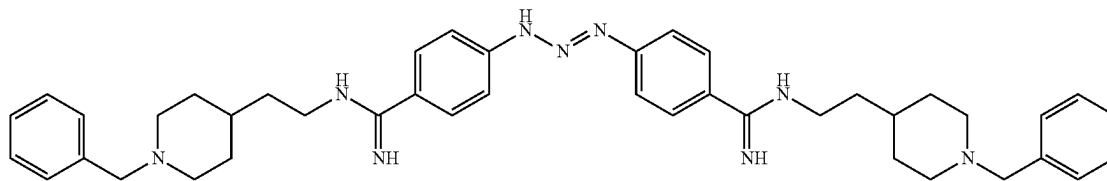
(DMZ-P8)

[0143] Disclosed herein is a compound of formula

(DMZ-P9)

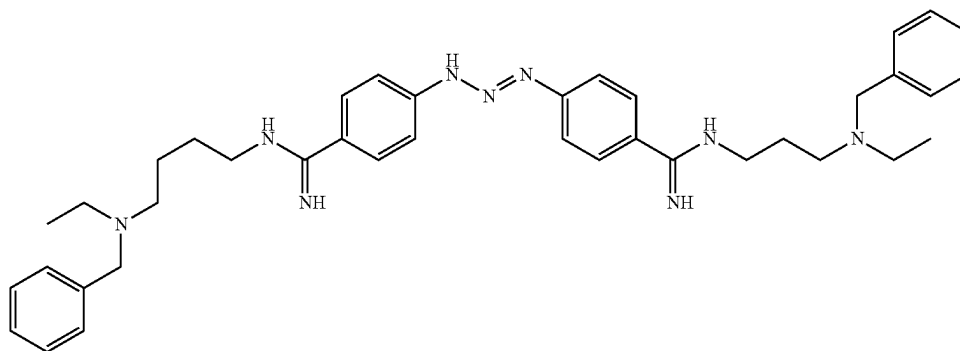
[0144] Disclosed herein is a compound of formula

(DMZ-P13)



[0145] Disclosed herein is a compound of formula

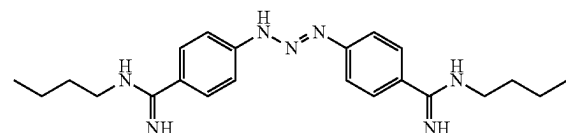
(DMZ-P17)



[0146] Disclosed herein is a compound of formula

[0148] Disclosed herein is a compound of formula

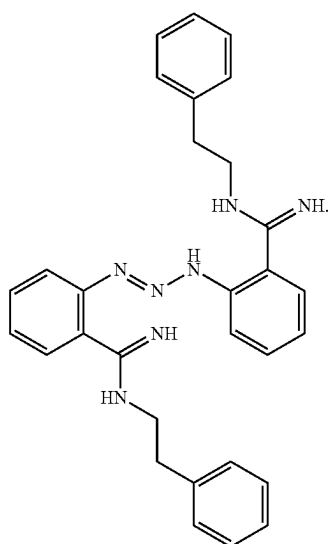
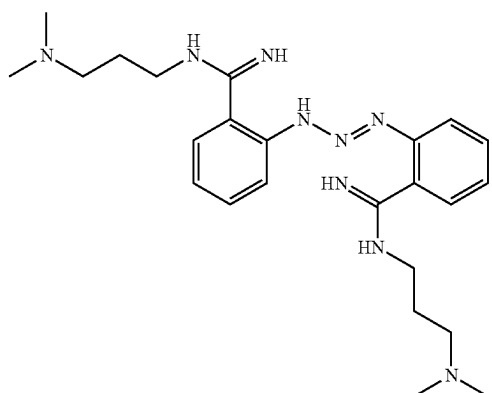
(DMZ-P17)



(DMZ-O2)

[0147] Disclosed herein is a compound of formula

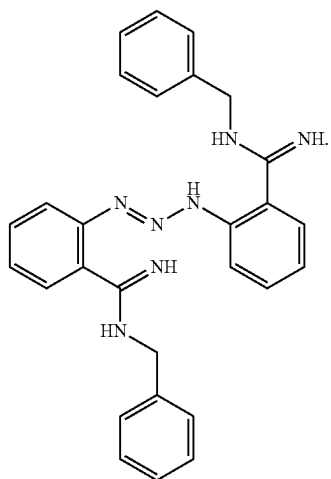
(DMZ-O1)



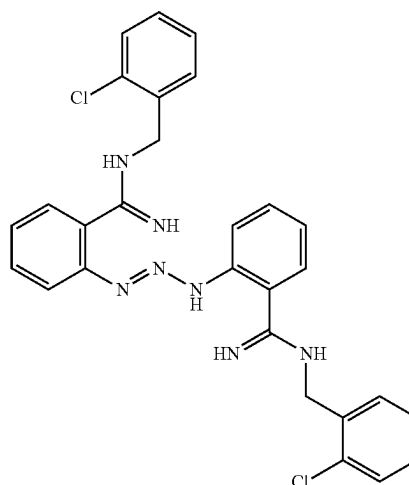
[0149] Disclosed herein is a compound of formula

[0151] Disclosed herein is a compound of formula

(DMZ-O4)

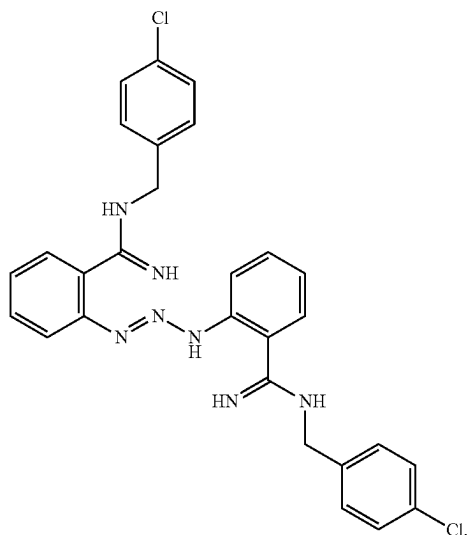


(DMZ-O6)



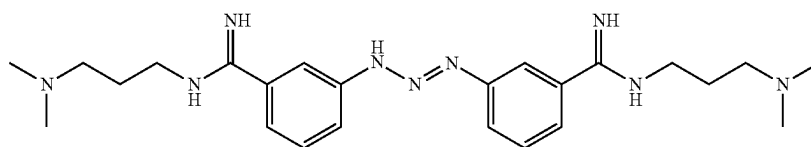
[0150] Disclosed herein is a compound of formula

(DMZ-O5)



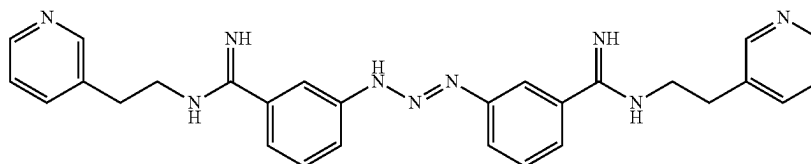
[0152] Disclosed herein is a compound of formula

(DMZ-M1)



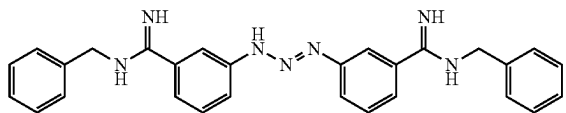
[0153] Disclosed herein is a compound of formula

(DMZ-M3)



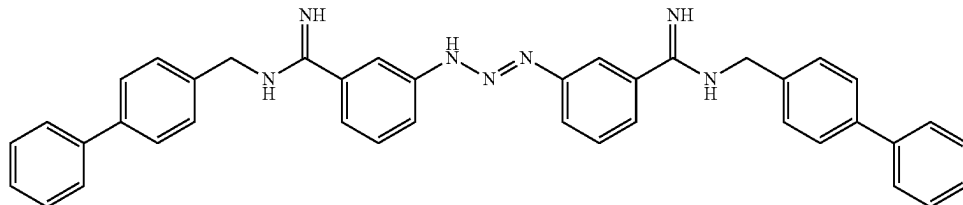
[0154] Disclosed herein is a compound of formula

(DMZ-M4)



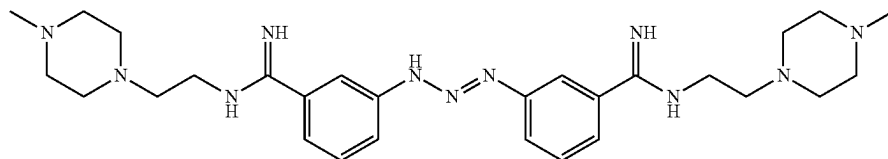
[0155] Disclosed herein is a compound of formula

(DMZ-M7)



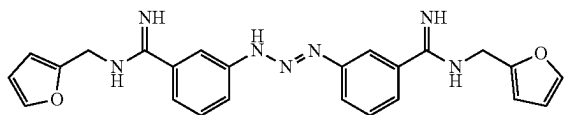
[0156] Disclosed herein is a compound of formula

(DMZ-M10)



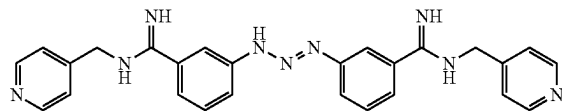
[0157] Disclosed herein is a compound of formula

(DMZ-M15)



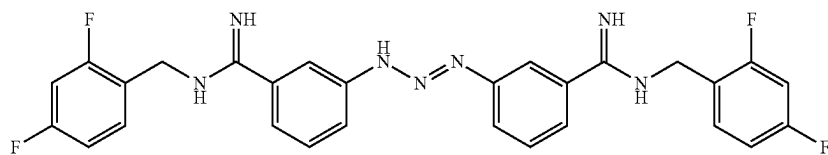
[0158] Disclosed herein is a compound of formula

(DMZ-M22)



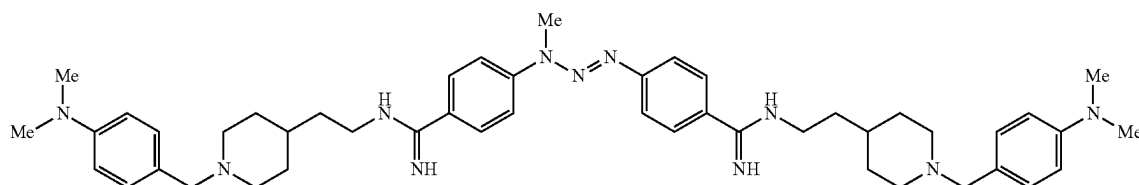
[0159] Disclosed herein is a compound of formula

(DMZ-M24)



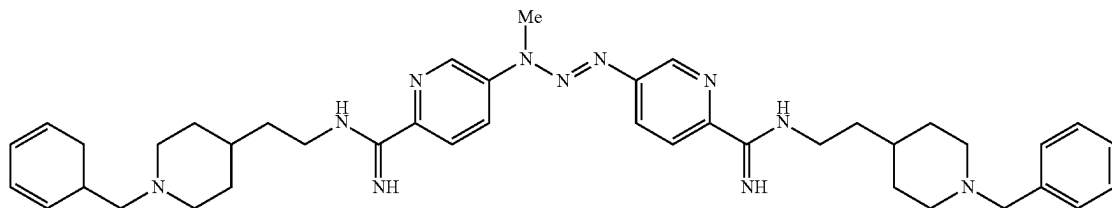
[0160] Disclosed herein is a compound of formula

(DMZ-N-Me-M9)



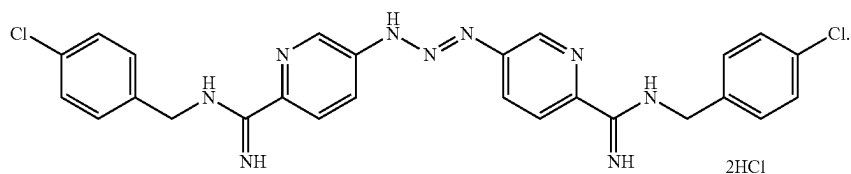
[0161] Disclosed herein is a compound of formula

(DMZ-N-Me-mPy-P13)



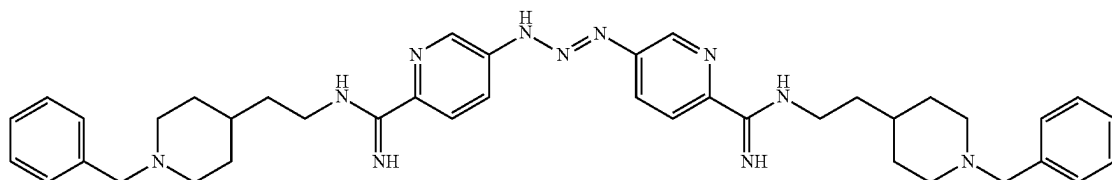
[0162] Disclosed herein is a compound of formula

(DMZ-mPy-P5-2HCl)



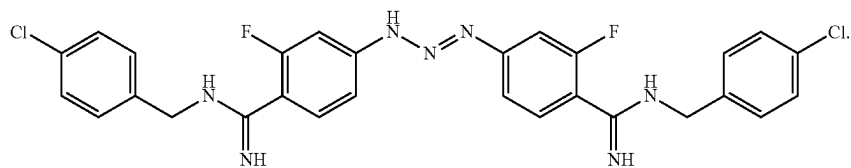
[0163] Disclosed herein is a compound of formula

(DMZ-mPy-P13)



[0164] Disclosed herein is a compound of formula

(DMZ-mF-P5)



N#Nc1ccc(cc1)/N=C(NCc2ccc(Cl)cc2)c3ccc(N)cc3

[0166] Disclosed herein is a compound of formula

c1ccccc1CN2CCCCC2CCNC(=O)c3ccc(cc3F)NN=Nc4ccc(cc4F)C(=O)NCC5CCCCC5N6CCCCC6Cc7ccccc7

[0167] Disclosed herein is a compound of formula

N#Cc1cc(F)ccc1N=Nc2ccc(F)c(c2)C(=O)NCCCN(Cc3ccccc3)CC4

[0168] Disclosed herein is a compound of formula

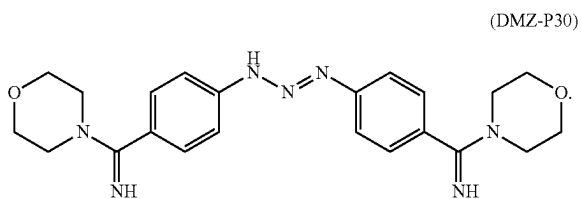
N#Nc1ccc(cc1C(=O)N2CCN(C2)c3ccccc3)N=Nc4ccc(cc4C(=O)N5CCN(C5)c6ccccc6)

[0169] Disclosed herein is a compound of formula

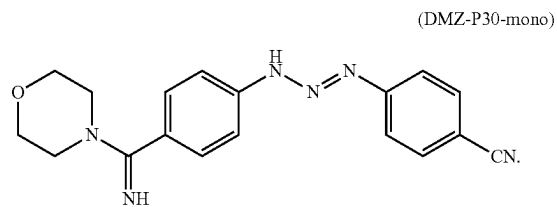
[0170] Disclosed herein is a compound of formula

N#CC1=CC=C(C=C1)/N=N/c2ccc(cc2)C(=O)N3CCc4ccccc4N3N=C(N)c1ccc(cc1)/N=N/c2ccc(cc2)C(=N)NCCc3ccc(Cl)cc3.[Cl-].[Cl-]

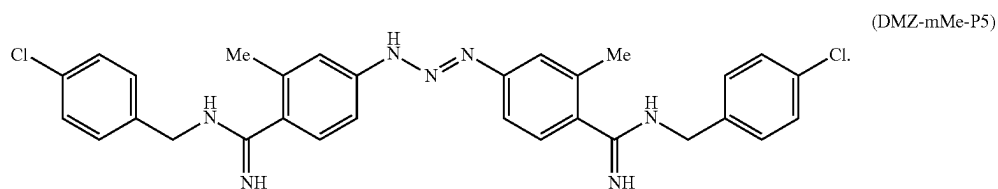
[0171] Disclosed herein is a compound of formula



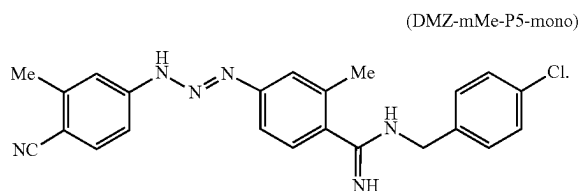
[0172] Disclosed herein is a compound of formula



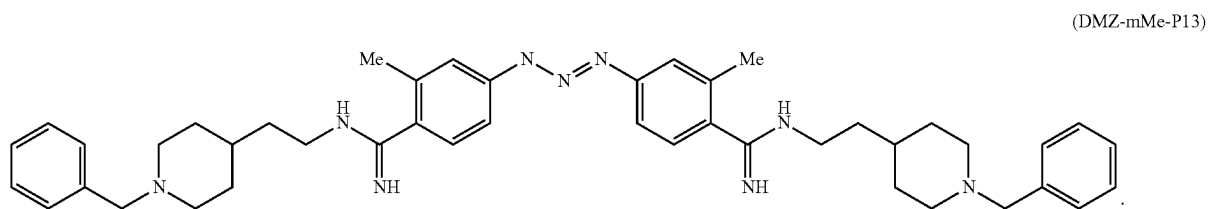
[0173] Disclosed herein is a compound of formula



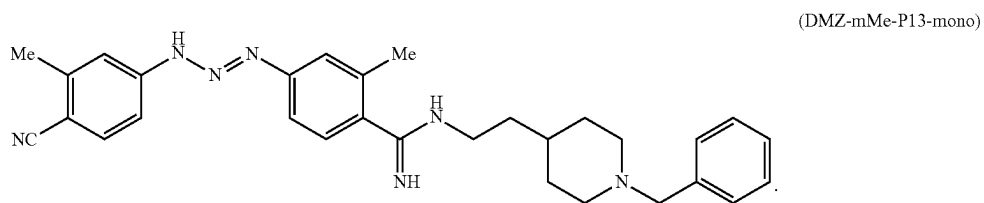
[0174] Disclosed herein is a compound of formula



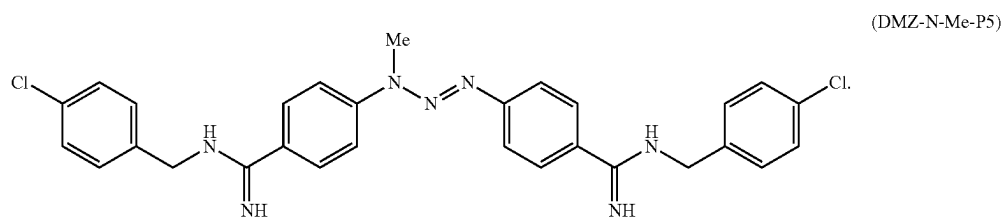
[0175] Disclosed herein is a compound of formula



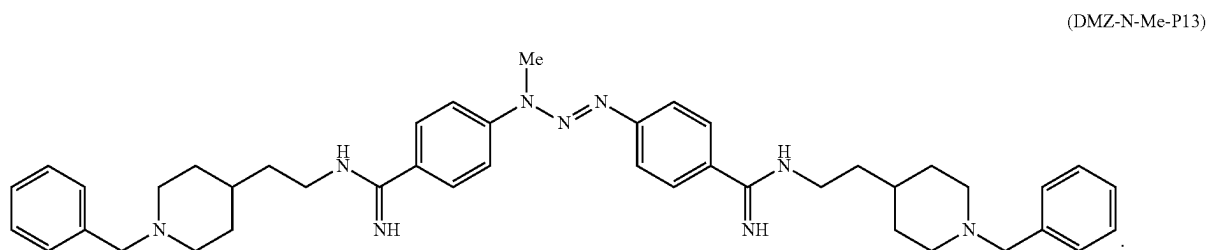
[0176] Disclosed herein is a compound of formula



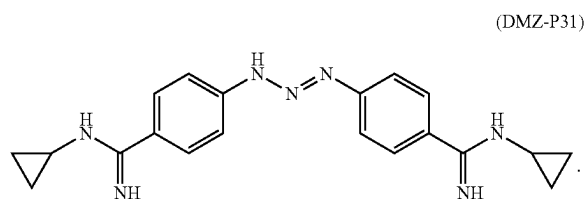
[0177] Disclosed herein is a compound of formula



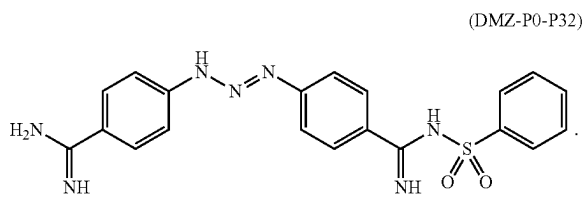
[0178] Disclosed herein is a compound of formula



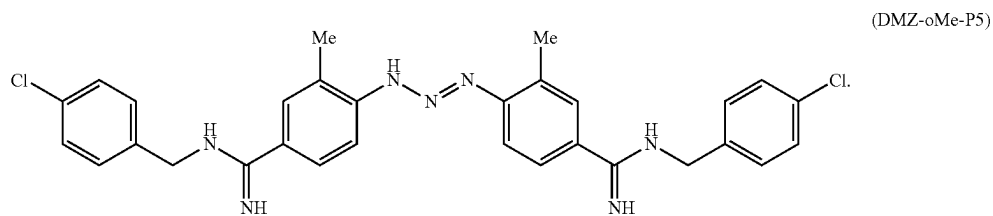
[0179] Disclosed herein is a compound of formula



[0180] Disclosed herein is a compound of formula

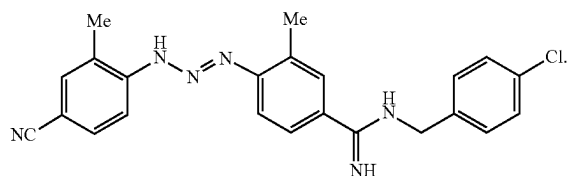


[0181] Disclosed herein is a compound of formula



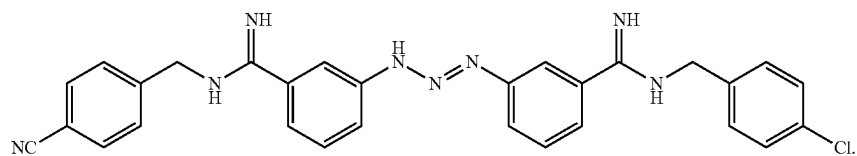
[0182] Disclosed herein is a compound of formula

(DMZ-oMe-P5-mono)



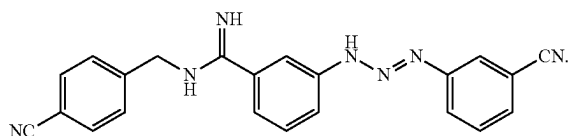
[0183] Disclosed herein is a compound of formula

(DMZ-M5)



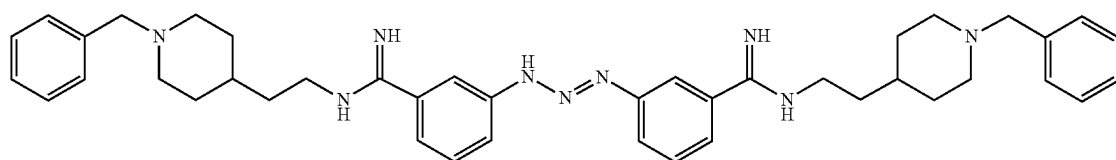
[0184] Disclosed herein is a compound of formula

(DMZ-M5-mono)



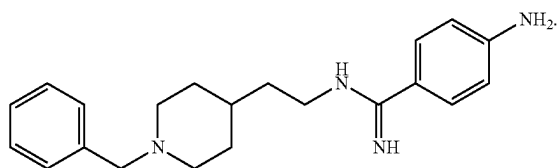
[0185] Disclosed herein is a compound of formula

(DMZ-M13)



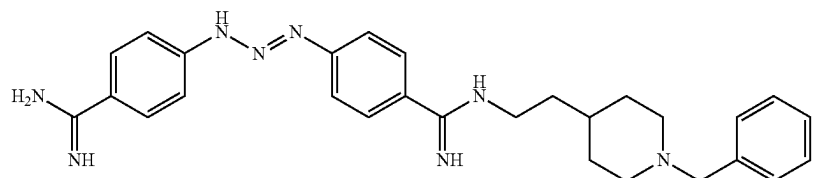
[0186] Disclosed herein is a compound of formula

(Aniline P13)

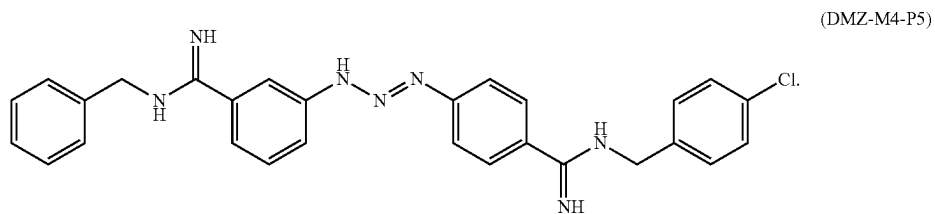


[0187] Disclosed herein is a compound of formula

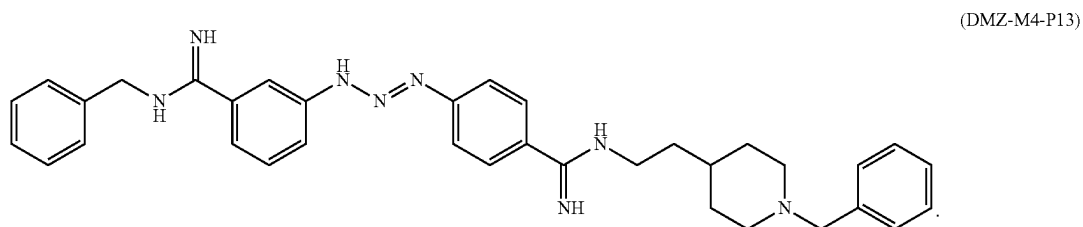
(DMZ-P0-P13)



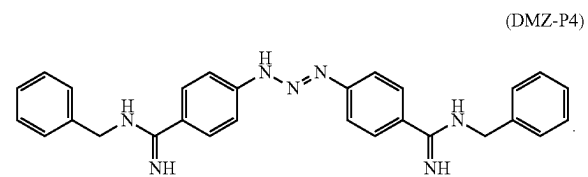
[0188] Disclosed herein is a compound of formula



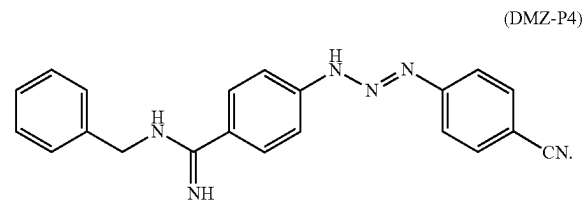
[0189] Disclosed herein is a compound of formula



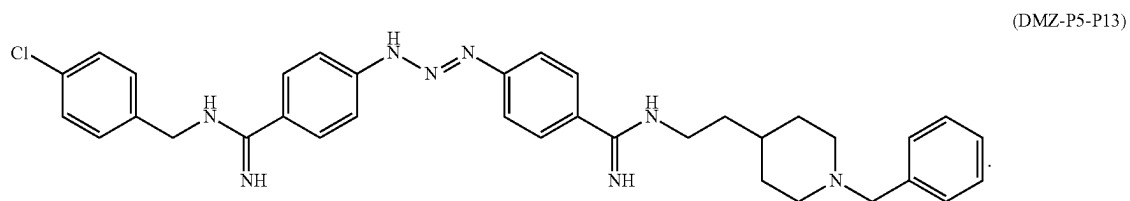
[0190] Disclosed herein is a compound of formula



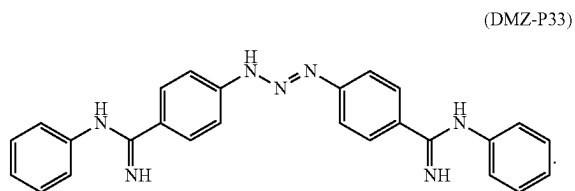
[0191] Disclosed herein is a compound of formula



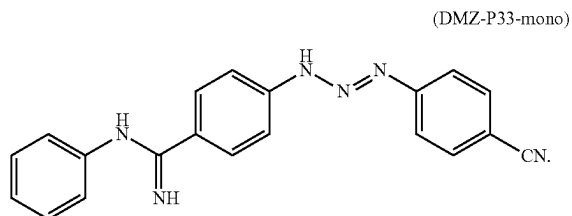
[0192] Disclosed herein is a compound of formula



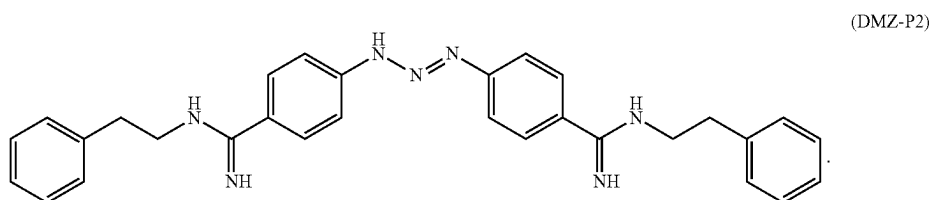
[0193] Disclosed herein is a compound of formula



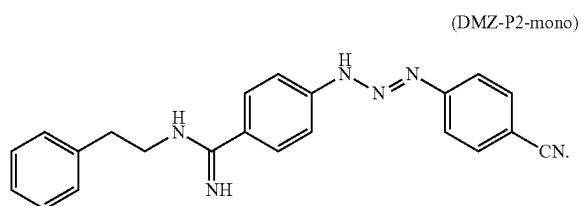
[0194] Disclosed herein is a compound of formula



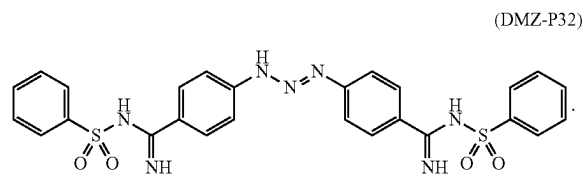
[0195] Disclosed herein is a compound of formula



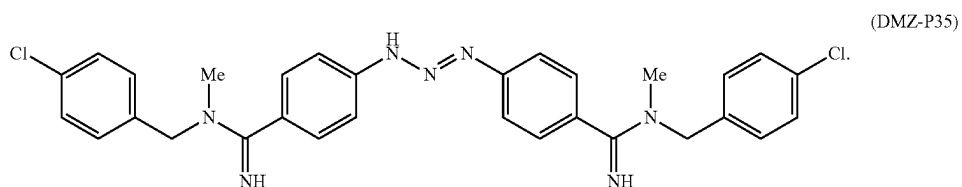
[0196] Disclosed herein is a compound of formula



[0197] Disclosed herein is a compound of formula

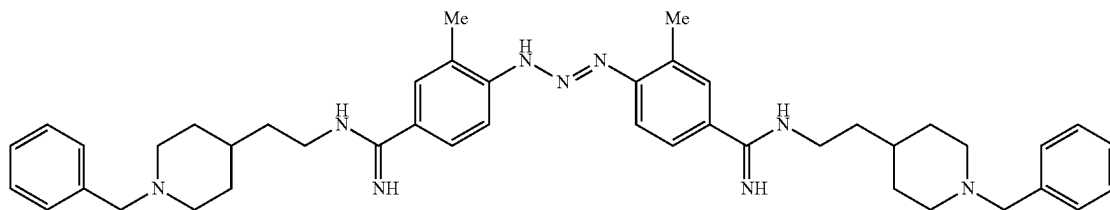


[0198] Disclosed herein is a compound of formula



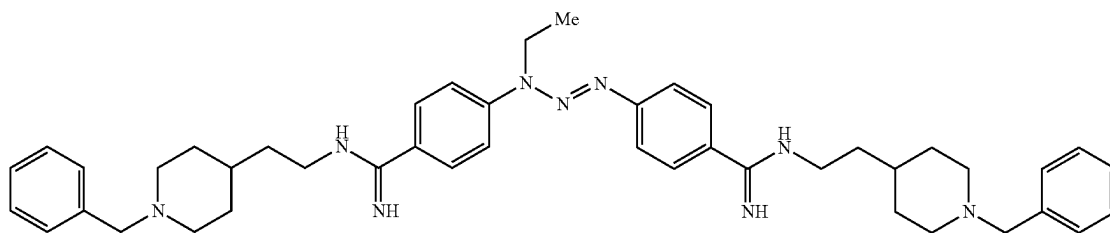
[0199] Disclosed herein is a compound of formula

(DMZ-oMe-P13)



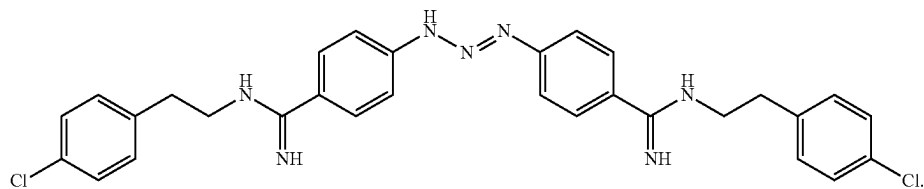
[0200] Disclosed herein is a compound of formula

(DMZ-N-Et-P13)



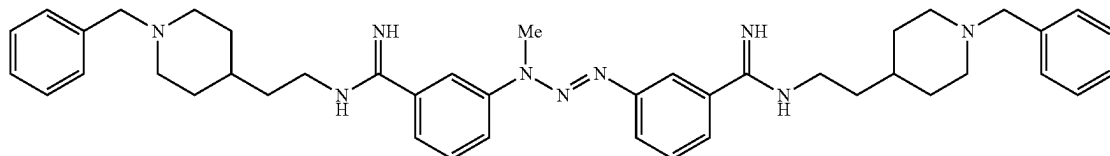
[0201] Disclosed herein is a compound of formula

(DMZ-P34)



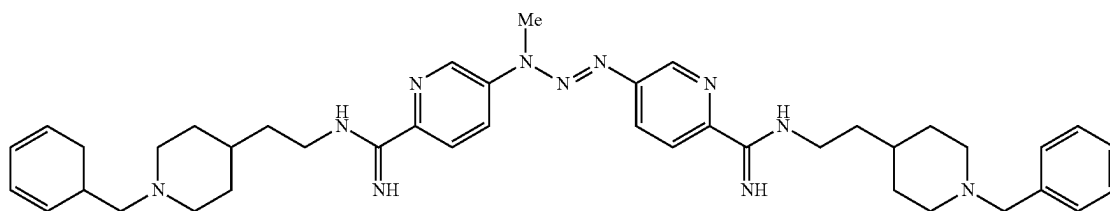
[0202] Disclosed herein is a compound of formula

(DMZ-N-Me-M13)

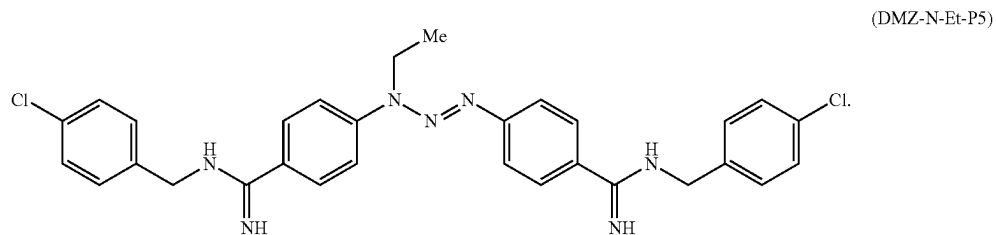


[0203] Disclosed herein is a compound of formula

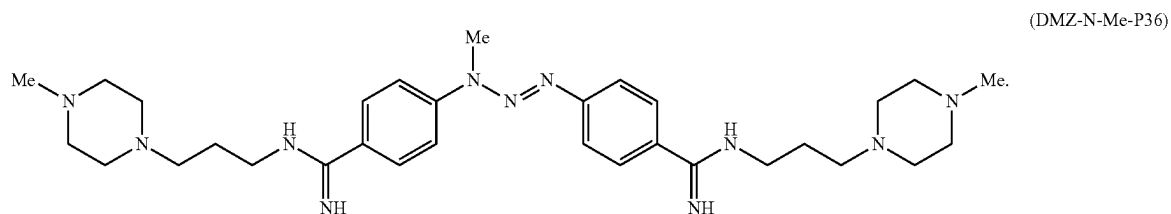
(DMZ-N-Me-mPy-P13)



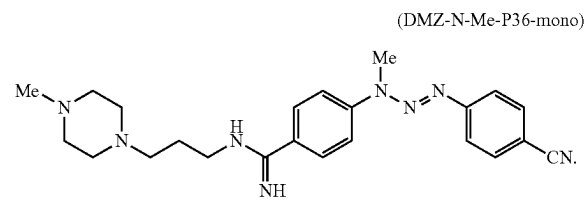
[0204] Disclosed herein is a compound of formula



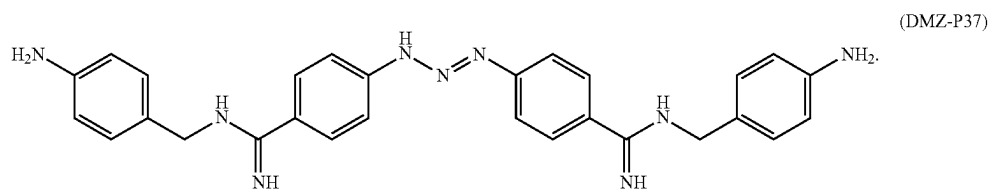
[0205] Disclosed herein is a compound of formula



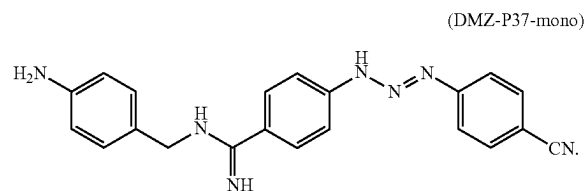
[0206] Disclosed herein is a compound of formula



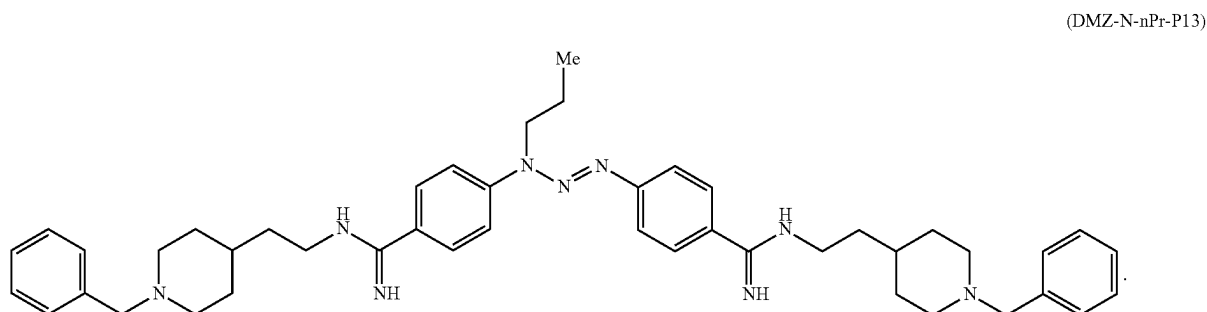
[0207] Disclosed herein is a compound of formula



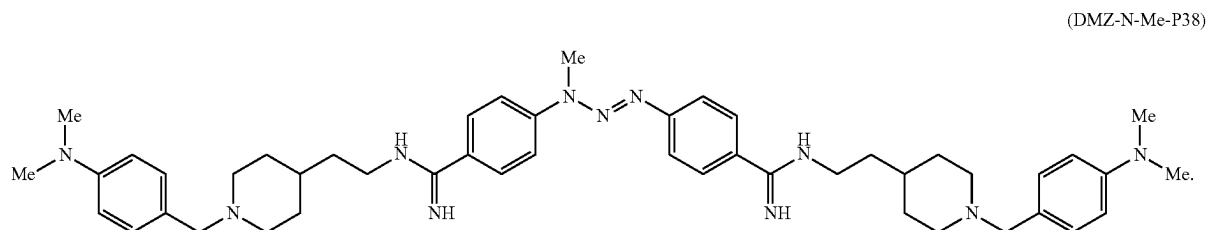
[0208] Disclosed herein is a compound of formula



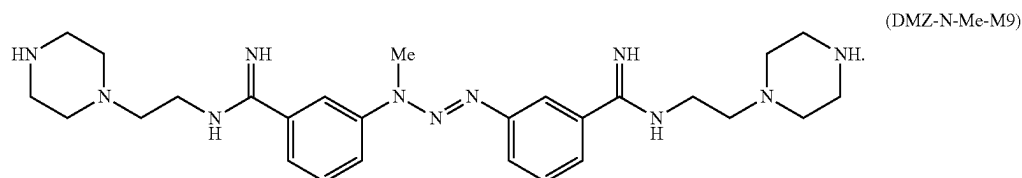
[0209] Disclosed herein is a compound of formula



[0210] Disclosed herein is a compound of formula



[0211] Disclosed herein is a compound of formula



[0212] Disclosed herein is a compound referred to as DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-

P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or a pharmaceutically acceptable salt, a hydrate, a prodrug, an ester, or a derivative thereof.

[02113] In an aspect, a disclosed DMZ compound can be a pharmaceutical acceptable salt, a hydrate, a prodrug, an ester, or a derivative thereof.

[02114] In an aspect, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P17, DMZ-O1, DMZ-O4, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M7, DMZ-M10, DMZ-NMe-P38, DMZ-NMe-mPy-P1, or any combination thereof can be destabilizing. In an aspect, DMZ-P0, DMZ-P1, DMZ-P5, DMZ-P14, DMZ-O2, DMZ-O5, DMZ-M4, DMZ-M15, DMZ-M22, or any combination thereof can be stabilizing.

[02115] In an aspect, a disclosed compound can target a disclosed RNA or a disclosed RNA structure. In an aspect, a disclosed RNA or a disclosed RNA structure can comprise a functionally important RNA.

[02116] In an aspect, a disclosed compound can target a bulge, a S-turn, an internal loop, a 3-multiloop, a hairpin loop, a pseudoknot, an apical loop, a kissing loop, a coaxial helix, a stacking helix, a two-way junction, a three-way junction, a four-way junction, or any combination thereof.

[02117] In an aspect, a disclosed compound can target a GNAA tetraloop, a GNGA loop, a T-loop, a C-loop, a E-loop, a Sarcin-ricin loop, a Kink-turn, a Reverse kink-turn, a Hook-turn, a Tandem shear, a Tetraloop-receptor motifs, an interacting tetraloop (e.g., GNRA, GANC, and GAAA tetraloops), a triple-stranded RNA, or any combination thereof.

[02118] In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed secondary RNA structure. In an aspect, a disclosed compound can increase the expression and/or level of a disclosed secondary RNA structure. In an aspect, a disclosed compound can modulate a disclosed secondary RNA structure. In an aspect, a disclosed compound can stabilize a disclosed secondary RNA structure. In an aspect, a disclosed compound can destabilize a disclosed secondary RNA structure.

[02119] In an aspect, stabilizing a disclosed secondary RNA structure can comprise measuring less degradation of the disclosed secondary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, stabilizing a disclosed secondary RNA structure can comprise measuring less degradation of the disclosed secondary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample.

[0220] In an aspect, destabilizing a disclosed secondary RNA structure can comprise measuring more degradation of the disclosed secondary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, destabilizing a disclosed secondary RNA structure can comprise measuring more degradation of the disclosed secondary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample.

[0221] In an aspect, a disclosed compound can destabilize a disclosed secondary RNA structure, can destabilize a disclosed secondary RNA structure, can be toxic to one or more cell types, can disrupt and/or impair the functionality of one or more cell types, can initiate, induce, promote, elicit, hasten, and/or cause the death of a cancer cell or a tumor cell, can reduce and/or minimize the likelihood of metastasis, or any combination thereof.

[0222] In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed tertiary RNA structure. In an aspect, a disclosed compound can increase the expression and/or level of a disclosed tertiary RNA structure. In an aspect, a disclosed compound can modulate a disclosed tertiary RNA structure. In an aspect, a disclosed compound can stabilize a disclosed tertiary RNA structure. In an aspect, a disclosed compound can destabilize a disclosed tertiary RNA structure.

[0223] In an aspect, stabilizing a disclosed tertiary RNA structure can comprise measuring less degradation of the disclosed tertiary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, stabilizing a disclosed tertiary RNA structure can comprise measuring less degradation of the disclosed tertiary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample.

[0224] In an aspect, destabilizing a disclosed tertiary RNA structure can comprise measuring more degradation of the disclosed tertiary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, destabilizing a disclosed tertiary RNA structure can comprise measuring more degradation of the disclosed tertiary RNA structure when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample.

[0225] In an aspect, a disclosed compound can destabilize a disclosed tertiary RNA structure, can destabilize a disclosed tertiary RNA structure, can be toxic to one or more cell types, can disrupt and/or impair the functionality of one or more cell types, can initiate, induce, promote, elicit, hasten, and/or cause the death of a cancer cell or a tumor cell, can reduce and/or minimize the likelihood of metastasis, or any combination thereof.

[0226] In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof. In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed lncRNA. In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed SARS-CoV-2 pseudoknot, a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof. In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof. In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response

element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof. In an aspect, a disclosed coronavirus can be 229E (alpha coronavirus), NL63 (alpha coronavirus), OC43 (beta coronavirus), HKU1 (beta coronavirus), MERS-COV (beta coronavirus), SARS-COV (beta coronavirus), and SARS-CoV-2, or a combination thereof.

[0227] In an aspect, a disclosed compound can increase the expression and/or level of a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof. In an aspect, a disclosed compound can increase the expression and/or level of a disclosed lncRNA. In an aspect, a disclosed compound can increase the expression and/or level of a disclosed SARS-CoV-2 pseudoknot, a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof. In an aspect, a disclosed compound can increase the expression and/or level of a disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof. In an aspect, a disclosed compound can increase the expression and/or level of a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof.

[0228] In an aspect, a disclosed compound can modulate a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof. In an aspect, a disclosed compound can modulate a disclosed lncRNA. In an aspect, a disclosed compound can modulate a disclosed SARS-CoV-2 pseudoknot, a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof. In an aspect, a disclosed compound can modulate a disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof. In an aspect, a disclosed compound can modulate a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end

triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof.

[0229] In an aspect, a disclosed compound can stabilize a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof. In an aspect, a disclosed compound can stabilize a disclosed lncRNA. In an aspect, a disclosed compound can stabilize a disclosed SARS-CoV-2 pseudoknot, a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof. In an aspect, a disclosed compound can stabilize a disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof. In an aspect, a disclosed compound can stabilize a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof.

[0230] In an aspect, a disclosed compound can destabilize a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof. In an aspect, a disclosed compound can destabilize a disclosed lncRNA. In an aspect, a disclosed compound can destabilize a disclosed SARS-CoV-2 pseudoknot, a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof. In an aspect, a disclosed compound can destabilize a disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof. In an aspect, a disclosed compound

can destabilize a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof.

[0231] In an aspect, stabilizing a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof can comprise measuring less degradation of the disclosed MALAT1 triple helix, the disclosed NEAT1 triplex, the disclosed PAN triplex, or any combination thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, stabilizing a disclosed lncRNA can comprise measuring less degradation of the lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, stabilizing a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof can comprise measuring less degradation of the disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation.

[0232] In an aspect, stabilizing a LMO7 functional secondary and/or tertiary RNA structure (including splice sites), an AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof can comprise measuring less degradation of the disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a the disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, stabilizing a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary struc-

ture in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof can comprise measuring less degradation of the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, the disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, the disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, the disclosed HIV element (e.g., a trans-activation response (TAR) element, the Rev response element (RRE), EESV, the frameshift site, the packing signal (interaction between (ψ) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), the disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, the disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation.

[0233] In an aspect, destabilizing a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof can comprise measuring more degradation of the disclosed MALAT1 triple helix, the disclosed NEAT1 triplex, the disclosed PAN triplex, or any combination thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, destabilizing a disclosed lncRNA can comprise measuring more degradation of the lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, destabilizing a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof can comprise measuring more degradation of the disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation.

[0234] In an aspect, destabilizing a LMO7 functional secondary and/or tertiary RNA structure (including splice sites), an AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof can comprise measuring more degradation of the disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a the disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, destabilizing a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) func-

tional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof can comprise measuring more degradation of the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, the disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, the disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, the disclosed HIV element (e.g., a trans-activation response (TAR) element, the Rev response element (RRE), EESV, the frameshift site, the packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), the disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, the disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation.

[0235] In an aspect, stabilizing a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof can comprise measuring less degradation of the disclosed MALAT1 triple helix, the disclosed NEAT1 triplex, the disclosed PAN triplex, or any combination thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, stabilizing a disclosed lncRNA thereof can comprise measuring less degradation of the disclosed lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, stabilizing a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof can comprise measuring less degradation of the disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, stabilizing a LMO7 functional secondary and/or tertiary RNA structure (including splice sites), an AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof can comprise measuring less degradation of the disclosed LMO7 functional second-

ary and/or tertiary RNA structure (including splice sites), a the disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, stabilizing a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof can comprise measuring less degradation of the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, the disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, the disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, the disclosed HIV element (e.g., a trans-activation response (TAR) element, the Rev response element (RRE), EESV, the frameshift site, the packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), the disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, the disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample.

[0236] In an aspect, destabilizing a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof can comprise measuring more degradation of the disclosed MALAT1 triple helix, the disclosed NEAT1 triplex, the disclosed PAN triplex, or any combination thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, destabilizing a disclosed lncRNA thereof can comprise measuring more degradation of the disclosed lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, destabilizing a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SCHLAP1, and combinations thereof can comprise measuring more degradation of

the disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, destabilizing a LMO7 functional secondary and/or tertiary RNA structure (including splice sites), an AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof can comprise measuring more degradation of the disclosed LMO7 functional secondary and/or tertiary RNA structure (including splice sites), a the disclosed AR functional secondary and/or tertiary RNA structure (including splice sites), or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, destabilizing a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof can comprise measuring more degradation of the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, the disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, the disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, the disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, the disclosed HIV element (e.g., a trans-activation response (TAR) element, the Rev response element (RRE), EESV, the frameshift site, the packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), the disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, the disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample.

[0237] In an aspect, a disclosed lncRNA can comprise any disclosed lncRNA (e.g., for example, those listed in one or more databases such as LincSNP, LncVar, LncRNADisease, EVLncRNAs, CLC, Lnc2Cancer, LncBook, NONCODE, LncRNAWiki, or any combination thereof).

[0238] In an aspect, a disclosed lncRNA can comprise ALAL-1, ANRIL, ATXN80S, CCAT2, CCR5AS, CUPID1/2, DA125942, DBE-T, DIRC3, DISC2, FALI, Gas5, HOTAIR, HULC, H19, IPW, IFNG-AS, Lnc-NR2F1, Lnc13, LINC00237, LINC00305, Linc-HELLP, LOC285194, MIAT, MIR2052HG, NEAT1, OVAL, PCAN-R1, PCAN-R2, PCAT1, PCAT1, PRAL, PTCSC3, PVT1, RMRP, RMST, RUNXOR, SAMMSON, SNHG17, or any combination thereof.

[0239] In an aspect, modulating a disclosed lncRNA can comprise stabilizing a disclosed lncRNA. In an aspect, modulating a disclosed lncRNA can comprise destabilizing a disclosed lncRNA. In an aspect, stabilizing a disclosed lncRNA can comprise measuring less degradation of a disclosed lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, stabilizing a disclosed lncRNA can comprise measuring less degradation of a disclosed lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, destabilizing a disclosed lncRNA can comprise measuring more degradation of a disclosed lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than a control or reference level of degradation. In an aspect, destabilizing a disclosed lncRNA can comprise measuring more degradation of a disclosed lncRNA when subjected to an enzymatic assay in the presence of the disclosed compound than the level of degradation in a control or reference sample. In an aspect, a disclosed lncRNA can comprise any disclosed lncRNA. In an aspect, modulating a disclosed lncRNA can comprise stabilizing a disclosed lncRNA. In an aspect, modulating a disclosed lncRNA can comprise destabilizing a disclosed lncRNA. In an aspect, a disclosed compound can decrease the expression and/or level of a disclosed lncRNA.

[0240] In an aspect, a disclosed enzymatic assay can comprise using an RNase. In an aspect, a disclosed enzymatic assay can comprise using one or more RNases. In an aspect, a disclosed RNase can comprise an endoribonuclease or an exoribonuclease. In an aspect, a disclosed endoribonuclease can comprise RNase A, RNase P, RNase H, RNase I, RNase III, RNase T1, RNase T2, RNase U2, RNase V1, RNase PhyM, RNase V, or any combination thereof. In an aspect, a disclosed RNase can comprise RNase A. In an aspect, a disclosed exoribonuclease can comprise RNase PH, RNase II, RNase R, RNase D, RNA T, or any combination thereof.

[0241] In an aspect, a disclosed control or reference level of degradation can be obtained. In an aspect, obtaining the control or reference level of degradation can comprise subjecting the disclosed tertiary RNA structure to a RNase in the absence of the compound. In an aspect, a disclosed level of degradation in a control or reference sample can be obtained. In an aspect, a disclosed level of degradation in a control or reference sample can be obtained by subjecting the tertiary RNA structure to a RNase in the absence of the disclosed compound. In an aspect, a disclosed control or reference level of degradation can be obtained. In an aspect, a disclosed control or reference level of degradation can be obtained by subjecting the RNA triplex to a RNase in the absence of the compound. In an aspect, a disclosed level of degradation in a control or reference sample can be obtained. In an aspect, a disclosed level of degradation in a control or

reference sample can be obtained subjecting the RNA triplex to a RNase in the absence of the disclosed compound.

[0242] In an aspect, a disclosed tertiary RNA structure can comprise one or more functionally important RNAs. In an aspect, disclosed functionally important RNAs can comprise human RNAs, viral RNAs, bacterial RNAs, fungal RNAs, yeast RNAs, or RNAs of an organism. In an aspect, a disclosed targeted RNA or a disclosed targeted RNA structure can comprise a bulge, a S-turn, an internal loop, a 3-multiloop, a hairpin loop, a pseudoknot, an apical loop, a kissing loop, a coaxial helix, a stacking helix, a two-way junction, a three-way junction, a four-way junction, or any combination thereof. In an aspect, a disclosed targeted RNA or a disclosed targeted RNA structure can comprise a GNAA tetraloop, a GNGA loop, a T-loop, a C-loop, a E-loop, a Sarcin-ricin loop, a Kink-turn, a Reverse kink-turn, a Hook-turn, a Tandem shear, a Tetraloop-receptor motifs, an interacting tetraloop (e.g., GNRA, GANC, and GAAA tetraloops), a triple-stranded RNA, or any combination thereof.

[0243] In an aspect, a disclosed tertiary RNA structure can comprise one or more disease-associated RNAs or disease relevant RNAs. In an aspect, a disclosed tertiary RNA structure can comprise a disclosed tertiary RNA structure. In an aspect, a disclosed tertiary RNA structure can comprise a NEAT1 triplex, a MALAT1 triplex, a PAN triplex, or any combination thereof. In an aspect, a disclosed tertiary RNA structure can comprise a SARS-CoV-2 pseudoknot, the HIV packaging signal, a complex junction in the long noncoding RNA SchLAP1, or any combination thereof. In an aspect, a disclosed tertiary RNA structure can comprise a MALAT1 triplex. In an aspect, a disclosed tertiary RNA structure can comprise a RNA triplex. In an aspect, a disclosed RNA triplex can comprise one or more functionally important RNAs. In an aspect, a disclosed RNA triplex can comprise one or more disease-associated RNA triplexes or one or more disease relevant RNA triplexes. In an aspect, a disclosed RNA triplex can comprise a NEAT1 triplex, a MALAT1 triplex, a PAN triplex, or any combination thereof. In an aspect, a disclosed RNA triplex can comprise a MALAT1 triplex. In an aspect, a disclosed tertiary RNA structure can comprise LMO7 tertiary RNA structure (including splice sites), an AR tertiary RNA structure (including splice sites), or any combination thereof. In an aspect, a disclosed secondary RNA structure can comprise LMO7 tertiary RNA structure (including splice sites), an AR secondary RNA structure (including splice sites), or any combination thereof.

[0244] In an aspect, a disclosed secondary structure and/or a disclosed tertiary structure can comprise a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix, a disclosed Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures, a disclosed SARS-CoV-2 5'-untranslated region structure and/or 3'-untranslated region, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot, a disclosed HIV element (e.g., a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof), a disclosed secondary structure or a disclosed tertiary structure in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-

associated herpesvirus, a hepatitis A virus, or a hepatitis C virus, a disclosed secondary or tertiary RNA structure in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus, or any combinations thereof.

[0245] In an aspect, a disclosed compound can physically disturb and/or disrupt and/or interrupt the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule. In an aspect, a disclosed compound can physically prevent the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule. In an aspect, a disclosed compound can be toxic to one or more cell types. In an aspect, a disclosed compound can disrupt and/or impair the functionality of one or more cell types. In an aspect, a disclosed cell type can comprise cancer cells or tumor cells.

[0246] In an aspect, disclosed cancer cells or disclosed tumor cells can comprise cells associated with adenocarcinoma of the ileum; adenocarcinoma; AIDS associated leukemias and adult T-cell leukemia lymphoma; basal cell carcinoma; basocellular cancer; bile duct cancer; biliary tract cancer; bladder cancer; brain cancer including glioblastomas and medulloblastomas; breast cancer; carcinosarcoma; cervical cancer; choriocarcinoma; clear cell renal cell carcinoma; colon cancer; embryonal carcinoma; embryonic testicular cancer; endometrial cancer; esophageal cancer; gastric cancer; head and neck cancer; hematological neoplasms (e.g., acute lymphocytic and myelogenous leukemia); hepatic cancer; intraepithelial neoplasms (e.g., Bowen's disease and Paget's disease); Kaposi's sarcoma; kidney cancer; liver cancer; lung cancer; lymphomas including Hodgkin's disease and lymphocytic lymphomas; malignant melanoma; malignant pleomorphic adenoma; melanoma; multiple myeloma; neuroblastomas; non-small cell lung cancer (NSCLC); oral cancer (e.g., squamous cell carcinoma); ovarian adenocarcinoma; ovarian cancer (e.g., including those arising and/or affect epithelial cells; stromal cells; germ cells and mesenchymal cells); ovarian teratocarcinoma; pancreatic cancer; papillary carcinoma; papillary renal cell carcinoma; placental choriocarcinoma; prostate cancer; rectal cancer; renal cell carcinoma; renal cancer including adenocarcinoma and Wilms tumor; sarcomas (e.g., leiomyosarcoma, rhabdomyosarcoma, liposarcoma, fibrosarcoma, osteosarcoma, etc.); skin cancer; small bowel adenocarcinoma; small bowel cancer; small cell lung cancer (SCLC); squamous cell carcinoma; squamous cell lung carcinoma; synovial sarcoma; teratocarcinoma; testicular embryonal carcinoma; testicular seminoma; testicular teratoma; testicular cancer including germinal tumors (seminoma, non-seminoma[teratomas; choriocarcinomas]), stromal tumors and germ cell tumors); thyroid cancer including thyroid adenocarcinoma and medullar carcinoma; transitional cell carcinoma; uterine cancer; or any combination thereof.

[0247] In an aspect, a disclosed compound can initiate, induce, promote, elicit, hasten, and/or cause the death of the cancer cell or tumor cell. In an aspect, a disclosed compound can reduce and/or minimize the likelihood of metastasis. In an aspect, cell death can be apoptotic and/or necrotic.

H. Inventive Compositions

[0248] Disclosed herein are compositions comprising one or more compounds made by a method of assessing a library

of shape diverse small molecules. Disclosed herein are compositions comprising one or more compounds made by a method of assessing a library of shape diverse small molecules. Disclosed herein are compositions comprising one or more compounds made by a disclosed method. Disclosed herein is a composition comprising a disclosed compound, and one or more carriers and/or excipients. Disclosed herein is a composition comprising a disclosed compound comprising a small molecule having a diminazene scaffold, and one or more carriers and/or excipients.

[0249] Disclosed herein is a composition comprising DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or any combination thereof, and one or more carriers and/or excipients.

[0250] In an aspect, a disclosed composition can target a disclosed RNA or a disclosed RNA structure. In an aspect, a disclosed RNA or a disclosed RNA structure can comprise a functionally important RNA.

[0251] In an aspect, a disclosed composition can target a bulge, a S-turn, an internal loop, a 3-multiloop, a hairpin loop, a pseudoknot, an apical loop, a kissing loop, a coaxial helix, a stacking helix, a two-way junction, a three-way junction, a four-way junction, or any combination thereof.

[0252] In an aspect, a disclosed composition can target a GNAA tetraloop, a GNGA loop, a T-loop, a C-loop, a E-loop, a Sarcin-ricin loop, a Kink-turn, a Reverse kink-turn, a Hook-turn, a Tandem shear, a Tetraloop-receptor motifs, an interacting tetraloop (e.g., GNRA, GANC, and GAAA tetraloops), a triple-stranded RNA, or any combination thereof.

[0253] In an aspect, a disclosed composition can modulate a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed composition can decrease the expression and/or activity level of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed composition can increase the expression and/or activity level of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed composition can increase the stability of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a dis-

closed composition can decrease the stability of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof.

[0254] Disclosed secondary RNA structures are discussed supra. Disclosed tertiary RNA structures are discussed supra. Disclosed RNA triplexes are discussed supra. Disclosed lncRNA are discussed supra.

[0255] In an aspect, stabilizing a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA can comprise measuring less degradation of the disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof in the presence of the disclosed composition when subjected to an enzymatic assay in the presence of the composition than a control or reference level of degradation. In an aspect, destabilizing a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA can comprise measuring more degradation of the disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof in the presence of the disclosed composition when subjected to an enzymatic assay in the presence of the composition than a control or reference level of degradation.

[0256] In an aspect, a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof can comprise one or more functionally important RNAs. In an aspect, a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof can comprise one or more disease-associated RNAs or disease relevant RNAs.

[0257] In an aspect, a disclosed composition can be toxic to one or more cell types. In an aspect, a disclosed composition can disrupt and/or impair the functionality of one or more cell types. In an aspect, a disclosed cell type can comprise cancer cells or tumor cells. Cancer cells and tumor cells are known to the art and are discussed supra. In an aspect, a disclosed composition can initiate, induce, promote, elicit, hasten, and/or cause the death of the cancer cell or tumor cell. In an aspect, a disclosed composition can reduce and/or minimize the likelihood of metastasis. In an aspect, cell death can be apoptotic and/or necrotic.

[0258] In an aspect, a disclosed composition can physically disturb and/or disrupt and/or interrupt the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule. In an aspect, a disclosed composition can physically prevent the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule.

I. Inventive Pharmaceutical Formulations

[0259] Disclosed herein is a pharmaceutical formulation comprising a disclosed compound and one or more pharmaceutically acceptable carriers. Disclosed herein is pharmaceutical formulation comprising a disclosed composition and one or more pharmaceutically acceptable carriers. Disclosed herein is a pharmaceutical formulation comprising one or more disclosed compounds and one or more pharmaceutically acceptable carriers. Disclosed herein is phar-

maceutical formulation comprising one or more disclosed compositions and one or more pharmaceutically acceptable carriers.

[0260] Disclosed herein is a pharmaceutical formulation comprising DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or any combination thereof, and one or more carriers and/or excipients.

[0261] In an aspect, a disclosed pharmaceutical formulation can target a disclosed RNA or a disclosed RNA structure. In an aspect, a disclosed RNA or a disclosed RNA structure can comprise a functionally important RNA.

[0262] In an aspect, a disclosed pharmaceutical formulation can target a bulge, a S-turn, an internal loop, a 3-multiloop, a hairpin loop, a pseudoknot, an apical loop, a kissing loop, a coaxial helix, a stacking helix, a two-way junction, a three-way junction, a four-way junction, or any combination thereof. In an aspect, a disclosed pharmaceutical formulation can target a GNAA tetraloop, a GNGA loop, a T-loop, a C-loop, a E-loop, a Sarcin-ricin loop, a Kink-turn, a Reverse kink-turn, a Hook-turn, a Tandem shear, a Tetraloop-receptor motifs, an interacting tetraloop (e.g., GNRA, GANC, and GAAA tetraloops), a triple-stranded RNA, or any combination thereof.

[0263] In an aspect, a disclosed pharmaceutical formulation can modulate a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed pharmaceutical formulation can decrease the expression and/or activity level of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed pharmaceutical formulation composition can increase the expression and/or activity level of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed pharmaceutical formulation can increase the stability of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. In an aspect, a disclosed pharmaceutical formulation can decrease the stability of a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof. Disclosed secondary RNA structures are discussed supra. Disclosed tertiary RNA

structures are discussed supra. Disclosed RNA triplexes are discussed supra. Disclosed lncRNA are discussed supra.

[0264] In an aspect, stabilizing a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA can comprise measuring less degradation of the disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof when subjected to an enzymatic assay in the presence of the disclosed pharmaceutical formulation a control or reference level of degradation. In an aspect, destabilizing a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA can comprise measuring more degradation of the disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof when subjected to an enzymatic assay in the presence of the disclosed pharmaceutical formulation than a control or reference level of degradation.

[0265] In an aspect, a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof can comprise one or more functionally important RNAs. In an aspect, a disclosed secondary RNA structure, a disclosed tertiary RNA structure, a disclosed RNA triplex, a disclosed lncRNA, or any combination thereof can comprise one or more disease-associated RNAs or disease relevant RNAs.

[0266] In an aspect, a disclosed pharmaceutical formulation can disrupt and/or impair the functionality of one or more cell types.

[0267] In an aspect, a disclosed pharmaceutical formulation composition can initiate, induce, promote, elicit, hasten, and/or cause the death of the cancer cell or tumor cell.

[0268] In an aspect, a disclosed pharmaceutical formulation can reduce and/or minimize the likelihood of metastasis. In an aspect, cell death can be apoptotic and/or necrotic.

[0269] In an aspect, a disclosed pharmaceutical formulation can physically disturb and/or disrupt and/or interrupt the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule. In an aspect, a disclosed pharmaceutical formulation can physically prevent the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule.

[0270] In an aspect, a disclosed pharmaceutical formulation can be toxic to one or more cell types. In an aspect, a disclosed composition can disrupt and/or impair the functionality of one or more cell types. In an aspect, a disclosed cell type can comprise cancer cells or tumor cells. Cancer cells and tumor cells are known to the art and are discussed supra. In an aspect, a disclosed pharmaceutical formulation can initiate, induce, promote, elicit, hasten, and/or cause the death of the cancer cell or tumor cell. In an aspect, a disclosed pharmaceutical formulation can reduce and/or minimize the likelihood of metastasis. In an aspect, cell death can be apoptotic and/or necrotic.

[0271] In an aspect, a disclosed pharmaceutical formulation can comprise (i) one or more active agents, (ii) biologically active agents, (iii) one or more pharmaceutically active agents, (iv) one or more immune-based therapeutic agents, (v) one or more clinically approved agents, or (vi) a combination thereof.

[0272] Disclosed herein are pharmaceutical formulations one or more compounds made by a method of assessing a library of shape diverse small molecules.

J. Inventive Library

[0273] Disclosed herein is a library of compounds made by a method of assessing a library of shape diverse small molecules.

[0274] Disclosed herein is a library comprising one or more disclosed small molecules having a diminazene scaffold. Disclosed herein is a library comprising one or more disclosed shape diverse small molecules having a diminazene scaffold. Disclosed herein is library comprising one or more disclosed small molecules having a diminazene scaffold and a diamidine moiety.

[0275] Disclosed herein is a library comprising one or more of DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or any combination thereof.

K. Methods of Treating a Subject

[0276] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations.

[0277] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, and treating a disease, condition, or disorder in the subject. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, and treating a disease, condition, or disorder in the subject. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, and treating a disease, condition, or disorder in the subject.

[0278] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein one or more secondary RNA structures are modulated. Disclosed

herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein one or more secondary RNA structures are modulated. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein one or more secondary RNA structures are modulated.

[0279] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein one or more tertiary RNA structures are modulated. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein one or more tertiary RNA structures are modulated. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein one or more tertiary RNA structures are modulated.

[0280] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein one or more secondary RNA structures are stabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein one or more secondary RNA structures are stabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein one or more secondary RNA structures are stabilized.

[0281] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein one or more tertiary RNA structures are stabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein one or more tertiary RNA structures are stabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein one or more tertiary RNA structures are stabilized.

[0282] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein one or more secondary RNA structures are destabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein one or more secondary RNA structures are destabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein one or more secondary RNA structures are destabilized.

[0283] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein one or more tertiary RNA structures are destabilized. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein one or more tertiary RNA structures are destabilized. Disclosed herein is a method of

treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein one or more tertiary RNA structures are destabilized.

[0284] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein the expression and/or activity level of one or more secondary RNA structures are decreased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein the expression and/or activity level of one or more secondary RNA structures are decreased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein the expression and/or activity level of one or more secondary RNA structures are decreased.

[0285] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein the expression and/or activity level of one or more tertiary RNA structures are decreased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein the expression and/or activity level of one or more tertiary RNA structures are decreased.

[0286] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein the expression and/or activity level of one or more secondary RNA structures are increased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein the expression and/or activity level of one or more secondary RNA structures are increased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein the expression and/or activity level of one or more secondary RNA structures are increased.

[0287] Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compounds, wherein the expression and/or activity level of one or more tertiary RNA structures are decreased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed compositions, wherein the expression and/or activity level of one or more tertiary RNA structures are decreased. Disclosed herein is a method of treating a subject in need thereof, the method comprising administering to a subject one or more disclosed pharmaceutical formulations, wherein the expression and/or activity level

[0288] In an aspect, a disclosed lncRNA can comprise any disclosed lncRNA (e.g., for example, those listed in one or more databases such as LincSNP, LncVar, LncRNADisease, EVLncRNAs, CLC, Lnc2Cancer, LncBook, NONCODE, LncRNAWiki, or any combination thereof).

[0289] In an aspect, a disclosed lncRNA can comprise ALAL-1, ANRIL, ATXN80S, CCAT2, CCR5AS, CUPID1/2, DA125942, DBE-T, DIRC3, DISC2, FALI, Gas5, HOTAIR, HULC, H19, IPW, IFNG-AS, Lnc-NR2F1, Lnc13, LINC00237, LINC00305, Linc-HELLP, LOC285194, MIAT, MIR2052HG, NEAT1, OVAL, PCAN-R1, PCAN-R2, PCAT1, PCAT1, PRAL, PTCSC3, PVT1, RMRP, RMST, RUNXOR, SAMMSON, SNHG17, or any combination thereof.

[0290] In an aspect, a disclosed RNA triplex can comprise a disclosed MALAT1 triple helix, a disclosed NEAT1 triplex, a disclosed PAN triplex, or any combination thereof.

[0291] In an aspect, a disclosed tertiary RNA structure can be a disclosed SARS-CoV-2 pseudoknot, a disclosed HIV packaging signal, a disclosed complex junction in the long noncoding RNA SchLAP1, and combinations thereof.

[0292] In an aspect of a disclosed method, (i) a disease, condition, or disorder in the subject can be treated, (ii) one or more symptoms of a disease, condition, or disorder in the subject can be attenuated, ameliorated, or eliminated, (iii) the onset of one or more symptoms of a disease, condition, or disorder in the subject can be delayed, (iv) the progression of the disease, condition, or disorder in the subject can be slowed and/or diminished, or any combination thereof.

[0293] In an aspect, a disclosed subject can be a human, can be male or female, and can be of any age (e.g., adult, adolescent, child, baby, etc.). In an aspect of a disclosed method, a disclosed subject can be suspected of having or can be diagnosed with having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, a disclosed subject can be symptomatic or asymptomatic. In an aspect of a disclosed method, a disclosed subject can be suspected of having or can be diagnosed with having a disease, condition, or disorder caused by, related to, and/or exacerbated by the absence of, lack of expression of, and/or lack of activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, a disclosed subject can be symptomatic or asymptomatic.

[0294] In an aspect of a disclosed method, the subject can have one or more cancers and/or one or more tumors. Cancers and tumors are known to the skilled person. In an aspect, a disclosed cancer can comprise adenocarcinoma of the ileum; adenocarcinoma; AIDS associated leukemias and adult T-cell leukemia lymphoma; basal cell carcinoma; basocellular cancer; bile duct cancer; biliary tract cancer; bladder cancer; brain cancer including glioblastomas and medulloblastomas; breast cancer; carcinosarcoma; cervical cancer; choriocarcinoma; clear cell renal cell carcinoma; colon cancer; embryonal carcinoma; embryonic testicular cancer; endometrial cancer; esophageal cancer; gastric cancer; head and neck cancer; hematological neoplasms (e.g., acute lymphocytic and myelogenous leukemia); hepatic cancer; intraepithelial neoplasms (e.g., Bowen's disease and Paget's disease); Kaposi's sarcoma; kidney cancer; liver cancer; lung cancer; lymphomas including Hodgkin's disease and lymphocytic lymphomas; malignant melanoma; malignant pleomorphic adenoma; melanoma; multiple myeloma; neuroblastomas; non-small cell lung cancer (NSCLC); oral cancer (e.g., squamous cell carcinoma); ovarian adenocarcinoma; ovarian cancer (e.g., including those arising and/or affect epithelial cells; stromal cells;

germ cells and mesenchymal cells); ovarian teratocarcinoma; pancreatic cancer; papillary carcinoma; papillary renal cell carcinoma; placental choriocarcinoma; prostate cancer; rectal cancer; renal cell carcinoma; renal cancer including adenocarcinoma and Wilms tumor; sarcomas (e.g., leiomyosarcoma, rhabdomyosarcoma, liposarcoma, fibrosarcoma, osteosarcoma, etc.); skin cancer; small bowel adenocarcinoma; small bowel cancer; small cell lung cancer (SCLC); squamous cell carcinoma; squamous cell lung carcinoma; synovial sarcoma; teratocarcinoma; testicular embryonal carcinoma; testicular seminoma; testicular teratoma; testicular cancer including germinal tumors (seminoma, non-seminoma[teratomas; choriocarcinomas]), stromal tumors and germ cell tumors); thyroid cancer including thyroid adenocarcinoma and medullar carcinoma; transitional cell carcinoma; uterine cancer; or any combination thereof.

[0295] In an aspect of a disclosed method, administering a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can comprise systemic administration and/or local administration. Methods of administration of known to the art. For example, in an aspect, administration can comprise oral administration, transdermal administration, administration by inhalation, nasal administration, topical administration, in utero administration, intrahepatic administration, intravaginal administration, epidural administration (such as epidural injection), intracerebroventricular (ICV) administration, ophthalmic administration, intraaural administration, depot administration, topical (skin) administration, otic administration, intra-articular (such as joint or vertebrate injection), intracerebral administration, rectal administration, sublingual administration, buccal administration, and parenteral administration, including injectable such as intravenous administration, intra-CSF administration, intra-cistern magna (ICM) administration, intra-arterial administration, intrathecal (ITH) administration, intramuscular administration, and subcutaneous administration.

[0296] Administration of a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can comprise administration directly into the CNS or the PNS. In an aspect, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can be administered via LNP administration.

[0297] In an aspect, administration can be continuous or intermittent. Administration can comprise a combination of one or more routes. In an aspect, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can be concurrently and/or serially administered to a subject via multiple routes of administration. In an aspect, aspect of administration can be informed on the likelihood of the disclosed compound, the disclosed composition, the disclosed pharmaceutical formulation, or any combination thereof achieving a pharmacological effect. In an aspect, concomitant administration does not require a single composition or pharmaceutical formulation, in the same dosage form, by the same route of administration, or at the same time. Rather, in an aspect, the effects of both compounds, compositions, and/or pharmaceutical formulations need not manifest themselves at the same time (e.g., effects need only be overlapping for a period and need not be coextensive). In an aspect, concomi-

tant administration or co-administration can comprise administration in parallel or sequentially.

[0298] In an aspect, administering to the subject can comprise contacting one or more cells affected by the targeted RNA or targeted RNA structure. In an aspect, after the administering step, one or more cells affected by the targeted RNA or targeted RNA structure in one or more of the subject's body systems are contacted by a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof.

[0299] In an aspect, local administration can comprise delivery to one or more of the subject's body systems affected by the targeted RNA or targeted RNA structure. In an aspect, the one or more body systems affected by the targeted RNA or targeted RNA structure can comprise the cardiovascular system, the digestive system, the endocrine system, the lymphatic system, the muscular system, the nervous system, the reproductive system, the respiratory system, the skeletal system, the urinary system, the integumentary system, or any combination thereof.

[0300] In an aspect of a disclosed method, following the administering step, the subject's symptoms can be diminished and/or decreased.

[0301] In an aspect of a disclosed method, following the administering step, the subject's quality of life can be improved and/or enhanced. In an aspect of a disclosed method, following the administering step, one or more of the subject's body systems can experience and/or show signs of normal physiology and/or cellular homeostasis. In an aspect, a disclosed method can comprise administering to the subject one or more therapeutic agents and/or active agents. In an aspect, a disclosed therapeutic agent and/or active agent can be any disclosed agent that effects a desired clinical outcome. In an aspect, one or more disclosed therapeutic agents and/or active agents comprises an anti-cancer agent.

[0302] In an aspect, a disclosed method can comprise repeating an administering step one or more times. In an aspect, a disclosed method can comprise monitoring the subject for adverse effects. In an aspect, in the absence of adverse effects, the method can comprise continuing to treat the subject and/or continuing to monitor the subject. In an aspect, in the presence of adverse effects, the method can comprise modifying one or more steps of the method.

[0303] In an aspect, a disclosed method can comprise reducing the pathological phenotype associated with a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0304] In an aspect, a disclosed method can comprise diagnosing the subject as having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof. In an aspect, a disclosed method can further treat one or more symptoms of the subject.

[0305] In an aspect, a disclosed method can restore one or more aspects of cellular homeostasis and/or cellular functionality and/or metabolic dysregulation. In an aspect, restoring one or more aspects of cellular homeostasis and/or cellular functionality and/or metabolic dysregulation can comprise reducing the expression and/or activity level of one or more secondary RNA structures, one or more tertiary

RNA structures, or any combination thereof that causes, relates to, elicits, and/or exacerbated a disease, disorder, and/or condition in the subject.

[0306] In an aspect, restoring one or more aspects of cellular homeostasis and/or cellular functionality can comprise one or more of the following: (i) correcting cell starvation in one or more cell types; (ii) normalizing aspects of the autophagy pathway (such as, for example, correcting, preventing, reducing, and/or ameliorating autophagy); (iii) improving, enhancing, restoring, and/or preserving mitochondrial functionality and/or structural integrity; (iv) improving, enhancing, restoring, and/or preserving organelle functionality and/or structural integrity; (v) correcting enzyme dysregulation; (vi) reversing, inhibiting, preventing, stabilizing, and/or slowing the rate of progression of the multi-systemic manifestations of a genetic disease or disorder; (vii) reversing, inhibiting, preventing, stabilizing, and/or slowing the rate of progression of a genetic disease or disorder, or (viii) any combination thereof. In an aspect, restoring one or more aspects of cellular homeostasis can comprise improving, enhancing, restoring, and/or preserving one or more aspects of cellular structural and/or functional integrity.

[0307] In an aspect, a disclosed method can restore the functionality and/or structural integrity of a missing, affected, deficient, and/or mutant protein or enzyme. In an aspect, restoring the activity and/or functionality of a missing, affected, deficient, and/or mutant protein or enzyme can comprise a 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, or any amount of restoration when compared to a pre-existing level such as, for example, a pre-treatment level. In an aspect, the amount of restoration can be 10-20%, 20-30%, 30-40%, 40-50%, 50-60%, 60-70%, 70-80%, 80-90%, or 90-100% more than a pre-existing level such as, for example, a pre-treatment level. In an aspect, restoration can be measured against a control level or a reference level (e.g., determined, for example, using one or more subjects not having a missing, deficient, and/or mutant protein or enzyme). In an aspect, restoration can be a partial or incomplete restoration. In an aspect, restoration can be complete or near complete restoration such that the level of expression, activity, and/or functionality is like that of a wild-type or control level.

[0308] In an aspect, restoring the activity and/or functionality of an affected gene of interest can comprise increasing or enhancing the expression and/or activity level of that affected gene. In an aspect, increasing or enhancing can comprise an elevation of at least about 5%, 10%, 15%, 20%, 25%, 35%, 50%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, 100%, 200%, 300%, 400%, 500%, or more as compared to a control (such as a pre-treatment level). In an aspect, restoring the activity and/or functionality of an affected gene of interest can comprise decreasing or reducing the expression and/or activity level of that affected gene. In an aspect, decreasing or reducing can comprise a decrease of at least about 5%, 10%, 15%, 20%, 25%, 35%, 50%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, 100%, 200%, 300%, 400%, 500%, or more as compared to a control (such as a pre-treatment level).

[0309] In an aspect of a disclosed method of treating a subject, techniques to monitor, measure, and/or assess the restoring one or more aspects of cellular homeostasis and/or cellular functionality can comprise qualitative (or subjective) means as well as quantitative (or objective) means.

These means are known to the skilled person. For example, representative regulated variables and sensors relating to systemic homeostasis are discussed supra.

[0310] In an aspect, a disclosed method can comprise measuring and/or ascertaining the expression and/or activity level and/or stability of one or more disclosed secondary RNA structures, one or more disclosed tertiary RNA structures, or any combination thereof in the subject prior to treatment or prior to the administering of a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof. In an aspect, a disclosed method can comprise measuring and/or ascertaining the expression and/or activity level and/or stability of one or more disclosed secondary RNA structures, one or more disclosed tertiary RNA structures, or any combination thereof in the subject after treatment or after the administering of a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof. In an aspect, a disclosed method can comprise measuring and/or ascertaining the expression and/or activity level and/or stability of one or more disclosed secondary RNA structures, one or more disclosed tertiary RNA structures, or any combination thereof in the subject one or more times.

[0311] In an aspect, a disclosed subject can be symptomatic or asymptomatic.

[0312] In an aspect, the disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can be administered in a single dose, or in multiple doses (such as 2, 3, 4, 5, 6, 7, 8, 9 or 10 doses) as needed for the desired therapeutic results.

[0313] In an aspect, a disclosed method of treating a subject can further comprise administering to the subject a therapeutically effective amount of an active agent and/or therapeutic that can correct one or more aspects of a dysregulated metabolic or enzymatic pathway caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0314] In an aspect, a disclosed method can comprise repeating one or more steps of the method and/or modifying one or more steps of the method (such as, for example, an administering step). In an aspect, a disclosed method of treating a subject can comprise modifying one or more of the disclosed steps. For example, modifying one or more of steps of a disclosed method can comprise modifying or changing one or more features or aspects of one or more steps of a disclosed method. For example, in an aspect, a method can be altered by changing the amount of a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof administered to a subject, or by changing the frequency of administration of a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof to a subject, or by changing the duration of time a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof is administered to a subject.

[0315] In an aspect, a disclosed method of treating a subject can be altered by changing the amount of a disclosed therapeutic agent and/or active agent administered to a subject, or by changing the frequency of administration of a disclosed therapeutic agent and/or active agent administered

to a subject, or by changing the duration of time a disclosed therapeutic agent and/or active agent is administered to the subject.

[0316] In an aspect, a disclosed method of treating a subject can comprise generating and/or validating one or more of disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof. Methods of generating and/or validating disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof are discussed infra. In an aspect, a disclosed method of treating a subject can further comprise administering to the subject one or more additional disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof.

[0317] In an aspect, a disclosed therapeutic agent and/or active agent can comprise (i) one or more active agents, (ii) biologically active agents, (iii) one or more pharmaceutically active agents, (iv) one or more immune-based therapeutic agents, (v) one or more clinically approved agents, or (vi) a combination thereof.

[0318] In an aspect of a disclosed method, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can physically disturb and/or disrupt and/or interrupt the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule.

[0319] In an aspect of a disclosed method, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can physically prevent the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule.

[0320] In an aspect of a disclosed method, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can be toxic to one or more cell types. In an aspect, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can disrupt and/or impair the functionality of one or more cell types. In an aspect, a disclosed cell type can comprise cancer cells or tumor cells. Cancer cells and tumor cells are known to the art and are discussed supra. In an aspect of a disclosed method, a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any combination thereof can initiate, induce, promote, elicit, hasten, and/or cause the death of the cancer cell or tumor cell. In an aspect, a disclosed pharmaceutical formulation can reduce and/or minimize the likelihood of metastasis. In an aspect, cell death can be apoptotic and/or necrotic.

[0321] In an aspect, a disclosed method can comprise administering to the subject DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-

P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or a pharmaceutically acceptable salt, a hydrate, a prodrug, an ester, or a derivative thereof, or a composition thereof, or a pharmaceutical formulation thereof.

L. Methods of Modulating Secondary and/or Tertiary RNA Structure

[0322] Disclosed herein is a method of modulating one or more secondary RNA structures and/or tertiary RNA structures, the method comprising contacting one or more RNA structures with one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof.

[0323] In an aspect of a disclosed method, the one or more RNA structures are in a cell or in cells. In an aspect, the cell or cells are in a subject. In an aspect, a subject has been diagnosed with or is suspected of having a disease, condition, or disorder caused by, related to, and/or exacerbated by the presence of, expression of, and/or activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof.

[0324] In an aspect, the one or more disclosed secondary RNA structures are modulated. In an aspect, the one or more disclosed tertiary RNA structures are modulated. In an aspect, the one or more disclosed secondary RNA structures are stabilized. In an aspect, the one or more disclosed secondary RNA structures are destabilized. In an aspect, the one or more tertiary RNA structures are stabilized. In an aspect, the one or more tertiary RNA structures are destabilized. In an aspect, the expression and/or activity level of one or more disclosed secondary RNA structures are decreased. In an aspect, the expression and/or activity level of one or more disclosed tertiary RNA structures are decreased. In an aspect, the expression and/or activity level of one or more disclosed secondary RNA structures are increased. In an aspect, the expression and/or activity level of one or more disclosed tertiary RNA structures are decreased.

[0325] In an aspect, the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule can be physically disturbed and/or disrupted and/or interrupted. In an aspect, the binding of one or more disclosed secondary RNA structures, one or more tertiary RNA structures, or any combination thereof to another structure or molecule can be physically prevented.

[0326] In an aspect of a disclosed method, modulating one or more secondary RNA structures and/or tertiary RNA structures be toxic to one or more cell types. In an aspect, modulating one or more secondary RNA structures and/or tertiary RNA structures can disrupt and/or impair the functionality of one or more cell types (such as, for example, cancer cells or tumor cells). Cancer cells and tumor cells are known to the art and are discussed supra. In an aspect, modulating one or more secondary RNA structures and/or tertiary RNA structures can initiate, induce, promote, elicit, hasten, and/or cause the death of the cancer cell or tumor cell. In an aspect, modulating one or more secondary RNA

structures and/or tertiary RNA structures can reduce and/or minimize the likelihood of metastasis. In an aspect, cell death can be apoptotic and/or necrotic.

[0327] In an aspect, a disclosed method of modulating one or more secondary RNA structures and/or tertiary RNA structures can comprise administering to a subject (i) one or more active agents, (ii) biologically active agents, (iii) one or more pharmaceutically active agents, (iv) one or more immune-based therapeutic agents, (v) one or more clinically approved agents, or (vi) a combination thereof.

[0328] In an aspect, a disclosed method can comprise contacting the cells and/or administering to the subject DMZ-P1, DMZ-P5, DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P14, DMZ-P17, DMZ-O1, DMZ-O2, DMZ-O4, DMZ-O5, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M4, DMZ-M7, DMZ-M10, DMZ-M15, DMZ-M22, DMZ-M24, DMZ-N-Me-M9, DMZ-N-Me-mPy-P13, DMZ-mPy-P5-2HCl, DMZ-mPy-P13, DMZ-mF-P5, DMZ-P5-mono, DMZ-mF-P13, DMZ-mF-P13-mono, DMZ-P29, DMZ-P29-mono, DMZ-P0-P5, DMZ-P30, DMZ-P30-mono, DMZ-mMe-P5, DMZ-mMe-P5-mono, DMZ-mMe-P13, DMZ-mMe-P13-mono, DMZ-N-Me-P5, DMZ-N-Me-P13, DMZ-P31, DMZ-P0-P32, DMZ-oMe-P5, DMZ-oMe-P5-mono, DMZ-M5, DMZ-M5-mono, DMZ-M13, Aniline-P13, DMZ-P0-P13, DMZ-M4-P5, DMZ-M4-P13, DMZ-P4, DMZ-P4-mono, DMZ-P5-P13, DMZ-P33, DMZ-P33-mono, DMZ-P2, DMZ-P2-mono, DMZ-P32, DMZ-P35, DMZ-oMe-P13, DMZ-N-Et-P13, DMZ-P34, DMZ-N-Me-M13, DMZ-N-Me-mPy-P13, DMZ-N-Et-P5, DMZ-N-Me-P36, DMZ-N-Me-P36-mono, DMZ-P37, DMZ-P37-mono, DMZ-N-nPr-P13, DMZ-N-Me-P38, DMZ-N-Me-M9, or a pharmaceutically acceptable salt, a hydrate, a prodrug, an ester, or a derivative thereof, or a composition thereof, or a pharmaceutical formulation thereof.

M. Methods of Assessing a Library of Small Molecules

[0329] Disclosed herein is a method of assessing a library of small molecules that target RNA, the method comprising using a multi-dimensional approach comprising designing and optimizing affinity-based and structural stability-based assays.

[0330] Disclosed herein is a method of assessing a library of shape diverse small molecule analogs that modulate a targeted RNA or a targeted RNA structure, the method comprising choosing a candidate small molecule; generating a theoretical library of shape diverse small molecule analogs based on the candidate small molecule; evaluating the three-dimensional shape of one or more shape diverse small molecule analogs of the theoretical library; selecting one or more shape diverse small molecules for synthesis; synthesizing one or more shape diverse small molecule analogs in the theoretic library; and characterizing the one or more synthesized shape diverse small molecule analogs.

[0331] In an aspect of a disclosed method, characterizing the one or more synthesized shape diverse small molecule analogs can comprise employing one or more assays. In an aspect, one or more disclosed assays can comprise an indicator displacement assay (IDA), isothermal titration calorimetry (ITC), assay differential scanning fluorimetry (DSF), enzymatic degradation assays, NMR and HPLC analysis, or any combination thereof.

[0332] Disclosed herein is a method of assessing a library of shape diverse small molecule analogs that modulate a

targeted RNA or a targeted RNA structure, the method comprising choosing a candidate small molecule; generating a theoretical library of shape diverse small molecule analogs based on the candidate small molecule; evaluating the three-dimensional shape of one or more shape diverse small molecule analogs of the theoretical library; selecting one or more shape diverse small molecules for synthesis; synthesizing one or more shape diverse small molecule analogs in the theoretic library; and screening the one or more synthesized shape diverse small molecule analogs.

[0333] In an aspect of a disclosed method, generating the disclosed theoretical library of shape diverse small molecule can comprise substituting one or more subunits at the ortho, para, and meta-substituted scaffolds of the candidate small molecule. In an aspect, the subunits are amine subunits.

[0334] In an aspect of a disclosed method, evaluating the 3D shape can comprise determining the theoretical principal moments of inertia of the one or more shape diverse small molecule analogs of the theoretical library.

[0335] In an aspect, a disclosed method can further comprise correcting the protonation state of the one or more shape diverse small molecule analogs. In an aspect of a disclosed method, correcting the protonation state can comprise adjusting the SMILES strings of the one or more shape diverse small molecule analogs. In an aspect of a disclosed method, adjusting the SMILE strings can comprise using a python code. In an aspect, correcting the protonation state can comprise employing one or more computational software programs. In an aspect, the disclosed one or more computations software programs can comprise ChemAxon, Schrodinger, CREST, or any combination thereof. In an aspect, a disclosed method can further comprise correcting the tautomerization state of the one or more shape diverse small molecule analogs.

[0336] In an aspect, a disclosed method can further comprise normalizing the theoretical principal moments of inertia for the one or more shape diverse small molecule analogs of the library. In an aspect, a disclosed method can further comprise visualizing the data by plotting the normalized theoretical PMI for the one or more shape diverse small molecule analogs of the theoretical library on an envelope diagram. In an aspect of a disclosed method, selecting one or more shape diverse small molecule analogs can comprise using one or more computational software programs. In an aspect of a disclosed method, selecting one or more shape diverse small molecule analogs can comprise using R, Python, Java, MatLab, or any combination thereof to maximize shape representation. In an aspect of a disclosed method, selecting one or more shape diverse small molecule analogs can comprise using Kennard-Stone algorithm to maximize shape representation. In an aspect of a disclosed method, evaluating the three-dimensional shape of one or more shape diverse small molecule analogs of the theoretical library can comprise performing a QSAR analysis using multiple descriptors. In an aspect, a disclosed QSAR analysis can employ a commercially available computational program. In an aspect, a disclosed QSAR analysis can employ R, Python, Java, MatLab, or any combination thereof. In an aspect, a disclosed QSAR analysis can comprise using molecular operating environment (MOE), ChemAxon, Schrodinger, KNIME, Mordred, E-Dragon, or any combination thereof.

[0337] In an aspect, disclosed descriptors can comprise 2D and 3D descriptors. In an aspect, disclosed descriptors can

comprise numerical descriptors, constitutional descriptors, topological descriptors, electronic descriptors, hybrid descriptors, geometric descriptors, QM descriptors, or any combination thereof. Descriptors include, but are not limited to the following: FID, rgyr, glob, AM1_dipole, AM1_E, AM1_Eele, AM1_HF, AM1_HOMO, AM1_IP, AM1_LUMO, apol, ASA, ASA+, ASA-, ASA_H, ASA_P, ast_violation, ast_violation_ext, a_acc, a_aro, a_base, a_count, a_donacc, a_heavy, a_hyd, a_IC, a_ICM, a_nC, a_nCl, a_nH, a_nN, a_nO, balabanJ, BCUT_PEOE_0, BCUT_PEOE_1, BCUT_PEOE_2, BCUT_PEOE_3, BCUT_SLOGP_0, BCUT_SLOGP_1, BCUT_SLOGP_2, BCUT_SLOGP_3, BCUT_SMR_0, BCUT_SMR_1, BCUT_SMR_2, BCUT_SMR_3, bpol, b_1rotN, b_1rotR, b_ar, b_count, b_heavy, b_maxllen, b_rotN, b_rotR, b_single, CASA+, CASA-, chi0, chi0v, chi0v_C, chi0_C, chi1, chi1v, chi1v_C, chi1_C, DASA, DCASA, dens, density, diameter, dipole, dipoleX, dipoleY, dipoleZ, E_ang, E_ele, E_nb, E_oop, E_rnb, E_rsol, E_sol, E_stb, E_str, E_strain, E_tor, E_vdw, FASA+, FASA-, FASA_H, FASA_P, FCASA+, FCASA-, FCharge, GCUT_PEOE_0, GCUT_PEOE_1, GCUT_PEOE_2, GCUT_PEOE_3, GCUT_SLOGP_0, GCUT_SLOGP_1, GCUT_SLOGP_2, GCUT_SLOGP_3, GCUT_SMR_0, GCUT_SMR_1, GCUT_SMR_2, GCUT_SMR_3, h_ema, h_emd, h_emd_C, h_logD, h_logP, h_logS, h_logdbo, h_logpbo, h_mr, h_pavgQ, h_pKa, h_pKb, h_pstates, h_pstrain, Kier1, Kier2, Kier3, KierA1, KierA2, KierA3, KierFlex, lip_acc, lip_don, lip_druglike, lip_violation, logP(o/w), logS, MNDO_dipole, MNDO_E, MNDO_Eele, MNDO_HF, MNDO_HOMO, MNDO_IP, MNDO_LUMO, mr, mutagenic, nmol, npr1, npr2, opr_brigid, opr_leadlike, opr_nring, opr_nrot, opr_violation, PC+, PC-, PEOE_PC+, PEOE_PC-, PEOE_RPC+, PEOE_RPC-, PEOE_VSA+0, PEOE_VSA+1, PEOE_VSA+2, PEOE_VSA+3, PEOE_VSA+4, PEOE_VSA+5, PEOE_VSA+6, PEOE_VSA-0, PEOE_VSA-1, PEOE_VSA-2, PEOE_VSA-3, PEOE_VSA-4, PEOE_VSA-5, PEOE_VSA-6, PEOE_VSA_FHYD, PEOE_VSA_FNEG, PEOE_VSA_FPNEG, PEOE_VSA_FPOL, PEOE_VSA_FPOS, PEOE_VSA_FPPOS, PEOE_VSA_HYD, PEOE_VSA_NEG, PEOE_VSA_PNEG, PEOE_VSA_POL, PEOE_VSA_POS, PEOE_VSA_PPOS, petitjean, petitjeanSC, PM3_dipole, PM3_E, PM3_Eele, PM3_HF, PM3_HOMO, PM3_IP, PM3_LUMO, pmi, pmi1, pmi2, pmi3, pmiX, pmiY, pmiZ, Q_PC+, Q_PC-, Q_RPC+, Q_RPC-, Q_VSA_FHYD, Q_VSA_FNEG, Q_VSA_FPNEG, Q_VSA_FPOL, Q_VSA_FPOS, Q_VSA_FPPOS, Q_VSA_HYD, Q_VSA_NEG, Q_VSA_PNEG, Q_VSA_POL, Q_VSA_POS, Q_VSA_PPOS, radius, reactive, rings, RPC+, RPC-, rsynth, SlogP, SlogP_VSA0, SlogP_VSA1, SlogP_VSA2, SlogP_VSA3, SlogP_VSA4, SlogP_VSA5, SlogP_VSA6, SlogP_VSA7, SlogP_VSA8, SlogP_VSA9, SMR, SMR_VSA0, SMR_VSA1, SMR_VSA2, SMR_VSA3, SMR_VSA4, SMR_VSA5, SMR_VSA6, SMR_VSA7, std_dim1, std_dim2, std_dim3, TPSA, VAdjEq, VAdjMa, VDistEq, VDistMa, vdw_area, vdw_vol, vol, VSA, vsa_acc, vsa_acid, vsa_base, vsa_don, vsa_hyd, vsa_other, vsa_pol, vsurf_A, vsurf_CP, vsurf_CW1, vsurf_CW2, vsurf_CW3, vsurf_CW4, vsurf_CW5, vsurf_CW6, vsurf_CW7, vsurf_CW8, vsurf_D1, vsurf_D2, vsurf_D3, vsurf_D4, vsurf_D5, vsurf_D6, vsurf_D7, vsurf_D8, vsurf_DD12, vsurf_DD13, vsurf_DD23, vsurf_DW12, vsurf_DW13, vsurf_DW23, vsurf_EDmin1, vsurf_EDmin2, vsurf_EDmin3, vsurf_EWmin1, vsurf_EWmin2, vsurf_EWmin3, vsurf_G, vsurf_HB1,

vsurf_HB2, vsurf_HB3, vsurf_HB4, vsurf_HB5, vsurf_HB6, vsurf_HB7, vsurf_HB8, vsurf_HL1, vsurf_HL2, vsurf_ID1, vsurf_ID2, vsurf_ID3, vsurf_ID4, vsurf_ID5, vsurf_ID6, vsurf_ID7, vsurf_ID8, vsurf_IW1, vsurf_IW2, vsurf_IW3, vsurf_IW4, vsurf_IW5, vsurf_IW6, vsurf_IW7, vsurf_IW8, vsurf_R, vsurf_S, vsurf_V, vsurf_W1, vsurf_W2, vsurf_W3, vsurf_W4, vsurf_W5, vsurf_W6, vsurf_W7, vsurf_W8, vsurf_Wp1, vsurf_Wp2, vsurf_Wp3, vsurf_Wp4, vsurf_Wp5, vsurf_Wp6, vsurf_Wp7, vsurf_Wp8, Weight, weinerPath, weinerPol, Zagreb, or any combination thereof. In an aspect, a disclosed QSAR analysis using multiple descriptors can generate and/or can provide a predictive model. In an aspect, disclosed 2D descriptors can consider atoms and connectivity information. In an aspect, disclosed 2D descriptors can comprise Q_VSA_PNEG, Q_VSA_POL, SMR_VSA7, or any combination thereof. In an aspect, disclosed 3D descriptor can consider coordinates and individual conformations. In an aspect, disclosed 3D descriptors can comprise glob, vsurf_EDmin2, vsurf_EDmin3, vsurf_DD23, E_stb, or any combination thereof. In an aspect, a disclosed QSAR analysis can comprise about 50-100 descriptors, about 100-150 descriptors, about 150-200 descriptors, about 200-250 descriptors, about 250-300 descriptors, about 300-350 descriptors, or more than 350 descriptors.

[0338] In an aspect of a disclosed method, screening the one or more shape diverse small molecule analogs can comprise performing indicator displacement assay (IDA), isothermal titration calorimetry (ITC), differential scanning fluorimetry (DSF), or any combination thereof. In an aspect of a disclosed method, screening the one or more shape diverse small molecule analogs can comprise obtaining the targeted RNA or targeted RNA structure. In an aspect of a disclosed method, obtaining the targeted RNA or targeted RNA structure can comprise synthesizing and/or purifying the targeted RNA or targeted RNA structure. In an aspect of a disclosed method, screening the one or more shape diverse small molecule analogs can comprise measuring the affinity of the one or more shape diverse small molecule analogs against the targeted RNA or targeted RNA structure.

[0339] In an aspect of a disclosed method, measuring affinity of the one or more shape diverse small molecule analogs can comprise using an indicator displacement assay (IDA). In an aspect, a disclosed indicator displacement assay can comprise an enantioselective indicator displacement assay (eIDA), a fluorescent indicator displacement assay (FIDA), a reaction-based indicator displacement assay (RIA), a DimerDye disassembly assays (DDAs), an intramolecular indicator displacement assay (IIDA), an allosteric indicator displacement assay (AIDA), a mechanically controlled indicator displacement assay (MC-IDA), a quencher displacement assay (QDA), or any combination thereof. In an aspect of a disclosed method, measured affinity can comprise high affinity ($CD_{50} < 5 \mu M$), medium affinity ($5 \mu M < CD_{50} < 30 \mu M$), or low affinity ($CD_{50} > 30 \mu M$).

[0340] In an aspect, a disclosed method can further comprise measuring one or more thermodynamic parameters of the biomolecular interaction between the one or more synthesized shape diverse small molecule analogs and the targeted RNA or targeting RNA structure using isothermal titration calorimetry. In an aspect, disclosed thermodynamic parameters of the one or more biomolecular interactions can comprise affinity (KA), enthalpy (ΔH), entropy (ΔS), stoichiometry (n), or any combination thereof.

[0341] In an aspect, a disclosed method can further comprise comparing the affinity measured by IDA to the affinity measured by ITC. In an aspect, a disclosed method can further comprise assessing thermal stability of the one or more shape diverse small molecule analogs and the targeted RNA or targeted RNA structure. In an aspect of a disclosed method, wherein assessing the thermal stability can comprise differential screen fluorimetry (DSF) of (i) the one or more synthesized shape diverse small molecule analogs and the targeted RNA structure, and (ii) the one or more synthesized shape diverse small molecule analogs and a control or vehicle.

[0342] In an aspect, a disclosed method can further comprise performing QSAR modeling to test whether the same parameters that yielded a predictive model for IDA are the same that govern thermal stabilization interactions as determined by DSF. In an aspect, a disclosed QSAR analysis using both 2D and 3D descriptors can generate and/or produce a predictive model. In an aspect, disclosed 2D descriptors can consider atoms and connectivity information. In an aspect, disclosed 2D descriptors can comprise Q_VSA_PNEG, Q_VSA_POL, SMR_VSA7, or any combination thereof. In an aspect, disclosed 3D descriptions can consider coordinates and individual conformations. In an aspect, disclosed 3D descriptors can comprise glob, vsurf_EDmin2, vsurf_EDmin3, vsurf_DD23, E_stb, or any combination thereof.

[0343] Descriptors include, but are not limited to, the following: FID, rgyr, glob, AM1_dipole, AM1_E, AM1_Eele, AM1_HF, AM1_HOMO, AM1_IP, AM1_LUMO, apol, ASA, ASA+, ASA-, ASA_H, ASA_P, ast_violation, ast_violation_ext, a_acc, a_aro, a_base, a_count, a_donacc, a_heavy, a_hyd, a_IC, a_ICM, a_nC, a_nCl, a_nH, a_nN, a_nO, balabanJ, BCUT_PEOE_0, BCUT_PEOE_1, BCUT_PEOE_2, BCUT_PEOE_3, BCUT_SLOGP_0, BCUT_SLOGP_1, BCUT_SLOGP_2, BCUT_SLOGP_3, BCUT_SMR_0, BCUT_SMR_1, BCUT_SMR_2, BCUT_SMR_3, bpol, b_1rotN, b_1rotR, b_ar, b_count, b_heavy, b_maxllen, b_rotN, b_rotR, b_single, CASA+, CASA-, chi0, chi0v, chi0v_C, chi0_C, chi1, chi1v, chi1v_C, chi1_C, DASA, DCASA, dens, density, diameter, dipole, dipoleX, dipoleY, dipoleZ, E_ang, E_ele, E_nb, E_oop, E_mnb, E_rsol, E_sol, E_stb, E_str, E_strain, E_tor, E_vdw, FASA+, FASA-, FASA_H, FASA_P, FCASA+, FCASA-, FCharge, GCUT_PEOE_0, GCUT_PEOE_1, GCUT_PEOE_2, GCUT_PEOE_3, GCUT_SLOGP_0, GCUT_SLOGP_1, GCUT_SLOGP_2, GCUT_SLOGP_3, GCUT_SMR_0, GCUT_SMR_1, GCUT_SMR_2, GCUT_SMR_3, h_ema, h_emd, h_emd_C, h_logD, h_logP, h_logS, h_log_dbo, h_log_pbo, h_mr, h_pavgQ, h_pKa, h_pKb, h_pstates, h_pstrain, Kier1, Kier2, Kier3, KierA1, KierA2, KierA3, KierFlex, lip_acc, lip_don, lip_druglike, lip_violation, log P(o/w), log S, MNDO_dipole, MNDO_E, MNDO_Eele, MNDO_HF, MNDO_HOMO, MNDO_IP, MNDO_LUMO, mr, mutagenic, nmol, npr1, npr2, opr_brigid, opr_leadlike, opr_nring, opr_nrot, opr_violation, PC+, PC-, PEOE_PC+, PEOE_PC-, PEOE_RPC+, PEOE_RPC-, PEOE_VSA+0, PEOE_VSA+1, PEOE_VSA+2, PEOE_VSA+3, PEOE_VSA+4, PEOE_VSA+5, PEOE_VSA+6, PEOE_VSA-0, PEOE_VSA-1, PEOE_VSA-2, PEOE_VSA-3, PEOE_VSA-4, PEOE_VSA-5, PEOE_VSA-6, PEOE_VSA_FHYD, PEOE_VSA_FNEG, PEOE_VSA_FPNEG, PEOE_VSA_FPOL, PEOE_VSA_FPOS, PEOE_VSA_FPPOS, PEOE_VSA_HYD, PEOE_VSA_NEG, PEOE_VSA_PNEG, PEOE_VSA_POL,

PEOE_VSA_POS, PEOE_VSA_PPOS, petitjean, petitjeanSC, PM3_dipole, PM3_E, PM3_Eele, PM3_HF, PM3_HOMO, PM3_IP, PM3_LUMO, pmi, pmi1, pmi2, pmi3, pmiX, pmiY, pmiZ, Q_PC+, Q_PC-, Q_RPC+, Q_RPC-, Q_VSA_FHYD, Q_VSA_FNEG, Q_VSA_FPNEG, Q_VSA_FPOL, Q_VSA_FPOS, Q_VSA_FPPOS, Q_VSA_HYD, Q_VSA_NEG, Q_VSA_PNEG, Q_VSA_POL, Q_VSA_POS, Q_VSA_PPOS, radius, reactive, rings, RPC+, RPC-, rsynth, SlogP, SlogP_VSA0, SlogP_VSA1, SlogP_VSA2, SlogP_VSA3, SlogP_VSA4, SlogP_VSA5, SlogP_VSA6, SlogP_VSA7, SlogP_VSA8, SlogP_VSA9, SMR, SMR_VSA0, SMR_VSA1, SMR_VSA2, SMR_VSA3, SMR_VSA4, SMR_VSA5, SMR_VSA6, SMR_VSA7, std_dim1, std_dim2, std_dim3, TPSA, VAdjEq, VAdjMa, VDistEq, VDistMa, vdw_area, vdw_vol, vol, VSA, vsa_acc, vsa_acid, vsa_base, vsa_don, vsa_hyd, vsa_other, vsa_pol, vsurf_A, vsurf_CP, vsurf_CW1, vsurf_CW2, vsurf_CW3, vsurf_CW4, vsurf_CW5, vsurf_CW6, vsurf_CW7, vsurf_CW8, vsurf_D1, vsurf_D2, vsurf_D3, vsurf_D4, vsurf_D5, vsurf_D6, vsurf_D7, vsurf_D8, vsurf_DD12, vsurf_DD13, vsurf_DD23, vsurf_DW12, vsurf_DW13, vsurf_DW23, vsurf_EDmin1, vsurf_EDmin2, vsurf_EDmin3, vsurf_EWmin1, vsurf_EWmin2, vsurf_EWmin3, vsurf_G, vsurf_HB1, vsurf_HB2, vsurf_HB3, vsurf_HB4, vsurf_HB5, vsurf_HB6, vsurf_HB7, vsurf_HB8, vsurf_HL1, vsurf_HL2, vsurf_ID1, vsurf_ID2, vsurf_ID3, vsurf_ID4, vsurf_ID5, vsurf_ID6, vsurf_ID7, vsurf_ID8, vsurf_IW1, vsurf_IW2, vsurf_IW3, vsurf_IW4, vsurf_IW5, vsurf_IW6, vsurf_IW7, vsurf_IW8, vsurf_R, vsurf_S, vsurf_V, vsurf_W1, vsurf_W2, vsurf_W3, vsurf_W4, vsurf_W5, vsurf_W6, vsurf_W7, vsurf_W8, vsurf_Wp1, vsurf_Wp2, vsurf_Wp3, vsurf_Wp4, vsurf_Wp5, vsurf_Wp6, vsurf_Wp7, vsurf_Wp8, Weight, weinerPath, weinerPol, Zagreb, or any combination thereof.

[0344] In an aspect, a disclosed QSAR analysis can comprise about 50-100 descriptors, about 100-150 descriptors, about 150-200 descriptors, about 200-250 descriptors, about 250-300 descriptors, about 300-350 descriptors, or more than 350 descriptors. In an aspect of a disclosed method, wherein performing QSAR modeling can comprise using the change in RNA degradation induced by the one or more 3D shape diverse small molecules analogs and identifying one or more 3D descriptors as contributing factors.

[0345] In an aspect of a disclosed method, screening the one or more shape diverse small molecule analogs can comprise subjecting the one or more molecule analogs to an enzymatic degradation assay. In an aspect, a disclosed enzymatic degradation assay can comprise a RNase. In an aspect, a disclosed RNase can comprise an endoribonuclease or an exoribonuclease. In an aspect, a disclosed endoribonuclease can comprise RNase A, RNase P, RNase H, RNase I, RNase III, RNase T1, RNase T2, RNase U2, RNase V1, RNase PhyM, RNase V, or any combination thereof.

[0346] In an aspect, a disclosed RNase can comprise RNase A. In an aspect, a disclosed exoribonuclease can comprise RNase PH, RNase II, RNase R, RNase D, RNA T, or any combination thereof. In an aspect, a disclosed targeted RNA or disclosed targeted RNA structure can comprise a tertiary RNA structure.

[0347] In an aspect, a disclosed enzymatic degradation assay can measure the stabilizing or destabilizing effect on the tertiary RNA structure asserted by the one or more shape diverse small molecule analogs. In an aspect of a disclosed method, wherein stabilizing the tertiary RNA structure can

comprise measuring less degradation of the tertiary RNA structure when subjected to an enzymatic assay in the presence of the compound than a control or reference level of degradation. In an aspect of a disclosed method, wherein stabilizing tertiary RNA structure can comprise measuring less degradation of the tertiary RNA structure when subjected to an enzymatic assay in the presence of the compound than the level of degradation in a control or reference sample. In an aspect of a disclosed method, wherein destabilizing tertiary RNA structure can comprise measuring more degradation of the tertiary RNA structure when subjected to an enzymatic assay in the presence of the compound than a control or reference level of degradation. In an aspect of a disclosed method, wherein destabilizing tertiary RNA structure can comprise measuring more degradation of the tertiary RNA structure when subjected to an enzymatic assay in the presence of the compound than the level of degradation in a control or reference sample.

[0348] In an aspect, a disclosed method can further comprise obtaining the control or reference level of degradation. In an aspect of a disclosed method, wherein obtaining the control or reference level of degradation can comprise subjecting the tertiary RNA structure to a RNase in the absence of the compound. In an aspect, a disclosed method can further comprise obtaining the level of degradation in a control or reference sample. In an aspect of a disclosed method, wherein obtaining the level of degradation in a control or reference sample can comprise subjecting the tertiary RNA structure to a RNase in the absence of the compound. In an aspect, a disclosed targeted RNA or a disclosed targeted RNA structure can comprise a RNA triplex. In an aspect of a disclosed method, wherein stabilizing the RNA triplex can comprise measuring less degradation of the RNA triplex when subjected to an enzymatic assay than a control or reference level of degradation. In an aspect of a disclosed method, wherein stabilizing the RNA triplex can comprise measuring less degradation of the RNA triplex when subjected to an enzymatic assay than the level of degradation in a control or reference sample.

[0349] In an aspect of a disclosed method, wherein destabilizing the RNA triplex can comprise measuring more degradation of the RNA triplex when subjected to an enzymatic assay than a control or reference level of degradation. In an aspect of a disclosed method, wherein destabilizing the RNA triplex can comprise measuring more degradation of the RNA triplex when subjected to an enzymatic assay than the level of degradation in a control or reference sample. In an aspect, a disclosed method can further comprise obtaining the control or reference level of degradation. In an aspect of a disclosed method, wherein obtaining the control or reference level of degradation can comprise subjecting the RNA triplex to a RNase in the absence of the compound. In an aspect, a disclosed method can further comprise obtaining the level of degradation in a control or reference sample. In an aspect of a disclosed method, wherein obtaining the level of degradation in a control or reference sample can comprise subjecting the RNA triplex to a RNase in the absence of the compound. In an aspect, disclosed shape diverse small molecules that stabilize the targeted RNA or targeted RNA structure can show less endonuclease driven degradation. In an aspect, disclosed shape diverse small molecules that stabilize non-targeted targeted RNA or targeted RNA can show more endonuclease driven degradation.

[0350] In an aspect, a disclosed method can further comprise performing a QSAR analysis using the level of degradation generated by the one or more shape diverse small molecule analogs.

[0351] In an aspect of a disclosed method, wherein screening the one or more shape diverse small molecule analogs further can comprise measuring the ^1H and ^{13}C NMR spectra of the one or more shape diverse small molecule analogs.

[0352] In an aspect, a disclosed method can further comprise characterizing the one or more shape diverse small molecule analogs as a stabilizer or destabilizer of the targeted RNA or targeted RNA structure. In an aspect, a disclosed method can further comprise using one or more shape diverse small molecule analogs in a method of stabilizing the targeted RNA or targeted RNA structure. In an aspect, a disclosed method can further comprise using one or more shape diverse small molecule analogs in a method of destabilizing the targeted RNA or targeted RNA structure.

[0353] In an aspect, a disclosed targeted RNA or disclosed targeted RNA structure can comprise a human, viral, bacterial, fungal, or yeast targeted RNA or targeted RNA structure. In an aspect, a disclosed targeted RNA or disclosed targeted RNA structure can comprise a targeted RNA or targeted RNA structure in a human, a non-human mammal, a non-mammal, a virus, a bacterium, a fungus, or some other organism. In an aspect, a disclosed targeted RNA or disclosed targeted RNA structure can comprise a synthetically designed RNA, an engineered RNA, or a recombinant RNA. In an aspect, a disclosed targeted RNA or disclosed targeted RNA structure can comprise a secondary or tertiary RNA structure in humans. In an aspect, a disclosed compound can target one or more RNA structural motifs. In an aspect, one or more disclosed RNA structural motifs can comprise a secondary RNA structural motif, a tertiary RNA structural motif, or any combination thereof. In an aspect, disclosed secondary RNA structural motif can comprise the scaffold of the tertiary RNA structural motif. In an aspect, disclosed secondary RNA structural motif can comprise collection of RNA base pairs in a tertiary RNA structural motif. In an aspect, disclosed RNA base pairs can comprise AU, GU, GC, UA, UG, or CG.

[0354] In an aspect, a disclosed tertiary RNA structural motif can comprise a bulge, a S-turn, an internal loop, a 3-multiloop, a hairpin loop, a pseudoknot, an apical loop, a kissing loop, a coaxial helix, a stacking helix, a two-way junction, a three-way junction, a four-way junction, or any combination thereof. In an aspect, a disclosed tertiary RNA structural motif can comprise a GNAA tetraloop, a GNGA loop, a T-loop, a C-loop, a E-loop, a Sarcin-ricin loop, a Kink-turn, a Reverse kink-turn, a Hook-turn, a Tandem shear, a Tetraloop-receptor motifs, an interacting tetraloop (e.g., GNRA, GANC, and GAAA tetraloops), a triple-stranded RNA, or any combination thereof. In an aspect, a disclosed RNA structural motif can comprise a RNA triplex structure.

[0355] In an aspect, a disclosed RNA triplex structure can concern one or more functionally important RNAs. In an aspect, disclosed functionally important RNAs can comprise human RNAs, viral RNAs, bacterial RNAs, fungal RNAs, yeast RNAs, RNAs of an organism, or any combination thereof. In an aspect, a disclosed targeted RNA or targeted RNA structure can comprise a targeted RNA or targeted RNA structure in a human, a non-human mammal, a non-

mammal, a virus, a bacterium, a fungus, or some other organism. In an aspect, disclosed functionally important RNAs can comprise telomerase RNAs, group I and II introns, long noncoding RNAs (lncRNAs), ribosomal RNAs, or any combination thereof. In an aspect, disclosed lncRNAs can comprise TTF-I, FENDRR, MEG3, HIF1 α -AS1PARTICLE, KHPS1, HOTAIR, MIR100HG, CHD4/NuRD, or any combination thereof.

[0356] In an aspect, functionally important human RNA can comprise NEAT1 ncRNA 3'-end triple helix, G-quadruplexes, other functional secondary and tertiary structures, or any combination thereof. In an aspect, disclosed functionally important human RNA can comprise MALAT1 ncRNA 3'-end triple helix, G-quadruplexes, other functional secondary and tertiary structures, or any combination thereof. In an aspect, disclosed functionally important human RNA can comprise SChLAP1 ncRNA functional secondary and tertiary structures. In an aspect, disclosed SChLAP1 ncRNA functional secondary structures and tertiary structures can comprise stem loops and/or complex junctions near the 3'-end. In an aspect, disclosed functionally important human RNA can comprise AR mRNA functional secondary structures and tertiary structures including splice sites. In an aspect, disclosed AR mRNA functional secondary structures and tertiary structures can comprise a cryptic exon 3. In an aspect, disclosed functionally important human RNA can comprise LMO7 mRNA functional secondary structures and tertiary structures. In an aspect, disclosed LMO7 mRNA functional secondary and tertiary structures can comprise splice sites associated with prostate cancer.

[0357] In an aspect, disclosed functionally important viral RNAs can comprise Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) 3'-end triple helix. In an aspect, disclosed functionally important viral RNAs can comprise Kaposi's sarcoma-associated herpesvirus (KSHV) polyadenylated nuclear (PAN) functional secondary structures and/or tertiary structures. In an aspect, disclosed functionally important viral RNAs can comprise SARS-CoV-2 5'-untranslated region structures and/or 3'-untranslated regions. In an aspect, a disclosed SARS-CoV-2 5'-untranslated region structure can comprise a frameshifting pseudoknot. In an aspect, disclosed functionally important viral RNAs can comprise one or more HIV elements. In an aspect, the HIV elements can comprise a trans-activation response (TAR) element, a Rev response element (RRE), EESV, a frameshift site, a packing signal (interaction between (w) stem-loop 3 (SL3) RNA and Gag), or any combination thereof. In an aspect, disclosed HIV elements can comprise other functional secondary structures and/or functional tertiary structures.

[0358] In an aspect, disclosed functionally important viral RNAs can comprise one or more enterovirus elements. In an aspect, disclosed enterovirus elements can comprise the 5'-untranslated region structures, internal ribosomal entry sites (IRES), functional coding region structures, a frameshift pseudoknot, a 3'-untranslated region structures, or any combination thereof.

[0359] In an aspect, disclosed functionally important viral RNAs can comprise one or more viral secondary structures or tertiary structures. In an aspect, disclosed functionally important viral RNAs can comprise one or more viral secondary structures or tertiary structures in a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpes-

virus, a hepatitis A virus, or a hepatitis C virus. In an aspect, disclosed functionally important viral RNAs can comprise a secondary structure or a tertiary structure in the 5'-untranslated region or the internal ribosome entry site of a poliovirus, a rhinovirus, an encephalomyocarditis virus, a foot-and-mouth disease virus, a Kaposi's sarcoma-associated herpesvirus, a hepatitis A virus, or a hepatitis C virus. In an aspect, disclosed functionally important viral RNAs can comprise one or more viral secondary structures or tertiary structures in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus. In an aspect, disclosed functionally important viral RNAs can comprise a secondary structure or a tertiary structure in the 5'-untranslated region, coding region, or the 3'-untranslated region in a coronavirus, a flavivirus, alphaviruses, a picornavirus, or a positive sense RNA virus.

[0360] In an aspect, disclosed RNA structural motifs can comprise a RNA quadruplex structure. In an aspect, disclosed RNA quadruplex structures can comprise telomeric-repeat containing RNA (TERRA RNA).

[0361] In an aspect, one or more disclosed functionally important RNAs can comprise telomerase RNAs, group I and II introns, long noncoding RNAs (lncRNAs), ribosomal RNAs, or any combination thereof.

[0362] In an aspect of a disclosed method, a disclosed destabilizing compound can comprise DMZ-P0, DMZ-P1, DMZ-P5, DMZ-P14, DMZ-O2, DMZ-O5, DMZ-M4, DMZ-M15, DMZ-M22, DMZ-NMe-P38, DMZ-NMe-mPy-P1, or any combination thereof. In an aspect of a disclosed method, a disclosed stabilizing compound can comprise DMZ-P8, DMZ-P9, DMZ-P13, DMZ-P17, DMZ-O1, DMZ-O4, DMZ-O6, DMZ-M1, DMZ-M3, DMZ-M7, DMZ-M10, or any combination thereof.

[0363] In an aspect, disclosed method can further comprise treating a subject in need thereof by administering to the subject a disclosed compound, a disclosed composition, a disclosed pharmaceutical formulation, or any other combination thereof.

[0364] In an aspect, a disclosed method can further comprise generating and/or validating one or more of disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof. Methods of generating and/or validating disclosed compounds, disclosed compositions, disclosed pharmaceutical formulations, or any combination thereof are discussed infra.

N. Kits

[0365] Disclosed herein is a kit comprising one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof. Disclosed herein is a kit comprising one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof with or without additional therapeutic agents to effect a disclosed method of treating a subject. Disclosed herein is a kit comprising one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof with or without additional therapeutic agents to effect a disclosed method of modulating secondary RNA structures and/or tertiary RNA structures. Disclosed herein is a kit comprising one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formula-

tion, or any combination thereof with or without additional therapeutic agents to effect a disclosed method of assessing a library of small molecules.

[0366] In an aspect, a disclosed kit can comprise at least two components constituting the kit. Together, the components constitute a functional unit for a given purpose (such as, for example, treating a subject diagnosed with or suspected of having or can be diagnosed with having a disease, condition, or disorder caused by, related to, and/or exacerbated by the absence of, lack of expression of, and/or lack of activity of one or more secondary RNA structures, one or more tertiary RNA structures, or any combination thereof).

[0367] Individual member components may be physically packaged together or separately. For example, a kit comprising an instruction for using the kit may or may not physically include the instruction with other individual member components. Instead, the instruction can be supplied as a separate member component, either in a paper form or an electronic form which may be supplied on computer readable memory device or downloaded from an internet website, or as recorded presentation. In an aspect, a kit for use in a disclosed method can comprise one or more containers holding one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof, and a label or package insert with instructions for use. In an aspect, suitable containers include, for example, bottles, vials, syringes, blister pack, etc. The containers can be formed from a variety of materials such as glass or plastic. The container can hold, for example, one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof, and can have a sterile access port (for example the container may be an intravenous solution bag or a vial having a stopper pierceable by a hypodermic injection needle). The label or package insert can indicate that one or more disclosed compounds, disclosed compositions, disclosed pharmaceutical formulation, or any combination thereof can be used for treating a subject or modulating secondary RNA structures and/or tertiary RNA structures. In an aspect, a disclosed kit can comprise additional components necessary for administration such as, for example, other buffers, diluents, filters, needles, and syringes. In an aspect, a disclosed kit can comprise those components (e.g., primers) necessary to measure one or more times the level of expression and/or the level of activity of one or more disclosed secondary RNA structures and/or disclosed tertiary RNA structures.

TABLE 1

Sequence Information.		
SEQ ID NO	Description	
1	MALAT1 TH - DNA Template	
2	MALAT1 TH - Fwd primer	
3	MALAT1 TH - Rev primer	
4	Stem Loop DNA Template	
5	Stem Loop Fwd primer	
6	Stem Loop Rev primer	
7	Upper Stem RNA Sequence	
8	MALAT1 TH RNA Cy5 Labelled	
9	MALAT1 (Metastasis Associated Lung Adenocarcinoma Transcript 1) (NR_002819.4)	
10	MALAT1 (Metastasis Associated Lung Adenocarcinoma Transcript 1) (NR_144567.1)	

TABLE 1-continued

Sequence Information.		
SEQ ID NO	Description	
11	MALAT1 (Metastasis Associated Lung Adenocarcinoma Transcript 1) (NR_144568.1)	
12	NEAT1 (nuclear paraspeckle assembly transcript 1) (NR_028272.1)	
13	NEAT1 (nuclear paraspeckle assembly transcript 1) (NR_131012.1)	
14	LIM domain 7 (isoform 1) (NM_005358.5)	
15	LIM domain 7 (isoform 2) (NM_001366632.2)	
16	LIM domain 7 (isoform 3) (NM_001306080.2)	
17	LIM domain 7 (isoform 1) (NP_005349.3)	
18	LIM domain 7 (isoform 2) (NP_001353561.1)	
19	LIM domain 7 (isoform 3) (NP_001293009.1)	
20	SWI/SNF complex antagonist associated with prostate cancer 1 (SCHLAP1), transcript variant 1 (NR_104319.1)	
21	SWI/SNF complex antagonist associated with prostate cancer 1 (SCHLAP1), transcript variant 2 (NR_104320.1)	
22	SWI/SNF complex antagonist associated with prostate cancer 1 (SCHLAP1), transcript variant 3 (NR_104321.1)	
23	SWI/SNF complex antagonist associated with prostate cancer 1 (SCHLAP1), transcript variant 4 (NR_104323.1)	
24	Androgen Receptor (AR), transcript variant 3 (NM_001348061.1)	
25	Androgen Receptor (AR), transcript variant 3 (NP_001334990.1)	

VII. EXAMPLES

[0368] The exponential increase in identification of both coding and non-coding RNAs actively involved in human diseases has opened the door to novel therapeutic targets of all shapes, sizes, and levels of structural complexity. (Zafferani M, et al. (2021) Cell Chem Biol. 28(5):594-609; Fedorova O, et al. (2018) Nat Chem Biol. 14(12):1073-1078). RNA targeting efforts to date have revealed that traditional drug discovery approaches built for protein targets may need to be optimized or re-designed to accommodate the chemical and conformational differences of RNA. (Warner K D, et al. (2018) Nat Rev Drug Discov. 17(8): 547-558). Recent surveys of known RNA-binding small molecules indicate that these probes may indeed exist in a privileged chemical space. (Padroni G, et al. (2020) RSC Med Chem. 11(7):802-813; Rizvi N F, et al. (2020) SLAS Discov. 25(4):384-396). One specific feature that emerged from these studies is that RNA-binding small molecules more often form rod-like 3D shapes when compared to FDA-approved protein binders. (Morgan B S, et al. (2017) Angew Chem Int Ed Eng. 56(43):13498-13502). However, the relatively small size of the surveyed library and the vastly different approaches that led to ligand discovery make the generalizability of this trend unclear. Furthermore, the majority of successfully targeted RNAs are secondary structures, which may be more challenging to selectively engage with small molecules due to their structural simplicity. (Warner K D, et al. (2018) Nat Rev Drug Discov. 17(8): 547-558; Carothers J M, et al. (2004) J Am Chem Soc. 126(16):5130-5137).

[0369] Consequently, experts in the field have proposed that ligand development should focus on disease-relevant targets with higher 'bits' of information, a unit-measure of the uniqueness of a given RNA target found to correspond with a higher probability of possessing classically-defined druggable sites and increased structural complexity. (Warner K D, et al. (2018) Nat Rev Drug Discov. 17(8):547-558).

[0370] There is thus a clear need to test the applicability of identified RNA-targeted chemical space to more complex RNAs and to develop new methods to better understand these molecular recognition events.

[0371] In this study, a multi-dimensional assessment of a synthetic library targeting a single RNA tertiary structure motif, namely the RNA triple helix present within the long non-coding RNA (lncRNA) metastasis-associated lung adenocarcinoma transcript 1 (MALAT1), was employed.

[0372] Recognition of the pervasiveness of RNA triple helices in the transcriptome of multiple species led to coining of the word 'triplexome' and underlined the need to expand the currently limited set of chemical probes aimed at targeting and modulating RNA triple helices. (Brown J A, (2020) WIREs RNA. 11(6):e1598; Conrad N K. (2014) WIREs RNA. 5(1):15-29).

[0373] The MALAT1 triplex, present at the 3'-end of a 6.7 kb mature lncRNA, was chosen as the case study for this report due to its established biological relevance and the precedented, though limited, small molecule ligands that can be used for comparison. (Donlic A, et al. (2020) Nucleic Acids Res. 48(14):7653-7664; Donlic A, et al. (2018) Angew Chem Int Ed Eng. 57(40):13242-13247; Abulwerdi F A, et al. (2019) ACS Chem Biol. 14(2):223-235). MALAT1 has been found to be overexpressed in a plethora of diseases, including cancer and diabetes. (Abdulle L E, et al. (2019) Int J Med Sci. 16(4):548-555; Sun Y, et al. (2019) Cancers (Basel). 11(2):216; Yan Y, et al. (2020) IUBMB Life. 72(3):334-342; Amodio N, et al. (2018) J Hematol Oncol. 11(1):63).

[0374] While all its functions are still being elucidated and the structure of the entire transcript remains elusive, the 3'-end blunt formation of an A-U rich RNA triple helix has been found to play an active role in the cellular accumulation of the transcript. (Wilusz J E, et al. (2012) Genes Dev. 26(21):2392-2407). One of the first identified and best characterized RNA triple helices, the MALAT1 triplex forms through the sequestration of its genomically encoded poly(A) tail in the major groove of a uridine-rich stem loop, ultimately protecting the transcript from RNase-mediated digestion and increasing its half-life. (Wilusz J E, et al. (2012) Genes Dev. 26(21):2392-2407; Brown J A, et al. (2014) Nat Struct Mol Biol. 21(7):633-640). Consequently, the MALAT1 triple helix has been the subject of small molecule targeting aimed at modulating the stability of the triplex. (Donlic A, et al. (2020) Nucleic Acids Res. 48(14):7653-7664; Donlic A, et al. (2018) Angew Chem Int Ed Eng. 57(40):13242-13247; Abulwerdi F A, et al. (2019) ACS Chem Biol. 14(2):223-235).

[0375] To elucidate fundamental recognition properties of triplex modulators, a focused library was built around the diminazene scaffold due to its synthetic accessibility and known nucleic acid binding properties. The scaffold architecture and subunit chemical identity were varied to maximize diversity in 3D shape of the analogues. For a holistic approach, a 21-member library was evaluated using four in vitro assays designed and/or optimized within this work to specifically evaluate small molecule:RNA triplex interactions.

[0376] Rod-like 3D shape trends held true for the newly synthesized focused library with regards to affinity for the MALAT1 triplex. While affinity also correlated with the effect of small molecules on triplex thermal stability, vastly different trends were found in enzymatic degradation assays,

showcasing the importance of assessing RNA-targeted small molecules using multiple complementary methods.

[0377] Furthermore, employment of quantitative structure activity relationships (QSAR) afforded predictive models that revealed the importance of 3D shape small molecule descriptors in relationship to enzymatic degradation, yielding a single computational model that can predict both stabilizing and destabilizing effects. The diminazene scaffold utilized in this work established the feasibility of bidirectional modulation of RNA tertiary structure degradation with rational synthetic tuning.

Materials and Methods

Theoretical Library and Principal Moments of Inertia

[0378] A theoretical 90-member library was built by coupling commercially available amine subunits to the ortho, para, and meta-substituted scaffolds. The SMILES strings of each small molecule were then adjusted for protonation at pH=7.4 using a python code (described supra). The corrected SMILES were then loaded on Molecular Operating Environment (MOE, 2018.01) to perform a conformational search. (Donlic A, et al. (2018) Angew Chem Int Ed Eng. 57(40):13242-13247). Small molecule descriptors were calculated on the lowest energy conformer of each small molecule for a total of 339 parameters. Small molecule shape was evaluated using normalized principal moments of inertia (npr1, npr2) and plotted on an envelope diagram. A 20 small molecule sub-set was chosen for synthesis using the Kennard Stone algorithm (presented above) to ensure maximum representation of small molecule 3D shape diversity (FIG. 8A). Extensively conjugated anilines resulted in poor reactivity under a plethora of conditions and with various Lewis acids. Due to this, small molecules closest in Euclidean distance were chosen for synthetic investigation. To quantify possible alteration in 3D shape representation of the small molecule sub-set, the envelope diagram was further divided in 16 sub-triangles. Except for sub-triangle #14 which contained a single molecule, the remaining sub-triangles retained at least one representative small molecule (FIG. 8B).

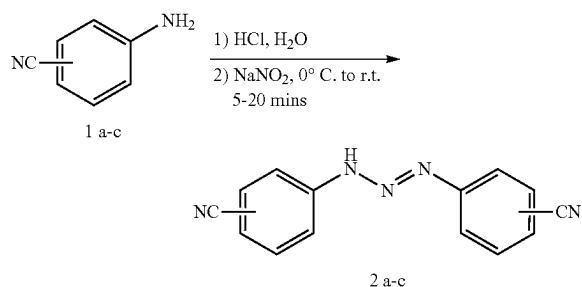
Library Synthesis and Purification

[0379] Reagents were purchased from commercial suppliers and were used without further purification. All solvents were ACS grade or better and used without further purification. Anhydrous toluene was obtained by storing ACS-grade toluene over 4 Å molecular sieves while anhydrous THF was dispensed from VWR SureSeal bottles and kept under argon. All microwave reactions were run on a Biotage Initiator+ reactor from Biotage Inc. and under argon inert atmosphere. All chromatographic purifications were conducted via flash chromatography using ultra-pure silica gel (230-400 mesh, 60 Å) purchased from Silicycle as the stationary phase. Thin Layer Chromatography was performed with glass-backed silica gel plates purchased from VWR and visualized with 254 nm UV light. All deuterated solvents for NMR experiments were purchased from Cambridge Isotope Laboratories. All ¹H and ¹³C NMR spectra were recorded using a 500 MHz Bruker spectrometer. The corresponding ¹³C resonance frequencies were 100 MHz and 125 MHz, respectively. Chemical shifts are expressed as

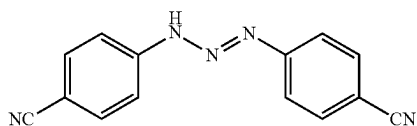
parts per million from tetramethylsilane. ^1H chemical shifts were referenced with that of the solvent (7.26 for CDCl_3 , 3.31 for CD_3OD and 4.87 for D_2O) and coupling constants (J values) are reported in units of Hertz (Hz). Splitting patterns have been designated as follows: s (singlet), d (doublet), t (triplet) and m (multiplet), br (broad). Low and high-resolution electrospray ionization (ESI) and mass spectra were recorded on an Agilent MSD-trap Spectrometer. HPLC spectra were recorded using a Shimadzu SIL-20AHT Prominence instrument. All HPLC experiments S19 were run at room temperature using gradients or isocratic mixtures of 0.1% TFA in water and acetonitrile as solvents A and B, respectively. Yields refer to $\geq 95\%$ spectroscopically and chromatographically pure compounds.

Synthesis of Bis-Cyano Scaffold

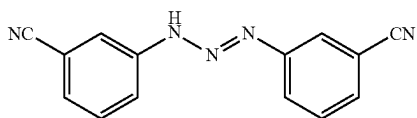
[0380] The synthesis of the bis-cyano scaffold is described below.



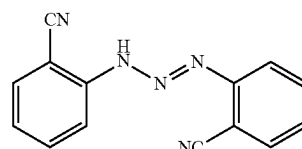
[0381] 4-(1a), 3-(1b), or 2-aminobenzonitrile (1c) (16.9 mmol) was dissolved in 0.5 M HCl (80 mL) in a 200 mL round bottom flask and cooled to 0° C. on an ice bath. Upon complete dissolution of the aniline, 2 mL of a 4.5 M solution of sodium nitrite was added dropwise. A precipitate formed immediately, and each coupling was allowed to react at room temperature until a dense solution formed. The reaction was then filtered through a frit funnel and solid washed with ice-cold deuterated water. Precipitate was left drying overnight under vacuum and used in the scaffold decoration (the procedure was adapted from previous methodology). (Cappoen D, et al. (2014) Eur J Med. Chem. 77:193-203).



[0382] Bright yellow solid; Yield=88%; ^1H NMR (500 MHz, D_2O) δ 13.28 (s, 1H), 7.97-7.54 (m, 9H). ^{13}C NMR (126 MHz, D_2O) δ 134.90, 133.58, 119.47. Calcd for $\text{C}_{14}\text{H}_9\text{N}_5$ ($[\text{M}+\text{H}]^+$): 248.0; found: 247.1(± 2.1 ppm).



[0383] Pale beige solid; Yield=53%; ^1H NMR (500 MHz, $\text{DMSO}-d_6$) δ 12.98 (s, 1H), 7.96 (s, 1H), 7.79 (d, $J=8.3$ Hz, 2H), 7.61 (q, $J=8.7, 7.5$ Hz, 4H), 6.88-6.82 (m, 1H). ^{13}C NMR (126 MHz, DMSO) δ 132.03, 130.50, 119.09, 112.76. Calcd for $\text{C}_{14}\text{H}_9\text{N}_5$ ($[\text{M}+\text{H}]^+$): 248.0; found: 249.1(± 1.8 ppm).



[0384] Bright yellow solid; Yield=74%; ^1H NMR (500 MHz, $\text{DMSO}-d_6$) δ 13.41 (s, 1H), 7.96 (d, $J=8.2$ Hz, 1H), 7.82 (dd, $J=37.5, 8.0$ Hz, 2H), 7.69 (dt, $J=21.9, 8.5$ Hz, 2H), 7.48-7.40 (m, 2H), 7.27-7.19 (m, 1H). ^{13}C NMR (126 MHz, DMSO) δ 152.05-151.93, 142.93, 135.56-132.83, 128.36, 124.26, 119.00-108.57, 98.40, 93.96, 55.32, 40.50-39.50, 29.99. Calcd for $\text{C}_{14}\text{H}_9\text{N}_5$ ($[\text{M}+\text{H}]^+$): 248.0; found: 248.1 (± 1.6 ppm).

General Procedure for Amidine Formation

[0385] 2a-c (0.41 mmol) and DABAL-Me3 (1.2 mmol) was added to a 5 mL over-dried pressure vial under argon. The solids were dissolved in anhydrous THF or toluene (2.5 mL) and a primary amine (1 mmol) was added drop-wise in a 5 mL over-dried pressure vial under argon and heated to 105° C. for 4.5 hours. After running, the reaction was diluted in dichloromethane and quenched with acetonitrile dropwise while stirring. The solution was then evaporated under vacuum. The solid was re-dissolved in methanol and a 5:1 ratio of celite:starting material was added. Compounds were purified using silica column chromatography in a gradient 95:4:1 dichloromethane:methanol:ammonium hydroxide to 85:14:1 dichloromethane:methanol:ammonium hydroxide to yield the final compounds. Procedure was adapted from previously published synthesis. (Donlic A, et al. (2018) Angew Chem Int Ed Eng. 57(40):13242-13247).

[0386] Extensively conjugated anilines have been found to be unreactive with various Lewis acids and under various conditions as listed below (FIG. 9). Specifically, no diamidine formation was observed in the presence of R-19, R-25, R-27, and R-28 (FIG. 2).

Indicator Displacement Assay (DIA)

[0387] The protocol was adapted from a previously reported procedure. (Donlic A, et al. (2020) Nucleic Acids Res. 48(14):7653-7664). A serial dilution of the MALAT1 triple helix RNA was performed in HEPES buffer (20 mM HEPES-KOH, 52.6 mM KCl, 0.1 mM MgCl_2 , pH=7.4) in a 96 well plate in triplicate. 8 mL of each dilution were transferred to a 384 well plate followed by 8 mL of a 500 nM solution of RiboGreen dye in the same buffer (Invitrogen™). The plates were excited at 487 nm (8 nm slit) and emission was recorded at 530 nm (8 nm slit, focal height 11.3 mm) using a CLARIOstar plate reader (BMG labtech). The affinity of the dye for the RNA construct was determined by fitting the raw fluorescence in GraphPad Prism version 8.3.1 for Macintosh (GraphPad Software, La Jolla California

USA, (www.graphpad.com)) by fitting to the Log[Agonist] vs. response—variable slope model that using Equation (1) below.

$$Y = \text{Bottom} + (X^{\wedge \text{HillSlope}} * (\text{Top} - \text{Bottom})) / (X^{\wedge \text{HillSlope}} + \text{LogCD50}^{\wedge \text{HillSlope}}) \quad (\text{Equation } 1)$$

where Y is normalized % change in fluorescence intensity, X is RNA concentration, Bottom is lowest fluorescence % change and Top is highest fluorescence % change. Affinity of the dye for the RNA construct was used as the ideal RNA concentration (174 nM) for small molecule titrations.

[0388] A serial dilution of small molecule (0, 0.5 μM , 1.0 μM , 2.5 μM , 5 μM , 7.5 μM , 10.0 μM , 12.5 μM , 25 μM , 50 μM , 75 μM , 100 μM , 125 μM , 150 μM , 250 μM , 500 μM) was performed in HEPES buffer (20 mM HEPES-KOH, 52.6 mM KCl, 0.1 mM MgCl_2 , pH=7.4) in a 96 well plate in triplicate. Then, 8 μL of each dilution were transferred to a 384 well plate, followed by 8 μL of a 174 nM solution of MALAT1 RNA and 0.5 mM of RiboGreenTM (InvitrogenTM). The RNA was previously annealed by heating to 95° C. for 5 minutes and cooling to 0° C. on ice for 30 minutes. The 384 well plates were shaken for 5 minutes, centrifuged at 4000 rpm for 1 minute, and incubated in the dark for 20 minutes. The plates were excited at 487 nm (8 nm slit) and emission was read at 525 nm (8 nm slit, focal height 11.3 nm) using a CLARIOstar plate reader (BMG Labtech). Percent fluorescence indicator displacement (% FID) was calculated by subtracting and, subsequently, dividing by the blank wells with RNA-dye complex and no small molecule as shown in Equation (2).

$$\% \text{ FID} = \left(\frac{F_0 - F}{F_0} \right) * 100 \quad (\text{Equation } 2)$$

[0389] Where F_0 is the fluorescence of the blank well with RNA+dye and no small molecule and F is the fluorescence of the well with all three components (RNA+dye+small molecule).

[0390] Each technical triplicate was averaged, and the resulting FID values were averaged between three independent experiments. The binding curve and Log (CD_{50}) value was obtained by using a non-linear fit curve (Log[Agonist] vs. response) with variable slope at four parameters (Graph-Pad Prism Software version for Macintosh 8.3.1, La Jolla, California, USA (www.graphpadprism.com)) as shown in Equation 1. Reported values are averages of three independent experiments=standard deviation.

Differential Scanning Fluorimetry

[0391] Differential scanning fluorimetry experiments were performed as previously reported. (Donlic A, et al. (2020) Nucleic Acids Res. 48(14):7653-7664). All experiments were performed in white 96 well plates in a LightCycler® 96 (Roche). In a typical experiment, 4 μM of RNA was annealed in the same buffer used for indicator displacement assays (20 mM HEPES-KOH, 52.6 mM KCl, 0.1 mM MgCl_2 , pH=7.4) by heating at 95° C. for 3 minutes, snap-cooling on ice for 10 minutes, and leaving the solution at room temperature for 1 hr. Then, 42 μL of this solution was

aliquoted to wells and 5 μM of ligand was added (2.5 μL of water, 2.0 μL of DMSO, 0.5 μL of 500 μM small molecule stock). The mixture was left to incubate at room temperature for 5 minutes, after which 3 μL of a 20 μM Quant-iTTM RiboGreenTM (ThermoFisher Scientific) dye stock solution in appropriate buffer was added to each well. The 96 well plate was then sealed with an optically clear foil and centrifuged for 1 minute at 4000 rpm prior to being placed in the instrument. The light cycler program was created by selecting the melting curve method; fluorescence intensity was monitored using the SYBR Green I/HRM dye filter combination (465-510 nm) from 37-98° C. at a ramp rate of 0.01° C./second with 150 acquisitions per° C. Melting profiles were obtained by T_m analysis in the LightCycler® 96 software (SW 1.1).

RNA Synthesis and Purification

[0392] DNA template sequence was purchased from Dharmacon and forward and reverse primers were purchased from Integrated DNA Technologies (IDT). For PCR amplification the following reagents were added to for a given 50 mL final reaction volume. First, the entire working space was sprayed with RNase Zap to prevent any contamination. Next, in the desired amount of sterile PCR tubes (ThermoFisher), 12.5 mL of RNase free water and 12.5 mL of Q5 reaction buffer were added, followed by 1 mL of dNTPs (10 mM), 2.5 mL of forward and reverse primer (10 mM), 1 mL of DNA template (50 ng/mL) and lastly 0.5 mL of Q5 polymerase (New England Biolabs). The DNA template was then amplified for 30 cycles in an Eppendorf Echo thermocycler. A Zymo DNA-clean-up kit was then utilized to clean-up the desired DNA sequence. A solution of amplified DNA in water was made to reach 28-35 ng/mL. The sequence was then in vitro transcribed (IVT) using the following protocol. For a given 50 mL IVT reaction the following were added in respective order: 31.75 mL of RNase free water, 1.25 mL MgCl_2 (1 M), 2 mL Tris-HCl (1 M at pH 8.0), 1.25 mL of spermidine (0.1 M), 0.5 mL of Triton-X (0.1%), 0.5 mL of DTT (1 M), 0.2 mL of pyrophosphatase enzyme (100 U/mL), 5 mL of DNA template and 2.5 mL of T7 polymerase enzyme kindly provided by the Tolbert lab at Case Western University.

[0393] The reaction was then incubated at 37° C. for 12 hours. Following incubation, the reaction was treated with DNase I buffer and DNase I (New England Biolabs) twice in intervals of 30 minutes, followed by addition of 10% of the reaction volume of EDTA. The desired RNA was then extracted using phenol chloroform extraction and further purified via ethanol precipitation. Purity and size of the RNA construct was confirmed by 15% TBE denaturing gel (see RNase A section) and by Small RNA chip on an Agilent Bioanalyzer. Upper stem RNA was purchased from Dharmacon (Horizon Discovery), Cy5-labelled MALAT1 Triple Helix was ordered from Dharmacon (Horizon Discovery), and upper stem loop DNA sequence was purchased from IDT.

TABLE 2

Sequences of the DNA Template and Primers used in In Vitro Synthesis of MALAT1 Triple Helix and Stem Loop	
Description	Sequence
MALAT1 TH DNA Template	GGAAGGTTTTCTTTTCCTGAGAAAACAACACGTATTGTTT TCTCAGGTTTTGCTTTTGGCCTTTTCTAGCTTAAAAAAA AAAAAGCAAAA (SEQ ID NO: 01)
MALAT1 TH Fwd primer	GAAATTAATACGACTCACTATAGGAAGTTTTCTTTTCCT GAGAAAACAACACGTATT (SEQ ID NO: 02)
MALAT1 TH Rev primer	TmU*TTGCTTTTTTTTTTTTAAGC (SEQ ID NO: 03)
Stem Loop DNA Template	GGAAGGTTTTCTTTTCCTGAGAAAACAACACGTATTGTT TTCTCAGGTTTTGCTTTTGGCCTTTT (SEQ ID NO: 04)
Stem Loop Fwd primer	GAAATTAATACGACTCACTATAGGAAGTTTTCTTTTCCT GAGAAAACAACACGTATT (SEQ ID NO: 05)
Stem Loop Rev primer	A[mA]AAGGCCAAAAGCAA (SEQ ID NO: 06)
Upper Stem RNA Sequence	CCUGAGAAAACAACAGUAUUUUUCUCAGG (SEQ ID NO: 07)
MALAT1 TH RNA Cy5 Labelled	Cy5GGAAGGUUUUUUCUUUCCUGAGAAAACAACAGUAU UGUUUUUCUCAGGUUUUGCUUUUUGGCCUUUUUCUAGCUU AAAAAAAAAAAAAGCAAAA (SEQ ID NO: 08)

RNase A Enzymatic Degradation Assay

[0394] The procedure was adapted from a published protocol for exonucleolytic enzymatic degradation assay. (Donlic A, et al. (2020) *Nucleic Acids Res.* 48(14):7653-7664). The MALAT1 destabilizing small molecules published by Le Grice (Abulwerdi F A, et al. (2019) *ACS Chem Biol.* 14(2):223-235) and co-workers SM5 and SM16 were purchased from ChemBridge. For a typical 70 μ L reaction, 0.2 mM of MALAT1 triple helix RNA was annealed in the same buffer used for IDA and DSF experiments (20 mM HEPES-KOH, 52 mM KCl, 0.1 mM MgCl₂, pH=7.4) by heating at 95° C. for 3 minutes, snap-cooling on ice for 10 minutes, and leaving the solution at room temperature for 1 hr.

[0395] RNase A enzyme (ThermoFisher) was diluted to 1.46 nM and incubated with 1 M NaCl for 30-35 minutes to achieve ssRNA selectivity as reported. (Struhl K, et al. (1989) *Current Protocols in Molecular Biology.* 8(1):3.13.1-3.13.3). Next, the RNA was mixed with 0.2 μ M of a GC-rich DNA duplex, 0.2 mM of small molecule/DMSO and incubated at room temperature for 20 minutes. Following the incubation, 1 mL of RNase A was added, and the reaction was incubated at 37° C. for 1 hour in a thermal cycler with a heated lid (Eppendorf). 4 μ L timepoint aliquots were taken at 0, 5, 10, 30, and 60 minutes, 4 μ L of 2 $\times$ RNA loading dye was added to each aliquot (ThermoFischer Scientific), and samples were stored at -80° C. prior to gel analysis. Reactions were analyzed on pre-cast 15% PAGE-Urea gels (Invitrogen) in 1 $\times$ TBE buffer. Aliquots were thawed from -80° C. storage and heated to 95° C. for 15 minutes prior to gel loading and 6 mL were loaded in each well. Gels were run for 45 minutes at 180 V, stained with Diamond™ Nucleic Acid dye (Promega), and visualized on an iBright 500 (ThermoFisher). DMSO controls were ran with every set of 4 small molecules to ensure consistency of enzyme activity and RNA integrity. Gel band intensity of the

MALAT1 triple helix and DNA loading control were quantified using ImageJ software. (Rueden C T, et al. (2017) *BMC Bioinformatics.* 18(1):529).

[0396] To quantify time-dependent endonucleolytic degradation for each sample, the intensity was normalized using Equation (3), and fold-difference was calculated as listed in Equation (4) as previously reported. (Donlic A, et al. (2020) *Nucleic Acids Res.* 48(14):7653-7664).

$$y = (TH \text{ Area}(tx)) \frac{LC \text{ Area}(t0)}{LC \text{ Area}(tx)}$$

(Equation 3)

where y is normalized RNA area for a specific time point and the sample (tx), TH (triple helix). Area is the raw band intensity for the RNA in the lane corresponding to time point x for that sample, LC (loading control). Area (t0) is the raw band intensity for the DNA in the lane corresponding to the 0 minute time point for the sample, and LC Area (tx) is the raw band intensity for the DNA in the lane corresponding to time point x for that sample.

$$y = \frac{\text{Normalized Area}(tx)}{\text{Normalized Area}(t0)}$$

(Equation 4)

where y is fold-difference of the exonucleolytic degradation for the specific sample, with normalized areas for its specific time point x and time point 0 calculated as described in Equation (3). Percent RNA degraded was calculated for every time point using Equation (5).

$$\% \text{ RNA degraded} = (\text{Normalized area}(t0) - \text{Normalized area}(tx)) * 100$$

(Equation 5)

[0397] Experimental results shown in FIG. 6C are plotted as three independent replicates and standard deviation between replicates as error bars. Relative fold change between treated condition and DMSO was calculated from Equation (6).

Protonation Correction

Fold change at 5 min =

(Equation 6)

$$\frac{\% \text{ RNA Degraded}_{\text{Treated}} - \% \text{ RNA Degraded}_{\text{DMSO}}}{\% \text{ RNA Degraded}_{\text{DMSO}}}$$

[0398] A python code was used to adjust the SMILES strings of each small molecule for protonation at pH=7.4. The python code is presented below.

```
#!/bin/bash
evaluate      | grep "version "
cxcalc -o 01-CanRes.sdf Input.txt -N ih canonicalResonant -f sdf
cxcalc -o 02-MMcorrected.txt 01-CanRes.sdf -N ih -g majormicrospecies -H 7.4 ### -M for
notaut comment-out -M
cxcalc      | grep "version "
evaluate -g 02-MMcorrected.txt -e
"molString('smiles');mass( );acceptorCount( );donorCount( );logPKLOP( );rotatableBondCount(
);PSA( );logD('7.4');atomCount('7');atomCount('8');ringCount( );aromaticRingCount( );heteroRi
ngCount( );ringSystemCount( );ringCount( );ringSystemCount( );count(filter('atno'=='6&&conn
ections('==4'))/atomCount('6');chiralCenterCount( );ASA( );PSA( );vanDerWaalsSurfaceArea(
);totalCharge( );vanDerWaalsSurfaceArea( )" -o 03-OutputCalcs.txt 2> 03-output-err.txt
cat 03-OutputCalcs.txt | tr -d, | tr ';>'> 03-OutputCalcs.csv ## translate semi-colon in
output.txt to comma for output.csv
cat 03-OutputCalcs.csv | awk -F, '{print $ 1}' > 03-OutputCalcs.smi ## create file with just
first column of smiles
[ -s 03-output-err.txt ] || rm 03-output-err.txt ## remove err if empty
evaluate      | grep "version "
wc *
echo -n "Finished" && date
```

Small Molecule Selection

[0399] The Kennard Stone algorithm was used during selection of the small molecules for synthesis to ensure maximum representation of small molecule 3D shape diversity (FIG. 8A).

```
function [model,test]=kenstone(X,k)
[m,n]=size(X);
if k>=m | k<=0
    h=errorlg('Wrongly specified number of objects to be selected to model set.','Error');
    model=[];
    if nargout==2
        test=[];
    end
    waitfor(h)
    return
end
x=[1:size(X,1)]'X';
n=size(x,2);
[i1,ind1]=min(fastdist(mean(x(:,2:n)),x(:,2:n)))
model(1)=x(ind1,1);
x(ind1,:)=[];
[i2,ind2]=max(fastdist(X(model(1,:),:),x(:,2:n)))
model(2)=x(ind2,1);
x(ind2,:)=[];
for d=3:k
    [ii, ww]=max(min(fastdist(x(:,2:n),X(model,:)))));
    model(d)=x(ww,1);
    x(ww,:)=[];
end
if nargout==2
    test=1:size(X,1);
    test(model)=[];
end
```

QSAR Modeling

[0400] Before calculation, all the ligands were tuned to the correct protonation and tautomerization states using above protonation correction steps. The optimized protonation and tautomerization states were sent to conformational search in Molecular Operating Environment (MOE, Chemical Computing Group, 2018.01) individually to account for the flexibility of the ligand. Low energy conformations of each molecule were calculated using the Conformation Search algorithm in MOE. The Conformation Search function was performed using the stochastic method with the MMFF94 force field and generalized Born solvation model. The input for each parameter is listed in Table 3, and the following options were checked: hydrogens.

TABLE 3

Parameters Used for Conformation Search.	
Parameter	Input
Rejection limit	100
Iteration limit	10000
RMS gradient	0.005
MM iteration limit	500
RMSD limit	0.15
Energy window	3
Conformation limit	10000

[0401] The 3 kcal/mol energy window was selected to survey biologically relevant conformation space and to

obtain a representative population of conformers at equilibrium (>99%) as described by Equation S1.

$$\frac{N_1}{N_o} = e^{-\Delta E/RT} \quad (\text{Equation S1})$$

where N_1/N_o is the ratio of the number of molecules in the relative energy states, ΔE is the energy difference between N_o and N_1 (3 kcal/mol), R is the ideal gas constant (0.00198588 kcal/K mol), and T is the temperature (298 K). After the conformation search was complete, the 339 descriptors (2D and 3D descriptors), ranging from electrostatic properties to topological terms, were calculated for each conformation and averaged using the Boltzmann weighted equation (Equation S2).

$$A = \frac{\sum_i A_i e^{-\frac{E_i}{k_B T}}}{\sum_i e^{-\frac{E_i}{k_B T}}} \quad (\text{Equation S2})$$

MatLab Modeling and Scripts

[0402] Modeling was performed on MATLAB (R2020a). Model search, selection, LOOCV, and weights calculation.

```
load('FIDdata.mat'); % this matrix contains the 1st row as the index of the variable names, 0 for
response variable
% find features with constant entry>=80%, delete such features resulting new
% dataset called:data_nonconst
data=FIDdata;
for i=2:size(data,2)
    Y(i)=max(sum(data(:,i))==data(:,i)));
end
idx_const=find(Y(:)>=0.8*(size(data,1)-1));
data_nonconst=data;
data_nonconst(:,idx_const)=[];
% find multicollinearity (abs(rho)>0.95) between features, delete ones with more than 1
% multicollinearity, based on the max number of multicollinearity, saved the
% refined data in the data_refine.
data_refine=data_nonconst;
cor=corrcoef(data_refine(2:size(data_refine,1),2:size(data_refine,2)));
cor=abs(cor);
[a,b]=find(cor>0.95);
A=[b,a];
id=find(b>=a);
A(id,:)=[];
uni=unique(A(:,1));
num=zeros(size(uni,1),1);
for i=1:size(uni,1)
    idx=find(uni(i)==A(:,1));
    num(i)=size(idx,1);
end
n=0;
while max(num)>1 % repeat until only one more feature is correlated
    id_max=find(num==max(num));
    if size(id_max,1)>1
        id_max=id_max(1,1);
    else
        id_max=id_max;
    end
    del_col=find(A(:,1)==uni(id_max));
    id_del=A(del_col,2);
    data_refine(:,id_del+1)=[];
    cor=corrcoef(data_refine(2:size(data_refine,1),2:size(data_refine,2)));
    cor=abs(cor);
```


-continued

```

[a,b]=find(cor>0.95);
A=[b,a];
id=find(b>=a);
A(id,:)=[];
uni=unique(A(:,1));
num=zeros(size(uni,1),1);
for i=1:size(uni,1)
    idx=find(uni(i)==A(:,1));
    num(i)=size(idx,1);
end
n=n+1; % record how many steps take to complete this task
end
% in a pair of multicorrelation, delete the one with lower cor to the y
% variable
m=0;
while size(A,1)>0
    cor1=abs(corrcoef(data_refine(:,1),data_refine(:,A(1,1)+1)));
    cor1=cor1(1,end);
    cor2=abs(corrcoef(data_refine(:,1),data_refine(:,A(1,2)+1)));
    cor2=cor2(1,end);
    if cor1>=cor2
        id_del=A(1,2);
        data_refine(:,id_del+1)=[];
    else
        id_del=A(1,1);
        data_refine(:,id_del+1)=[];
    end
    cor=corrcoef(data_refine(2:size(data_refine,1),2:size(data_refine,2)));
    cor=abs(cor);
    [a,b]=find(cor>0.95);
    A=[b,a];
    id=find(b>=a);
    A(id,:)=[];
    m=m+1;
end
save('FID_data_refine.mat','data_refine');
%% two-parameter linear model
% T_FID.mat is a table contains variable names from data_refine.
idx_keep=data_refine(1,:)+1;
T_FID=readtable('FIDdata.csv','PreserveVariableNames',true);
T_FID=T_FID(:,idx_keep);
save('T_FID.mat','T_FID');
x=T_FID(:,2:size(T_FID,2));
y=T_FID(:,1);
Results=[];
for i=1:size(x,2)-1
    for j=(i+1):size(x,2)
        X=[x(:,i),x(:,j)];
        mdl=fitlm(X,y);
        r_A=mdl.Rsquared.Adjusted;
        r_O=mdl.Rsquared.Ordinary;
        Results=[Results;r_A r_O i j];
        X=sprintf('This is the run of %d, %d', i,j);
        disp(X)
    end
end
end
%sort the R2 from high to low. Chose the best model based on the R2 and Q2, here pick the first
one as the only qualifiable one (with feature index 11 and 91) for loocv.
Results=sortrows(Results,1,'descend');
% summarzie overall R2 and model statistics
data_mdl=[T_FID(:,11),T_FID(:,91),T_FID(:,1)];
mdl_overall=fitlm(data_mdl)
% LOOCV for the top model
yhat=zeros(size(data_mdl,1),1);
for i=1:size(data_mdl,1)
    test=data_mdl(i,:);
    train=data_mdl;
    train(i,:)=[];
    mdl=fitlm(train);
    yhat(i)=predict(mdl,test);
end
end
%calculate Q2_loocv
Rsquare(table2array(data_mdl(:,3)),yhat)
% calculate the weight of top features, threshold as >=0.6 R2_ordinary
idx_top=find(Results(:,2)>=0.6);
feature_top=Results(idx_top,3:4);

```

-continued

```

%% Occur is an*1 vector containing all the descriptors appeared in the well fitted models
%% Y =n*2 matrix, first column is the descriptor ID, the second column is their occurrence.
Occur=[feature_top(:,1);feature_top(:,2)];
Y=(sum(Occur==Occur));
Y=[Occur, Y];
[~,idu]=unique(Occur(:,1));
Y=Y(idu,:);
histogram(Occur,'BinWidth',1);
title('Occurrence - FID',FontSize,18);
xlabel('Descriptor ID',FontSize,18);
ylabel('Counts',FontSize,18);
% plot the figure
scatter(table2array(T_FID(:,1)),mdl_overall.Fitted,100,'MarkerEdgeColor',[0.00 0.45 0.74], ...
'MarkerFaceColor','none','LineWidth',2)
ax=gca;
ax.FontSize=15;
hold on
scatter(table2array(T_FID(:,1)),yhat,100,'x','MarkerEdgeColor',[1 0 0], ...
'MarkerFaceColor','none','LineWidth',2)
x=linspace(0,2.5);
y=x;
plot(x,y,'LineWidth',2,'LineStyle','--','Color','k')
legend('Fitted','LOOCV',FontSize,15,TextColor,'k','Location','southeast','FontWeight','bold')
title('Pred. vs Observed log(FID)',FontSize,20,'FontWeight','bold')
xlabel('Observed',FontSize,20,'FontWeight','bold')
ylabel('Predicted',FontSize,20,'FontWeight','bold')
annotation('textbox',[0.15,0.8,0.45,0.1],'FontSize',12,'String', ...
'streat(log(FID) = 1.7 - 0.007 ASA\_\_P + 0.036 PEOE\_\_VSA-5;',\{'32'\},\{'it R^2=0.69';'\,'32'\},
...
'Q^2_{LOOCV}=0.60'),EdgeColor','none');

```

[0403] For the descriptor weight calculation, the threshold for the top models were set for $R^2_{ordinary}$. IDA Data was 0.69, RNase was 0.75, and T_m was 0.90. All the models (31375), then $R^2 \geq 0.6$, then good models, then $R^2_{LOOCV} \geq 0.40$ and $Q^2_{F2} \geq 0.5$, then top models.

QSAR Modeling

[0404] The MTT reagent (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-2H-tetrazolium bromide) is a mono-tetrazolium salt that consists of a positively charged quaternary tetrazole ring core containing four nitrogen atoms surrounded by three aromatic rings including two phenyl moieties and one thiazolyl ring. Reduction of MTT results in disruption of the core tetrazole ring and the formation of a violet-blue water-insoluble molecule called formazan. The MTT reagent can pass through the cell membrane as well as the mitochondrial inner membrane of viable cells presumably due to its positive charge as well as its lipophilic structure and is reduced to formazan by metabolically active cells. The chromogenic nature of this redox chemical reaction provides a colorimetric-based measurement of intracellular formazan production based on which the MTT assay was developed. Consequently, the assay has extensive utility as a cell metabolic activity assay.

Example 1

Design and Synthesis of Diminazene Centric Library

[0405] The diminazene scaffold was selected due to a number of reported properties, including bioavailability, solubility, ease of synthetic tuning, and reported nucleic acid binding. Also known as Berenil®, diminazene is an FDA approved (Rajapaksha I G, et al. (2018) Sci Re. 8(1):10175) antiparasitic agent and a known nucleic acid binder.

(Nguyen B, et al. (2009) Acc Chem Res. 42(1):11-21; Pilch D S, et al. (1995) Biochemistry. 34(49):16107-16124; Nguyen B, et al. (2004) Biophys J. 86(2):1028-1041). It has also been shown to preferentially bind A-U rich RNA triple helices, (Pilch D S, et al. (1995) Biochemistry. 34(49): 16107-16124) making it an attractive case study for targeting the A-U rich MALAT1 triple helix (FIG. 1A). In addition, the synthetic versatility of this scaffold allowed maximization of 3D shape diversity while minimizing confounding variables that would arise from using different chemotypes and might bias shape-based analyses. This scaffold, and the scaffold-based library approach in general, allows us to identify the key moieties needed for engagement of the intended RNA target while revealing cheminformatic trends that can guide future synthetic efforts.

[0406] First, to ensure the suitability of the diminazene scaffold to yield 3D shape diverse analogues, a 90-member theoretical library was created by coupling commercially available amines with the three possible regioisomers of the scaffold precursor (FIG. 1B, white, grey, black). All analogues were equipped with the diamidine moiety as it appears to be necessary for small molecule interactions in the minor groove of A-U rich sequences. (Pilch D S, et al. (1994) Proc Natl Acad Sci USA. 91(20):9332-9336).

[0407] To afford a direct comparison with known RNA-binding small molecules, the 3D shape of the analogues was evaluated by calculating the principal moment of inertia (PMI) on the pH-adjusted lowest energy conformers of each theoretical ligand. (Morgan B S, et al. (2017) Angew Chem Int Ed Eng. 56(43):13498-13502).

[0408] Normalized principal moments of inertia (npr1: I_1/I_3 , npr2: I_2/I_3) were calculated and plotted on the envelope diagram commonly used for 2D visualization of the data. Notably, most of the para-substituted analogues clustered in the rod-like sub-triangle while the ortho- and meta-substi-

tuted subsets were scattered across the hybrid, sphere, and disc sub-triangles, affording a suitable coverage of 3D shape diversity.

[0409] To maximize shape representation within the synthesized subset, the Kennard-Stone algorithm (Morais C L M, et al. (2019) *Bioinformatics*. 35(24):5257-5263) was used to select 21 small molecules for synthetic evaluation (FIG. 1B, red).

[0410] The synthetic route designed for the focused library capitalized on the oxidation of a 2-, 3-, or 4-aminobenzonitrile (1 a-c) to form a diazonium ion intermediate that is subsequently coupled with another equivalent of the same aniline (FIG. 1C). (Hill D T, et al. (1983) *J Med Chem*. 26(6):865-869; Rastogi S K, et al. (2018) *Eur J Med Chem*. 143:1-7). Purified intermediate (2 a-c) of each biscyano regioisomer was then submitted to diamidine formation through activation of the nitrile functional group by the Lewis acid DABAL-Me₃, optimizing a previously reported procedure (3). (Donlic A, et al. (2018) *Angew Chem Int Ed Eng*. 57(40):13242-13247).

[0411] Three of the initially selected subunits contained extensively conjugated anilines that were found to be unreactive in the presence of several Lewis acids and under various reaction conditions (FIG. 9).

[0412] To choose replacement library members, Euclidean distance was used to identify the analogues in the theoretical library closest in distance on the PMI envelope diagram to each initially selected untractable ligand, thus preserving diversity in the final 21-member library (FIG. 2B, FIG. 8A-FIG. 8B). In total, 21 symmetrical analogues were afforded in good yield and purity.

[0413] A series of affinity-based methods were used as screening and selection benchmarks to identify small molecule leads that were further assessed via thermal stability and enzymatic degradation assays. (Wicks S L, et al. (2019) *Methods*. 167:3-14). This range of in vitro methods was chosen to assess 3D shape trends as a function of a variety of outputs, allowing assessment of discrepancies, if any, between measured affinity, thermal stability, and profiles of enzymatic degradation.

Example 2

Indicator Displacement Assay (IDA)

[0414] IDAs are valuable high throughput screening (HTS) tools in RNA ligand discovery, yet the vast majority of currently reported IDAs have been optimized for RNA secondary structures utilizing peptides and fluorescent indicators reported to bind bulges and/or internal loops. (Wicks S L, et al. (2019) *Methods*. 167:3-14).

[0415] Given the structural differences between secondary and tertiary structures, where more continuous and complex base-pairing occurs, the design and optimization of an RNA triple helix IDA amenable to HTS was the first focused on. The utility of RiboGreen™ dye was confirmed as an indicator. The RiboGreen™ structure is proprietary, and stoichiometry is thus unconfirmed. Upon assay optimization, the 21-member library was screened using a 16-point titration in three independent replicates (FIG. 2A, FIG. 12A-FIG. 12U).

[0416] The data showed a range of affinities from low to high micromolar, indicating the potential tunability of the diminazene scaffold for targeting the MALAT1 triple helix (FIG. 2C). Aliphatic subunits with protonated nitrogens

(FIG. 2B; see R-1, R-9, R-10) were among the best binders regardless of the regioisomer, indicating a charge-driven binding event.

[0417] In contrast, regioisomer discrimination was observed for aromatic subunit R-4, which showed no affinity in the ortho analogue DMZ-O4 but displayed a CD₅₀ (50% competitive displacement dose) of 36.6 mM when coupled to the meta scaffold as DMZ-M4.

[0418] A similar pattern was observed for R-5, where a 13-fold higher affinity was observed with the para scaffold (DMZ-P5) versus the ortho scaffold (DMZ-O5).

[0419] To qualitatively analyze possible 3D shape trends, the principal moment of inertia diagram for the 21 synthesized small molecules was color-coded based on high affinity (CD₅₀ < 5 mM, red), medium affinity (5 mM < CD₅₀ < 30 mM, orange), and weak affinity ligands (CD₅₀ > 30 mM, grey, FIG. 3A).

[0420] Notably, all the high affinity ligands clustered in the rod-like sub-triangle, while the medium affinity binders are spread between the rod-like and hybrid sub-triangle. The most hybrid, disc-like, and sphere-like ligands had weak or no affinity.

[0421] The diminazene scaffold has a flexible three nitrogen core, which might allow analogues with subunits that have a high propensity for binding to adopt conformations distinct from the lowest energy conformer in order to allow RNA:small molecule interactions. (Nguyen B, et al. (2004) *Biophys J*. 86(2):1028-1041; Spychala J, et al. (1994) *Eur J Med Chem*. 29(5):363-367). Furthermore, comparable affinity of the medium diminazene binders from different PMI sub-triangles might be indicative of binding to a different triplex conformer. (Pilch D S, et al. (1995) *Biochemistry*. 34(49):16107-16124).

[0422] Despite being some of the most commonly used descriptors, PMIs are not the only way to define small molecule 3D properties (Kumar A, et al. (2018) *Front Chem*. 6(315)) and 2D parameters might also correlate to binding trends. To further expand the set of possible predictive parameters for small molecule affinity, a quantitative structure activity relationship (QSAR) method to gain insight into small molecule descriptors that might be driving molecular recognition events was employed. (Cai Z, et al. (2022) *J Med Chem*. 65(10) 7262-7277).

[0423] Multiple linear regression models were iteratively generated by fitting combinations of small molecule descriptors to the small molecule's response to the RNA target which, in this case, is affinity. (Nantasenamat C, et al. (2010) *Expert Opin Drug Discov*. 5(7):633-654; Gramatica P. (2007) *QSAR & Combinatorial Science*. 26(5):694-701; Maciagiewicz I, et al. (2011) *Bioorg Med Chem Lett*. 21(15):4524-4527).

[0424] In this QSAR study, both 2D and 3D parameters were calculated for a total of 339 descriptors for each ligand. 2D descriptors only use the atoms and connectivity information of the molecule for calculations, while 3D descriptors consider coordinates and individual conformations. Model fitness was calculated based on the accuracy of predicted affinity of the molecules (R²) and the model robustness was evaluated by leave-one-out cross validation (LOOCV, Q²). A final predictive model with R²=0.87 and Q²_{LOOCV}=0.74 was obtained, supporting the affinity predictions for the given small molecule set (FIG. 3B). (Cai Z, et al. (2021) *J Med Chem*. 65(10):7262-7277).

[0425] One of the advantages of the chosen QSAR method over other machine-learning based approaches is the open framework of model construction that allows identification of parameters that directly contribute to the binding event. (Cai Z, et al. (2021) J Med Chem. 65(10):7262-7277).

[0426] For example, in the above IDA model, 3D parameters (molecular globularity, glob and contact distances of vsurf_EDmin2 and vsurf_EDmin3, vsurf_DD23) (Cruciani G, et al. (2000) J Mol Structure: THEOCHEM. 503(1):17-30), were identified to contribute positively to the binding affinity, along with a 2D descriptor (total negative polar van der Waals surface area, Q_VSA_PNEG). (Cruciani G, et al. (2000) J Mol Structure: THEOCHEM. 503(1):17-30). The QSAR analysis for the IDA assay indicates that even in an unbiased set of molecular descriptors, 3D properties are critical to predictive models of small molecule affinity to the MALAT1 triple helix.

Example 3

Isothermal Titration Calorimetry of Library of Small Molecule

[0427] Prior to isothermal titration calorimetry (ITC), RNA was heated at 95° C. and snap cooled on ice for 10 minutes. Then, 0.5 mL of MALAT1 triple helix RNA was loaded in a 3,000 MWKO dialysis cassette and dialyzed for 12 hours at 4° C. in a Tris-based buffer chosen to maximize the small molecule concentration range of solubility (10 mM Tris-HCl, 25 mM NaCl, 3 mM MgCl₂ pH=7.4). The dialysis cassette was suspended in 400 mL of buffer and 2 total buffer changes were performed over the course of the 12 hours of dialysis. The dialysate and final buffer were stored for ITC experiments. For each experiment 5 mM or 10 mM of RNA were used in the sample cell and 1-3 mM of small molecule were used in the syringe of a PEAQ-ITC (Malvern).

[0428] After thermal equilibration at 25° C., and initial 150 seconds delay and one initial 0.4 mL injection, 18 serial injections of 2.0 mL were performed at intervals of 150 seconds at a stirring speed of 750 rpm on high feedback mode and 10 kcal reference power. Raw data outlined below was recorded at mcal/s over time (minutes). Small molecules tested were dissolved in dialyzed buffer from powder for every experiment. Each small molecule was screened in three independent experiments and every batch of small molecule and RNA was tested by running a control experiment with buffer. For each RNA control in buffer alone a baseline DP of no more than 0.1 mcal/sec was recorded. A maximum of 3% DMSO was used with small molecules not soluble in buffer alone. In these cases, the % DMSO was matched in RNase free water in the reference cell and control experiments never exceeded 0.2 mcal/s baseline. The dissociation constant K_d was extracted from a curve fit calculated with the PEAQ-ITC (Malvern) software. K_d values reported below are the average of the three independent experiments ± the error derived from the standard deviation of the three trials.

TABLE 4

Thermodynamic Parameters Derived from ITC Curves of the Diminazene Analogues against the MALAT1 Triple Helix.				
	K_D (mM)	DG (kJ/mol)	DH (kJ/mol)	-TDS (kJ/mol)
DMZ-P0	34.1	-6.10	-7.81	1.71
DMZ-P1	10.9	-6.10	-6.77	-0.67
DMZ-P5	36.2	-6.06	-6.76	0.70

TABLE 4-continued

Thermodynamic Parameters Derived from ITC Curves of the Diminazene Analogues against the MALAT1 Triple Helix.				
	K_D (mM)	DG (kJ/mol)	DH (kJ/mol)	-TDS (kJ/mol)
DMZ-P8	9.87	-6.83	-3.64	-3.19
DMZ-P9	25.9	-6.26	-1.53	-4.74
DMZ-P13	16.6	-6.53	-2.72	-3.80
DMZ-M1	5.01	-7.23	-6.75	-0.48
DMZ-M10	31.2	-6.15	-1.33	-4.82
Furamidine	8.46	-6.92	-10.50	3.55

Example 4

Isothermal Titration Calorimetry of a Small Molecule Sub-Set

[0429] To directly assess the correlation of the IDA assay with affinity, isothermal titration calorimetry (ITC) was performed in three independent replicates for the eight diminazene small molecules that showed high enough solubility to be compatible with ITC, along with furamidine as a reference for DPFs (FIG. 4A, FIG. 21A-FIG. 21C, and FIG. 22A-FIG. 22J). The CD_{50} values obtained via Ribogreen™ IDA and the K_d values obtained through ITC of the eight diminazene analogues are in good agreement, with minor discrepancies (DMZ-P9, DMZ-P0, DMZ-M10) that could be due to the differences in buffers required for the two methods or differential indicator displacement modes by the small molecules (FIG. 4B). All the DMZs screened via ITC reported a stoichiometry of $n=1$ by a simplex fit where all variables were allowed to float.

[0430] The necessary number of base pairs for binding was found to be proportional to the length of the ligands, with 8 base-pairs needed for the longest diminazene analogue in that study. (Nguyen B, et al. (2004) Biophys J. 86(2):1028-1041).

[0431] Given that the DMZs studied here are substantially longer than those in the previous study, a stoichiometry of $n=1$ for the 10-base triples of MALAT1 is consistent. (Pilch D S, et al. (1995) Biochemistry. 34(49):16107-16124; Nguyen B, et al. (2004) Biophys J. 86(2):1028-1041).

[0432] Interestingly, direct fluorescence titrations of furamidine with the MALAT1 triple helix yielded a CD_{50} of 560 nM while ITC reported a K_d of 8.35 mM, indicating a possible higher stoichiometry in the direct fluorescence titration conditions.

[0433] Importantly, the reasonable correlation between the IDA assay optimized herein and the non-competitive ITC assay supports the utility of this rapid, cost-effective and general IDA method in high-throughput screens against RNA triple helices.

Example 5

Effects of Small Molecules on MALAT1 Triplex Thermal Stability

[0434] The MALAT1 triple helix is known to confer stability to the full-length transcript and protect it from RNase-mediated degradation, ultimately resulting in transcript accumulation and aggregation in nuclear speckles. (Brown J A, (2020) WIREs RNA. 11(6):e1598; Wilusz J E, et al. (2012) Genes Dev. 26(21):2392-2407; Brown J A, et

al. (2014) *Nat Struct Mol Biol.* 21(7):633-640; Tycowski K T, et al. (2016) *Cell Rep.* 15(6):1266-1276). To better understand how the diminazene focused library affects the MALAT1 triplex stability, the entire library was evaluated via differential scanning fluorimetry (DSF) (FIG. 5A). This method allows for rapid assessment of small molecule effects on nucleic acid thermal stability (Silvers R, et al. (2015) *Chembiochem.* 16(7):1109-1114) in a 96-well plate format, substantially reducing the time and cost associated with traditional UV-melting experiments.

[0435] The bi-phasic melting profile distinctive of triple helices is attributed to an initial melting of the Hoogsteen triple base pairs followed by dissociation of Watson-Crick-Franklin base pairs (FIG. 16A-FIG. 16E). (Pilch D S, et al. (1995) *Biochemistry.* 34(49):16107-16124). At an equimolar ratio of RNA:small molecule, a range of thermal stabilization of the first (T_{m1}) was observed, but not the second (T_{m2}) melting peak (FIG. 5A, FIG. 13A-FIG. 13Z). At increasing small molecule concentration, a dose-dependent increase in ΔT_{m1} but not in ΔT_{m2} was observed, indicating small molecule interactions in the triplex region and not in the upper stem region (FIG. 15).

[0436] To further explore this hypothesis, a sub-set of small molecules that showed a range of changes in T_{m1} was tested against two truncated MALAT1 constructs: the upper stem of MALAT1 and the stem loop proxy obtained from deletion of the poly(A) tail (FIG. 11, FIG. 16A

[0437] FIG. 16E, FIG. 17A-FIG. 17E). Interestingly, the melting profile of the upper stem of MALAT1 directly overlapped with the second peak of the triplex bi-phasic melting curve, consistent with correspondence of T_{m1} to the melting of the triplex tract and T_{m2} to the melting of the upper stem (FIG. 5B). None of the small molecules caused a substantial change in the melting profiles of either the upper stem or the stem loop proxy, confirming a triplex-selective interaction.

[0438] DMZ effects on T_{m1} ranged from negligible ($<1^\circ$ C.) for the biscyano-scaffolds (DMZ-PS, DMZ-MS, DMZ-OS) to approximately 6.1° C. for DMZ-P1. The relative affinity values obtained via the IDA assay was plotted versus the ΔT_{m1} values (FIG. 14) and found reasonable correlations at both high and low affinity values (Spearman's correlation $R=-0.89$, p value <0.0001). Conformity between these two assays comes as a surprise as there was little to no correlation between small molecule affinity and changes in MALAT1 thermal stability. Despite the overall agreement between the two methods, a few discrepancies emerged. Specifically, DMZ-M3, DMZ-P17 and DMZ-M15 showed moderate affinity but negligible thermal stabilization. This discrepancy could be due to different binding modes and/or engagement of a less populated triplex conformer. To gain further insight, two commercially available small molecule leads published by Le Grice (Abulwerdi F A, et al. (2019) *ACS Chem Biol.* 14(2):223-235) and co-workers were examined using DSF. In that study, the small molecules were screened using a two-body differential scanning FRET (DS-FRET) thermal melting assay and revealed a destabilization of the triple helix complex. (Abulwerdi F A, et al. (2019) *ACS Chem Biol.* 14(2):223-235). Neither small molecule caused a change in thermal stability using the single body triplex in DSF, indicating that this system might be limited to primarily identifying stabilizing small molecules rather than destabilizing small molecules (FIG. 13A-FIG. 13Z).

[0439] The overall agreement between IDA and DSF data prompted us to employ QSAR to test whether the same parameters that yielded a predictive model for IDA are the same that govern thermal stabilization interactions. The model obtained for DSF has higher R^2 than the one obtained for IDA ($R^2=0.95$, $Q^2=0.91$) indicating an increased robustness of the model (FIG. 5C). The contributive factors identified by the best model are E_{stb} (bond stretch-bend cross-term potential energy, 3D descriptor), Q_VSA_POL (total polar van der Waals surface area, 2D descriptor) and SMR_VSA7 (sum of van der Waals surface area calculated based on the molar refractivity, 2D descriptor). Polarity, albeit in the form of a different descriptor, is a common factor contributing to IDA and DSF. Additionally, both assays have a combination of 2D and 3D descriptors present in the best models. The combination of E_{stb} , Q_VSA_POL and SMR_VSA7 indicate that molecular flexibility and the polarizability of the small molecules can lead to stronger complexations that are not disrupted by the addition of heat in DSF experiments and, ultimately, result in stabilization of the triple helix melting event.

Example 6

Small Molecule Screening Using an Enzymatic Assay

[0440] The library was screened using a more biologically relevant in vitro enzymatic degradation assay. Previous work evaluated the effect on triplex degradation of various ionic buffer conditions (Ageeli A A, et al. (2018) *Nucleic Acids Res.* 47(3):1468-1481) and a sub-set of DPF small molecules (Donlic A, et al. (2020) *Nucleic Acids Res.* 48(14):7653-7664) using *E. coli* RNase R exonuclease as it degrades from 3' to 5', reporting on the availability of the poly(A) tail normally sequestered in the triplex construct. While helpful, RNase R does not have a human paralogue and the data collection is time intensive (5 hours/experiment). To overcome these limitations, a fast one-hour enzymatic degradation assay utilizing endonuclease RNase A designed and optimized (FIG. 6A), which is one of the most used, readily available, and best studied members of the RNase superfamilies. (Aryani A, et al. (2015) *BMC Res Notes.* 8:164; Cuchillo C M, et al. (2011) *Biochemistry.* 50(37):7835-7841; Kennell D. (2002) *J Bacteriol.* 184(17):4645-4665). Specificity for ssRNA endonuclease activity was achieved by pre-incubating the enzyme at high salt conditions as previously reported. (Yakovlev G I, et al. (1995) *Nucleic Acids Symp Ser.* (33):106-108; Struhl K, et al. (1989) *Current Protocols in Molecular Biology.* 8(1):3.13.1-3.13.3). Aliquot time-points were collected at 0 minutes, 5 minutes, 10 minutes, 30 minutes, and 60 minutes, ultimately yielding a degradation curve for each small molecule while avoiding the non-specific RNA degradation that can occur over longer experiments (FIG. 6B). Small molecules that stabilize the triplex conformer are expected to lead to a decrease in degradation over time, as ssRNA regions are less available to the enzyme. Small molecules that stabilize non-triplex conformations are expected to result in more degradation over time as more ssRNA regions become accessible to RNase A.

[0441] The assay was first validated by screening two previously reported stabilizing and destabilizing small molecules, respectively, evaluated with RNase R. Both small molecules displayed the same effect on triplex degradation

irrespective of enzyme, confirming the ability of the RNase A assay to similarly reflect small molecule impacts on MALAT1 triplex enzymatic stability.

[0442] The 21 synthesized DMZ small molecules were screened in three independent replicates (FIG. 18A-FIG. 18W). The ratio of RNA remaining at 5 minutes relative to time 0 was used to classify stabilizing and destabilizing small molecules based on the assumption that the 5-minute time-point falls in the linear phase of decay (FIG. 6C).

[0443] Strikingly, 8 out of 21 total small molecules were found to increase the amount of MALAT1 triple helix degraded and were therefore classified as destabilizers. Interestingly, three of the small molecules that showed increases in thermal stability via DSF were found to be destabilizers via RNase A assay (DMZ-P0, DMZ-P1, DMZ-M4), with DMZ-M4 resulting in an increase of the amount of MALAT1 degraded at 5 minutes by 2-fold. Aliphatic subunits with positively charged nitrogens (DMZ-P1, DMZ-O1, DMZ-M1, DMZ-P9, and DMZ-M10), which have con-

sistently shown similar affinity and thermal stability profiles, have vastly different effects on enzymatic degradation of the triplex (Table 5). Notably, DMZ-O1 and DMZ-M1 are strong stabilizers ($\geq 50\%$ decrease in degradation) while DMZ-P1 is a weak destabilizer ($< 50\%$ increase in degradation). RNase A degradation is the only assay in this study that discriminates between the R-1 regioisomers, indicating that the regioisomers can be engaging different MALAT1 conformers or binding sites. Another example of differential modulation is DMZ-O5, which showed a much stronger destabilizing effect than DMZ-P5. DMZ-O6, on the other hand, is a strong stabilizer and differs from DMZ-O5 by just the position of the chlorine atom on the amine subunit, which is at the ortho-position in DMZ-O6 and at the para-position in DMZ-O5. In addition, subunits that have small architectural differences, like DMZ-O2 and DMZ-O4, which differ by a single carbon atom and have very similar 3D shape as calculated by PMIs, show opposite effects on triplex degradation.

TABLE 5

One-way ANOVA Test for Statistical Significance of RNase A Normalized Values for the 5-Minute Time Point					
Zidek's Multiple Comparisons Test	Mean Diff.	95.00% CI of diff.	Below threshold?	Summary	Adjusted P Value
p0 vs. p5	2.019	0.6813 to 3.356	Yes	****	<0.0001
p0 vs. p8	2.878	1.540 to 4.215	Yes	****	<0.0001
p0 vs. p9	2.649	1.311 to 3.986	Yes	****	<0.0001
p0 vs. p13	2.735	1.397 to 4.072	Yes	****	<0.0001
p0 vs. p14	1.564	0.2261 to 2.901	Yes	**	0.0061
p0 vs. p17	2.656	1.318 to 3.993	Yes	****	<0.0001
p0 vs. o1	2.572	1.235 to 3.910	Yes	****	<0.0001
p0 vs. o2	1.594	0.2568 to 2.932	Yes	**	0.0045
p0 vs. o4	2.38	1.043 to 3.718	Yes	****	<0.0001
p0 vs. o5	1.591	0.2537 to 2.929	Yes	**	0.0047
p0 vs. o6	2.639	1.302 to 3.976	Yes	****	<0.0001
p0 vs. m1	2.881	1.544 to 4.219	Yes	****	<0.0001
p0 vs. m3	2.474	1.137 to 3.812	Yes	****	<0.0001
p0 vs. m7	3.054	1.716 to 4.391	Yes	****	<0.0001
p0 vs. m10	2.886	1.549 to 4.224	Yes	****	<0.0001
p0 vs. m15	1.702	0.3642 to 3.039	Yes	**	0.0016
p0 vs. m22	2.13	0.7921 to 3.467	Yes	****	<0.0001
p0 vs. m24	2.314	0.9762 to 3.651	Yes	****	<0.0001
p1 vs. p8	1.559	0.2211 to 2.896	Yes	**	0.0064
p1 vs. p13	1.415	0.07774 to 2.753	Yes	*	0.0247
p1 vs. m1	1.562	0.2244 to 2.899	Yes	**	0.0062
p1 vs. m7	1.734	0.3968 to 3.072	Yes	**	0.0012
p1 vs. m10	1.567	0.2295 to 2.904	Yes	**	0.0059
p5 vs. m4	-1.685	-3.023 to -0.3479	Yes	**	0.0019
p8 vs. m4	-2.544	-3.882 to -1.207	Yes	****	<0.0001
p9 vs. m4	-2.315	-3.653 to -0.9777	Yes	****	<0.0001
p13 vs. m4	-2.401	-3.739 to -1.064	Yes	****	<0.0001
p14 vs. m7	1.49	0.1526 to 2.828	Yes	*	0.0123
p17 vs. m4	-2.322	-3.660 to -0.9848	Yes	****	<0.0001
o1 vs. m4	-2.239	-3.576 to -0.9015	Yes	****	<0.0001
o2 vs. m7	1.459	0.1219 to 2.797	Yes	*	0.0164
o4 vs. m4	-2.047	-3.384 to -0.7096	Yes	****	<0.0001
o5 vs. m7	1.462	0.1250 to 2.800	Yes	*	0.0159
o6 vs. m4	-2.306	-3.643 to -0.9681	Yes	****	<0.0001
m1 vs. m4	-2.548	-3.885 to -1.210	Yes	****	<0.0001
m3 vs. m4	-2.141	-3.478 to -0.8035	Yes	****	<0.0001
m4 vs. m7	2.72	1.383 to 4.058	Yes	****	<0.0001
m4 vs. m10	2.553	1.215 to 3.890	Yes	****	<0.0001
m4 vs. m15	1.368	0.03079 to 2.706	Yes	*	0.0379
m4 vs. m22	1.796	0.4586 to 3.134	Yes	****	0.0006
m4 vs. m24	1.98	0.6428 to 3.318	Yes	****	0.0001
m7 vs. m15	-1.352	-2.689 to -0.01453	Yes	*	0.0439

[0444] As a control, protein DSF was conducted on mixtures of the RNase A enzyme and small molecules to assess whether small molecules that resulted in decreased degradation of the RNA triplex over time were binding to the enzyme, which would provide enzyme inhibition as an alternative mechanism. DSF has been used as a tool to identify RNase A inhibitors and have shown that small molecule binders cause a shift in thermal melting of the enzyme, which is tracked by association of the dye SYPRO Orange® to non-solvent exposed hydrophobic regions. (Lang B E, et al. (2017) *Biotechnol Prog.* 33(3):677-686; Ortiz-Alcantara J, et al. (2010) *Virus Adapt Treat* 2(1):125-133).

[0445] In these studies, the enzyme's melting profile agreed with previous studies and none of the small molecules tested caused an appreciable ($>0.5^{\circ}\text{C}$.) change in thermal melting, indicating that it is unlikely that the DMZ small molecules act via enzyme inhibition (FIG. 20A-FIG. 20H).

[0446] Then, QSAR was performed utilizing the fold-change in small molecule-induced changes in RNA degradation at the 5-minute timepoint of the RNase A assay. QSAR yielded a predictive model with $R^2=0.74$, $Q^2=0.61$

and revealed multiple 3D descriptors as contributing factors (FIG. 6D). For instance, CASA-represents negative charge weighted surface area. Since it has a positive coefficient, the model indicates that increasing this descriptor value will increase the triplex's susceptibility to enzymatic degradation, indicating that small molecules with more negatively charged surface area will destabilize the RNA fold, which is consistent with the nature of negatively charged RNA backbone. Overall, these data underline the RNase A assay's feasibility for efficient assessment of small molecule impacts on the susceptibility of RNA structures to degradation.

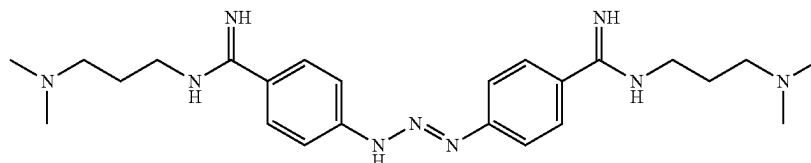
[0447] This is the first report of dual modulation of an RNA triple helix within the same scaffold class, supporting diminazene as a valuable probe for future studies of MALAT1 conformational dynamics and modulation.

Example 7

Small Molecule Characterization

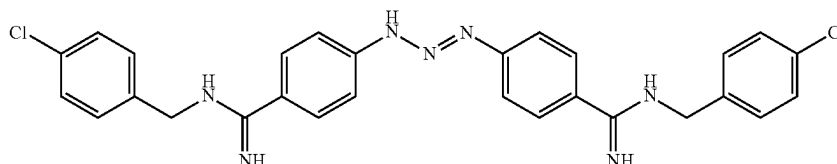
[0448] Spectral characterization for the library of small molecules was performed.

DMZ-P1



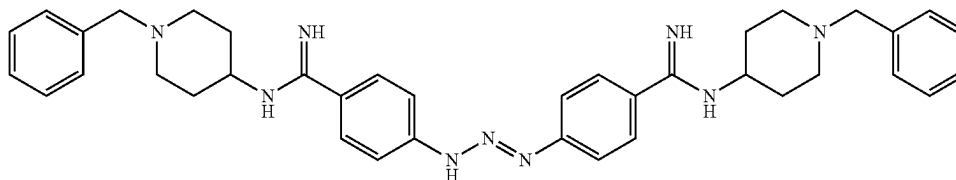
[0449] Bright red solid; Yield=36%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.62-7.59 (m, 4H), 7.48-7.44 (m, 4H), 3.31 (t, $J=6.8$ Hz, 4H), 2.35 (t, $J=7.2$ Hz, 4H), 2.18 (s, 12H), 1.79 (p, $J=7.0$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 163.49, 150.80, 128.24, 127.72, 117.81, 56.34, 44.49, 44.02, 43.95, 41.15, 25.89. HRMS-ESI (m/z) Calcd for $\text{C}_{24}\text{H}_{37}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 452.3; found: 452.3(± 0.4 ppm).

DMZ-P5



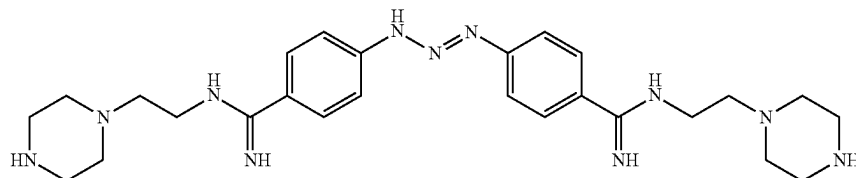
[0450] Bright yellow solid; Yield=51%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.65-7.61 (m, 4H), 7.43 (dd, $J=8.6$, 2.1 Hz, 4H), 7.29-7.23 (m, 8H), 4.39 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 162.78, 148.62, 137.23, 132.58, 130.72, 129.06, 128.85, 128.25, 128.08, 117.45, 46.39. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{25}\text{Cl}_2\text{N}_7$ ($[\text{M}+\text{H}]^+$): 530.2; found: 530.2(± 0.7 ppm).

DMZ-P8



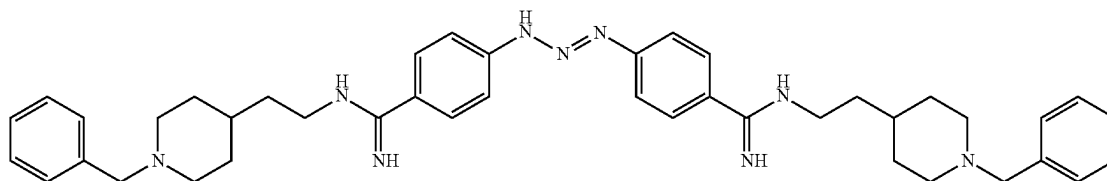
[0451] Bright orange solid; Yield=36%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.79 (d, $J=8.4$ Hz, 4H), 7.68 (d, $J=8.4$ Hz, 4H), 7.37-7.34 (m, 8H), 7.30 (dd, $J=5.9, 2.8$ Hz, 2H), 3.74 (ddd, $J=11.2, 6.8, 4.1$ Hz, 2H), 3.60 (s, 4H), 3.02 (dt, $J=12.6, 3.6$ Hz, 4H), 2.27-2.21 (m, 4H), 2.10-2.06 (m, 4H), 1.80 (dt, $J=12.1, 6.0$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 129.23, 127.98, 127.11, 62.39, 51.48, 48.12, 47.94, 47.77, 47.09, 30.01. HRMS-ESI (m/z) Calcd for $\text{C}_{38}\text{H}_{45}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 628.4; found: 628.4(± 1.0 ppm).

DMZ-P9



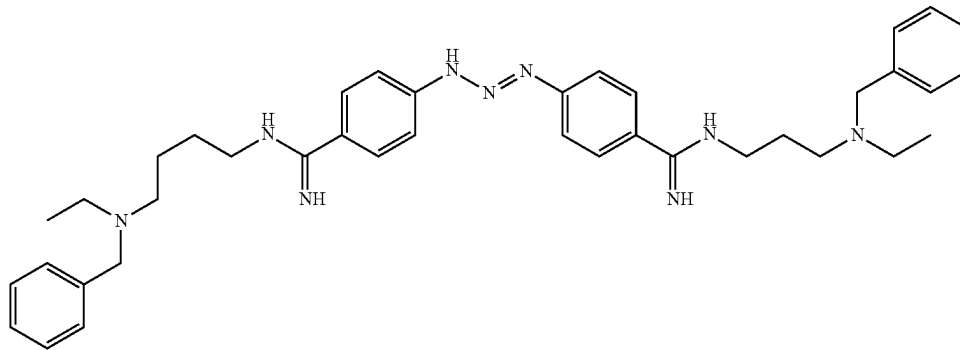
[0452] Bright yellow solid; Yield=21%; ^1H NMR (500 MHz, Deuterium Oxide) δ 7.69 (d, $J=8.2$ Hz, 4H), 7.54 (d, $J=8.3$ Hz, 4H), 3.55 (t, $J=6.3$ Hz, 4H), 3.25 (t, $J=5.2$ Hz, 8H), 2.81 (d, $J=6.2$ Hz, 12H). ^{13}C NMR (126 MHz, D_2O) δ 163.85, 129.29, 54.44, 49.15, 43.19, 39.97. HRMS-ESI (m/z) Calcd for $\text{C}_{26}\text{H}_{39}\text{N}_{11}$ ($[\text{M}+\text{H}]^+$): 506.3; found: 506.3 (± 1.2 ppm).

DMZ-P13



[0453] Bright orange solid; Yield=27%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.82 (d, $J=8.3$ Hz, 4H), 7.68 (d, $J=8.3$ Hz, 4H), 7.55-7.45 (m, 10H), 4.28 (s, 4H), 3.56-3.44 (m, 8H), 3.00 (t, $J=12.6$ Hz, 4H), 2.04 (d, $J=14.1$ Hz, 4H), 1.78 (q, $J=6.5$ Hz, 6H), 1.59 (q, $J=11.6, 10.9$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 163.89, 161.77, 161.49, 130.85, 129.61, 129.13, 128.87, 118.03, 115.70, 51.89, 47.45, 47.28, 47.11, 40.34, 32.75, 31.15, 28.66. HRMS-ESI (m/z) Calcd for $\text{C}_{42}\text{H}_{53}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 684.45; found: 684.45(± 3.3 ppm).

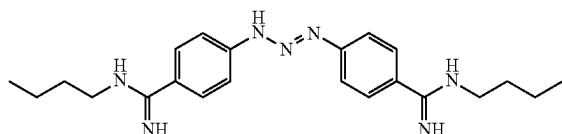
DMZ-P14



[0454] Bright orange solid; Yield=32%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.62-7.58 (m, 4H), 7.55 (d, $J=8.7$ Hz, 4H), 7.28-7.25 (m, 4H), 7.24-7.20 (m, 4H), 7.19-7.16 (m, 2H), 3.73 (s, 4H), 2.70 (d, $J=7.7$ Hz, 6H), 1.92 (p, $J=6.9$ Hz, 4H), 1.09 (t, $J=7.1$ Hz, 6H). ^{13}C NMR (126 MHz, MeOD)

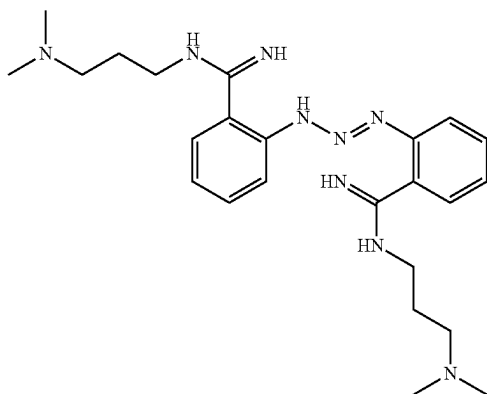
δ 163.87, 161.78, 129.50, 129.08, 128.30, 118.03, 115.70, 57.17, 49.69, 48.12, 46.97, 44.39, 41.02, 23.78, 9.35. HRMS-ESI (m/z) Calcd for $\text{C}_{39}\text{H}_{51}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 645.4; found: 645.8(± 2.0 ppm).

DMZ-P17



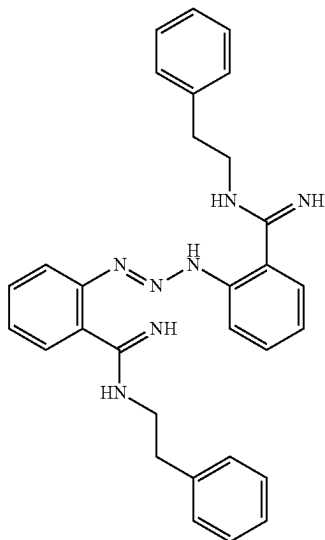
[0455] Bright red solid; Yield=82%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.68-7.64 (m, 4H), 7.52 (d, $J=8.4$ Hz, 4H), 3.31 (d, $J=7.3$ Hz, 4H), 1.70 (tt, $J=7.8, 6.6$ Hz, 4H), 1.49 (h, $J=7.4$ Hz, 4H), 1.02 (t, $J=7.4$ Hz, 6H). ^{13}C NMR (126 MHz, MeOD) δ 163.48, 150.10, 130.22, 127.75, 117.56, 47.11, 45.79, 42.49, 30.68, 20.05, 12.86. HRMS-ESI (m/z) Calcd for $\text{C}_{22}\text{H}_{39}\text{N}_{11}$ ($[\text{M}+\text{H}]^+$): 645.4; found: 645.8(± 2.0 ppm).

DMZ-O1



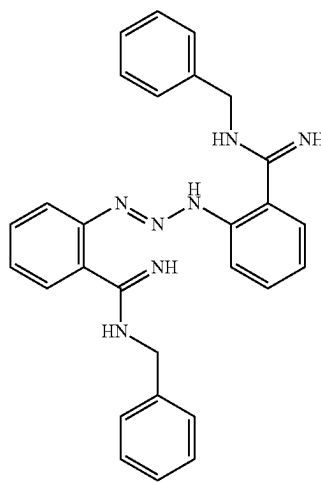
[0456] Bright red solid; Yield=45%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.77 (d, $J=8.3$ Hz, 2H), 7.61 (d, $J=7.9$ Hz, 2H), 7.51 (ddd, $J=8.5, 7.2, 1.4$ Hz, 2H), 7.17-7.13 (m, 2H), 3.43 (t, $J=6.6$ Hz, 4H), 2.43 (t, $J=7.2$ Hz, 4H), 2.26 (s, 12H), 1.91-1.86 (m, 4H). ^{13}C NMR (126 MHz, MeOD) δ 150.85, 132.16, 127.98, 122.38, 118.18, 56.13, 43.90, 40.84, 25.71. HRMS-ESI (m/z) Calcd for $\text{C}_{24}\text{H}_{37}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 452.3; found: 452.3(± 0.4 ppm).

DMZ-O2



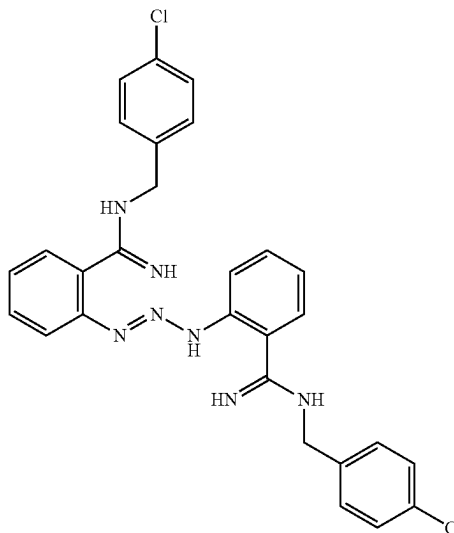
[0457] Pale yellow solid; Yield=42%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.82 (d, $J=8.3$ Hz, 2H), 7.53 (ddd, $J=9.5, 6.9, 2.6$ Hz, 4H), 7.30-7.25 (m, 10H), 7.17-7.14 (m, 2H), 2.89 (t, $J=7.1$ Hz, 4H), 2.77 (t, $J=7.3$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 138.26, 132.32, 128.60, 128.33, 126.41, 123.36, 119.60, 117.57, 115.77, 48.15, 47.13, 45.80, 34.05. HRMS-ESI (m/z) Calcd for $\text{C}_{30}\text{H}_{31}\text{N}_7$ ($[\text{M}+\text{H}]^+$): 490.3; found: 490.3(± 0.9 ppm).

DMZ-O4



[0458] Pale yellow solid; Yield=46%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.78 (d, $J=8.2$ Hz, 2H), 7.63 (d, $J=7.8$ Hz, 2H), 7.48-7.44 (m, 2H), 7.39 (d, $J=7.7$ Hz, 4H), 7.31 (t, $J=7.6$ Hz, 4H), 7.23-7.19 (m, 4H), 4.44 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 145.61, 138.77, 130.58, 128.17, 128.15, 127.23, 126.61, 123.76, 116.80, 45.79, 44.46. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{27}\text{N}_7$ ($[\text{M}+\text{H}]^+$): 462.2; found: 462.2(± 2.6 ppm).

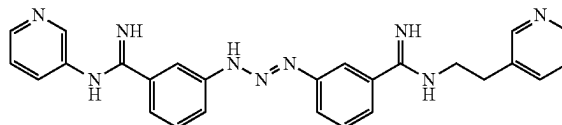
DMZ-O5



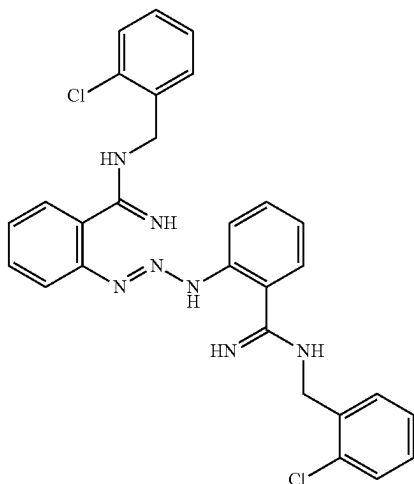
[0459] Bright yellow solid; Yield=31%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.79-7.76 (m, 2H), 7.63 (d, $J=7.8$ Hz, 2H), 7.48 (ddd, $J=8.5$, 7.3, 1.5 Hz, 2H), 7.37-7.34 (m, 4H), 7.28 (d, $J=8.4$ Hz, 4H), 7.22 (td, $J=7.6$, 1.2 Hz, 2H), 4.43 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 161.66, 145.60, 137.54, 132.25, 130.76, 128.88, 128.80, 128.18, 128.13, 123.87, 116.87, 48.48, 45.79. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{25}\text{Cl}_2\text{N}_7$ ($[\text{M}+\text{H}]^+$): 530.2; found: 530.2(± 0.6 ppm).

[0461] ^1H NMR (500 MHz, Methanol- d_4) δ 7.77 (t, $J=2.0$ Hz, 2H), 7.73 (dd, $J=8.4$, 2.0 Hz, 2H), 7.53 (t, $J=7.9$ Hz, 2H), 7.46-7.43 (m, 2H), 3.46 (t, $J=6.8$ Hz, 4H), 2.47 (t, $J=7.0$ Hz, 4H), 2.27 (s, 12H), 1.88 (p, $J=7.0$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 164.55, 130.40, 130.18, 123.77, 122.05, 116.84, 55.61, 43.68, 41.04, 24.91. HRMS-ESI (m/z) Calcd for $\text{C}_{24}\text{H}_{37}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 452.3; found: 452.3(± 1.9 ppm).

DMZ-M3

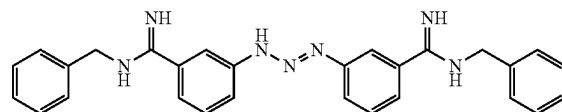


DMZ-O6



[0462] Beige solid; Yield=32% ^1H NMR (500 MHz, Methanol- d_4) δ 8.39 (s, 2H), 8.29 (d, $J=4.7$ Hz, 2H), 7.73 (d, $J=7.8$ Hz, 2H), 7.61 (s, 2H), 7.53 (d, $J=8.1$ Hz, 2H), 7.39 (t, $J=7.9$ Hz, 2H), 7.28 (dd, $J=7.9$, 4.2 Hz, 4H), 3.54 (t, $J=7.3$ Hz, 4H), 2.97 (t, $J=7.2$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 163.53, 149.17, 146.83, 137.45, 135.55, 134.73, 129.57, 123.89, 123.11, 120.60, 116.07, 47.62, 47.45, 47.28, 47.11, 43.96, 31.37. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{29}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 492.6; found: 246.3(± 1.1 ppm) for $1/2[\text{M}+\text{H}]^+$.

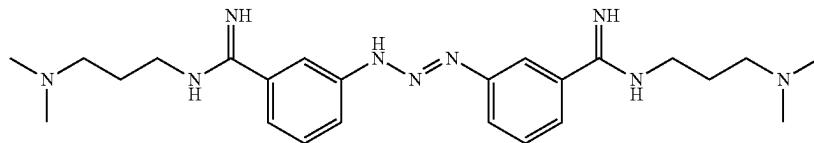
DMZ-M4

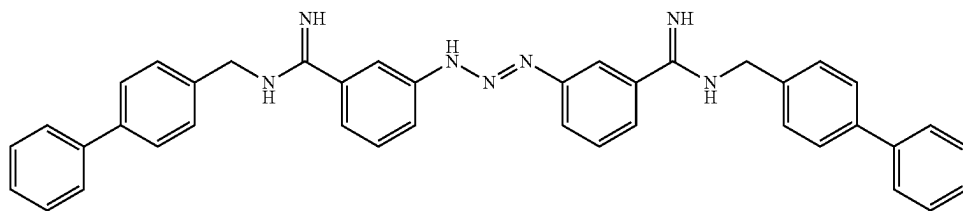


[0460] Bright yellow solid; Yield=11%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.78 (d, $J=8.2$ Hz, 2H), 7.64 (d, $J=7.7$ Hz, 2H), 7.53 (d, $J=7.5$ Hz, 2H), 7.47 (ddd, $J=8.6$, 7.3, 1.5 Hz, 2H), 7.39-7.34 (m, 2H), 7.26-7.19 (m, 6H), 4.49 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 130.43, 128.93, 128.87, 128.24, 127.95, 126.76, 53.46, 47.58, 47.44, 47.42, 47.27, 47.10. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{25}\text{C}_{12}\text{N}_7$ ($[\text{M}+\text{H}]^+$): 530.1621; found: 530.1624 ($=0.5$ ppm).

[0463] Beige solid; Yield=21%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.66 (t, $J=1.9$ Hz, 2H), 7.44 (dt, $J=7.4$, 1.9 Hz, 2H), 7.34-7.30 (m, 4H), 7.29-7.25 (m, 6H), 7.21 (dd, $J=8.5$, 6.9 Hz, 5H), 7.13-7.10 (m, 2H), 4.36 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 163.02, 146.29, 139.18, 137.88, 129.15, 128.14, 127.17, 126.61, 122.84, 119.26, 115.97, 45.79. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{29}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 492.6; found: 246.3(± 1.1 ppm) for $[\text{M}+\text{H}]^+$.

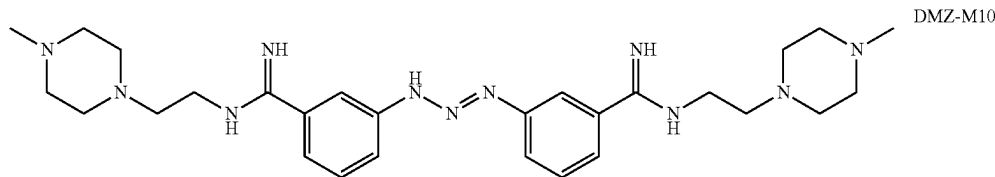
DMZ-M1





DMZ-M7

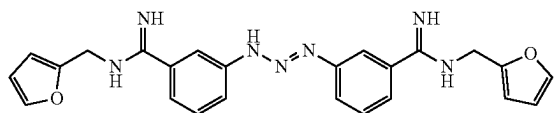
[0464] Yellow solid; Yield=12%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.91 (s, 2H), 7.81 (d, $J=8.0$ Hz, 2H), 7.71-7.68 (m, 4H), 7.63 (d, $J=4.8$ Hz, 6H), 7.53 (d, $J=7.9$ Hz, 4H), 7.45 (d, $J=7.7$ Hz, 4H), 7.37 (t, $J=7.5$ Hz, 2H), 3.33 (d, $J=2.9$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 164.24, 141.03, 140.34, 134.22, 131.52, 130.02, 128.56, 127.88, 127.23, 127.16, 126.56, 123.71, 121.74, 116.75, 46.11, 45.77. HRMS-ESI (m/z) Calcd for $\text{C}_{40}\text{H}_{35}\text{N}_7$ ($[\text{M}+\text{H}]^+$): 614.3; found: 614.3(± 1.9 ppm) for $[\text{M}+\text{H}]^+$.



DMZ-M10

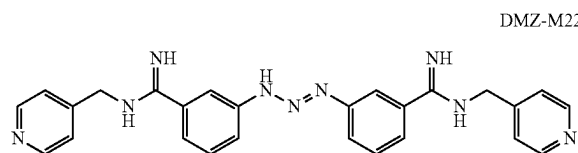
[0465] Bright yellow solid; Yield=44%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.96 (t, $J=2.0$ Hz, 1H), 7.79-7.75 (m, 1H), 7.63-7.61 (m, 2H), 7.47 (dd, $J=17.0, 7.6$ Hz, 4H), 3.58 (t, $J=6.8$ Hz, 2H), 3.48 (t, $J=6.5$ Hz, 2H), 2.86 (t, $J=5.1$ Hz, 12H), 2.76 (t, $J=6.6$ Hz, 6H), 2.45 (t, $J=6.7$ Hz, 8H). ^{13}C NMR (126 MHz, MeOD) δ 168.42, 163.41, 146.46, 135.49, 129.32, 123.07, 120.52, 116.38, 60.73, 57.42, 53.66, 44.70, 37.43, 36.45. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{23}\text{N}_7\text{O}_2$ ($[\text{M}+\text{H}]^+$): 534.4; found: 534.4 (± 1.3 ppm).

4H), 4.37 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 162.67, 151.94, 146.27, 141.95, 136.32, 129.21, 123.09, 119.79, 116.15, 110.00, 106.92, 40.93. HRMS-ESI (m/z) Calcd for $\text{C}_{24}\text{H}_{23}\text{N}_{11}$ ($[\text{M}+\text{H}]^+$): 442.2; found: 442.2(± 2.9 ppm).



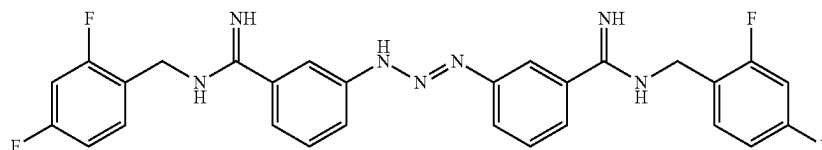
DMZ-M15

[0466] Bright yellow solid; Yield=23%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.65 (s, 2H), 7.50-7.44 (m, 2H), 7.35 (s, 2H), 7.33 (d, $J=6.8$ Hz, 4H), 6.26 (dd, $J=11.6, 2.7$ Hz,



DMZ-M22

[0467] Pale rose solid; Yield=22%; ^1H NMR (500 MHz, Methanol- d_4) δ 8.96 (s, 4H), 8.29-8.15 (m, 4H), 8.01-7.67 (m, 8H), 5.18 (d, $J=11.1$ Hz, 4H). ^{13}C NMR (126 MHz, MeOD) δ 156.90, 150.88, 148.39, 133.73, 122.95, 122.35, 119.92, 118.36, 117.48, 117.38, 37.30. HRMS-ESI (m/z) Calcd for $\text{C}_{26}\text{H}_{25}\text{N}_9$ ($[\text{M}+\text{H}]^+$): 464.2; found: 464.2(± 0.6 ppm).



DMZ-M24

[0468] Pale brown solid; Yield=11%; ^1H NMR (500 MHz, Methanol- d_4) δ 7.67 (s, 2H), 7.45 (dd, $J=7.3, 2.2$ Hz, 2H), 7.34 (dt, $J=11.7, 7.1$ Hz, 6H), 6.85-6.80 (m, 4H), 4.36 (s, 4H). ^{13}C NMR (126 MHz, MeOD) δ 163.05, 162.24, 161.79, 161.08, 159.73, 146.06, 137.60, 130.30, 128.94, 122.76, 122.39, 119.07, 115.87, 110.63, 102.64, 41.01. HRMS-ESI (m/z) Calcd for $\text{C}_{28}\text{H}_{23}\text{F}_4\text{N}_7$ ($[\text{M}+\text{H}]^+$): 534.2; found: 534.2(± 1.5 ppm).

Example 8

RNase A Differential Scanning Fluorimetry (DSF)

[0469] All experiments were performed in Roche 96 well plates in a LightCycler® (Roche). In a typical experiment, 10 μM RNase A was diluted in the same buffer used for RNase A degradation assays (20 mM HEPES-KOH, 52.6 mM KCl, 0.1 mM MgCl_2 , pH=7.4). 5000 $\times$ SYPRO™ Orange Protein Gel Stain (ThermoFisher Scientific) dye stock solution was diluted to 10 $\times$ in the same buffer with the RNase A enzyme. The mixture was then aliquoted in the light cycler plates by adding 49 μL per well. A 2.5 mM final concentration of ligand was made by adding 1 mL of DMSO or small molecule to each well. The mixture was left to incubate at room temperature for 10 minutes with a black lid, the 96 well plate was then sealed with an optically clear foil and centrifuged for 1 minute at 4000 rpm prior to being placed in the instrument. A high-resolution melting profile was obtained by monitoring fluorescence intensity using the VIC dye filter combination (577-620 nm) from 37-98° C. at a ramp rate of 0.01° C./second with 120 acquisitions per° C. Melting temperatures were obtained by T_m analysis in the LightCycler® software (W 1.1) of the derivative of fluorescence over time.

Example 9

In Vivo Toxicity in LNCaP Cells

[0470] LNCaP is a cell line derived from a metastatic lymph node lesion of human prostate cancer which is androgen receptor (AR) positive, exhibits androgen-sensitive growth. In the first set of in vivo experiments, four (4) members of the small molecule library were examined in a MTT assay. The IC_{50} at 48 hours (FIG. 23A) and 72 hours (FIG. 23B) was determined for DMZ-P0, DMZ-P5, DMZ-P13, and DMZ-P17. At 48 hours, DMZ-P13 had the lowest IC_{50} value of 4.1 μM . At 72 hours, DMZ-P13 had the lowest IC_{50} value of 2.7 μM . The IC_{50} values and trends were similar to the apparent small molecule affinities for MALAT-1, supporting a MALAT-1 related mechanism for the toxicity. DMZ-P0 was an exception, with a much higher IC_{50} , possibly indicating a lack of cell uptake or off target engagement. Ongoing cytotoxicity studies using other DMZ small molecule analogs (as well as DMZ-5, DMZ-P13, and DMZ-P17) continue to illuminate mechanism of action regarding toxicity to cancer cells.

[0471] In a second set of in vivo experiments, four (4) additional members (DMZ-M1, DMZ-M4, DMZ-M7, and DMZ-M10) of the small molecular library were examined in a MTT assay. At 48 hours, DMZ-M7 had the lowest IC_{50} of 2.9 μM . The IC_{50} values and trends are similar to the apparent small molecule affinities for MALAT-1, supporting a MALAT-1 related mechanism for the toxicity. Here, DMZ-M10 was an exception, with a higher IC_{50} , possibly indicating a lack of cell uptake or off target engagement.

Ongoing cytotoxicity studies using other DMZ small molecule analogs (as well as DMZ-M7, DMZ-M1, and DMZ-M4) continue to illuminate mechanism of action regarding toxicity to cancer cells.

Summary of Experiments Example 1-Example 9

[0472] In this study, a 3D-shape diverse focused library was synthesized from an under-explored scaffold to elucidate molecular recognition properties of modulators of MALAT1 triplex stability. Following the design of a focused theoretical library, 21 representative members were synthesized based on Berenil®, an FDA-approved antiparasitic scaffold with limited synthetic exploration but reported nucleic acid binding properties. To gain a holistic perspective on small molecule trends as a function of the assay, the library was assessed against the MALAT1 triple helix using a multi-dimensional approach that involved designing and optimizing both affinity-based and structural-stability based assays.

[0473] Evaluation of the library using an IDA assay reported a range of affinities from low to high micromolar that were in good agreement with ITC affinity values. Analysis of ligand 3D shape confirmed rod-likeness as a predictor of small molecule affinity for the triple helix. The diminazene molecules that showed the highest affinity measured via IDA also caused the greatest temperature increase in thermal melting via DSF. RNase A trends were vastly different, and in some cases opposing, relative to trends observed in the IDA and DSF assays. The discrepancies observed between the affinity, thermal stability, and enzymatic degradation might be due to differences in the triplex conformation reported on by each assay. Indeed, affinity and thermal stability assays are equilibrium-based and thus may report on the more populated triplex conformers. On the other hand, binding to a less populated conformer might shift the equilibrium enough to allow irreversible degradation of the RNA construct. This hypothesis could explain why, for example, DMZ-M4 showed mediocre affinity and thermal stabilization but resulted in >2-fold increase in MALAT1 triplex degraded after 5 minutes. Structural studies are underway to better understand the RNA:small molecule conformational landscape in order to gain insight into the conformers engaged by both stabilizing and destabilizing small molecules.

[0474] To gain deeper understanding of these trends, QSAR was employed. For affinity and effect on thermal stability, the best predicting models included a combination of 3D and 2D parameters. Application of QSAR on RNase A data, on the other hand, resulted in a model composed of 3D parameters only. It is noteworthy that despite the limited size of the focused library, a model with $R^2=0.74$ was generated and was informative of the classification of stabilizers and destabilizers, with predictive classification accuracy of 0.62 (8 out of 13) and 1 (8 out of 8), respectively.

[0475] The data presented herein provide essential insight into the development of small molecule modulators of the MALAT1 triple helix. The diminazene scaffold has revealed that both stabilization and destabilization of the triple helix can be achieved through synthetic tuning of a single scaffold, indicating that DMZs may make an excellent chemical probe to study the effects of small molecules on transcript half-life and phenotypic traits in MALAT1-overexpressing cell lines.

[0476] The interrogation of the relationship between small molecule engagement of RNA triple helices and the effect on triplex conformational dynamics lays the foundation for small molecule-mediated spatiotemporal control of association of triple helix motifs with their cellular interactome.

[0477] The RNase A assay as a preliminary stepping stone toward addressing a largely unmet need in the field of RNA:small molecule studies, namely the design of functional high throughput and cost efficient in vitro assays that report on the effect of small molecules on RNA conformational dynamics. Such in vitro functional assays would be the missing bridge between affinity- and activity-based screenings to ultimately allow function-based ligand design and a better understanding of the relationship between RNA conformation and cellular function. This workflow illustrates the potential of QSAR-mediated trend analysis in search of guiding principles. Library sizes can be increased, which in turns, increase chemical diversity—all ensuring that more accurate models are obtained.

[0478] From these studies, a few points emerged: (a) small molecule targeting of the MALAT1 triple helix in vitro and in cellulo was feasible; (b) a focused library approach led to a higher hit rate than high-throughput-based approaches; (c) a possible rod-like trend was identified amongst the best small molecule binders; (d) small molecules both increased or decreased MALAT1 susceptibility to degradation; and (e) binding strength and impact on degradation may not be correlated.

[0479] Collectively, the findings and novel tools presented herein raise important considerations for ligand and assay design for RNA triple helices. The evaluation of the complete focused library using a multi-assay approach and subsequent employment of computational tools provided a better understanding of the parameters that impact different assessments and paves the way for future work to evaluate both selectivity and correlation with biological activity. The discovery of a predictive model for small molecule impacts on stabilization and destabilization of an RNA triple helix

opens the door for function-based rational design of small molecule triple helix modulators, which are uniquely poised to provide essential insight into the relationship between biological outcome and modulation of a triple helix dynamic landscape.

SUMMARY OF EXPERIMENTS

[0480] The methodical approach described herein, which uses computational parameters (e.g., shape) and are derived from analysis of bioactive small molecules that target RNA, is new and exciting. The approach is both unique and non-obvious because currently the standard in industry (as detailed in numerous publication) relies on simple binding rather than bioactivity. By considering bioactivity, this methodological approach considers uptake, selectivity, and modulation of function. Thus, this approach is significantly more effective for identifying small molecule and library design than what is currently available in the industry.

[0481] Moreover, the uses described herein are also new and exciting. The approach again differs from the industry standard in that the industry relies on the process of optimizing binding and binding-based selectivity prior to examining function in the cell. At this stage, many RNA targeting efforts in the industry fail because binding in and of itself does not provide information about the modulation of conformation. The lack of the industry's consideration of modulation of conformation is significant. The methodology described here, however, does consider modulation of conformation and it can be examined using the disclosed function-proximal assays.

[0482] Finally, as only described herein, the use of quantitative structure activity relationship to predict small molecule binding and functional modulation of RNA has not been done. Nor has there been any use or attempt to use that relationship to target RNA.

[0483] The studies discussed herein represent the first methodical approach employing rational design and optimization of RNA targeted ligands.

SEQUENCE LISTING

```
Sequence total quantity: 25
SEQ ID NO: 1          moltype = DNA length = 94
FEATURE              Location/Qualifiers
source                1..94
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 1
ggaaggtttt ttttttctg agaaaacaac acgtattgtt ttctcaggtt ttgctttttg 60
gcctttttct agcttaaaaa aaaaaaaagc aaaa                               94

SEQ ID NO: 2          moltype = DNA length = 59
FEATURE              Location/Qualifiers
source                1..59
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 2
gaaattaata cgactcacta taggaaggtt tttctttttcc tgagaaaaca acacgtatt 59

SEQ ID NO: 3          moltype = DNA length = 21
FEATURE              Location/Qualifiers
source                1..21
                     mol_type = other DNA
                     organism = synthetic construct
misc_binding          1
                     bound_moiety = Tmt 2- OMe modified nucleotide

SEQUENCE: 3
```

-continued

```

ttgctttttt tttttttaag c 21

SEQ ID NO: 4      moltype = DNA length = 67
FEATURE          Location/Qualifiers
source           1..67
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 4
ggaagggtttt tcttttctcg agaaaacaac acgtattgtt ttctcaggtt ttgctttttg 60
gcctttt 67

SEQ ID NO: 5      moltype = DNA length = 59
FEATURE          Location/Qualifiers
source           1..59
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 5
gaaattaata cgactcacta taggaagatt tttcttttcc tgagaaaaa acacgtatt 59

SEQ ID NO: 6      moltype = DNA length = 17
FEATURE          Location/Qualifiers
source           1..17
                 mol_type = other DNA
                 organism = synthetic construct

misc_binding     2
                 bound_moiety = modified by a sugar moiety

SEQUENCE: 6
aaaaggccaa aaagcaa 17

SEQ ID NO: 7      moltype = RNA length = 32
FEATURE          Location/Qualifiers
source           1..32
                 mol_type = other RNA
                 organism = synthetic construct

SEQUENCE: 7
cctgagaaaa caacacgtat tgtttttctca gg 32

SEQ ID NO: 8      moltype = RNA length = 95
FEATURE          Location/Qualifiers
source           1..95
                 mol_type = other RNA
                 organism = synthetic construct

misc_binding     1
                 bound_moiety = Cy5

SEQUENCE: 8
tggaagggtt ttttttctcg gagaaaaaca cagctattgt ttctcaggt tttgcttttt 60
ggccttttct tagcttaaaa aaaaaaaaaaag caaaa 95

SEQ ID NO: 9      moltype = RNA length = 8779
FEATURE          Location/Qualifiers
source           1..8779
                 mol_type = mRNA
                 organism = Homo sapiens

SEQUENCE: 9
cgcagcctgc agcccgagac tttgttaaag gactggggcc cgcgaactgg cctctcctgc 60
cctcttaagc gcagcgccat ttttagcaacg cagaagcccg gcgcggggaa gcctcagctc 120
gcctgaaggc aggtcccttc tgacgcctcc gggagcccag gttccccaga gtccttgga 180
cgcagcgacg agttgtgctg ctatcttagc tgtccttata ggctggccat tccaggtgg 240
ggtattttaga taaaaccact caaactctgc agtttggctc tgggggtttg aggaaagct 300
ttatttttct tctgtctccg gttcagaagg tctgaagctc atacctaacc aggcataaca 360
cagaatctgc aaaaacaaaaa cccctaaaaa agcagaccga gagcagtgta aacacttctg 420
ggtgtgtccc tgactggctg cccaagggtc ctgtgtcttc ggagacaaaag ccattcgctt 480
agttgggtcta ctttaaaagg ccacttgaac tcgctttcca tggcgatttg ccttgtgagc 540
actttcagga gagcctggaa gctgaaaaac ggtagaaaaa ttccgtgctg ggcctggtgg 600
ggctggcgcg aactgggggg cgcgagatca gagtggggca ctggcagcca acggccccc 660
gggctcaggc ggggagcagc tctgtgggtg gggattgagg cgttttccaa gagtgggttt 720
tcacgtttct aagatttccc aagcagacag cccgtgtctg tccgatttct cgaacaaaaa 780
agcaaaacgt gtggctgtct tgggagcaag tcgcaggact gcaagcagtt gggggagaaa 840
gtccgccatt ttgccacttc tcaaccgtcc ctgcaaggct ggggctcagt tgcgtaatgg 900
aaagtaaaag cctgaactat cacactttaa tcttcttcca aaagggtgta aactatacct 960
actgtccctc aagagaacac aagaagtgtc ttaagaggtt ttttaaaagt tccgggggtt 1020
ttgtgaggtg ttgatgacc cgtttaaaaa atgattttcca tgtttctttt gtctaaagtt 1080
tgcagctcaa atctttccac acgctagtaa ttaagtatt tctgcatgtg tagtttgcat 1140
tcaagttcca taagctgtta agaaaaatct agaaaaagta aactagaacc tatttttaac 1200
cgaagaacta ctttttgcct ccttcacaaa ggcggcggaa ggtgatcgaa ttccggtgat 1260
gcgagttgtt ctccgtctat aaatacgctc cgcccagact gtgcggtagg cattgaggca 1320

```

-continued

gccagcgcag	gggcttctgc	tgagggggca	ggcggagctt	gaggaaaccc	cagataagtt	1380
tttttctctt	tgaagatag	agattaatac	aactacttaa	aaaatatagt	caatagggtta	1440
ctaagatatt	gcttagcggt	aagtttttaa	cgtaatttta	atagcttaag	atttttaagag	1500
aaaatatgaa	gacttagaag	agtagcatga	ggaaggaaaa	gataaaaggt	ttctaaaaca	1560
tgacggaggt	tgagatgaag	cttcttcctg	gagtaaaaaa	tgtatttaaa	agaaaattga	1620
gagaaaggac	tacagagccc	cgaattaata	ccaatagaag	ggcaatgctt	ttagattaaa	1680
atgaagggtga	cttaaacagc	ttaaagttta	gtttaaaagt	tgtagggtgat	taaaataatt	1740
tgaaggcgat	cttttaaaaa	gagattaaaac	cgaagggtgat	taaaagacct	tgaatccat	1800
gacgcaggga	gaattgcgtc	atttaaagcc	tagttaacgc	atttactaaa	cgcagacgaa	1860
aatggaaaga	ttaattggga	gtggtaggat	gaaacaattt	ggagaagata	gaagtttgaa	1920
gtggaaaact	ggaagacaga	agtacgggaa	ggcgaagaaa	agaatagaga	agatagggaa	1980
attagaagat	aaaaacatac	ttttagaaga	aaaaagataa	atttaaacct	gaaaagtagg	2040
aagcagaaga	aaaaagacaa	gctaggaaac	aaaaagctaa	gggcaaaatg	tacaaactta	2100
gaagaaaatt	ggaagataga	aacaagatag	aaaatgaaaa	tattgtcaag	agtttcagat	2160
agaaaatgaa	aaacaagcta	agacaagtat	tggagaagta	tagaagatag	aaaaatataa	2220
agccaaaaat	tggataaaat	agcactgaaa	aaatgaggaa	attattggta	accaatttat	2280
tttaaaagcc	catcaattta	atttctgggtg	gtgcagaagt	tagaaggtaa	agtttcagaa	2340
gatgagggtg	tttacgtaga	ccagaaccaa	tttagaagaa	tacttgaagc	tagaagggga	2400
agttgggttaa	aaatcacatc	aaaaagctac	taaaaggact	ggtgtaattt	aaaaaaaact	2460
aaggcagaag	gcttttggaa	gagtttagaag	aatttgggaag	gccttaaaat	tagtagctta	2520
gtttgaaaaa	tgtgaaggac	tttcgtaacg	gaagtaattc	aagatcaaga	gtaattacca	2580
acttaagtgt	tttgcattgg	actttgagtt	aagattattt	tttaaatcct	gaggactagc	2640
attaattgac	agctgaccca	ggtgctacac	agaagtggat	tcagtgaatc	taggaagaca	2700
gcagcagaca	ggattccagg	aaccagtggt	tgatgaagct	aggactgagg	agcaagcgag	2760
caagcagcag	tctgtggtga	agataggaaa	agagtcacag	agccagtgcg	atttggtgaa	2820
ggaagctagg	aagaaggaaag	gagcgctaac	gatttgggtg	tgaagctagg	aaaaaggatt	2880
ccaggaaagga	gcgagtgcaa	tttggtagtg	aaggtagcag	gcggcttgcc	ttggcaacca	2940
cacggaggag	gcgagcaggc	gttgtgcgta	gaggatccta	gaccagcatg	ccagtggtgc	3000
aaggccacag	ggaaaagcgag	tggttggtaa	aaatccgtga	ggtcgccaat	atgtttgttt	3060
tctggaactt	acttatggta	accttttatt	tattttctaa	tataatgggg	gagtttcgta	3120
ctgaggtgta	aaaggattta	tatggggacg	tagggcgatt	tccgggtggt	gtagggttct	3180
ctttttcagg	cttatactca	tgaatcttgt	ctgaagcttt	tgagggcaga	ctgccaaagt	3240
ctggagaaat	agttagtggt	aagtttgggt	gttttttttt	ttcacacgaa	tttgaggaaa	3300
accaaattgaa	tttgatagcc	aaattgagac	aatttcagca	aactctgaag	cagtttgtat	3360
gtttagtgtg	ggtaaatgaag	tatttcagtt	ttgtgaatag	atgacctgtt	tttacttctt	3420
acccttgaa	tctgtttgta	aatgtagagt	ttggatgtgt	aactgaggcg	ggggggaggt	3480
ttcagttatt	ttttttgtgg	gggtgggggc	aaaatatggt	ttcagttctt	tttcccttag	3540
gtctgtctag	aatcctaaag	gcaaatgact	caagggtgta	cagaaaacaa	gaaaaatccaa	3600
tatcaggata	atcagaccac	cacaggttta	cagtttatag	aaactagagc	agttctcacc	3660
ttgaggtctg	tggaaagagt	gtccattgga	gaaatggctg	gtagttactc	ttttttcccc	3720
ccaccctctt	aatcagactt	taaaagtgtc	taaccctcta	aaactgttat	tttttacttg	3780
aagcattttg	ggatggctct	aacagggaa	agagaggggt	ggggagaaaa	tgtttttttc	3840
taagattttc	cacagatgct	atagtactat	tgacaaaactg	ggttagagaa	ggagtgtacc	3900
gctgtgctgt	tggcacgaac	accttcaggg	actggagctg	cttttatcct	tggaaagata	3960
ttccagttg	aagctgaaaa	gtcacagaca	gtgcagcttt	gggtcatatt	cagtcactct	4020
agggaagact	cagaagagct	tgaataggcc	aaatggtgaa	gttaagtttt	ccaataatgt	4080
gacttcttaa	aagttttatt	aaagggggag	ggcaaatatt	ggcaattagt	tggcagtggt	4140
ctgttacggt	tgggattggt	gggggtgggt	taggttaattg	tttagtttat	gattgcagat	4200
aaactcatgc	cagagagact	aaagtcttag	aatggaaaaa	gtaaagaaat	atcaacttcc	4260
aagttggcaa	gtaactccca	atgatttagt	ttttttcccc	ccagtttgaa	ttgggaagct	4320
gggggaagtt	aaatatgagc	cactgggtgt	accagtgcat	taatttgggc	aagggaagtg	4380
tcataaattg	atactgtatc	tgttttctct	caaagtatag	agcttttggg	gaaggaaagt	4440
attgaaactg	gggttgggtc	ggcctactgg	gctgacatta	actacaatta	tgggaaatgc	4500
aaaagttggt	tggatatggt	agtggtggtg	tctcttttgg	aatttttttc	aggtgattta	4560
ataataattt	aaaactacta	tagaaaactgc	agagcaaaag	aagtggctta	atgatcctga	4620
agggattttc	tctgatggta	gcttttgtat	tatcaagtaa	gattctattt	tcagttgtgt	4680
gtaagcaagt	ttttttttag	tgtaggagaa	atacttttcc	attgtttaac	tgcaaaaaca	4740
gatgttaagg	tatgattcaa	aaattttgta	aattgtttat	tttaaaactta	tctgtttgta	4800
aattgttaact	gattaagaat	tgtgatagtt	cagcttgaat	gtctcttaga	gggtgggctt	4860
ttgttgatga	gggaggggaa	actttttttt	tttctataga	cttttttcag	ataacatctt	4920
ctgagtcata	accagctctg	cagtatgatg	gcctagatgc	agagaaaaaca	gctccttggt	4980
gaattgataa	gtaaaaggcag	aaaagattat	atgtcatacc	tccattgggg	aataagcata	5040
accctgagat	tcttactact	gatgagaaca	ttatctgcat	atgcaaaaaa	atttttaagca	5100
aatgaaagct	accaatttaa	agttacggaa	tctaccattt	taaaagttat	tgcttgtaa	5160
gctataacca	caaaaataat	gaattgatga	gaaatacaat	gaagaggcaa	tgctcatctc	5220
aaaataactgc	ttttacaaaa	gcagaataaa	agcgaaaaga	aatgaaaatg	ttacactaca	5280
ttaatcctgg	aataaaaaga	gcgaaataa	atgagagatg	agttggggtc	aagttggattg	5340
aggaggctgt	gctgtgtgcc	aatgtttcgt	ttgcctcaga	caggtatctc	ttcgttatca	5400
gaagagttgc	ttcattttcat	ctgggagcag	aaaacagcag	gcagctgtta	acagataagt	5460
ttaacttgca	tctgcagtat	tgcattgttag	ggataagtcg	ttatttttaa	gagctgtgga	5520
gttcttaaat	atcaaacatg	gcactttctc	ctgaccctct	ccctagggga	tttcaggatt	5580
gagaaatttt	tccatcgagc	cttttttaaa	ttgtaggact	tgttctctgt	ggcttcagtg	5640
atgggatagt	acacttcaact	cagaggcatt	tgcactctta	aataatttct	taaaagcctc	5700
taaagtgatc	agtgccctga	tgccaactaa	ggaaatttgt	ttagcattga	atctctgaag	5760
gctctatgaa	aggaatagca	tgatgtgctg	ttagaatcag	atgttaactgc	taaaatttac	5820
atgttgtgat	gtaaatgtgt	tagaaaaaca	ttaaatcatt	caaaataata	aactattttt	5880

-continued

```

attagagaat gtatactttt agaaagctgt ctccttattt aaataaaata gtgtttgtct 5940
gtagttcagt gttgggggcaa tcttgggggg gattcttctc taatctttca gaaactttgt 6000
ctgcgaacac tctttaatgg accagatcag gatttgagcg gaagaacgaa tgtaacttta 6060
aggcagggaaa gacaaattttt atctctcata aagtgatgag catataataa ttccaggcac 6120
atggcaatag aggcocctcta aataagggaat aaataacctc ttagacaggt gggagattat 6180
gatcagagta aaaggtaatt acacatttta ttccagaaa gtcaggggtc tataaatgta 6240
cagtgattag agtaataactt ttccacattt ccaaagtttg catgttaact ttaaatgctt 6300
acaatcttag agtggtaggc aatgttttac actattgacc ttatataggg aagggagggg 6360
gtgctctgtg ggttttaaa aattttccct tgcagaggca ttctatcctt catgaagcca 6420
ttcaggattt tgaattgcatt atgagtgcct ggctcttctc tctgttctag tgagtgtatg 6480
agaccttgca gtgagtttat cagcatactc aaaatttttt tctgggaatt tggaggggatg 6540
ggaggagggg gtggggctta cttgtgttag cttttttttt ttttacagac ttacagaga 6600
atgcagttgt cttgacttta ggtctgtctg ttctgttggc aagtaaatgc agtactgttc 6660
tgatcccgct gctattagaa tgcattgtga aacgactgga gtatgattaa aagtttgtgt 6720
cccaaatgct tggagtagtg attgttgaag gaaaaaatcc agctgagtgta taaaggctga 6780
gtgttgagga aatttctgca gttttaagca gtctgatttg tgattgaagc tgagtacatt 6840
ttgctgggtg attttttagg aaaaatgctt ttgttcattt ctggtgggtg gaggggagctg 6900
aagcctttag tcttttccag atgcaacctt aaaatcagtg acaagaaaca ttccaacaa 6960
gcaacagctc tcaagaaatt aaactggcaa gtggaaatgt ttaaacagtt cagtgcattt 7020
tagtgcattg tttatgtgtg ggttttctct tcccctccct tggcttaaat tcttacatgc 7080
aggaacactc agcagacaca cgtatgcgaa gggccagaga agccagaccc agtaagaaaa 7140
aatagcctat ttacttttaa taaaccaaac attccatttt aaatgtgggg attgggaacc 7200
actagttctt tcagatggta ttcttcagac tatagaagga gcttccagt gaattcacca 7260
gtggacaaaa tgaggaaaac aggtgaacaa gctttttctg tatttacata caaagtcaga 7320
tcagttatgg gacaatagta ttgaatagat ttcagcttta tgctggagta actggcatgt 7380
gagcaaacctg tgttggcctg tgggtggagg ggtgaggtgg gcgctaagcc tttttttaag 7440
atttttcagg taccctcac taaaggcacc gaaggcttaa agtaggacaa ccatggagcc 7500
ttcctgtggc aggagagaca acaaacgcgt attatcctaa ggtcaagaga agtgtcagcc 7560
tcacctgatt tttattagta atgaggactt gcttcaactc cctctttctg gagtgaagca 7620
tccgaaggaa tgcttgaagt acccctgggc ttctcttaac atttaagcaa gctgttttta 7680
tagcagctct taataataaa gcccaaatct caagcgggtg tgaaggggga gggaaagggg 7740
gaaagcgggc aaccactttt ccttagcttt tccagaagcc tgttaaaagc aagggtctcc 7800
cacaagcaac ttctctgcca catcgccacc ccgtgccttt tgatctagca cagacccttc 7860
accctcacc tcgattgacg catgactgtg gatccttgtg ggcagatgcc ataactggtt 7920
tcaaggtaac gatgggtgct aggtcttttg tgggttgaac tatgttagaa aaggccatta 7980
atttgcctgc aaattgttaa cagaagggtg ttaaaaccac agctaagtag ctctattata 8040
atacttatcc agtgactaaa accaacttaa accagtaagt ggagaaataa catgttcaag 8100
aactgtaatg tgggggggga acatgtaact tgtagactgg agaagatagg catttgagtg 8160
gctgagaggg cttttgggtg ggaatgcaaa aattctctgc taagactttt tcagggtgaac 8220
ataacagact tggccaagct agcatcttag cggaaagctga tctccaatgc tcttcagtag 8280
ggtcatgaag gtttttcttt tccagagaaa acaacacgta ttgtttctc aggttttctt 8340
ttttggcctt tttctagctt aaaaaaaaaa aaagcaaaag atgtctgttg ttggcactcc 8400
tgggttccag gacgggggtc aaatccctgc ggctctttt ctttgactac taactctgtc 8460
tcaggactct tctgtatatt ctccttttct ctgcaggtgc tagttcttgg agttttgggg 8520
aggtgggagg taacagcaca atacttttga actatataca tccctgatgt ataatttgtc 8580
aggagcttga cttgatttga tattcatatt tacacgagaa cctaataataa ctgccttgtc 8640
tttttcagggt aatagcctgc agctggtgtt ttgagaagcc ctactgctga aaacttaaca 8700
atttgtgtga ataaaaatgg agaagctcta aattgttgtg gttcttttgt gaataaaaaa 8760
atcttgattg gggaaaaaa 8779

```

SEQ ID NO: 10 moltype = RNA length = 8545

FEATURE Location/Qualifiers

source 1..8545

mol_type = mRNA

organism = Homo sapiens

SEQUENCE: 10

```

cgcagcctgc agcccgagac tttgttaaag gactggggcc ccgcaactgg cctctcctgc 60
cctcttaagc gcagcgccat tttagcaacg cagaagcccg gcgcggggaa gcctcagctc 120
gcctgaaggc aggtcccctc tgacgcctcc gggagcccag gtttccaga gtcttggga 180
cgcagcgacg agttgtgctg ctatcttagc tgccttata ggctggccat tccaggtggt 240
ggattttaga taaaaccaact caaactctgc agtttggctc tgggggttgg aggaaagctt 300
ttatttttct tctgtctccg gttcagaagg tctgaagctc atacctaacc aggcataaca 360
cagaatctgc aaaacaaaaa cccctaaaaa agcagaccga gagcagtgta aacacttctg 420
gggtgtgtcc tgactggctg ccaagggtct ctgtgtcttc ggagacaaag ccattcgctt 480
agttgggtcta ctttaaaagg ccaactgaac tcgctttcca tggcgatttg ccttgtgagc 540
actttcagga gagcctggaa gctgaaaaac ggtagaaaaa tttccgtgcg ggcctgtggg 600
ggctgggggc aactgggggg ccgcagatca gagtgggcca ctggcagcca acggcccccg 660
gggtcagcgc ggggagcagc tctgtggtgt gggattgagg cgttttccaa gagtgggttt 720
tcacgtttct aagatttccc aagcagacag cccgtgctgc tccgatttct cgaacaaaaa 780
agcaaaacgt gtggctgtct tgggagcaag tgcgaggact gcaagcagtt gggggagaaa 840
gtccgccatt ttgccacttc tcaaccgtcc ctgcaaggct ggggctcagt tgcgtaatgg 900
aaagtaaagc cctgaactat cacactttaa tcttcttcca aaagggtgga aactatacct 960
actgtccctc aagagaacac aagaagtgtc ttaagaggcg gcggaagggt atcgaaatcc 1020
ggtgatgcga gttgttctcc gtctataaat acgcctcgcc cgagctgtgc ggtaggcatt 1080
gaggcagcca gcgcaggggc ttctgtctag ggggcaggcg gagcttgagg aaaccgcaga 1140
taagtttttt tctctttgaa agatagagat taatacaact acttaaaaaa tatagtcatt 1200

```


-continued

aggttactaa	gatattgctt	agcgttaagt	ttttaacgta	attttaatag	cttaagattt	1260
taagagaaaa	tatgaagact	tagaagagta	gcatgaggaa	ggaaaagata	aaaggtttct	1320
aaaacatgac	ggagggtgag	atgaagcctc	ttcatggagt	aaaaaatgta	tttaaaagaa	1380
aattgagaga	aaggactaca	gagccccgaa	ttaataccaa	tagaagggca	atgcttttag	1440
attaaaaatga	agggtgactta	aacagcttaa	agtttagttt	aaaagttgta	ggtgattaaa	1500
ataaatttgaa	ggcgatcttt	taaaaagaga	ttaaaccgaa	ggtgattaaa	agaccttgaa	1560
atccatgacg	caggggagaat	tgcgtcattt	aaagcctagt	taacgcattt	actaaacgca	1620
gacgaaaatg	gaaagattaa	ttgggagtg	taggatgaaa	caattttggag	aagatagaag	1680
tttgaaagtgg	aaaactggaa	gacagaagta	cgggaagcgc	aagaaaagaa	tagagaagat	1740
agggaaatta	gaagataaaa	acatactttt	agaagaaaaa	agataaaatt	aaacctgaaa	1800
agtagggaagc	agaagaaaaa	agacaagcta	ggaaacaaaa	agctaagggc	aaaatgtaca	1860
aacttagaag	aaaattggaa	gatagaaaca	agatagaaaa	tgaaaatatt	gtcaagagtt	1920
tcagatagaa	aatgaaaaac	aagctaagac	aagtattgga	gaagtataga	agatagaaaa	1980
atataaaagcc	aaaaattgga	taaaaatgca	ctgaaaaaat	gaggaaatta	ttggtaacca	2040
attttatttta	aaagcccatc	aatttaattt	ctggtggtgc	agaagttaga	aggtaaaagt	2100
tgagaagatg	agggtgttta	cgtagaccag	aaccaattta	gaagaatact	tgaagctaga	2160
aggggaagt	ggttaaaaaa	cacatcaaaa	agctactaaa	aggactgggt	taatttaaaa	2220
aaaactaagg	cagaaggctt	ttggaagagt	tagaagaatt	tggaaggcct	taaatatagt	2280
agcttagttt	gaaaaatgtg	aaggactttc	gtaacggaag	taattcaaga	tcaagagtaa	2340
ttaccaactt	aatgtttttg	cattggactt	tgagttaaga	ttatttttta	aatcctgagg	2400
actagcatta	attgacagct	gacccaggtg	ctacacagaa	gtggattcag	tgaatctagg	2460
aagacagcag	cagacaggat	tcacgaacc	agtgtttgat	gaagctagga	ctgaggagca	2520
agcgagcaag	cagcaggttc	tggtagaat	agggaaaagag	tccaggagcc	agtgcgattt	2580
ggtgaaggaa	gctaggaaga	aggaaggagc	gctaacgatt	tgggtggtga	gctaggaaaa	2640
aggattccag	gaaggagcga	gtgcaatttg	gtgatgaagg	tagcaggcgg	cttggtctgg	2700
caaccacacg	gaggaggcga	gcaggcgttg	tcgctagagg	atcctagacc	agcatgccag	2760
tgtgcccaagg	ccacagggaa	agcagtggtt	tggtaaaaaa	ccgtgaggtc	ggcaatatgt	2820
tgtttttctg	gaacttaact	atggtaacct	tttattttat	ttctaataa	atgggggagt	2880
ttcgtactga	ggtgtaaagg	gatttatatg	gggacgtagg	ccgatttccg	ggtgtgttag	2940
gtttctcttt	ttcaggctta	tactcatgaa	tctgtctga	agcttttgag	ggcagactgc	3000
caagtctcgg	agaaatagta	gatggcaagt	ttgtgggttt	ttttttttta	cacgaatttg	3060
agggaaaacca	aatgaatttg	atagccaaat	tgagacaatt	tcagcaaatc	tgttaagcagt	3120
ttgtatgttt	agtgggggtg	atgaagtatt	tcagttttgt	gaatagatga	cctgttttta	3180
cttctccacc	ctgaattcgt	ttgttaaag	tagagttttg	atgtgtaact	gaggcggggg	3240
ggagttttca	gtattttttt	ttgtgggggt	gggggcaaaa	tatgttttca	gttctttttc	3300
ccttaggtct	gtctagaatc	ctaaaggcaa	atgactcaag	gtgtaacaga	aaacaagaaa	3360
atccaatatc	aggataatca	gaccaccaca	ggtttacagt	ttatagaaac	tagagcagtt	3420
ctcacgttga	ggctctgtga	agagatgtcc	attggagaaa	tggtcgttag	ttactctttt	3480
ttccccccac	ccccttaatc	agactttaaa	agtgcctaac	cccttaaaat	tgttattttt	3540
tacttgaagc	attttggggt	ggtcttaaca	gggaagagag	aggggtgggg	agaaaatgtt	3600
tttttcaaa	attttccaca	gatgctatag	tactattgac	aaactgggtt	agagaaggag	3660
tgtaccgctg	tgtctgtggc	acgaacacct	tcagggaactg	gagctgcttt	tatccttgga	3720
agagtattcc	cagttgaagc	tgaaaagtag	agcacagctgc	agctttgggt	catattcagt	3780
catctcagga	gaacttcaga	agagcttgag	tagggccaaat	gttgaagtta	agttttccaa	3840
taatgtgact	tcttaaaaag	tttattaaag	gggaggggca	aatattggca	attagttggc	3900
agtgccctgt	tacgggtggg	attggtgggg	tgggttttag	taattgttta	gtttatgatt	3960
gcagataaac	tcatgccaga	gaacttaaa	tcttagaatg	gaaaaagtaa	agaaatatca	4020
acttccaagt	tggcaagtta	ctcccaatga	tttagttttt	ttccccccag	tttgaattgg	4080
gaagctgggg	gaagttaaat	atgagccact	gggtgtacca	gtgcattaat	ttgggcaagg	4140
aaagtgtcat	aaatttgatc	tgtatctgtt	ttccttcaaa	gtatagagct	tttggggaag	4200
gaaagtattg	aactgggggt	tgtgtctggc	tactgggctg	acatttaacta	caattatggg	4260
aaatgcaaaa	gttgtttgga	tatggtagt	tgtgtgtctc	ttttggaatt	tttttcaggt	4320
gatttaataa	taattttaaa	ctactataga	aactgcagag	caaaggaagt	ggcttaatga	4380
tcctgaaggg	atttctctctg	atggtagctt	ttgtattatc	aagtaagatt	ctattttcag	4440
ttgtgtgtaa	gcaagttttt	ttttagtgt	ggagaaaatac	ttttccattg	tttaactgca	4500
aaacaagatg	ttaaggtatg	cttcaaaaat	tttgtaaaat	gtttatttta	aaacttatctg	4560
tttgtaaatt	tgaactgatt	aagaatttg	atagttcagc	ttgaattgct	cttagagggt	4620
gggtttttgt	tgatgaggga	ggggaaactt	tttttttttc	tatagacttt	tttcagataa	4680
catcttctga	gtcataacca	gcctggcagt	atgatggcct	agatgcagag	aaaacagctc	4740
cttggtgaat	tgataagtaa	agggcagaaa	gatttatatg	catacctcca	ttggggaata	4800
agcataaacc	tgagattcct	actactgatg	agaacattat	ctgcataatg	caaaaaattt	4860
taagcaaatg	aaagctacca	attttaaagt	acggaaatcta	ccatttttaa	gttaattgct	4920
tgtcaagcta	taaccacaaa	aataatgaat	tgatgagaaa	tacaatgaag	aggcaatgtc	4980
catctcaaaa	tactgctttt	acaaaagcag	aataaaaagc	aaaagaaatg	aaaatgttac	5040
actacattaa	tcttggaata	aaagaagcgc	aaataaatga	gagatgagtt	gggatcaagt	5100
ggattgagga	ggctgtgctg	tgtgccaatg	tttcgtttgc	ctcagacagg	tatctctctg	5160
ttatcagaag	agtgcttcca	tttcatctgg	gagcagaaaa	cagcaggcag	ctgttaacag	5220
ataagtttaa	cttgcactctg	cagtatgtga	tgttagggat	aagtgcctat	ttttaagagc	5280
tgtggagtgc	ttaaaatatc	accatggcac	tttctcctga	cccttccctc	aggggatttc	5340
aggattgaga	aatttttcca	tcgagccttt	ttaaaaattg	aggacttggt	cctgtgggct	5400
tcagtgtagg	gatagtacac	ttcaactcaga	ggcattttgca	tctttaaata	atttcttaaa	5460
agcctctaaa	gtgatcagtg	ccttgatgcc	aactaaggaa	atttgttttag	cattgaaatc	5520
ctgaaggctc	tatgaaagga	atagcatgat	gtgctgttag	aatcagatgt	tactgtctaa	5580
atttcatagt	tgtgatgtaa	attgtgtaga	aaaccattaa	atcattctaa	ataataaact	5640
atttttatta	gagaatgtat	acttttagaa	agctgtctcc	ttattttaat	aaaatagtgt	5700
ttgtctgtag	ttcagtgttg	gggcaatctt	gggggggatt	cttctctaat	ctttcagaaa	5760

-continued

```

ctttgtctgc gaacactctt taatggacca gatcaggatt tgagcggaag aacgaatgta 5820
actttaagggc aggaaagaca aattttattc ttcataaagt gatgagcata taataattcc 5880
aggcacatggc caatagaggc cctcctaata aggaataaat aacctcttag acagggtgga 5940
gattatgacg agagtaaaag gtaattacac attttatttc cagaaagtca ggggtctata 6000
aattgacagt gattagagta atactttttc acatttccaa agtttgcatg ttaactttaa 6060
atgcttacaa tcttagagtg gtaggcaatg ttttacacta ttgaccttat atagggaagg 6120
gaggggggtgc ctgtgggggt ttcaagaatt ttcccttgca gaggcatttc atccttcatt 6180
aagccattca ggatttttga ttgcataatg gtgcttggtc cttcctctcg tctagttag 6240
tgtatgagac cttgcagtg gtttatcagc atactcaaaa ttttttccct ggaatttgga 6300
gggatgggag gaggggggtg ggcctacttg ttgtagcttt tttttttttt acagacttca 6360
cagagaatgc agttgtcttg acttcaggtc tgtctgttct gttggcaggt aaatgcagta 6420
ctgttctgat cccgctgcta ttagaatgca ttgtgaaacg actggagtat gattaaaagt 6480
tgtgttcccc aatgcttgga gtagtgattg ttgaaggaaa aaatccagct gagtataaaa 6540
ggctgagtg ttgaggaatt tctgcagttt taagcagtcg tatttgtag tgaagctgag 6600
tacattttgc tgggtgattt ttaggtaaaa tgccttttgt tcatttcttg tgggtggagg 6660
ggactgaagc ctttagtctt ttccagatgc aaccttaaaa tcagtgcaca gaaacattcc 6720
aaacaagcaa cagtcttcaa gaaattaaac tggcaagtgg aaatgtttaa acagttcagt 6780
gatctttagt gcattgttta tgtgtgggtt tctctctccc ctcctctggt ctttaattctt 6840
acatgcagga acactcagca gacacacgta tgcaaggggc cagagaagcc agaccagta 6900
agaaaaataa ccctatttac tttaaataaa ccaaacattc cttttttaa gtggggattg 6960
ggaaccacta gttctttcag atgggtattc tcagactata gaaggagctt ccagttgaat 7020
tcaccagtgg acaaaatgag gaaaaacagg gaacaagctt tttctgtatt tacatacaaa 7080
gtcagatcag ttatgggaca atagatttga gctttatgct ggagtaactg 7140
gcatgtgagc aaactgtggt ggcgtggggg tggagggggt aggtgggcgc taagcctttt 7200
tttaagattt ttcaggtaac cctcactaaa ggcaccgaag gcttaaagta ggacaacat 7260
ggagccttcc tgtggcagga gagacaacaa agcgctatta tcctaaggtc aagagaagtg 7320
tcagcctcac ctgattttta tttagtaatg gaacttgcct caactccctc tttctggagt 7380
gaagcatccg aaggaatgct tgaagtacct ctgggcttct cttaacattt aagcaagctg 7440
tttttatagc agctcttaat aataaaagccc aaatctcaag cgggtgctga aggggagggg 7500
aagggggaaa gcggggcaac acttttccct agcttttcca gaagcctgtt aaaagcaagg 7560
tctccccaca agcaacttct ctgccacatc gccaccocgt gccttttgat ctagcacaga 7620
cccttcaccc ctacactega tgcagccagt agcttggatc cttgtgggca tgatccataa 7680
tcggtttcaa ggtaacgatg gtgtcgaggt ctttggtggg ttgaactatg ttagaaaagg 7740
ccattaattt gcctgcaaat tgttaacaga agggatttaa aaccacagct aagtagctct 7800
attataatc ttatccagtg actaaaacca acttaaacca gtaagtggag aaataacatg 7860
ttcaagaact gtaatgtcgg gtgggaacat gtaacttgta gactggagaa gataggcatt 7920
tgagtggctg agagggctg tgggtgggaa tgcaaaaatt ctctgctaag actttttcag 7980
gtgaacataa cagacttggc caagctagca tcttagcgga agctgatctc caatgctctt 8040
cagtaggggtc atgaaggttt ttcttttctc gagaaaaaaa cactgtattg tttctcaggt 8100
tttgcttttt ggcctttttc tagcttaaaa aaaaaaaaag caaagatgct tgggtggttg 8160
cactcctggt ttccaggagc ggggttcaaat cccctcgggc tctttgcttt gactactaat 8220
ctgtcttcag gactcttctc gtatttctcc tttctctcgc aggtgctagt tcttgaggt 8280
ttggggaggt gggaggtaac agcacaatat ctttgaacta tatacatcct tgatgataa 8340
ttgtcagga gcttgacttg attgtatatt catatttaca cgagaaccta atataactgc 8400
cttgtctttt tcaggtaata gctgcagct ggtgttttga gaagccctac tgcgtgaaac 8460
ttaacaattt tgtgtaataa aaatggagaa gctctaaatt gttgtggttc tttgtgaat 8520
aaaaaatctc tgattgggga aaaaaa 8545

```

```

SEQ ID NO: 11      moltype = RNA length = 8302
FEATURE            Location/Qualifiers
source              1..8302
                    mol_type = mRNA
                    organism = Homo sapiens

```

```

SEQUENCE: 11
cgcagcctgc agcccgagac tttctgaaag gactggggcc ccgcaactgg cctctcctgc 60
cctcttaagc gcagcgccat tttagcaacg cagaagcccg gcgcccggaa gcctcagctc 120
gcctgaaggc aggtcccccct tgacgcctcc gggagcccag gtttcccaga gtccctggga 180
cgcagcgacg agttgtgctg ctatcttagc tgtccttata ggctggccat tccaggtggt 240
ggtattttaga taaaaccact caaactctgc agtttggctc tgggggtttg aggaaagctt 300
ttatttttct tccgtgctcg gtccagaagg tctgaagctc atacctaacc aggcataaca 360
cagaatctgc aaaaacaaaaa cccctaaaaa agcagaccca gagcagtgta aacacttctg 420
ggtgtgtccc tgactggctg cccaaggtct ctgtgtcttc ggagacaaaag ccattcgctt 480
agttggtcta ctttaaaagg ccacttgaac tcgctttcca tggcgatttg ccttgtgagc 540
actttcagga gagcctggaa gctgaaaaac ggtagaaaaa tttccgtgcg ggcctgggg 600
ggctggcgcg aactgggggg ccgcagatca gagtggggca ctggcagcca acggccccc 660
gggctcaggc ggggagcagc tctgtggtgt gggattgagg cgttttccaa gagtgggttt 720
tcacgtttct aagatttccc aagcagacag cccgtgctgc tccgatttct cgaacaaaaa 780
agcaaaacgt gtggctgtct tgggagcaag tcgcaggact gcaagcagtt gggggagaaa 840
gtccgccatt ttgccacttc tcaaccgtcc ctgcaaggct ggggctcagt tgcgtaatgg 900
aaagtaaacg cctgaactat cacactttaa tcttcttcca aaagggtgga aactataact 960
actgtccctc aagagaacac aagaagtgtc ttaagaggcg gcggaagggt atcgaaattcc 1020
ggtgatgcga gttgttctcc gtctataaat acgcctcgcc cgagctgtgc ggttaggcatt 1080
gaggcagcca gcgcaggggc ttctgctgag ggggcaggcg gagcttgagg aaaccgcaga 1140
taagtttttt tctctttgaa agatagagat taatacaact acttaaaaaa tatagtcaat 1200
aggttactaa gatattgctt acggttaagt ttttaacgta attttaatag ctttaagattt 1260
taagagaaaa tatgaagact tagaagagta gcatgaggaa ggaaaagata aaaggtttct 1320

```

-continued

aaaacatgac	ggaggttgag	atgaagcttc	ttcatggagt	aaaaaatgta	tttaaaagaa	1380
aattgagaga	aaggactaca	gagccccgaa	ttaataccaa	tagaaggcca	atgcttttag	1440
atttaaatga	aggtgactta	aacagcttaa	agtttagttt	aaaagttgta	ggtgattaaa	1500
ataatttgaa	ggcgcatttt	taaaaagaga	ttaaacccgaa	ggtgattaaa	agaccttgaa	1560
atccatgacg	caggagagaat	tgcgtcattt	aaagcctagt	taacgcattt	actaaacgca	1620
gacgaaaatg	gaaagattaa	ttgggagtg	taggatgaaa	caatttggag	aagatagaag	1680
tttgaagtgg	aaaactggaa	gacagaagta	cgggaaggcg	aagaaaagaa	tagagaagat	1740
agggaaatta	gaagataaaa	acatactttt	agaagaaaaa	agataaaatt	aaacctgaaa	1800
agtaggaagc	agaagaaaaa	agacaagcta	ggaaacaaaa	agctaaggcg	aaaatgtaca	1860
aaacttagaag	aaaattggaa	gatagaaaca	agatagaaaa	tgaaaatatt	gtcaagagtt	1920
tcagatagaa	aatgaaaaac	aagctaagac	aagtattgga	gaagtataga	agatagaaaa	1980
atataaagcc	aaaaattgga	taaaatagca	ctgaaaaaat	gaggaaaatta	ttggtaacca	2040
ttttatttta	aaagcccatc	aatttaattt	ctggtgggtg	agaagttaga	aggtaaaagt	2100
tgagaagatg	aggggtgtta	cgtagaccag	aaccaattta	gaagaatact	tgagctaga	2160
aggggaagtt	gggtaaaaat	cacatcaaaa	agctactaaa	aggactgggt	taatttataa	2220
aaaactaagg	cagaaggctt	ttggaagagt	tagaagaatt	tggaaggcct	taatatagt	2280
agcttagttt	gaaaaattgt	aaggactttc	gtaacggag	taattcaaga	tcaagagtaa	2340
ttaccaactt	aatgtttttg	cattggactt	tgagttaa	ttatttttta	aatcctgagg	2400
actagcatta	attgacagct	gacccaggtg	ctacacagaa	gtggattcag	tgaatctagg	2460
aagacagcag	cagacaggat	tcagggaacc	agtgtttgat	gaagctagga	ctgaggagca	2520
agcagagca	cagcagttcg	tggtgaagat	aggaaaagag	tcaggagcc	agtgcgattt	2580
ggtgaaggaa	gctaggaaga	aggaaggagc	gctaacgatt	tggtgggtgaa	gctaggaaaa	2640
aggattccag	gaaggagcga	gtgcaatttg	gtgatgaagg	tagcaggcgg	cttggcttgg	2700
caaccacacg	gaggaggcga	gcaggcggtg	tgcgtagagg	atcctagacc	agcatgccag	2760
tgtgccaaag	ccacagggaa	agcagtggtt	tggtaaaaat	ccgtgaggtc	ggcaatatgt	2820
tgtttttctg	gaacttaact	atggtaacct	tttattttat	ttctaataa	atgggggagt	2880
ttcgtactga	gggtgaaag	gatttatatg	gggacgtagg	ccgatttccg	ggtgtgttag	2940
gtttctcttt	tcaggcttta	tactcatgaa	tcttgtctga	agcttttgag	ggcagactgc	3000
caagtccctg	agaaaatagta	gatggcaagt	ttgtgggttt	ttttttttta	cacgaatttg	3060
aggaaaacca	aatgaatttg	atagccaaat	tgagacaatt	tcagcaaatc	tgtaaagcagt	3120
ttgtatgttt	agttggggta	tgaagtattt	tcagttttgt	gaatagatga	cctgttttta	3180
cttctctacc	ctgaattcgt	tttgtaaatg	tagagtttgg	atgtgtaact	gaggcggggg	3240
ggagttttca	gtattttttt	ttgtgggggt	gggggcaaaa	tatgttttca	gttctttttc	3300
ccttaggtct	gtctagaatc	ataaggcaca	atgactcaag	gtgtaacaga	aaacaagaaa	3360
atccaatatc	aggataatca	gaccaccaca	ggtttacagt	ttatagaaac	tagagcagtt	3420
ctcacgttga	ggctctggga	agagatgtcc	attggagaaa	tggtcggtag	ttactctttt	3480
ttccccccac	cccccttaac	agactttaaa	agtgcctaac	cccttaaaat	tgttattttt	3540
tacttgaagc	attttgggat	ggctcttaaca	gggaagagag	aggggtgggg	agaaaatggt	3600
tttttctaag	attttcacac	gatgctatag	tactattgac	aaactgggtt	agagaaggag	3660
tgtaccgctg	tgcgtgtggc	acgaaacact	tcagggactg	gagctgcttt	tatccttggg	3720
agagtattcc	agcttgaaag	tgaaaagtac	agcacagtcg	agctttgggt	catattcagt	3780
catctcagga	gaacttcaga	agagcttgag	tagggcaaat	gttgaagtta	agttttccaa	3840
taatgtgact	tctttaaagc	tttattaaag	gggaggggca	aatattggca	attagttggc	3900
agtggcctgt	tacgggtggg	attgggtggg	tggggtttagg	taattgttta	gtttatgatt	3960
gcagataaac	tcatgcccga	gaacttaaa	tcttagaatg	gaaaagataa	agaaatatca	4020
acttccaagt	tgccaagtta	ctcccaatga	tttagttttt	ttccccccac	tttgaattgg	4080
gaagctgggg	gaagttaaat	atgagccact	gggtgtacca	gtgcattaat	ttgggcaagg	4140
aaagtgtcat	aatttgatag	tgtatctgtt	ttccttcaaa	gtatagagct	tttgggggag	4200
gaaagtattg	aactgggggt	tggtctggcc	tactgggctg	acatttaacta	caattatggg	4260
aaatgcaaaa	gtttgttggg	tatggtagtg	tgtggtttct	ttttggaatt	tttttcagg	4320
gatttaataa	taatttataa	ctactataga	aactgcagag	caaagggaag	ggcttaataa	4380
tcctgaagg	atttctctct	ttgtattatc	aaactttttt	cagataaacat		4440
ctttctgagtc	ataaccagcc	tggcagtagt	atggcctaga	tgccagagaa	acagctcctt	4500
ggtagaattga	taagttaaag	cagaaaagat	tatatgtcat	acctccattg	gggaataaag	4560
ataaccctga	gattcttact	actgatgaga	acattatctg	catatgccaa	aaaattttta	4620
gcaaatgaaa	gctaccaatt	taaagttacg	gaatctacca	ttttaaagt	aattgcttgt	4680
caagctataa	ccacaaaaat	aatgaattga	tgagaaatac	aatgaagagg	caatgtccat	4740
ctcaaaatag	tgtttttaca	aaagcagaat	aaaagcgaag	agaaatgaaa	atgttacact	4800
acattaatcc	tggaataaaa	gaagccgaaa	taaatgagag	atgagttggg	atcaagtggg	4860
ttgaggaggc	tgtgctgtgt	gccaatgttt	cgtttgcctc	agacaggtag	ctcttcgtta	4920
tcagaagagt	tgcttcattt	catctgggag	cagaaaaacag	caggcagctg	ttaacagata	4980
agtttaactt	gcactctcag	tattgcatgt	tagggataag	tgcttatttt	taagagctgt	5040
ggagttctta	aatatcaacc	atggcacttt	ctcctgaccc	cttccctagg	ggatttcagg	5100
attgagaaat	ttttccatcg	agccttttta	aaattgtagg	acttggtcct	gtgggcttca	5160
tgtaggggat	agtcaccttc	actcagaggg	atttgtcatc	ttaataaatt	tcttaaaagc	5220
ctctaaagt	atcagtgcct	tgatgccaac	taaggaaatt	tgtttagcat	tgaatctctg	5280
aaggctctat	gaaaggaata	gcattgatgt	ctgttagaat	cagatgttac	tgctaaaaat	5340
tacatgttgt	gatgttaatt	gtgtagaaaa	ccattaaatc	attcaaaata	ataaactatt	5400
tttattagag	aatgtatact	tttagaaagc	tgtctcctta	tttaaatata	atagtgtttg	5460
ctctgtagtc	agtggtgggg	caatcttggg	ggggattctt	ctctaactct	tcagaaactt	5520
tgtctgcgaa	cactctttta	tgaccagat	caggatttga	gcggaagaa	gaatgttaact	5580
tttaaggcagg	aaagacaaat	tttattcttc	ataaagtgtg	gagcatataa	taattccagg	5640
cacatggcaa	tagaggccct	ctaaataaag	aataaataac	ctcttagaca	ggtgggagat	5700
tatgatcaga	gtaaaaggta	attacacatt	ttatttccag	aaagtcaggg	gtctataaat	5760
tgacagtgtg	tagagtaata	cttttccaca	tttccaaagt	ttgcatgtta	acttttaaat	5820
cttacaatct	tagagtggta	ggcaatgttt	tacactattg	accttatata	gggaaggagg	5880

-continued

```

ggggtgacctg tgggggtttta aagaatatttc ctttgcagag gcaatttcac cttcatgaag 5940
ccattcaggga ttttgaatttg catatgagtg cttgggtcctt ccttctgttc tagtgagtggt 6000
atgagacacct gcagtgaggtt tatcagcata ctcaaaattt ttttctggga atttggagggt 6060
atgggaggag ggggtggggc ttacttggtg tagctttttt tttttttaca gacttcacag 6120
agaatgcagt tgtcttgact tcaggctctgt ctgttctgtt ggcaagtaaa tgcagtactg 6180
ttctgatccc gctgctatta gaatgcattg tgaacgactt ggagtgatgat taaaagtgtg 6240
gttccccaat gcttgagta gtgattgttg aaggaaaaaa tccagctgag tgataaaggc 6300
tgagtgttga ggaaatttct gcagttttaa gcagtcgtat ttgtgattga agctgagtac 6360
attttgctgg tgtattttta ggtaaaatgc tttttgttca tttctgttgg tgggagggtga 6420
ctgaagcctt tagtcttttc catatgcaac cttaaaatca gtgacaagaa acattccaaa 6480
caagcaacag tcttcaagaa attaaactgg caagtggaaa tgtttaaaca gttcagtgat 6540
ctttagtcca ttgtttatgt gtgggtttct ctctcccttc ccttggctt aattcttaca 6600
tgcagggaaca ctacgcagac acacgtatgc gaagggccag agaagccaga cccagtaaga 6660
aaaaatagcc ttattacttt aaataaacca aacattccat tttaaatgtg gggattggga 6720
accactagtt ctttcagatg gtattcttca gactatagaa ggagcttcca gttgaattca 6780
ccagtggaca aaatgaggaa aacaggtgaa caagcttttt ctgtatttac atcaaaagtc 6840
agatcagtta tgggacaata gtattgaata gatttcagct ttatgctgga gtaactggca 6900
tgtgagcaaa ctgtgttggc gtgggggtgg aggggtgagg tgggcgctaa gccttttttt 6960
aagatttttc aggtaccctt cactaaagcc accgaaggct taaagttaga caacctgga 7020
gccttctctg ggcaggagag acacaaaagc gctattatcc taagggtcaag agaagtgtca 7080
gcctcacctg atttttatta gtaatgagga ctgctctcaa ctccctcttt ctggagtgaa 7140
gcatccgaag gaatgcttga agtaccctg ggcttctctt aacatttaag caagctgttt 7200
ttatagcagc tcttaataat aaagcccaaa tctcaagcgg tcttgaagg ggagggaagg 7260
ggggaagcgg ggcaaccact ttccctagc ttttccagaa gcctgttaaa agcaaggctt 7320
ccccacaagc aacttctctg ccacatcgcc acccctgccc ttttgatcta gcacagacc 7380
ttcacccttc acctcgatgc agccagtagc ttggatcctt gtgggcatga tccataatcg 7440
gtttcaaggt aacgatgggt tcagaggtctt tgggtgggtg aactatgtta gaaaaggcca 7500
ttaatttgcc tgc aaattgtt taacagaagg gtattaaac cacagctaa tagctctatt 7560
ataatactta tccagtgact aaaaccaact taaccagta agtgagagaa taactgttc 7620
aagaactgta atgctgggtg ggaacatgta actttagtag tggagaagat aggcatttga 7680
gtggctgaga gggcttttgg gtgggaatgc aaaaattctc tgctaagact ttttcagggtg 7740
aacataacag acttggccaa gctagcatct tagcgggaag tgatctccaa tgcctctcag 7800
tagggctcatg aaggtttttt ttttctgag aaaaacaac gtattgtttt ctcaggtttt 7860
gcttttttgc ctttttctag cttaaaaaaa aaaaagcaa aagatgctgg tggttggcag 7920
tctgtgttgc caggacgggg ttcaaatccc tgcggcgtct ttgctttgac tactaatctg 7980
tcttcaggac tcttctgta ttctctctt tctctgcagg tgctagtctt tggagttttt 8040
gggaggtggg aggtaacagc acaatatctt tgaactatat acatccttga tgtataattt 8100
gtcaggagct tgacttgatt gtatatctat atttacacga gaacctaaat taactgcctt 8160
gtctttttca ggtaatagcc tgcagctggt gttttgagaa gccctactgc tgaaaactta 8220
acaattttgt gtaataaaaa ttgagaagct ctaaatgttt gtggttcttt tgtgaataaa 8280
aaaatcttga ttggggaaaa aa 8302

```

```

SEQ ID NO: 12      moltype = RNA length = 3756
FEATURE            Location/Qualifiers
source              1..3756
                    mol_type = mRNA
                    organism = Homo sapiens

```

```

SEQUENCE: 12
ggagttagcg acagggagggt atgcgcgcct ggggtgtagtt gtgggggagg aagtggctag 60
ctcaggggctt caggggacag acagggagag atgactgagt tagatgagac gagggggcgg 120
gctggggggtg cgagaaggaa gcttggcaag gagactaggt ctagggggag cacagtgggg 180
caggctgcag gga aaatc atc caggggtcc cccaggcaga acagccacgc tccaggccag 240
gctgtcccta ctgcctgggt gaggggggaa ttgacctctg ggagggcgcc gctcttgcat 300
agctgagcga gcccggtgct gctgggtctgt gtggaaggag gaaggcagg agaggtagaa 360
ggggtggagg agggctgcag gagggccggg agggctaact ttcaacttgt ccatgccagc 420
cagtgggtgc agggctgcag gagggccggg agggctaact ttcaacttgt ccatgccagc 480
agcccttttt tttccagacc aagggctgtg aaccgcctg gggatgaggc ctgggtctgt 540
ggaaactgaac ttatgctgac ggggctgacc gctctggccc aggggtggtat gtaattttcg 600
ctcgccctgg gacggggccc agggccgggc cagcctggtg gagcgtccag gctctgggtg 660
gaagccaggc ccctgggcgg aggtgagggg tggctgagag agtgatgtgg agttaaggcg 720
ccatcctcac cggtagctgg tgcggcacct agcatgtttg acagggcggg actgagggc 780
acgctgctcg ggtgttgggg acaacattga ccaacgcttt attttccagg tggcagtgct 840
ccttttgga ctttctctag gtttggcgtt aaactctctt tgtgagctca ctccaccct 900
tcttctctcc ttaacttat ccatcactt aaaaattac ctggtcatct ggtaagccc 960
ggacagtaag ccagtggtgt gttggagtcg gtattgttgg taatgggtga ggaagagag 1020
ccttcccgtc gaggtggggg tggggcggat cgggtgttgc tgcctgcaga gaggggtggg 1080
agtgaatgtg caccctggg tgggctgca gccatccagc tgaaagtta aaaaatgctt 1140
catggaccgt ggtttgttac ttatgttgc ctcattggcg gcagatggaa ccgggagaca 1200
tggagtcctt ggccagtggt agtcttagca ttgcaggagg ggagacctg gagggagag 1260
ccgcctcaa ttgatgctg cagattgaat ttccagagggc ttaggaggag gaagtctcc 1320
aatgttctgt ttccaggcct gctcaggaa gccctgatt caggaggcta ccatttaag 1380
tttcagatg agcttatggg gggcaattct aaaaagcca cagcagatgc atccggctcg 1440
aggggccatc agctttgaat aaatgcttgt tccagagccc atgaatgcca gcaggcacc 1500
ctcctttctt ggggtaagg ttttcagatg ctgcatcttc taaattgagc ctccggctcat 1560
actagtttgg tcttggaaac ctgcttcaa gaagatccct aagctgtaga acattttaac 1620
gttgatgcca caacgcagat tgatgccttg tagatggagc ttgcagatgg agcccgtaga 1680

```

-continued

```

cctctcacct acccacctgt ttgectgcct tcttgtgcgt ttctcggaga agttctttagc 1740
ctgatgaaat aacttggggc gttgaagagc tgtttaattt taaatgcctt agactgggga 1800
tatatttagag gaagcagatt gtcaaattaa ggggtgcatt gtgttgcctt aaacgcctggg 1860
agggtagaag ttggtcattc ctaaatctgt gtgtgagaaa tggcaggtct agtttgggca 1920
ttgtgattgc attgcagatt actaggagaa gggaaatgggt ggtacacggg tagtgctctt 1980
ttgttcttgc ttctgttttt taaacttgaa ctttacttcg ttagatttca taatactttc 2040
ttggcattct agtaagagga ccttgaggtg ggagttgttg gggacgggga gaaggggaca 2100
gcttggcacc ggtcccggtg gcgttgcagt gtgggggatg ggggtatgca gcttggcact 2160
ggtactggga gggatgaggg tgaagaaggg gagaggggtg gttagagata cagtgtgggt 2220
gggtgggggtg gtaggaaatg caggttgaag ggaattctct ggggcttctg ggaatttagt 2280
gcgtgggtga gccaaagaaa tactaattaa taatagtaag ttgttagtgt tggtaagt 2340
gttgcttgga agtgagaagt tgcttagaaa ctttccaaag tgcttagaac tttaatgtga 2400
aacagacaaa ctaacaaaaa aaaattgttt tgctttgcta caaggtgggg aagactgaag 2460
aagtgttaac tgaaaacagg tgacacagag tcaccagttt tccgagaacc aaagggaggg 2520
gtgtgtgatg ccatctcaca ggcaggggaa atgtctttac cagcttctct ctggtggcca 2580
agacagcctg tttcagaggg ttgttttgtt tggggtgttg gtgttatcaa gtgaattagt 2640
cacttgaag atgggcgtca gacttgcata cgcagcagat cagcatcctt cgtgcccc 2700
tagcaactta ggtggtgat ttgaaactgt gaaggtgtga ttttttcagg agctggaagt 2760
cttagaaaag cctgtaaaat gcctatattg tgggctttta acgtatttaa gggaccactt 2820
aagacgagat tagatgggct cttctggatt tgttctcat ttgtcacagg tgtcttgga 2880
ttgaaaatca tgagcgaagt gaaattgcat tgaatttcaa ggaatttag tatgtaaatc 2940
gtgccttaga aacacatctg ttgtcttttc tgtgtttgtt cgtatattaat aatggcaaaa 3000
tttttgccta tctagtatct tcaaatgtga gtctttgtaa caacaaaata accttttgtg 3060
gtcactgtaa aattaattat ttgtagacag aatccatgta cctttgctaa ggttagaatg 3120
aataatttat tgtattttta attgaaatgt ttgtgctttt taaatgagcc aagactagag 3180
gggaaactat cacctaaaat cagtttggaa aacaagacct aaaaagggaa ggggatgggg 3240
attgtgggga gagagtgagg gaggtgcctt tactacatgt gtgatctgaa aacctgctt 3300
ggttctgagc tgcgtctatt gaattggtaa agtaatacca atggcttttt atcatttctt 3360
tcttccccctt aagtttccact tgaattttta aaaaatcatgg ttatttttat cgttgggatc 3420
tttctgtctt ctgggttcca ttttttaaat gtttaaaaat atgttgacat ggtagtctag 3480
ttcttaacca atgacttggg gatgatgcaa acaattactg tcttgggat ttgagtgtga 3540
ttagtcacgc atgtatgggg aagtagtctc gggtagctg ttgtgaaatt gaaactgtaa 3600
aagtagatgg ttgaaagtac ttgtatgttg ctctgtatgg taagaactaa ttctgttacg 3660
tcatgtacat aattactaat cacttttctt cccctttaca gcacaaaata agtttgagtt 3720
ctaaactcat tagaaaaaaa aaaaaaaaaa aaaaaa 3756

```

```

SEQ ID NO: 13      moltype = RNA length = 22743
FEATURE            Location/Qualifiers
source              1..22743
                    mol_type = mRNA
                    organism = Homo sapiens

```

```

SEQUENCE: 13
ggagttagcg acaggggagg atgcgcgcct ggggtgtagt gtgggggagg aagtggctag 60
ctcagggcctt caggggacag acagggagag atgactgagt tagatgagac gagggggcgg 120
ctctgggggtg cgagaaggaa gcttgcaag gagactaggt ctagggggac cacagtgggg 180
caggctgcat ggaataatc cgcagggtcc cccaggcaga acagccacgc tccaggccag 240
gctgtcccta ctgcctgggt gaggggggaa ttgacctctg ggagggcgcc gctcttgcat 300
agctgagcga gcccggtgct gctggctctg gtggaaggag gaaggcagg agaggtagaa 360
ggggtggagg agtcaggagg aataggccgc agcagccctg gaaatgatca ggaaggcagg 420
cagtgggtgc agggctgcag gagggccggg agggctaact tcaacttgt ccatgccagc 480
agcccccttt ttccagacc aagggtctgt aacccgcctg gggatgaggg ctggtcttgt 540
ggaactgaac ttagctcgac ggggctgacc gctctggccc aggggtggtat gtaattttcg 600
ctcggcctgg gacggggccc agggccgggg cagcctgggt gagcgtccag gctctgggtg 660
gaagccaggc cctggggcgg aggtgagggg ttgctgaggg agtgatgtgg agttaaggcg 720
ccatcctcac cgggtgacct tgcggcacct agcatgttg acaggcgggg actgagggc 780
acgctgctcg ggtgttgggg acaacattga ccaacgcttt attttccagg tggcagtgct 840
ccttttgga tttctctag gtttggcgtt aaactctct tgtgagctca ctccacctc 900
tcttctctcc tttaaactat ccattcactt aaaacattac ctggtcatct ggttaagccc 960
ggacagtaag ccgagtggtt gttggagtcg gtattgttg taatgggtga ggaagagagg 1020
ccttcccgct gaggtgggg ttgggctggat cgggtgtgct tgctgcaga gaggggtggg 1080
agtgaatgtg cacccttggg ttggcctgca gccatccagc tgaaagtta aaaaatgctt 1140
catggaccgt ggtttgttac tatagtgttc ctcatggcga gcagatggaa cggggagaca 1200
tggagtccct ggccagtgtg agtcctagca ttgcaggagg ggagaccctg gagggagagag 1260
cccgcctcaa ttgatgcctg cagattgaat ttccagaggg ttaggaggag gaagtctctc 1320
aatgttctgt ttccaggcct tgctcaggaa gccctgattt caggaggcta ccatttaaag 1380
tttgagatg agcttatggg gggcaattct aaaaagtcca cagcagatgc atccggctcg 1440
aggggccatc agctttgaat aaatgcttgt tccagagccc atgaatgcca gcaggcacc 1500
ctcctttctt ggggtaaaag ttttcagatg ctgcatcttc taaattgagc ctccggtcat 1560
actagttttg tgcctggaac cttgcttcaa gaagatccct aagctgtaga acattttaac 1620
gttgatgcca caacgcagat tgatgccttg tagatggagc ttgcagatgg agccccgtga 1680
cctctcacct acccaactgt ttgctgcctt tcttgtgcgt ttctcggaga agttctttagc 1740
ctgatgaaat aacttggggc gttgaagagc tgtttaattt taaatgcctt agactgggga 1800
tatatttagag gaagcagatt gtcaaattaa ggggtgcatt gtgttgcctt aaacgcctggg 1860
agggtagaag ttggtcattc ctaaatctgt gtgtgagaaa tggcaggtct agtttgggca 1920
ttgtgattgc attgcagatt actaggagaa gggaaatgggt ggtacacggg tagtgctctt 1980
ttgttcttgc ttctgttttt taaacttgaa ctttacttcg ttagatttca taatactttc 2040

```

-continued

ttggcattct	agtaagagga	ccctgaggtg	ggagttgttg	gggacgggga	gaaggggaca	2100
gcttggcacc	ggtcccggtg	gcgttgcagt	gtgggggatg	ggggatgca	gcttggcact	2160
ggtactggga	gggatgaggg	tgaagaaggg	gagaggggtg	gttagagata	cagtgtgggt	2220
ggtgggggtg	gtaggaatg	caggttgaag	ggaattctct	ggggcttttg	ggaatttagt	2280
gcgtgggtga	gccaagaaaa	tactaattaa	taatagtaa	ttgttagtgt	tggttaagtt	2340
gttgcttgga	agtgaaga	tgcttagaaa	ctttccaaa	tgcttagaac	tttaagtga	2400
aacagacaaa	ctaacaaca	aaaattgttt	tgctttgcta	caaggtgggg	aagactgaag	2460
aagtgttaac	tgaaaaacgg	tgacacagag	tcaccagttt	tccgagaacc	aaagggaggg	2520
gtgtgtgatg	ccatctcaca	ggcaggggaa	atgtctttac	cagcttcctc	ctggtggcca	2580
agacagcctg	tttcagaggg	ttgttttgtt	tggggtgttg	gtgttatcaa	gtgaattagt	2640
cacttgaaa	atgggcgtca	gacttgcata	cgcagcagat	cagcatcctt	cgtgcccc	2700
tagcaactta	ggtggtgat	ttgaaactgt	gaaggtgtga	tttttcagg	agctggaagt	2760
cttagaaaa	ccttgtaaat	gcctatatgt	tgggctttta	acgtatttaa	gggaccactt	2820
aagacgagat	tagatgggct	cttctggatt	tgcttctcat	ttgtcacagg	tgctttgtga	2880
ttgaaaaatca	tgagcgga	gaaattgcat	tgaatttcaa	gggaatttag	tatgtaaatc	2940
gtgccttaga	aacacatctg	ttgtcttttc	tggttttgtt	cgatattaat	aatggcaaaa	3000
tttttgccca	ttagtatctt	tcaaatgtga	gtctttgtta	caacccaaat	acctttttgt	3060
gtcactgtaa	aattaaattt	tggttagacag	aatccatgta	cctttgtcaa	ggttagaatg	3120
aataatttat	tgtattttta	atttgaatgt	ttgtgctttt	taaatgagcc	aagactagag	3180
gggaaactat	cacctaata	cagtttgga	aacaagacct	aaaaagggaa	ggggatgggg	3240
attgtgggga	gagagtggtg	gaggtgcctt	tactacatgt	gtgatctgaa	aacctgtctt	3300
ggttctgagc	tgcgtctatt	gaattggtaa	agtaatacca	atggcttttt	atcatttctt	3360
tcttcccttt	aagtttctct	tgaattttta	aaaatcatgg	ttatttttat	cgttgggac	3420
tttctgtctt	ctgggttcca	ttttttta	gttttaaaat	atgttgacat	ggttagttcag	3480
ttcttaacca	atgacttggg	gatgatgcaa	acaattactg	tcgttgggat	ttagagtgtg	3540
ttagtcacgc	atgtatgtct	aagtagtctc	gggtatgctg	ttgtgaaat	gaaactgtaa	3600
aagtagatgg	ttgaaagtac	tggtatgttg	ctctgtatgg	taagaactaa	ttctgttacg	3660
tcatgtacat	aattactaat	cacttttctt	cccccttaca	gcacaaata	agtttgagtt	3720
ctaaactcat	tagaattgtt	gtattgctat	gttacatttc	tcgaccccta	tcacattgcc	3780
ttcataacga	ctttggatgt	atcttcatat	tgtagattta	ggcttagatt	tgctagctcc	3840
aagtaattaa	ggccatctag	gagagcatgg	taaccacaga	tagaactggt	attatcccaa	3900
gtggtctgca	gactgctgag	tggggatggg	atctgctctc	tggtgagagt	tggtaatcat	3960
tgggtttgaa	tgtgatgaaa	ccactcaagc	caatgaaggt	gggtgtgtag	gtggggagta	4020
ctttgcccata	atatttttaa	acattacctg	gttagagttc	taagtggtag	ttatttttgt	4080
ttggttaggg	gaaagcctga	ataaaacag	aaatggacac	ataatagca	tattccatag	4140
tctttgggag	gctggaatgt	gcctgggatt	tgggtctaa	tgtagcgta	attcttacct	4200
cactaaagaa	tttgccctgt	tttttctctt	ttggtgagtg	actaaaacgt	ctgggcttcc	4260
ctgtgtgcgt	gctacagtaa	gcaagcagag	gctgtgcaaa	gggtgtgaca	ggatcacgtg	4320
gaatctggag	gatacatctt	ggcttgcaaa	ctgctctgtt	ctctcgggtg	ggactgttct	4380
gtccttgca	tgtgttctgt	gtttacctct	tggggtgtta	ggttttgctt	acaggagaca	4440
aaacttgggc	gtagaatgga	agccactgcc	agcctctgtg	ctgagaagga	aggtgcttgt	4500
ttcaaaggga	gcagcaaggg	aggcttgttc	tactcacctg	ggcctgtttg	cctgagaagg	4560
ggagataagg	gctgaactgg	gactagccag	ggggaccaa	acaaatgggt	ggggatcatg	4620
acctgaaggga	ttcttctctt	cccatgagct	gcagggctgg	ttgctgtctt	tgcaactgtg	4680
tcttatttgc	ctgtgcgctt	atatcttgg	gacctctcca	cggtgacact	actgacaaac	4740
gggtggagtg	gtggggagaa	gtcactgtgc	gcgccacct	gtaaaccttc	tgtctgtgct	4800
catggcatct	ccaagatggg	gcactgctgt	gtgcagaatc	cagggtcctc	tttctgcttg	4860
caactccttt	ccctggatgc	ccagaaaaca	atccaggcct	cctttctcat	cttaccctct	4920
tgctttgctt	tttacccttg	accctctata	accgccttct	cttcttttca	gaactccttg	4980
tttctcgtcc	tgttttttat	gattacaaaa	ctcttgcttc	cacctgggaa	gataactgct	5040
atagatgcct	gtatgtaaat	ggtgctgtct	ccagcaactg	gcattgctgaa	gaagaattga	5100
ttcacgggg	ataaatgttg	gggattggaa	gtggggatga	aatggcactt	ggtgatacag	5160
gagcagagag	gtgaggccga	ctgctgaaga	cagctcgcca	ccctccttgc	ctccactcca	5220
atccaggggg	tgggggccaca	ttctttgcct	tcatttatcc	tcagatcagg	tgagatcgac	5280
aggaggtgtt	gatggcagtg	ccagcaatta	ttgctaatac	gtttgatccc	ttatgcatag	5340
atctgaatcc	agactttgtg	aatttccaga	ggtgtgggta	atataataga	attcagtgag	5400
tgggcatggc	tgatcttgtg	caaattaaaa	gttatggggc	ataagaatag	caaaagtgtg	5460
acttctttta	aaaaggaagg	taccctgaga	gccagtattg	gttgaggctc	ttcagtagtc	5520
ccaggttggc	agcactgaga	accgcaggaa	cggcctgttg	ttacaaaaag	gagattgact	5580
cagctgcctc	tggtgcatct	gactgactat	gactgctgag	agattccaag	gacctttaat	5640
gccagggcta	accctctcat	gtgcagtgag	accctgggag	gaagtgtcat	cctctggctt	5700
tgtgtggtag	tcattatggt	gcagtgccgg	catgaaatga	agacacccaa	ataggtttac	5760
agatacgata	tgttttta	gttcgtat	aacaaaaaca	tactgacact	ggttggaat	5820
ggcaacagga	agatagcaaa	atgaatacta	acattacgaa	aagatgaaca	ggtacatggt	5880
ccaagggcag	tggctgtgaa	cttctctg	gtgaaggcat	ccctccagc	accttccagc	5940
ctgctagtta	ggacgaccgc	ccgccaccct	ccaggacctc	cagccctgca	ctgcttctcc	6000
tctcttttaa	ataattcttc	attgagttct	aatatgtaaa	aaaaaaaaag	ttactgtaaa	6060
gtttgcaaat	aaggaaattt	tttttaaa	tcctcagtaa	tcctaccagt	aacaattggt	6120
atgggcatat	ttgcttttgg	aagatttctt	ttgtatgcat	gggataagta	cattttttaa	6180
caaaaatggg	attatgccat	aaattctatt	ttgtgacttt	aatatatagt	gaacaccttt	6240
tttaatgatg	acaggtatgt	ccctgcatg	gctgtatcaa	tttaacaat	cttgtttcaa	6300
tgggcataca	gggtattttt	tagttttttt	ttcctcttag	aaaataatac	ttgcgatgac	6360
tttcttgtga	gctcagactt	tttcacgtct	gttgttatct	ctttgggaat	gctgaataca	6420
tacatttcca	gaaggaaatg	actgttaaac	tcttaagact	tcaggttcat	attgctaaac	6480
tgccagcag	ggagggattt	tttcaattag	tggtctcact	ggtaggca	acctgatgcc	6540
ttccctctt	cctcagaacc	ggctttatca	cattgaaaac	ctttgctcct	ccgacggatc	6600

-continued

gagtcgtcgtt	tcctctcgga	tgtgagcatt	gctttgtctg	ctggtgactg	aacatctcta	6660
ccttggtgtca	attggccatt	tgtgggtgtg	gtgtgtgtgc	gtgtgtgtgt	gtgtgtgtgt	6720
gtatgatttt	ctaattcccta	gtcatttttc	tattgattgt	tttgcaaaag	ccattttacat	6780
cttaaggata	ttgataatct	tttgttatat	ttgatgcaaa	tatttttttc	cagttttatag	6840
gttgcccttt	aatttttgtt	ttcaggtaga	taaaagttaa	acgattttct	taggttagtt	6900
tatcaactgtg	gtttctgaac	ttgttatgtg	tagatctttt	ccaccccaag	agtacataaa	6960
tattaatcca	tactttctta	tggaacttgt	atggtttcgt	ttttacatt	taaaccttct	7020
tccccgtggt	gtgtgtgtgt	gaatctgtgt	ttgtgtgagg	aggggcatgg	tgctctcaga	7080
acccacctcc	tgtggccaga	gagccctgtc	ctgtgagggg	ggttgtcaca	gtggcagggt	7140
tcaattcaga	agaccttgag	ggcaggctga	tgtttctga	atgggcccc	ggttgttgc	7200
tgctccgtgac	tctccatttc	cccatctgag	tggattttga	cctaataggg	cactggagct	7260
ggttcgaatc	ctgactggac	tacttgga	ctttatgtct	gggagcaagt	tacttaacct	7320
ccccaagcct	gtgtctgtga	aatgcgggta	aatgaatgta	gatgtttggc	agcagctact	7380
ccttggttag	ctctcacagt	gaactctcct	gcctctgccc	tccttcccc	cctccccgg	7440
tgccctagcgt	caggtctagc	cacttctctc	tgggccccct	tcctttttct	gtggctggct	7500
gcctgcccgc	ctggcgctgg	acctttcatg	taacgggaat	cagcatgtat	attctggct	7560
ggctcgttct	tacacttaat	tttgtttcca	gtagtatttc	cctgtacogg	cagagttcac	7620
aaacacattt	gaagaggctt	ttctcagga	ttcttaacct	tcccaaaagg	agtccccatg	7680
atgggtttct	agaagcttat	aaatgctctg	aaattgtatt	ttctgtgga	aagcataact	7740
ttcatctgct	tgctcgtgct	caaaaaagat	catgaatgaa	tgattgcatg	attttatgcc	7800
atttgctcta	tactaaagga	tatgtagccc	atctcttgag	ctgttaaa	gttttgacta	7860
ctttaaatcg	tgcagctgtg	agcatctctg	taaaatttag	gtacacatgt	atccccgga	7920
gtggcattgc	ctgggcagtg	agcaacttat	gttttataac	tctcttcaca	gactcaaatg	7980
actccagaaa	gctacacttc	ctgtgtgtgag	tatatgatat	ccatttcccc	acatagccac	8040
taacatcagg	tttttacaat	tttatttatt	tcttgctact	ttaagaaatt	tttgtggtga	8100
aatacatata	atagaagttg	actatctgaa	tcatttttaa	gtatacattc	agtagtgta	8160
agtatgtcgc	cattgtgtga	caaccaatct	ccagaacttt	ttcatcttgc	aaaacaaact	8220
ctgtaccat	taataaact	taaacattcc	attccctcca	gcctcagcaa	ccccattcta	8280
cctttctgtt	ctgtgagttt	gactatttcca	agcaacttcat	atcagttaaa	tcatgaagta	8340
tttgtctgtc	tgtgactggc	ttattctctc	gagcacagtg	tcctcgagat	gcgtctatgt	8400
tgtagcatat	gtcagatatt	ccttcttttt	taaaagatcc	aaataatatt	cttattttat	8460
atcttttttt	tatccattca	tccatttagt	gacacttggg	ttgcttttgg	ctattgtaaa	8520
taatgggtct	atgtacaaat	atctatatta	ttgtatttac	aagtataatg	ctgtaatgta	8580
cacacatctt	tttgagattc	tactctcagt	tcttttgagt	atatagccag	aagtgggtatt	8640
actaaatctt	acgatatttc	tatttttaat	ttattgagga	accactgtag	tttttcatag	8700
caactgcacc	atttttagtt	ctcaccagga	gtgcacaagg	gttccgagg	tcccacatcc	8760
tcccacaacac	ttgttatttt	ctgctttttt	tagattgcag	ccatcatagt	gggtgtgagg	8820
tgacatttca	ttgtggtttt	gatttgcaat	tcctcaatga	ggagtgtatg	tgagcatctt	8880
ttcatatgct	tactggctcat	ttgtatgttg	tctttggaaa	aatgtctatt	caagtccctt	8940
gactatttta	aaaattgggt	tattagagtt	atcgttgttg	tgacttgta	ggagtttctt	9000
tctatattct	ggatattta	ccctatcag	atatatgatt	tgcaaatatc	ttctcttatt	9060
ccataaggtt	actttttcac	tttgttgatt	gtgttctttg	atgtatagaa	gttttttagt	9120
ttgaaatagt	ctaattttac	tttgtttact	tttgtgtct	gtgcttttgg	tgcatatcc	9180
aagaaatcct	tgccaaatcc	aacgttataa	ggtaacttta	aggtatttta	gttgtcttag	9240
tctatatttc	tgtactcacc	ttcttttatc	cactcatcag	ttgatgggca	tgtaggttgg	9300
ttccatattc	ttgcaattct	gaattgtgct	atgatcagg	gtcttttttag	tataatgatt	9360
tactctcctt	tgggtagata	cccagtagtg	ggattgctgg	atcgaatggt	ttttataatt	9420
ttctatttta	ccacagtttc	tctctgcaat	ttctctctt	gaccactaac	catgtgaaat	9480
ttctcatattg	accatttata	tgatcatgaa	ctcttagtat	cattgggaag	gccacatttg	9540
ccacttatga	ttgtaaacct	tatctctcat	ttttctctgt	attgtttggg	caaaaagcac	9600
ctattatacc	aggactttaa	aaatcagctc	gataagctct	tgataagctc	aataataata	9660
actgataagt	ccattgaaat	tccttctgat	tactttttct	ttagttagcta	aacatgtatg	9720
tactcctatg	attacaatga	acactcctct	ccattttaat	taattattta	cattgatgaa	9780
atagcaaaat	gttaatgact	aaatactgtc	ttgggttttt	cggtccagg	cagtcaatat	9840
taacttctta	taattttctt	tttttctctt	atgtgtgtgt	gtgtgtgtat	tttttttttt	9900
tttaatttcaa	tggtttttgg	ggtaacaaatg	gcttttggct	atatagatga	attctacagt	9960
agtgaagctc	gagattttac	tgaccgggtc	acctgagtag	tgtacattgt	acccaatag	10020
tgggttttta	tacctttacc	ccctcttacc	tgagttctcta	gtgtccatta		10080
tgtcactctg	tatacctttt	tgtaccata	agttagctct	cacttataag	tgagaacaca	10140
cagtatattgg	ttttccattc	ctgagttgct	tcacttagaa	taatatctct	cagctccatc	10200
caaaattgct	gcaaaaaaaa	aaaaaacacc	aaacattatt	ttgttctttt	ttattgctaa	10260
gtcatattcc	atgggtgtaga	gataccacat	tttattttac	cactcactgg	ttgatgggtt	10320
ggttccacat	cttgcaatt	gtgacttgta	ctgccatcaa	gtgtctttct	ggtataatga	10380
cttcttttcc	tttgggtaga	tacccaggag	tgggattgct	agatcaaatg	gttcttaaca	10440
ttttctctct	ggatctatct	ctggaaattt	taggctccag	ttttgtgtgt	tggtgtta	10500
aaaaatgcaat	ggaatgta	gatcatcaat	tttcattatg	ctttaaaaatc	tggttaaatg	10560
aggctagaac	actcctgtaa	ggcaagaata	ttctctctgt	tggaaactcaa	atacacagaa	10620
ctgggttaaat	ctcaatctta	atctttgatt	caggacacaa	catggctctc	ttttacttgc	10680
tttctttaat	tgttttttaa	taattgtgga	agcatttctg	aatctcctat	ccaatacaaa	10740
aactaggaca	atacagacag	taactcctat	ggttacaatg	aacactcctc	tccacttaaa	10800
tttaattattt	acactgtgaa	aattgaaata	gcaaaatttt	aatgactaaa	tactgtcttt	10860
gattttttgt	tccaggtctg	tcaatattaa	cttcttataa	ttttcttttt	ttttctttat	10920
gtgtgtgtgt	gtgtgtgtat	atatatatat	tttaatttcaa	tggcttttgg	ggtaacaaatg	10980
gcttttggct	atatatatga	gttctacagt	agtgaaagtct	gagattttac	tacaccttcc	11040
acttatgtgg	tcccacacca	cccgctctcc	ctgcccctcc	ctgcccctcc	ctaggcccaag	11100
gtaataatca	tctgtaatcc	tgggtttatc	tctcacttgc	tttcttttca	tataattttg	11160

-continued

caaaagaatc	tgatctaata	gtgtttttca	gagtatatat	ttatatattta	gctgttctta	11220
gagaaaaatt	attattttgc	atgtaatcct	atggaacatt	ctcatttaaat	accatggtaa	11280
gattcagccc	ttgcccaggg	gatagttcat	ttagttttgt	tactggatag	agctcatcat	11340
gtgactatac	ctcaggttag	ttatcagttc	tccatcccat	ggtgactagg	ttgectctca	11400
gcctctcaac	aacactgttt	ctcagtgccc	ttgtagaagt	gatatgtggg	tgttttctcc	11460
ttacacagag	ttgaaaagg	acgacaacaa	cggtggcact	accaatcccc	caccctccag	11520
aggggtaacc	agtgttacca	gtttgctgtg	ttctctgcta	cacctcgcc	tattcacttc	11580
catttgatc	tgaaaaacgt	gttgcatgg	ttcttttcta	tagaagtggt	aaaatgctat	11640
tgtgtcctgt	acattattga	ttactttttt	tcatttaaca	gtagggagat	gcctgggagt	11700
acacagagaa	ctgcccctac	tgttttcaac	ttctgcactg	tatgtctgtg	agtttagcca	11760
ttctgctgtt	aatggaaatt	tacagtattc	taactttttg	atattacaaa	cagttctgtg	11820
cgatcatcgt	catacacaac	cccttggtga	caatgcata	gtgtttctca	gggtaggtag	11880
caagaagtga	aattcctggg	tcatagggag	tgagtcggac	atttttctcc	attctgccct	11940
gttgccctcc	agagtgggtg	tccagctttg	catacctaa	tatgagagta	ctgtgtgttc	12000
atatcctcta	cgacgctcca	tatatgaaac	ttaaagttct	gctagttgcc	atctttgatc	12060
tatcatgtat	gcagtgacct	actaagactg	taattggtag	agtagattct	tgcatctgt	12120
gtgtgaattt	agcattcatg	ggcttaatgc	tgacaaggcc	cccagggtcc	aagacatata	12180
atcatgtata	attttgtcaa	ggtataat	tttaaatg	ttttgtcatg	tgctgtctgg	12240
tgatgcccaa	cccagtgctc	tgacccagg	tcacactgtg	gctttgtcct	ctgctatgc	12300
ctgcattgca	gcaactgtcc	tgaaagagcc	aaaattatgc	agattttagg	aagtcctatg	12360
ctaatgttat	tatatattgt	gctattgtaa	tggtggggg	tggtggagt	atgaatttat	12420
aaatcactgg	tcttgtaact	aaaattcaaa	cactatagaa	aaaggccatg	tagaagataa	12480
aagttctctc	ataatcccg	acccctaaga	taactactaa	tgacaacttc	atttatattc	12540
cttcagacat	ttctctggctg	tggtgttact	aaaatgtatc	ctattattct	ctgccctaaa	12600
atggaatcat	acaaggtgta	ctgttatttt	tatggctcta	taacatgtca	tattgtacgt	12660
gttggtagg	tcattttaac	catttttcta	gtgatggcct	tgagggtatt	tgcaagtttc	12720
tagccatctc	aaagtgtgct	gcggggatct	cttttgcac	cctctgggtg	cagagctgag	12780
gcacccagag	gcagtgtcca	gaggaggcag	catctgtagg	tgcttccacc	tgctctggct	12840
cttgccacat	ctggttggtg	acactgtttt	gtgagatggg	ttgaaagcac	gtgctgccaa	12900
aatagaataa	tggtggctct	ctcctcatgt	gccgtggaa	tggggtaaaa	ctgcgtagt	12960
gctgcagctg	ccgtgtcata	ccggaatcga	gtataacacg	gtgcctggct	tagcacaana	13020
cagtagtggg	tcctgcaggc	cccagagtct	aattcctgg	attcttccc	ctacacagat	13080
taataaacc	aaaaacaac	tattctagga	aagcgtctgt	gacatttgta	aaaagtggt	13140
tttaatgatc	ttttattcac	ttgtctgttt	agtttgttga	aatcttaagt	ggcatcctgg	13200
tctgggaagg	agtgtgtctc	gcgcctgccc	tcgcctgggc	acagcgtggc	tgcttcagg	13260
gctaagcaca	cactttctgt	cttctaaagg	gccgccacat	gccaggagct	caggtgtgag	13320
cccgctctg	gctcttaact	catagggtca	ctcatagggg	cacaggggagc	agaacattgt	13380
acacagcgag	gcaccaccgc	gcttggcacc	tgccctgggtg	gacttaactac	ctctagaagg	13440
aaatacctga	gttcctctgg	cctcagctcc	tagagtgtact	gggtgtgtgt	cctgtttact	13500
cttctgtcaa	ggtgtacaact	gtgtgaccca	tcatctgtgt	gtcaaaagcaa	ggcctgcct	13560
gggcctctgc	tcctgtgtgt	accccaagg	caaatgcttt	gctagtttcc	ttccagttta	13620
tttcacctat	gaatagatgt	gtgaaaactg	ttcaaaagca	tacctgcaca	tgtttgaa	13680
tcaaacctctg	tggtgtgatt	agtggcatct	ttctctaacc	cccagcctcc	cttcccacag	13740
aggccaccgt	catggccagt	tgctgcagtt	tctttccaga	gaacctgtgt	atgtgtaaag	13800
ctgtacaggg	tggtgtacac	cacacagcct	gtcttgcact	gtggactgtt	gagttactag	13860
tacatctagg	taagcacccg	atatctgtat	tcatgtctgc	cttgggtctt	tcaacatctg	13920
tgtggtagcc	gtgtttgaag	tacccattcc	ctttttgggg	aaccattaa	ttgtttcagc	13980
aatttttact	gtagataagg	ctataccgca	tatctgtgta	catgggtttt	tatgtacatg	14040
ggcaagtata	ctgtgtgag	aaaagtgtcc	tcaggaggaa	ttctggggcac	agcatgtgta	14100
aatttctaaa	tatgatggac	acccccagct	tcacactcaa	ggagggttgg	cccatgaca	14160
tttccccaca	ccttcaccca	ggctgtgccc	ttaaacttgg	ttatttgtca	atgtgagaag	14220
tggaataatg	tatttaattg	tagtttggat	attgggttgt	atactactg		14280
atttaataata	agagctcttt	acatatata	gaaattaacc	cttttcaaat	acatttctat	14340
ttctcactaa	tctttaagtt	ttattgtaat	attttgtctt	ttagtttata	tatatatgta	14400
tatatatata	tatgtatata	tatatatata	catatatata	tacatatata	tataactaatt	14460
ttcttttatg	gttctgggat	tttgtgagta	gtttgaaaag	gctaactccag	ctgaagattt	14520
tggtgtgtgt	gttaaacccc	atgttttctc	ctaactcttt	ttatttttat	tttggaggac	14580
tctatctaga	cttaattttta	gcataacaag	tgacagggtt	agttagcctg	ttgtccttac	14640
accattttct	ggctaataca	gctattaact	attgatctgt	ctattcacgt	gcccagttcct	14700
aatggtttta	catagtgtaa	ctgtcacttc	aaaatagcga	aggggaagccc	tacctcatta	14760
ttctactttt	ccagaattct	cctggctatt	tggtttacct	aaacctccct		14820
gtgatgtctt	catgccgttg	tcttcttatg	caagaataag	gtacgtcttt	ccatccactc	14880
acgtctattt	aatttgactt	tgattacac	agaaagctgg	tcttggctg	tctacctcgg	14940
catctagtgt	tctcactgc	cccctagccg	acccccccc	atctgactga	ctaccccatc	15000
acagagtact	tttatttacg	ttttgtctg	cctaaggtt	acttgatact	gtcacgccga	15060
cagtgtccag	ttcagtggtc	tttgacgttg	aaatgctccc	gtacacactg	tcttggttaa	15120
aatgccagta	agttcataca	aaaccagctt	gcacccaagg	tcacattcag	agagcgtagg	15180
gctgggagtg	gtgtgtttcc	aagcttctgc	cactgtgtgg	ctagctcttc	ccactgggaa	15240
gttctgtgta	cccgaatgt	cggagtgagg	tcctgttcta	gtgtccagca	cctgacctg	15300
tgcccaaccc	ctcaacagcc	tattctctgt	gtccacagcc	tgctggaact	ttttacaaaa	15360
tatgttgcca	tgctggagcc	ggggcactgg	acataagccc	cctggcagcc	tttttcatgt	15420
caaaaaagg	ggtaattgtc	ctactgggtg	tctgtaagat	gagttagggt	gacttgctaa	15480
tagacattgt	aaatctta	atttatgtat	gtattttatt	attaccggtt	ttccatttat	15540
gatggtaata	ttgtttcttc	taagaatatt	tatttttctt	tctaataatt	gagataaaat	15600
tcatgctttt	gaaatgttcc	attcagtgcc	tttttagtata	ttgtctatgt	tggtcaacca	15660
tcgacactat	ccattttctag	aactttttctg	tcattccaaa	cagacgtctc	gtattcataa	15720

-continued

aaaaataact	tectacettgt	ctctcccct	agtctttggt	aacctttggt	atactggtaa	15780
actttgttgt	gctctctgtc	tgtgtgaatt	tgccattct	aggggcctca	tataagtgtg	15840
atcatacagt	atttgtcttt	ttgggtctgt	ctgatttcac	ttagcgggtt	ttcagggttc	15900
attcatgttg	cagcatataa	cagtactgcg	ttcccttttc	tggetgaata	atattccact	15960
gtatggatag	accccatttt	gtttattcac	acatcatttg	gacatttgga	ttatttctgg	16020
tttttggtcta	ttatgaacaa	tggtgctatg	aacagttgcg	tacaagtttt	tgtgtgaaca	16080
tatgttttca	attctctcat	tatatacct	ggagtagaat	tactgggtca	tatggtaact	16140
gtataatttt	gaggaactgc	caaacatttt	tcccacgtcc	atgcaccatt	tcacattccc	16200
accagtaagt	aagagggttc	caatttctgc	gcattcttgc	caacactagt	tattatctga	16260
ctttctggtt	ataatcattc	taatgagtgt	gaagtagcct	ctggtgtcat	ttggatttgc	16320
atttctctga	tgagtgtatg	tatcaagcac	ctttgctggt	gctgttggtg	atatgtgtat	16380
gttcctctga	gaagtgtctg	tgctgagcct	tggtccactt	tttaattagg	cgtttgtctt	16440
tttattactg	agtgtgaaga	gttctttata	tattctggat	tctagaccct	tatcagatac	16500
atgggtttga	aataattttct	ccattctgtg	gggtgtgtgt	ttcactttat	cgataatgtc	16560
cttagacata	taataaaattt	gtattttaaa	agtgtactga	tttggtctgtg	caagggtggt	16620
cacgttgtga	atcccagcac	tttgggagac	tgaggtgggt	ggatcatatg	aggaggctag	16680
gagttcgagg	tcagccctggc	cagcatagcg	aaaactttgc	tctactaaaa	atacaaaaat	16740
tagtcaggca	tggtgggtgca	cgtctgtaat	accagcttct	caggaggctg	aggcacgagg	16800
atcacttgaa	cccaggagga	ggaggttgca	gtgagctgag	atcatgccag	ggcaacagaa	16860
tgagactttg	tttaaaaaaa	aaaaaaagtg	acttgattta	agggaaaaaa	tgactggcta	16920
tattcagcta	gatatggcaa	aaagtctcaa	ggtgttaatg	tgaatgatta	aggtcttggg	16980
gggggtgtcc	cctatcagac	tacaggtgtt	tagaggcaca	gaaaaagggtg	cagttgggtt	17040
cttaagtga	aatatgagaga	agcacaactc	cagtggtgtc	ctttgtgtag	aatgtcagca	17100
gacacccctc	gctagatgtg	ctggatcatg	ggaaagcatt	tccatttgtt	actagattgt	17160
tcagaagtgt	taattttatga	tggtgtgtgt	ggctcatgcc	tgtagtccca	gcactgtggg	17220
aggctgaggc	aggaggatca	tctgaggcca	agagttcaag	atcagcctgg	gcaacatagt	17280
gataccctat	ctcttaaaaa	agaagaagtt	tttaaatgtg	aaataataat	aggtactgga	17340
tttatgcaaa	tgtcttttct	gcgtcttttg	agatgagtat	caggtttttt	ttttctcttt	17400
tatcatctga	tgatgaaact	aatgtttcca	tttgtattaa	tggaatacta	agtccctctg	17460
gtatttctga	accaagctat	tccataggct	gagttttatt	ttgttgacac	agaaataaat	17520
tagaaggcca	agcgtggtgg	ctgtgcctg	tagtcctagt	tgctgaggta	agaggattgc	17580
ttgagcccag	gagttcaagg	ctgcagcaag	ctttgatgtc	gccactgcac	tcacagcctg	17640
gcgacagact	aagacgtgtg	ctcaaaaaaa	aacaaaaacg	acaaaaaaaa	aacaaaaacg	17700
aaaaaataaa	ctaaggcaat	gacagctcct	ggcaaatgct	gggaggggagg	cagcagtggt	17760
caggggaaggt	aacctgaag	caggacttgt	aaagcaataa	agattgggag	gccaagggtg	17820
gtggatcacg	aggtcaggag	ttcgagacca	gcctggccaa	catagtgaag	ccccgtcttt	17880
actaaaaata	ctaaaaaatt	agccaggtgt	ggtggtgggt	gcctgtagtc	ccagctactt	17940
gggaggctga	ggcaggagaa	tctcgaaacc	aggaggcgga	ggttacagtc	agctgagacc	18000
gcaccattgc	actccagcct	gggtgacaga	gcaagattcc	gtctcaaaaa	aaaaaaaaaa	18060
aaaaaaacca	aaagaaaaag	gaatgaatta	gaacttcttc	tgcttggaact	taagggcatac	18120
atcaggcagg	tttgggttag	gtagcagggt	gagggcagaga	catagtcggg	gtcagtggtc	18180
atgagtgtgt	ctttgagccc	aaaaacttgg	tttctgttcc	ctactttgcc	actcagtagt	18240
gcatgacttt	ggcacaattt	cttaaatcca	tgaagcaagt	ttccgggtga	atgaaatggg	18300
gataaaaaata	gtgttcaaac	ctatccgttg	gtttgtgtga	aactgaaatg	aatagtatcg	18360
tgccaggtact	tgtgagcaag	gggagctgct	gtttcctgtc	cctttatgat	gggaaatatc	18420
tagacaagtt	cccacccctc	tgcaactggc	gctgcatggc	acggagggtc	ttgtaacacc	18480
agctggggct	ggccttcttt	taggagcttc	agtggttctg	aaaactttta	tttgtttgtt	18540
tgtttttagta	gatgtggggg	ctttctgtgt	tgcccggact	gggtctcaaac	ttctggactc	18600
aatgtatcct	cctccgctca	acctcccaaa	gtgttgggat	tacaggtgtg	agccactgtg	18660
cccagccttg	aaaacttttt	caggttcttc	cagggttact	gggctattaa	atatttctat	18720
ttcattataa	gtcagttttt	caaagtataa	ttatcttaat	tacotttttt	atatgtatta	18780
gtgtagagta	gcatttttata	ttttgatatc	ctccttatgc	atagtttttc	actttttatt	18840
ctagtttttt	gcttttttaat	aagactttca	agaaatttat	tttatgggcc	ttttgaaaaa	18900
agcagcttta	gataaagttaa	gcagttctgc	tttcatttta	taattttatt	ctacttttgt	18960
ttcattaatc	ttttctctcg	gcattgtctg	gattttgttg	tgtaactctt	ttctagagg	19020
ctgcatttgt	gtgtctgggt	cacttatgat	cacgcttgcc	tacttttaag	aatggaagag	19080
gggagggtga	gggtggctgc	acagtcgagg	gtgtgaggca	gtcttgctct	agccccacca	19140
tgccctcagc	ccgctgtggc	cacgctggtt	cctcaattgc	tggggctgtg	agtgctgtga	19200
agggaggcta	ctgatgccat	ccgagggaaga	tgtaagggtt	cgtgtgggca	gcgagagcct	19260
agcaggcatg	tggggtgccc	agcaaaagggt	aacagtggaac	agttgtgtgc	tcattccaca	19320
gagttttgat	tttttttttt	tttttaattg	tcactccatc	aacatcccc	atggccagag	19380
cctgagctgg	tccccagaga	cacaggcatt	cagctgacag	cctcgcttcc	acgctgctgc	19440
tgttctcatg	ggggacaggc	ctcaggtggc	aatgcacaaa	tcatttagtta	agggcagttg	19500
tgacagttac	caaggagtgt	agtcccccgc	ccccgcacca	gtgaaaaacg	ccctaaccag	19560
gggtggggac	ctttgggctc	tgaccgaag	ggtaggagaa	gctggaagga	cagcattcct	19620
gtctgcgaag	gcaggagcaa	agctgccagg	ctatgaagga	aatggctgga	gcctgaagtc	19680
atgcaagctg	gggtctggcag	ggacagggcc	aacttccagg	cctggggggc	accatgagga	19740
ttcaggagct	gaccccagag	gcacatgaag	gccttccatc	tgattttaag	aaaagacttt	19800
atcagacgag	tatggtggct	cacgcctgaa	tcttagcact	tgaggaggct	gaggcaggtg	19860
gatcacgagg	tcaggagtgc	aataccagcc	tgcccaatat	ggtaaaaccc	catctctact	19920
aaaaactcaa	aaattagcca	ggcatgggtg	cgcacgcctg	tagtccagc	tactcgggag	19980
gctgaggcag	aagaatcact	tgaaccgggg	agggtggagg	tacagtgagc	caagatcgcg	20040
ccactacact	ccagcctggg	tgacagagtg	agactccgtc	tcaaaaaaac	caaaagactt	20100
tatcttattt	cctatatgtt	tgtggtttca	gtcctgatgt	ataatttgac	cctagttaga	20160
atgggttact	gagggaagtg	cctgtacgat	ttctgctttt	ttaaatgtgt	ggctcccttt	20220
cttcattgat	taacgtatga	ttatttttat	aaatgttcca	tggcagtggt	aagggtattct	20280

-continued

```

ctgtcacatt ccacatctgg atcagttcct ccccatthttg ttggtcaaat ccgatctgcc 20340
atatcctgtg taatgacaag tgagttgcat tctcaccgtc actcctgggg tctctccgct 20400
tccccctgagc tggctcagca gtctgtccca tgtgtthttga tgcagggtga cccattggta 20460
ttcccgcacac taacgcccc gctctgtggac tgettgctgc ttgggttca ctgtgtctgg 20520
tgttgacagt gcagacctaa aggtgtgcac acatgtgcac acacactccg ctgtcttctt 20580
gtttgcactg gacttaataa tctatgaggg ttattttcaa ctgtgtaatt tggaaatgatt 20640
tttatatctt tctgtctttc tgcccagta catgtgttta tttacactg ttgtgattgg 20700
tagttactat gtggggacac aattacttgg gctgaaataa tccacctgt gtggttgggg 20760
tctcttgggg cattccaggg tgagaggttg tcaactgccac ctgggccatg tgggccggca 20820
ccagcattht gtggttacga attctacagt cacaaatctc tttgggcaaa tccccctcta 20880
tacctcaagg cagctthttg tttgcaacc cactggccag aggggaaggc cagtcaactg 20940
gctctctcac tgccttgcgc cccagatggt tctagggtcg ctgtthttcc ttggccctgc 21000
caacaccact gthtttactt ctgctcattg gctgagtcca gtggttctcg gaagccagt 21060
gcacgtthttc ccgctgagct cgcttatccc acagcacaca ccaagggtt ctgttgcata 21120
cacgtgaat taattctttg ctcatcttac agagtgtgtt ttgactgccc ccatttctga 21180
ggccttgtaa ggcagagctt tgttgcttc atcgccaggt tgggacttag atggccgtga 21240
atgtttcttc tctgtctgtg cagtaagtaa gtgccgcac catagtgtgt ttggaggctg 21300
aagttgaagg gaggtctgtg ggggagatgg acgtgtgagg agggatgatg gggcttgagc 21360
aaagtggggg agggggcaca ggcagttggc ccaacacatt cccacccct ttgagaggtc 21420
tgaggcctgc agacctggct cggagcccac ctggtagtcc tcagactgtg tgtgtgtgtg 21480
tgtgtgtgtg tgtgtgtgtg tgtgtgtgtg tgtgtgtgtg tgtgtgtgtg agagaagtgt 21540
tggagaaatg gggggctgat tctgtcaga ttcacagga tgaatagaag gcaccagct 21600
ctcacccctg cctgacatgt gtgtccctga gcaggttaca gtctctctg agcctctgct 21660
tccccctgag accctgctgg gcagggtctc tgagctcctt agcactagca ggaggggctc 21720
cagggggcct cctccatgag cagccaggac aggaacttca aatgaggaca gcagagctcg 21780
tggggggctc cccagcagcc gcctggggcc cagggggagg agagcctgag ccaacagcag 21840
tgggtgctgt gaccgtggat cctgaggggt gcttggggca agtaccggct gaggggtccag 21900
gtgggctttg tgtacctttg ggtcctgggg ccttgggtgac ttggaactca ggttagagtc 21960
aagtgaagg agaaaggtgt gtggggccct gtgcttccga cttcatttcg agtgatggca 22020
gttcccagga aggaatccac agctgacggt ggtgtacaga tcagagaatg gaaggcgagg 22080
cagggcgggc tgtgcgtgac ctcaggtgct tggggccag cagaccaga gaaccatttc 22140
cactaggcca ggggtgcgga agtgctcaca ggtcttagat tccctgttca gatgaaaaga 22200
tttgtgcctt taatgataaa agtgatctgc atagagtaaa aaattcaagc catgggtata 22260
aaatgcaagt aaaaatccctg cctcaccta tcccacccta ctacacagag atgtctcttc 22320
gagtttctta gactcactct ggaattttct gtatacacac agaagcttgt gcctctgctc 22380
gtgaaggcag agggagggag agctgaaggg ccagcacctt ctcacctgtg gggcccttca 22440
tgctcggtc ccagagcatg caggactgtg cctcgtgttc agtttgctgg tctgacttca 22500
tgctccttgg gcaggatag catgtgccat gctaggagac atgtggatgt gaagctgggg 22560
gacaatgtcc cctggctatg cctttacaag ggaagtaagg aaggtaggag gtgagcctgg 22620
gagggaggga gggaggcgcg gagccgccg aggtgtttct tttactgagt gcagcccatg 22680
gcccactca ggttttgcct ttcaccttcc catctgtgaa agagtgaaga ggaaaaagca 22740
aaa 22743

```

```

SEQ ID NO: 14      moltype = RNA length = 7253
FEATURE            Location/Qualifiers
source              1..7253
                    mol_type = mRNA
                    organism = Homo sapiens

```

```

SEQUENCE: 14
ggaaagaagt ggaataatta ggaacctagg gtggggtagg gtagcaggac atttcaaaca 60
ttaatgagca tatgagattc caggtcttgt taaaatgcaa attctgattc agctggtagg 120
tgagggtctga gattgtgcat ttctaacaag cactcagata atcttaaggc tgttggcccc 180
agggtcacac ttatagtgat tttctagaac ccagttgggg aagtgaatct tgggcaggag 240
aaatacacac ctcttgcatg gagtttgagg atctcatctg atataacttt ttaagaaaga 300
aaaataaatt tccaaatctc caattgataa gctttcccac taagtggcct tcccactaag 360
tggtctgctt atgaaaattg cttcactttg aaacttctgg tcttggtaat atagaatttc 420
tgtgttctca cagtgttga ttgagaatat gatattgaga ttatggcata aaatatagtg 480
gctgtacaaa aaaaaatata ttattaggat ctctaacaat tatgtaaaag tcattgcttc 540
atgggttagag ctcaaaacttt ggtgtgagac ctggttttat tcttggcact tactctgagt 600
tgtcttaggc aaattaatac cttaagcaaaa aatattctca tgtacatttt acatgagaaat 660
tataaatgaa gtacataaag tccagcagtc acaaatgtta tctattatta ccatcgctct 720
aagactgcaa tcagctatag tgaagtagt ctcaaaagatt gtttcataaa tcatcagatt 780
cacctaattt tctaaagaat ttaataaagg agatggaatg aatagattgc attttgtttc 840
catgcacagg ggaactgtgc atatttcttc tgtgactcgg aaatgggtta acttttaaaa 900
atcccaaat agctgaagtt agcagacatg caatttacca aggatgattg gaatttttat 960
ctttctgtga ataactat acccaagcac actgctcatg aggaaaacat ttttatgtga 1020
atcttttact cttgggggca aagaatgtct tttttctttt tgataactat gtttatagaa 1080
tctaaatcac cctgagcaat tatttcaaca tctaagttta ttattaccat tcatgtttca 1140
tttatagcta tttgaatttt gatgaatttc aatgtgtgct tacagtgata gggcaagtgc 1200
aaataagttc aatatatggg tacggtctaa agctatttta atttttttat tacaactgct 1260
atgaagaaaa tttagtatgt ccatattttc acgttttaca gttggatgtc ctatgatgtt 1320
ctcttcagaa gaacagagct cggagctctg gaaatttggg ggcaactgat atgtgctcat 1380
gtctgcatct gtgtgggttg gctgtatctc agggacagag tctgcagcaa aaaagatata 1440
atthttgagga ctgaacaaaa ttcaggaagg actatttctca ttaaggcagt aacagagaag 1500
aattttgaaa caaaagattt tcgagcctct ctagaaaatg gtgttctgct gtgtgatttg 1560
attaataagc ttaaacctgg cgtcattaag aagatcaata gactgtctac accaatagca 1620

```

-continued

ggattggata	atataaacgt	ttctctgaaa	gcttgtgaac	agattggatt	gaaagaagcc	1680
cagctttttc	atcctggaga	tctacaggat	ttatcaaatc	gagtcactgt	caagcaagaa	1740
gagactgaca	ggagagtga	aaatgttttg	ataacattgt	actggctggg	aagaaaagca	1800
caaagcaacc	cgtactataa	tggtcccat	cttaatttga	aagcgtttga	gaatctttta	1860
ggacaagcac	tgacgaaggg	actcgaagac	tccagcttcc	tgaaaagaag	tggaaggac	1920
agtggctacg	gtgacatctg	gtgtcctgaa	cgtggagaat	ttcttgctcc	tccaaggcac	1980
cataagagag	aagattcctt	gaaagcttg	gaactctttg	gctcgaggtc	attgacaagc	2040
tgtcctctcg	atatcactgt	gagagggggg	cgtgaaggtt	ttgaaagtga	cacagattcg	2100
gaattttacat	ttaagatgca	ggattataat	aaagatgata	tgtcgtatcg	aaggattttcg	2160
gctgttgagc	caaagactgc	gttacccttc	aatcgttttt	taccaacaa	aagtagacag	2220
ccatcctatg	taccagcacc	tctgagaaa	aaaaagccag	acaaacatga	ggataacaga	2280
agaagtgggg	caagcccggg	ttatacagaa	gcagatggaa	cattttcaag	actctttcaa	2340
aagatttatg	gtgagaatgg	gagtaagtcc	atgagtgatg	tcagcgcaga	agatgttcaa	2400
aacttgctgc	agctgcgtta	cgaggagatg	cagaaaaata	aatcacatt	aaaagaacaa	2460
gatcagaaat	ggcaggatga	ccttgcaaaa	tggaagatc	gtcgaaaaag	ttacacttca	2520
gatctgcaga	agaaaaaaga	agagagagaa	gaaattgaaa	agcaggcact	tgagaagtct	2580
aagagaagct	ctaagacggt	taaggaaatg	ctgcaggaca	gggaatccca	aaatcaaaag	2640
tctacagtto	cgtcaagaag	gagaatgtat	tcttttgatg	atgtgctgga	ggaaggaaag	2700
cgacccctta	caatgactgt	gtcagaagca	agttaccaga	gtgagagagt	agaagagaag	2760
ggagcaactt	atccttcaga	aattcccaaa	gaagattcta	ccacttttgc	aaaaagagag	2820
gaccgtgtta	caactgaaat	tcagcttcct	tctcaaaagt	ctgtggaaga	acaaagccca	2880
gctcttttgt	cttctctgcg	ttcacggagc	acacaaatgg	aatcaactcg	tgtttcagct	2940
tctctcccca	gaagttaccg	gaaaaactgat	acagtcaggg	taacatctgt	ggtcacacca	3000
agaccctttg	gctctcagac	aaggggaaat	tcatcactcc	ccagatctta	cacgatggat	3060
gatgcttggg	agtataatgg	agatgttgaa	gacattaaga	gaactccaaa	caatgtgggtc	3120
agcacccctg	caccagccccc	ggacgcaagc	caactggcct	caagcttatc	tagccagaaa	3180
gaggtagcag	caacagaaga	agatgtgaca	agggtgccct	ctctacatc	ccccttctca	3240
tctctttccc	aagaccaggc	tgccacttct	aaagccacat	tgtcttcac	atctgggtctt	3300
gatttaaatg	ctgaaatctg	agaaggggaa	atctccccc	aaagagaagt	ctcaagatcc	3360
caggatcagt	tcagtgtatg	gagaatcagc	ataaaccaga	cgctgggaa	gagttcttgac	3420
tttggtttta	taataaaaatg	ggatattcct	gggatcttcg	tagcatcagt	tgaagcagggt	3480
agccacagag	aaattttctca	gtacaaagta	gatgatgaaa	ttattgtctat	taacaacacc	3540
aagttttcat	ataacgatcc	aaaagagtg	gaggaagcca	tggtcaaggc	tcaagaaact	3600
ggacacctag	tgatggatgt	gagggcctat	ggaaaggctg	gttcacctga	aacaaagtgg	3660
attgatgcaa	cttcttggaat	ttacaactca	gaaaaatctt	caaatctatc	tgttaacaact	3720
gattttctccg	aaagccttca	gagttctaat	attgaatcca	aagaatcaa	tggaatttcat	3780
gatgaaagca	atgcttttga	atcaaaaagca	tctgaatcca	tttctttgaa	aaacttaaaa	3840
agggatcac	aaatttttga	acaaggaagc	tctgattcgg	tggttctctga	tcttccagtt	3900
ccaacccatca	gtgcccgcag	tcgctgggtg	tgggatcaag	aggaggagcg	gaagcggcag	3960
gagaggtggc	agaaggagca	ggaccgcta	ctgcaggaaa	aatatcaacg	tgagcaggag	4020
aaactgaggg	aagagtggca	aagggccaaa	caggagggcag	agagagagaa	ttccaagtac	4080
ttggatgagg	aactgatggg	cctaagctca	aacagcatgt	ctctgaccac	acgggagccc	4140
tctcttgcca	atcgggaagc	tacctggagt	gaagggtcca	agtcttcaga	cagagaagga	4200
acccgagcag	gagaagagga	gaggagacag	ccacaagagg	aaagtgttca	tgaggaccaa	4260
ggaaagaagc	cgcaggatca	gcttgtttat	gagagagaga	ggaaatggga	gcaacagctt	4320
caggaagagc	aagagcaaaa	gcggcttcag	gctgaggctg	aggagcagaa	gcgtcctgcg	4380
gaggagcaga	agcggcaggc	agagatagag	cgggaaacat	cagtcagaat	ataccagtac	4440
aggagggcctg	ttgattccta	tgatatacca	aagacagaag	aagcatcttc	aggttttctt	4500
cctggtgaca	ggaataaaatc	cagatctact	actgaactgg	atgattactc	cacaaataaa	4560
aatggaaaca	ataaatattt	agaccaaaat	gggaacatga	cctcttcaca	gaggagatcc	4620
aagaaagaac	aagtaccatc	aggagcagaa	ttggagaggc	aacaaatcct	tcaggaaatg	4680
aggaagagaa	caccccttca	caatgacaac	agctggatcc	gacagcgag	tgccagtgtc	4740
aacaaagagc	ctgttagtct	tctctgggac	atgagaagag	gcgaatcttt	agataacctg	4800
gactcccccc	gatccaatct	ttggagacag	cctcctgggc	tcaatcagcg	cacaggattc	4860
tatgctctct	cctctgtgca	agactttagt	cgcccaccac	ctcagctggg	gtccacatca	4920
aaccgtgcct	acatgcggaa	cccctcctcc	agcgtgcccc	caccttcagc	tggtccgtg	4980
aagacctcca	ccacaggtgt	ggccaccaca	cagtccccca	cccgcagaag	ccattcccct	5040
tcagcttcc	agtcaggctc	tcagctgcgt	aacaggtcag	tcagtgggaa	gcgcataatg	5100
tctactgca	ataacattct	gggcaaaagga	gcgcacatga	tcacagagtc	cctgggtctt	5160
tgttatcatt	tgcatgtgtt	taagtgtgtt	gcctgtgagt	gtgacctcgg	aggctcttcc	5220
tcaggagctg	aagtcaggat	cagaaaccac	caactgtact	gcaacgactg	ctatctcaga	5280
ttcaaatctg	gacggccaac	cgccatgtga	tgtaagcctc	catacgaag	cactgttgca	5340
gataagaaga	gaggtggttg	ctgctcatgt	agatctataa	atatgtgttg	tatgtctttt	5400
ttgctttttt	tttaaaaaaa	agaataactt	tttttgcttc	tttagattac	atagaagcat	5460
tgtagtcttg	ttagaaccag	tatttttggt	gtttatttat	aaggttaattg	tgtgtgggga	5520
aaagtgcagt	atttacctgt	tgaattcagc	atcttgagag	cacaagggaa	aaaataagaa	5580
cctacgaata	tttttgaggc	agataatgat	ctagtttgac	tttctagtta	gtgggtgttt	5640
gaagagggtta	ttttatgttt	ttttaaaaaa	aggttcttaa	acattatttg	aaatagttaa	5700
tataaataca	taattgcatt	tgctctgttt	attgtaatgt	attctaatt	aatgcagaac	5760
catatggaaa	atttcattaa	aatctatccc	caaatgtgct	ttctgtatcc	ttcctctac	5820
ctattattct	gattttttaa	atgacagtta	atgtaccatt	tatttgcttg	atgaagggag	5880
ctctattttc	tttaccagaa	atgttgctaa	gtaattccca	atagaaagct	gcttattttc	5940
attaatgaaa	aataaccatg	gtttgtatac	tagaagtctt	cttcagaaac	tggtgagcct	6000
ttctgttcaa	ttgcatttgt	aaataaactt	gctgatgcac	ttaacgagtg	ggctgctctt	6060
ttcttaggtg	tatgtgtctg	acctcaggcc	ttttagccat	atttcagtat	gtggcctttt	6120
ttgatgttat	gttttatcca	gtagctttac	taaggtataa	ttgatgtaat	aaactgcata	6180

-continued

tatttaaagt	gtatactttg	acaaatthttg	acatgggtga	taccttcgaa	actatgccac	6240
agtctggatg	tgthttactga	aacatthttaa	taaggaaagt	tattthttgat	aaagtatatgt	6300
tttttgatag	aataatthttg	tatgggtgaga	gtgatgaatt	gttggtatcat	ttgaataaaa	6360
tctthttacta	accccatgat	aaaaggagaa	gacaacagt	agcttagaat	atctataaag	6420
caaaaaatgt	agctctctgt	ttaaaaaatc	tggaagcgga	atgcaaggat	acaaaacttt	6480
agcatgcttt	gagcaaaaat	ttaaacttac	tggaatcttt	tataataatg	taagtggaa	6540
ggagattct	aggaactgag	aactgtattg	gaatagggtc	aaaatgtga	agaaatgcta	6600
atgtgggaga	taaaaatthtt	atttagtact	tattctgatt	attattaaag	taataatgtg	6660
ttccttgagg	ataacttgtc	aaatgcccc	aagcataaag	aataataatc	tgaatcccaa	6720
attccaaaga	caagaactct	gtgtttgaat	tattctgca	tataattatt	tataagtata	6780
gattgtgaat	ttttccatgt	tcttaaaat	atthtttatct	tttttcattg	ttgcatagtg	6840
ctccattgtt	tggtcttggt	aataatttagt	tgataattcc	attactgtgt	atthtttcaact	6900
tgtttctaag	atcaaacatt	ttaatatgtg	catgttatat	ataaatatgt	aaattctgtg	6960
atactctatg	atcatctctt	tcttttatatt	atthttcatag	acatgaaata	gttgctcaga	7020
gattatgcat	tttaagacac	tcatagtata	tattgcccac	gtgtgttcca	gaaaggcact	7080
gctggcttcg	actcctataa	gcagcacgtg	ggcttgttca	tctcactgca	tgthttatgaa	7140
gatacagttc	ttttgccttg	ttctctgcct	gatgtgtatg	cagaggcagc	cctcaatatg	7200
cagtgggtga	ataaatgaat	gaagaaacca	ctatcaaaaa	aaaaaaaaa	aaa	7253

SEQ ID NO: 15

FEATURE

source

moltype = RNA length = 6729

Location/Qualifiers

1..6729

mol_type = mRNA

organism = Homo sapiens

SEQUENCE: 15

attccacagt	cgtttctgta	tttgaatca	tagaaacaga	atgttctctc	tcttatttct	60
tccctgccaa	cagtttctgg	gtgggtgaat	gtcagtagct	ggagtatttt	ttaaagagga	120
gaggtgaggc	ttgcctgtca	aaactggaaa	gagggcggtt	gtttgcagag	cagagctgac	180
atcaaatgtg	agattactgc	tcagtggtga	ggcactttgc	ctgtaaacag	ataatataaa	240
cgttttcttg	aaagctctgt	aacagattgg	attgaaagaa	gccagctttt	tccatcctgg	300
agatctacag	gatttatcaa	atcaggtcac	tgtcaagcaa	gaagagactg	acaggagagt	360
gaaaaatgtt	ttgataacat	tgtactggct	gggaagaaaa	gcacaaagca	accctacta	420
taatgggtccc	catcttaatt	tgaagcgttt	tgaagaatctt	ttaggacaag	cactgacgaa	480
ggcactcgaa	gactccagct	tctgaaaaag	aagtggcagg	gacagtggct	acggtgacat	540
ctgggtgtcct	gaacgtggag	aatttcttgc	tcttccaaag	caccataaga	gagaagattc	600
ctttgaaagc	ttggactctt	tggtgtcgag	gtcattgaca	agctgtcctc	ctgatatcac	660
gttgagaggg	gggctggaag	gttttgaaa	tgacacagat	tcggaattta	catttaagat	720
gcaggattat	aataaagatg	atattgtcga	tcgaaggatt	tcggtgtgtg	agccaaagac	780
tgctgtacc	ttcaatcggt	ttttacccaa	caaaagtaga	cagccatcct	atgtaccagc	840
acctctgaga	aagaaaaagc	gcacaaaaa	tgaggataac	agaagaagtt	gggcaagccc	900
ggtttataca	gaagcagatg	gaacattttc	aagtaatcag	aggaggattt	ggggcaccac	960
tgtggagaa	tgcccaactg	tacaaggaac	ttcaaagtc	tctgttatt	tggaagagga	1020
aaaagcaaa	acaagaagca	taccacaac	tgtaagagat	gatctttatg	tgccgaagct	1080
cagtccagtc	atgccaaac	cagggaaatgc	ttttgatcag	tttcttccca	aatgttggac	1140
cccagaagat	gtgaactgga	aaagaataaa	aagggaaact	tataagccat	ggataaaga	1200
atthtcaggga	ttcagttcag	ttttactgct	tcaggccctc	caaacatact	ctgatgacat	1260
cttgtcttct	gaaacacata	ccaaaattga	tccacttctc	ggcccaaggc	tcataaccgc	1320
caggaagaat	ctctcttatg	caccaggcta	tagaagagat	gacctcgaga	tgccagccct	1380
ggatcctgac	ttagagaagt	atgatttctt	tgctcagaag	actggggctt	tccatgcaaa	1440
tccatattgt	ctccgagctt	ttgaagactt	tagaaagttc	tctgagcaag	atgattctgt	1500
agagcgagat	ataattttac	agtgtagaga	agggtgaact	gtacttccgg	atttgaaaaa	1560
agatgatatt	atgtttcgcc	gaattccagc	acagaagaaa	gaagtgcgcg	tgtctggggc	1620
cccagataga	taccaccagc	tcccttttcc	cgaaccctgg	actcttcttc	cagaaattca	1680
agcaaaaatt	ctctgtgtac	ttgaaaggac	atgcccatcc	aaagaaaaaa	gtaatagctg	1740
tagaatatta	gttctctcat	atcggcagaa	gaaagatgac	atgctgacac	gtaagattca	1800
gtcttgga	gtgggaacta	cgtgtcctcc	catcagtttc	acccctggcc	cctgcagtga	1860
ggctgacttg	aagagatggg	aggccatccg	ggaggccagc	agacttaggc	acaagaaaag	1920
gctgatggtg	gagagactct	ttcaaaagat	ttatgggtgag	aatgggagta	agtcocatgag	1980
tgatgtcagc	gcagaagatg	ttcaaaactt	gcgtcagctg	cgttacgagg	agatgcagaa	2040
aataaaatca	caattaaaag	aacaagatca	gaaatggcag	gatgaccttg	caaaatggaa	2100
agatcgctga	aaaagttaca	cttcagatct	gcagaagaaa	aaagaagaga	gagaagaaat	2160
tgaaaagcag	gcacttgaga	agtctaagag	aagctctaag	acgtttaagg	aaatgctgca	2220
ggacagggaa	tcccaaaatc	aaaagtctac	agttccgtca	agaaggagaa	tgtattcttt	2280
tgatgatgtg	ctggaggaag	gaaagcgacc	ccctacaatg	actgtgtcag	aagcaagtta	2340
ccagagtga	agagtgaag	agaaggagc	aacttatcct	tcagaaattc	ccaaagaaga	2400
ttctaccact	tttgcaaaaa	gagaggaccg	tgtaacaact	gaaattcagc	ttcttctca	2460
aagtctgtg	gaagaacaaa	gccagccctc	tttgtcttct	ctgctgtcac	ggagcacaca	2520
aatggaatca	actcgtgttt	cagcttctct	cccagaagt	taccgaaaa	ctgatacagt	2580
caggttaaca	tctgtgtgta	caccaagacc	ctttggctct	cagacaaggg	gaatctcatc	2640
actccccaga	tcttacacga	tggtgatgac	ttggaagtat	aatggagatg	ttgaagacat	2700
taagagaact	ccaacaatg	gggtcagcac	ccctgcacca	agcccgagc	caagccaact	2760
ggcttcaagc	ttatctagcc	agaaaagagt	agcagcaaca	gaagaagatg	tgacaaggct	2820
gcccctctct	acatccccct	tctcatctct	ttcccaagac	caggctgcca	cttctaagac	2880
ccacattgtct	tccacatctg	gtcttgattt	aatgtctgaa	tctggagaag	gggaaatctc	2940
cccacaaaaga	gaagtctcaa	gatcccagga	tcagttcagt	gatatgagaa	tcagcataaa	3000
ccagagcct	gggaagagtc	ttgactttgg	gtttacaata	aaatgggata	ttctctgggat	3060

-continued

```

cttcgtagca tcaagttgaag caggtagccc agcagaatth tctcagctac aagtagatga 3120
tgaaattatt gctattaaca acaccaagtt ttcataatac gattcaaaag agtgggagga 3180
agccatggct aaggtcgaag aaactggaca cctagtgtat gatgtgaggg gctatggaaa 3240
ggctgggtca cctgaaacaa agtggattga tgcacttctt ggaatttaca actcagaaaa 3300
atcttcaaat ctatctgtaa caactgattt ctccgaaagc cttcagagtt ctaatatgta 3360
atccaaagaa atcaatggaa ttcatgatga aagcaatgct tttgaatcaa aagcatctga 3420
atccatttct ttgaaaaact taaaaaggcg atcacaatth tttgaacaag gaagctctga 3480
ttcgggtggt cctgatcttc cagttccaac catcagtgcc ccgagtcgct ggggtgtggga 3540
tcaaggaggag gagcggaagc ggccaggagag gtggcagaag gagcaggacc gcctactgca 3600
ggaaaaatat caacgtgagc aggagaaact gagggagag tggcaagggg ccaaacagga 3660
ggcagagaga gagaattcca agtacttgga tgaggaaact atggtcctaa gctcaaacag 3720
catgtctctg accacacggg agccctctct tgccacctgg gaagctacct ggagtgaagg 3780
gtccaagtct tcagacagag aaggaacccg agcaggagaa gaggagagga gacagccaca 3840
agaggaaagt gttcatgagg accaaggaaa gaagccgcag gatcagcttg ttattgagag 3900
agaggagaaa tgggagcaac agcttcagga agagcaagag caaaagcggc ttcaggctga 3960
ggctgaggag cagaagcgct ctgcggagga gcagaagcgc caggcagaga tagagcggga 4020
aacatcagtc agaataatcc agtacaggag gcctgttgat tctatgata taccaaagac 4080
agaagaagca tcttcagggt ttcttcctgg tgacaggaat aaatccagat ctactactga 4140
actggatgat tactccacaa ataaaaatgg aaacaataaa tatttagacc aaattgggaa 4200
catgacctct tcacagagga gatccaagaa agaacaagta ccatcaggag cagaattgga 4260
gaggcaacaa atccttcagg aaatgaggaa gagaacaccc cttcacaaatg acaacagctg 4320
gatccgacag cgcagtgcca gtgtcaacaa agagcctgtt agtcttctg ggatcatgag 4380
aagaggcgaa tctttagata acctggactc ccccgatcc aattcttgga gacagcctcc 4440
ttggctcaat cagcccacag gattctatgc ttcttctct gtgcaagact ttagtgcgcc 4500
accacctcag ctggtgtcca catcaaaccc tgccctacat cggaacccct cctccagcgt 4560
gccccaccct tcagctggct ccgtgaagac ctccaccaca ggtgtggcca ccacacagtc 4620
ccccaccctg agaagccatt ccccttcagc ttcacagtc ggctctcagc tgcgtaacag 4680
tgtgttgct gtgagtgtag cctcggaggg tcttctctc gagctgaagt caggatcaga 4740
aaccaccaac tgtactgcga cgaactgctat ctcagattca aatctggagc gccaaaccgc 4800
atgtgatgta agcctccata cgaagacact gttgcagata gaagaagagg tggttgctgc 4860
tcagttagat ctataaatat gtgtgtgatg tcttttttgc ttttttttta aaaaaagaa 4920
taactttttt tgctctttta gattacatag aagcattgta gtcttgtag aaccagttat 4980
tttgttgttt attataagg taattgtgtg tgggaaaaag tgcagttatt acctgttgaa 5040
ttcagcatct tgagagcaca agggaaaaaa taagaacctc cgaatatttt tgaggcagat 5100
aatgatctag tttgactttc tagtttagtg gtgtttgaag agggattttt attgtttttt 5160
aaaaaaaggt tcttaaacat tatttgaaat agttaatata aatacataat tgcatttgct 5220
ctgttttatt taatgtattc taaattaatg cagaacctata tggaaaaatt cattaaaaatc 5280
tatccccaaa tgtgtctttc gtatccttcc ttctacctat tattctgatt tttaaaaatg 5340
cagttaatgt accatttatt tgcttgatga agggagctct attttcttta ccagaaatgt 5400
tgctaaagta ttcccaatag aaagctgctt attttcatta atgaaaaata accatgggtt 5460
gtatactaga agtcttcttc agaaactggg gagcctttct gtccaattgc atttgtaaat 5520
aaacttgctg atgcatttaa cgaagtggtc gtctttttct taggtgtatg tgtctgacct 5580
caggcctttt agcatattt caglatgtgg ccttttttga tgtatgttt tatccagtag 5640
ctttactaag gtataattga tgaataaaac tgcatatatt taaagtgtat actttgacaa 5700
attttgacat ggtgtatacc ttcgaaacta tgccacagtc tggatgtgt tactgaaaca 5760
ttttaataag gaagtgtttt ttgtataaag ttatgttttt ggatacaata tatttgtatg 5820
gtgagagtga tgaattgttg gatcatttga ataaaaatct ttactaacc catgataaaa 5880
ggagaagaca acagttagct tagaatatct ataaagcaaa aaatgtagtc tctgttttaa 5940
aaaaatctga gcggaatgac aaggatacaa aactttagca tgctttgagc aaaaatttaa 6000
acttactgga actctttata ataagtgaag tggaaatggg gattctagga actgagaact 6060
gtattggaat aggttcaaaa tatgtaagaa atgctaagtgt gggagataaa aattttattt 6120
agtaacttatt ctgatttata ttaaaagtaaat aatgtgttcc ttgaggataa cttgtcaaat 6180
gccccaaagc ataagaata taattctgaa tcccaaatcc caaagacaag aactctgtgt 6240
tgaattcat tctgcatata attatttata agtatagatt gtgaattttt ccatgttctt 6300
aaaaattatt ttatcttttt tcatggttgc atagtgtcc attgtttggc cttggtaata 6360
tttagttgat aattccatta ctgtgtattt ttcaacttgt tctaagatca aacattttaa 6420
tatgtgcatg ttatatataa atagttaaatt tctgtgatac tctatgatca tctcttctt 6480
tatatttatt tcatagacat gaaatagttg ctcagagatt atgcatttta agacactcat 6540
agtatatatt gccaaagtgg ttccagaaaa ggcaactgct gcttcgactc ctataagcag 6600
cacgtgggct tgttcacttc actgcatgtt tatgaagata cagttctttt gcctgttct 6660
ctgcctgatg tgtatgcaga ggcagccctc aatatgcagt ggttgaataa atgaatgaag 6720
aaaccacta 6729

```

```

SEQ ID NO: 16      moltype = RNA length = 7147
FEATURE            Location/Qualifiers
source              1..7147
                    mol_type = mRNA
                    organism = Homo sapiens

```

```

SEQUENCE: 16
aaactgtcgt cggggcgggt gccgcgcgct gccgcgtgggc accgcgtctg cttcccgctg 60
cctgcctgac tgcgccgggt cccgcggggc ttgggtcgct ttcaggagtt tagagaaagc 120
caggtcttca cgttcggtga ggttcgagac cttaacgaac tgcagagcgc aacaaaggga 180
actagagccc cggcgccctc gcagcgggag cggaaagccg agttgtggga ggcccgctg 240
ccctccccgc ccgtgggggc cagacgcgcg gggacaaccc ctccctccca cgcattccga 300
gtctctctc cctcccttg tctagccat ggaagggtg gaggagcag aggccaaact 360
ctccgtggcg ttcgtgagg ctcagagatg ggtggaggca gtaacagaga agaattttga 420

```

-continued

aacaaaaagat	tttcgagcct	ctctagaaaa	tgggtgttctg	ctgtgtgtatt	tgattaataa	480
gcttaaacct	ggcggtcatta	agaagatcaa	tagactgtct	acaccaatag	caggatttga	540
taataataaac	gtttttcttga	aagcttgtga	acagatttga	ttgaaaagaag	cccagctttt	600
ccatccttga	gatctacagg	atttatcaaa	tcgagtcact	gtcaagcaag	aagagactga	660
caggagagtg	aaaaatgttt	tgataacatt	gtactggctg	ggaagaaaag	cacaaagcaa	720
cccgactat	aatgggtccc	atcttaattt	gaaagcggtt	gagaatcttt	taggacaagc	780
actgacgaag	gcactcgaag	actccagctt	cctgaaaaga	agtggcagg	acagtggcta	840
cgggtgacatc	tgggtgcctg	aacgtggaga	atttcttgct	cctccaaggc	accataagag	900
agaagattcc	tttgaaagct	tggactcttt	gggctcgagg	tcatttgcaa	gctgctcctc	960
tgatcacg	ttgagagggg	ggcgtgaagg	tttgaaagt	gacacagatt	cggaatttac	1020
atttaagatg	caggattata	ataaagatga	tatgtcgtat	cgaaggattt	cggctgttga	1080
gccaagact	gcgttaccct	tcaatcggtt	tttaccacac	aaaagtagac	agccatccta	1140
tgtaccagca	cctctgagaa	agaaaaagcc	agacaaacat	gaggataaca	gaagaagttg	1200
ggcaagcccg	gtttatacag	aagcagatgg	aacattttca	agtaatcaga	ggaggatttg	1260
gggcaccaat	gtggagaact	ggccaactgt	acaaggaact	tcaaagtcct	cttggtattt	1320
ggaagaggaa	aaagcaaaag	caagaagcat	acccaacatt	gtaaaggatg	atctttatgt	1380
gcgcaagctc	agtccagtc	tgccaaaccc	agggaaatgct	tttgatcagt	ttcttcccaa	1440
atgttggacc	ccagaagatg	tgaactggaa	aagaataaaa	agggaaactt	ataagccatg	1500
gtataaagaa	tttcagggat	tcagtcagtt	tttactgctt	caggccctcc	aaacatactc	1560
tgatgacatc	ttgtcttctg	aaacacatc	caaaattgat	cccacttctg	gccccaggct	1620
cataaccccg	agggaagatc	ttctcttatg	accaggctat	agaagagatg	acctcgagat	1680
ggcagccctg	gatcctgact	tagagaatga	tgatttcttt	gtcagaaaga	ctggggcttt	1740
ccatgcaaat	ccatattgtt	tcagagcttt	tgaagacttt	agaaagtctc	ctgagcaaga	1800
tgattctgt	gagcgagata	taattttaca	gtgtagagaa	gggtgaactt	tacttccgga	1860
tttgaaaaaa	gatgataga	ttgttcgctg	aattccagca	cagaagaaag	aagtgcctgt	1920
gtctggggcc	ccagatagat	accacccagt	cccttttccc	gaacccctgga	ctcttctctc	1980
agaaattcaa	gcaaaatttc	ttctgtgtact	tgaaggaca	tgcccatcca	aagaaaaaag	2040
taatagctgt	agaatattag	ttccttcata	tcgggcagaag	aaagatgaca	tgctgacacg	2100
taagattcag	tctcggaaac	tgggaaactac	cgtgcctccc	atcagtttca	ccctgggcc	2160
ctgcagtgag	gctgacttga	agagatggga	ggccatccgg	gaggccagca	gacttaggca	2220
caagaaaaag	ctgatggtg	agagactctt	tcaaaagatt	tatgggtgaga	atgggagtaa	2280
gtccatgagt	gatgtcagcg	cagaagatgt	tcaaaacttg	cgtcagctgc	gttacgagga	2340
gatgcagaaa	ataaaatcac	aattaaaaaga	acaagatcag	aaatggcagg	atgaccttgc	2400
aaaatggaaa	gactcgcgaa	aaagttacac	ttcagatctg	cagaagaaaa	aagaagagag	2460
agaagaaatt	gaaaagcagg	cacttgagaa	gtctaagaga	agctctaaga	cgtttaagga	2520
aatgctgcag	gacagggagt	cccaaaatca	aaagtctaca	gttcctgcaa	gaaggagaa	2580
gtattctttt	gatgatgtgc	tgagggaagg	aaagcgaccc	cctacaatga	ctgtgtcaga	2640
agcaagttac	cagagttaga	gagtagaaga	gaaggagca	acttatcctt	cagaaattcc	2700
caaaagagat	tctaccactt	ttgcaaaaag	agaggaccgt	gtaacaaactg	aaattcagct	2760
tccttctcaa	agtctctgtg	aagaaacaaag	cccagcctct	ttgtcttctc	tgctgtcacg	2820
gagcacacaa	atggaaatcaa	ctcgtgtttc	agcttctctc	cccagaagtt	accggaaaac	2880
tgatacagtc	aggttaacat	ctgtgtgtac	accaagaccc	tttggtctc	agacaagggg	2940
aatctcatca	ctccccagat	cttacacgat	ggatgatgct	tggagatata	atggagatgt	3000
tgaagacatt	aagagaactc	caaacaatgt	ggtcagcacc	cctgcacca	gcccggagcc	3060
aagccaactg	gcttcaagct	tatctagcca	gaaagaggt	gcagcaacag	aagaagatgt	3120
gacaaggctg	ccctctctca	catccccctt	ctcatctctt	tcccaagacc	aggctgccac	3180
ttctaaagcc	acattgtctt	ccacatctgg	tcttgattta	atgtctgaat	ctgggagaagg	3240
ggaaatctcc	ccacaaagag	aagtctcaag	atcccaggat	cagttcagtg	atatgagaat	3300
cagcataaac	cagacgcctg	ggaagagtct	tgactttggg	tttacaataa	aatgggatat	3360
tccctgggac	ttcgtagcat	cagttgaagc	aggtagccca	gcagaatttt	ctcagctaca	3420
agtagatgat	gaaattattg	ctattaacaa	caccaagttt	tcatataacg	attcaaaaga	3480
gtgggaggaa	gcttaggcta	aggtcaaga	aactggacac	ctagtgatgt	atgtgaggcg	3540
ctatggaaaag	gctgggtcac	ctgaaacaaa	gtggattgat	gcaacttctg	gaatttacaa	3600
ctcagaaaaa	tcttcaaatc	tatctgtaac	aactgatttc	tccgaaaagc	ttcagagttc	3660
taatattgaa	tccaaagaaa	tcaatggaat	tcatgatgaa	agcaatgctt	ttgaatcaaa	3720
agcatctgaa	tccatttctt	tgaaaaactt	aaaaaggcga	tcacaatttt	ttgaacaagg	3780
aagctctgat	tgggtgtgtc	ctgatcttcc	agttccaacc	atcagtgccc	cagctcgctg	3840
gggtgtggat	caagcaggag	agcggaaagc	gcaggagagg	tggcagaagg	agcaggaccg	3900
cctactgcag	gaaaaatctc	aacgtgagca	ggagaaaactg	aggggaagagt	ggcaaaagggc	3960
caaacaggag	gcagagagag	agaattccaa	gtacttggat	gaggaaactga	tggtcctaag	4020
ctcaaacagc	atgtctctga	ccacacggga	gccctctctt	gccacctggg	aagctacctg	4080
gagtgaaagg	tccaaagtctt	cagacagaga	aggaacccga	gcaggagaag	aggagaggag	4140
acagccacaa	gaggaaagt	ttcatgagga	ccaaggaag	aagcccgagg	atcagcttgt	4200
tattgagaga	gagaggaat	gggagcaaca	gcttcaggaa	gagcaagagc	aaaagcggct	4260
tcaggctgag	gctgaggagc	agaagcgtcc	tgccgaggag	cagaagcgcc	aggcagagat	4320
agagcgggaa	acatcagtc	gaatatacca	gtacaggagg	cctgttgatt	cctatgatat	4380
accaaagaca	gaagaagcat	cttcagggtt	tcttctcgtt	gacaggaata	aatccagatc	4440
tactactgaa	ctggatgatt	actccacaaa	taaaaatgga	aacaataaat	atttagacca	4500
aattgggaac	atgacctctt	cacagaggag	atccaagaaa	gaacaagtac	catcaggagc	4560
agaattggag	aggcaacaaa	tcttcaggga	aatgagggaag	agaacacccc	ttcacaatga	4620
caacagctgg	atccgacagc	gcagtgccag	tgtcaacaaa	gagcctgtta	gtcttctctg	4680
gatcatgaga	agaggcgaat	ctttagataa	cctggactcc	ccccgatcca	attcttggag	4740
acagcctcct	tggctcaatc	agcccacagg	attctatgct	tcttctctctg	tgcaagactt	4800
tagtcgccc	ccacctcagc	tgggtgtccac	atcaaacctg	gcctacatgc	ggaacccctc	4860
ctccagcgtg	ccccccactt	cagctggctc	cgtgaagacc	tccaccacag	gtgtggccac	4920
cacacagctc	cccccccgga	gaagccattc	cccttcagct	tcacagtcag	gctctcagct	4980

-continued

```

gcgtaacagc tcaatcagtg ggaagcgc atgctcctac tgcaataaca tctggggcaa 5040
aggagccgcc atgatcatcg agtccctggg tctttgttat catttgcaatt gttttaagtg 5100
tggttgctgt gagtgtagcc tccgaggctc ttcctcagga gctgaagtca ggatcagaaa 5160
ccaccaactg tactgcaacg atcgtatctc cagattcaaa tctggacggc caaccgccat 5220
gtgatgtaag cctccatcac aaagcactgt tgcagataga agaagagggtg gttgctgctc 5280
atgtagatct ataaatatgt gttgtatgtc ttttttgctt tttttttaa aaaaagaata 5340
actttttttg cctctttaga ttacatagaa gcattgtagt cttggtagaa ccagtatatt 5400
tgttgtttat ttataaggta attgtgtgtg gggaaaagtg cagtatttac ctgttgaatt 5460
cagcatcttg agagcacaa ggaaaaaata agaacctacg aatatttttg aggcagataa 5520
tgatctagtt tgactttcta gttagtgggtg ttttgaagag ggtattttat tgttttttaa 5580
aaaaagggtc ttaaacatta ttgaaatag ttaataataa tacataaattg cattttctct 5640
gtttattgta atgtattcta aattaatgca gaaccatag gaaaatttca ttaaaatcta 5700
tccccaaatg tgcctttctg atccttctct ctacctatta tctgtatttt taaaaatgca 5760
gttaatgtac catattattt ctgtatgaag ggagctctat tttctttacc agaaatgttg 5820
ctaagtaatt ccaatagaa agctgcttat tttcattaat gaaaaataac catggtttgt 5880
atactagaag tcttcttcag aaactgggtg gcctttctgt tcaattgcat ttgtaaaaa 5940
acttgctgat gcattttaac agtgggtcgt ctttttctta ggtgtatgtg tctgacctca 6000
ggccttttag ccatatttca gtatgtggcc ttttttgatg ttatgtttta tccagtagct 6060
ttactaaggt ataattgatg taataaactg catatattta aagtgtatac ttgacaaat 6120
tttgacatgg tgtatacctt cgaaaactat ccacagctct gatgtgttta ctgaaacatt 6180
ttaataagga agtttatttt tgataaagtt atgttttttg atacaataa ttgtgatggt 6240
gagagtgtat aattgttgga tcatttgaat aaaatctttt actaacccca tgataaaaag 6300
agaagacaac agtgagctta gaatatctat aaagcaaaaa atgtagctct ttgtttaaaa 6360
aatctggagc gggaaatgcaa ggatacaaaa ctttagcatg ctttgagcaa aaatttaaac 6420
ttactggaat tctttataat aatgtaagt gaaatggagg ttctaggaa tgagaactgt 6480
attggaatag gttcaaaata tgaagaatat gctaagtgtg gagataaaaa ttttatttag 6540
tacttattct gattattatt aaagtaataa tgtgttctct gaggataact tgtcaaatgc 6600
cccaaagcat aagaatata attctgaatc ccaaattcca aagacaagaa ctctgtgttt 6660
gaattcattc tgcataatat tatttataag tatagattgt gaatttttcc atgttcttaa 6720
aattattttt atcttttttc atgggtgcat agtgcctcat tgtttggcct tggtaatat 6780
tagttgataa ttccattact gtgtattttt cacttgtttc taagatcaaa cattttaata 6840
tgtgcatggt atataataat atgtaaatc tgtgatactc tatgatcatc tctttcttta 6900
tattattttc atagacatga aatagttgct cagagattat gcatttttaag acactcatag 6960
tataatttgc aagaatgggt tccagaaagg cactgctggc ttcgactcct atagcagca 7020
cgtgggcttg ttcactctac tgcattgtta tgaagataca gttcttttgc cttgttctct 7080
gcctgatgtg tatgcagagg cagccctcaa tatgcagtgg ttgaataaat gaatgaagaa 7140
accacta 7147

```

```

SEQ ID NO: 17      moltype = AA length = 1349
FEATURE           Location/Qualifiers
source            1..1349
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 17
MKKIRICHIF TFYSWMSYDV LFQRTGELGAL EIWRQLICAH VCICVGLWLYL RDRVCSKKDI 60
ILRTEQNSGR TILIKAVTEK NFETKDFRAS LENGVLLCDL INKLKPGVIK KINRLSTPIA 120
GLDNINVLFLK ACEQIGLKEA QLFHPGDLQD LSNRVTVKQE ETDRRVKNVL ITLYWLGKRA 180
QSNPYYPNGH LNLKAFENLL GQALTKALED SSFLKRSGRD SGYGDWCP E RGEFLAPPRH 240
HKREDSFESL DSLGSRSLTS CSSDITLRGG REGFESDSTS EFTFKMQDYN KDDMSYRRIS 300
AVEPKTALPF NRFLPNKSRQ PSYVPAPLRK KKPDKHEDNR RSWASPVYTE ADGTFSRRLFQ 360
KIYGENGSKS MSDVSAEDVQ NLRQLRYEEM QIKIKSLKEQ DQKWQDDLAK WKDRRKSYS 420
DLQKKKEERE EIEKQALEKS KRSSKTFKEM LQDRESQNK STVPSRRMY SFDVLEEGK 480
RPPTMTVSEA SYQSERVEEK GATYPSEIPK EDSTTFKRE DRVTTEIQLP SQSPVEEQSP 540
ASLSSLRSRS TMESTRVSA SLPRSRYKTD TVRLTSVVP RPFSGQTRGI SSLPRSYTMD 600
DAWKYNGDVE DIKRTPNNVV STPAPSPDAS QLASSLSQK EVAATEEDVT RLPSPSPFS 660
SLSQDQAATS KATLSSTSGL DLMSESSEGE ISPOREVSRS QDQFSDMRIS INQTPGKSLD 720
FGFTIKWDIP GIFVASVEAG SPAEFSQLQV DDEIIAINT KFSYNDKKEW EEAMAKAQET 780
GHLVMDVRRY GKAGSPETKW IDATSGIYNS EKSSNLSVTT DFESLQSSN IESKEINGIH 840
DESNAFESKA SESISLKNL RRSQFFEQGS SDSVVPDLPV PTISAPSRWV WQDEERKRQ 900
ERWQKEQDRL LQEKYQREQE KLREEWQRAK QEAERENSKY LDEELMVLSS NSMSLTREP 960
SLATWEATWS EGSKSDEREG TRAGEERRQ PQEEVVEDQ GKPKQDLVI ERERKWEQQL 1020
QEEQEQKRLQ AEAEQKQKRA EEQKQAEIE RETSVRIYQY RRPVDSYDIP KTEEASSGFL 1080
PGDRNKRST TELDDYSTNK NGNKNYLDQI GNMSTSSQRS KKEQVPSGAE LERQQILQEM 1140
RKRTPLHNDN SWIRQRSASV NKEPVSPLGI MRRGESLDNL DSPRSNSWRQ PPWLNQPTGF 1200
YASSSVQDFS RPPQLVSTS NRAYMRNPSS SVPPPSAGSV KTSSTGTATT QSPTRSHSP 1260
SASQSGSQLR NRSVSGKRIC SYCNNILGKG AAMIESLGL CYHLHCFKCV ACBCDLGGSS 1320
SGAEVRIRNH QLYCNDCYLR FKSGRP TAM 1349

```

```

SEQ ID NO: 18      moltype = AA length = 1385
FEATURE           Location/Qualifiers
source            1..1385
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 18
MQDYNKDDMS YRRISAVEPK TALPFNRFLP NKSQPSYVP APLRKKKPKD HEDNRRSWAS 60
PVYTEADGTF SSNQRIWGT NVENWPTVQG TSKSSCYLEE EKAKTRIPN IVKDDLYVRK 120

```

-continued

LSPVMPNPGN	AFDQFLPKCW	TPEDVNWKRI	KRETYKPWYK	EFQGFSSQFLL	LQALQTYSD	180
ILSSETHTKI	DPTSGPRLIT	RRKNLSYAPG	YRRDDLEMAA	LDPDLENDFF	FVRKTGAFHA	240
NPYVLRAPED	FRKFSEQDDS	VERDIIILQCR	EGELVLPDLE	KDDMIVRRIP	AQKKEVPLSG	300
APDRYHPVPF	PEPWTLPPEI	QAKFLCVLER	TCPSKEKSNS	CRILVPSYRQ	KKDDMLTRKI	360
QSWKLGTTVP	PISFTPGPCS	EADLKRWEAI	REASRLRHKK	RLMVERLFQK	IYGENGSKSM	420
SDVSAEDVQN	LRQLRYEEMQ	KIKSQLKEQD	QKWQDDLAKW	KDRRKSYS	TD LQKKKEERE	480
IEKQALEKSK	RSSKTFKEML	QDRESQNKQS	TVPSRRRMYS	FDDVLEEGR	PPTMTVSEAS	540
YQSERVEEK	ATYPSEIPKE	DSTTFAKRED	RVTTEIQLPS	QSPVEEQSPA	SLSSLSRST	600
QMESTRVSAS	LPRSRYKTD	T VRLTSVVT	PR PFGSQTRGIS	SLPRS	SYMDD AWKYN	660
IKRTPNNVVS	TPAPSPDASQ	LASSLSQKE	VAATEEDVTR	LPSPTSPFSS	LSQDQAATSK	720
ATLSSTSGLD	LMESGEGEI	SPQREVSRQ	DQFSDMRIS	I NQTPGKSL	DF GFTIKWD	780
IFVASVEAGS	PAEFSQLQVD	DEIIAINNTK	FSYNDSKEWE	EAMAKAQETG	HLVMDVRRYG	840
KAGSPETKWI	DATSGIYNSE	KSSNLSVTTD	FSLSQSSNI	ESKEINGIHD	ESNAFESKAS	900
ESISLKNLKR	RSQFFEQSS	DSVVPDLV	P TISAPSRWV	W DQEEERKR	QE RWQKEQD	960
QEKYQREQEK	LREWEQRAK	EAERENSKYL	DEELMVLSSN	SMSLT	TREPS LAT	1020
GKSSDREGT	RAGEEERRQP	QEEVVHEDQ	G KKPQDQLV	IE RERKWEQ	QLQ EEQE	1080
EAEQKRP	AE EQKRAE	IER ETSVRIY	QYR RPVDSYD	IPK TEEASSG	FLP GDRN	1140
ELDDYSTNKN	GNNKYLQIG	NMTSSQRRSK	KEQVP	SGAEL ERQQI	LQEMR KRTPL	1200
WIRQRSASN	KEPVS	LPN RRGESL	DNLD SPRSNS	WRQP PWNQPT	GFY ASSSV	1260
PPQVLVSTN	RAYMRNPSS	S VPPPSAGSV	K TSTTG	VATTQ SPTPR	SHSPS ASQSG	1320
SVLPVSVTSE	ALPQELKSGS	ETTNC	TATTA ISDSN	LGGP PCDVSL	HTKA LLQIEE	1380
AHVDL						1385

SEQ ID NO: 19 moltype = AA length = 1631
 FEATURE Location/Qualifiers
 source 1..1631
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 19

MEGLEEA	EA	AN	CSVAF	AA	AQR	WVEAV	TEKNF	ETKDF	RASLE	NGVLL	CDLIN	KLKPG	VIKKI	60
NRLSTPI	AGL	DNIN	VFLK	AC		EQIGL	KEAQL	FHPGD	LQDLS	NRVT	VKEET	DRRVK	NVLIT	120
LYWLGR	KAQS	NPYYN	GPHLN	LKAF	ENLLGQ	ALTKA	LEDSS	FLKRS	GRDSG	YGD	IWC	PERG		180
EPLAP	PRHHK	REDSF	ESLDS	LGSRL	TS	SCS	SDITL	RGGRE	GFESD	TDSEF	TFKM	QDYNK	D	240
DMSYR	RISAV	EPK	TAL	PFNR	FLPNK	SRQPS	YVPAP	LRKKK	PDKH	EDNRRS	WASPV	YTEAD		300
GTFSS	NQ	RRI	WG	TN	VEN	WPT	VQGT	SKSSCY	LEEEK	AKTRS	IPNIV	KDDL	VRKL	360
PGNAP	DQFLP	KCWT	PE	DVNW	KRIK	RETYK	WYKE	FQGSQ	FLLL	QALQTY	SDDIL	SSETH		420
TKIDPT	SGPR	LITR	RKNLSY	APGY	RRD	LE	MAAL	DPDLEN	DDFF	V	RTGA	FHANP	VVLR	480
FEDFR	KFSEQ	DDS	VERDII	L	QCRE	GELVLP	DLEK	DDMIVR	RIP	AQKKEVP	LSGAP	DRYHP		540
VFPPE	PWTL	P	EQAK	FLCV	LERT	CP	SKEK	SNSC	RILVPS	YRQK	KDMLT	RKIQ	SWKLGT	600
TVPPIS	FTPG	PCSE	ADLKR	W	EAIRE	ASRLR	HKKRL	MVERL	FQKI	YGENG	KSMSD	VSAED		660
VQNL	RQLRYE	EMQ	KIKSQLK		EQDQ	KWQDDL	AKWK	DRRKS	Y	TS	DLQKKKEE	REEIE	KQALE	720
KSKRS	SKTFK	EMLQ	DRESQ	N	KQST	VPSRRR	MYSF	DDVLEE	GKR	PPTMTVS	EASYQ	SERVE		780
EKGAT	YPSEI	PKED	ST	FAI	Y		Y							840
SASL	PRSYRK	TDT	VRLTSV	V	TPRP	FGSQTR	GISS	LPRS	Y	T	MDDAW	KYNGD	VEDIK	900
VVST	PAPSPD	ASQL	ASSLS	S	QKEV	AATEED	VTRL	PSPTSP	FSSLS	QDQAA	TSKAT	LSSTS		960
GLDL	MS	ESGE	GEIS	PQREV	S	RSQD	QFSDMR	I	SINQ	TPGKS	LDFF	GFTIKWD	IPGIF	1020
AGSPA	AEFSQL	QVD	DEII	AIN	N	TKFS	YNSK	EWEE	A	MAKAQ	ETGHL	VMDVR	RYGK	1080
KWIDAT	SGIY	NSEK	SSNLS	V	TTDF	SESLQS	SNIES	KEING	I	HDES	N	AFES	KASES	1140
LKRRS	QFFBQ	GSSD	SV	PDL	PVPT	I	APSR	WVWD	QEEERK	RQERW	KEQD	RLQEKYQRE		1200
QEKLR	REWQR	AKQ	EAERENS		KYL	DEELMVL	SSNS	MSLTTR	EPS	LATWEAT	WSEGS	KSSDR		1260
EGTR	AGEER	RQ	QEEVVHE		DQGK	KPDQL	VIER	ERKWEQ	QLQ	EEQEQKR	LQAE	AEEQKR		1320
PAEEQ	KRQAE	I	PERD	ST	SVRIY									1380
NKNGN	NKYLD	QIGN	MTSSQR		RSKKE	QVPSG	AELER	QILQ	EMR	KRTPLHN	DNSWIR	Q	SA	1440
SVNKE	PVSLP	GIM	RRGESLD		NLDS	SPRSNSW	RQPP	PWLNQPT	GFY	ASSSVQD	FSRPP	PQLVS		1500
TSN	RAYMRNP	SSSV	PPPSAG		SVK	TS	TGVA	TTQSP	TPRSH	SPSAS	QSGSQ	LRNRS	VSGKR	1560
ICS	YCNILG	KGA	AMIESL		GLCY	HLHCFK	CVACE	CDLGG	SSSG	AEVRIR	NHQLY	CNDCY		1620
LRFK	SGRPTA	M												1631

SEQ ID NO: 20 moltype = RNA length = 1436
 FEATURE Location/Qualifiers
 source 1..1436
 mol_type = mRNA
 organism = Homo sapiens

SEQUENCE: 20

gcttttatga	gctgtaacac	tcaccgcgaa	gggtccgcagc	ttcactcctg	aagccagcga	60
gaccacgagc	ctactgggag	gaacgaacaa	ctcccgcagc	gccgccttaa	gagctgtaac	120
actcaccgcg	aagggtctgca	gcttactctc	tgagccagcg	agaccacgaa	cccaccagaa	180
ggaaaaaact	ccgaacacat	ctgaacatca	gaagcaacaa	actccggaca	cgccgccttt	240
aagaactgta	acactcactg	cgagggtccg	cggcttcatt	cttgaagtga	gtgagaccaa	300
gaacccacca	gttctggaca	caatttcaag	tcctcaggtg	ccatcaatat	tctgaaaatg	360
gcagtgattt	ttattcaacc	gtataaaggc	actttcacca	tgtacctgga	agcaacatct	420
acatcttttt	cagcaatcta	gatgctgggg	acacaaggct	caccttcag	gaatatggcc	480
atgacaccag	aaatcacaaa	catgatgaga	atggaatgac	tggggaagaa	gtgccagatg	540
cttactctgt	aaatgaagac	ccagcctctg	gggatgcaga	taccacctcc	ctgaagaagc	600
tgaatatctg	cagataagtg	gagttcacca	atgatgagga	gcgggatgga	gaaaggaggt	660
aggagagatc	atccaaggaa	catgagcaac	atgttaaaag	ccaagtggtt	taatttctgtg	720

-continued

```
agatggtgaa cccaagaggc tctgctggga gacaacaaaa ataatgaaga attgaaccag 780
agtccgggtga atatcagcac tgggaccagt tagcagagga aaaggaaaga ataaaagcga 840
aaagaatgaa gagtcatatg attaccaact ttctcttttt catataaatt gagtgtatat 900
gggtctggaa caacctgaat tccatcaag tcctgggctaa cctcattatg tcctatgaat 960
atttttgact aatccactt tacattaatc tgtattgtga atgtggatat tgaattatat 1020
ttctttgtaa tccattatc caaaatccag ttcagagact attagttacc aatgttcact 1080
gtgaagggaaa aaaaaaaaaa aaaagctcag aggataaaca tgtgatattg tttggctgtg 1140
tccccacca aatatcatct tgaattgtag ctcccataat tcccacgtgt tgtgggaggg 1200
acccggtggg agataattgt atcatggggg tgggtccccc atactattct catagtagtg 1260
aataagtcct ccaaaatctg atggttttat gagggaaaac ccctttcacc tggttctcat 1320
tctcttctct gctctgtcgt catgtaagac atgcctttca ccttctccac catgactgtg 1380
aggctcccc agccacgtgg aactgtgagc ccattaaacc tctttcactt ataaat 1436
```

```
SEQ ID NO: 21      moltype = RNA length = 1367
FEATURE           Location/Qualifiers
source            1..1367
                  mol_type = mRNA
                  organism = Homo sapiens
```

```
SEQUENCE: 21
gcttttatga gctgtaaacac tcaccgcgaa ggtccgcagc ttcactcctg aagccagcga 60
gaccacgagc ctactgggag gaacgaacaa ctcccgcgcg gccgccttaa gagctgtaac 120
actcaccgcg aaggtctgca gcttcaactc tgagccagcg agaccacgaa cccaccagaa 180
ggaaaaaaact ccgaacacat ctgaacatca gaagcaacaa actccggaca cgcgccttt 240
aagaactgta acactcactg cgaggggtccg cgggttcatt cttgaagtga gtgagaccaa 300
gaaccacca gttctggaca caatttcaag tcctcaggtg ccatcaatat tctgaaaatg 360
gcagtgattt ttattcaacc tgtataaggc actttcacca tgtacctgga agcaacatct 420
acatcttttt cagaatcta gatgctgggg acacaaggtc caccttccag gaatatggcc 480
atgacaccag aaatcacaaa catgatgaga atggaatgac tggggaagaa gtgccagatg 540
cttcaactgt aaatgaagac ccagcctctg gggatgcaga taccacctcc ctgaagaagc 600
tgaatatctg cagataagtg gagttcacca atgatgagga gcgggatgga gaaaggaggt 660
agggagagtc atccaaggaa catgagcaac atgttaaaag aattgaacca ggtccgggtg 720
aatatcagca ctggggcagc tttagcagag aaaaggaaag aataaaagcg aaaagaatga 780
agagtcatat gattaccaac ttttcttttt tcatataaat tgagtgtata tgggtctgga 840
acaacctgaa tttccatcaa gtccctggcta acctcattat gtcctatgaa tatttttgac 900
taatccact ttacattaat ctgtattgtg aatgtggata ttgaattata tttctttgta 960
atcccattat caaaaatcca gttcagagac tattagttac caatgttcac tgtgaaggaa 1020
aaaaaaaaaa aaaaagctca gaggataaac atgtgatatg gtttggctgt gtccccacc 1080
aaatatcatc ttgaattgta gctcccataa ttcccacgtg ttgtgggagg gacccgggtg 1140
gagataattg tatcatgggg gtgggtcccc catactattc tcatagtagt gaataagtct 1200
cacaaaatct tcgtggtttta taggggaaaaa cccctttcac ctggttctca ttctcttctc 1260
tgggtctgtc gcatgtaaga catgcctttc accttctcca ccatgactgt gaggcctccc 1320
cagccacgtg gaactgtgag ccattaaac ctctttcact tataaat 1367
```

```
SEQ ID NO: 22      moltype = RNA length = 1100
FEATURE           Location/Qualifiers
source            1..1100
                  mol_type = mRNA
                  organism = Homo sapiens
```

```
SEQUENCE: 22
gcttttatga gctgtaaacac tcaccgcgaa ggtccgcagc ttcactcctg aagccagcga 60
gaccacgagc ctactgggag gaacgaacaa ctcccgcgcg gccgccttaa gagctgtaac 120
actcaccgcg aaggtctgca gcttcaactc tgagccagcg agaccacgaa cccaccagaa 180
ggaaaaaaact ccgaacacat ctgaacatca gaagcaacaa actccggaca cgcgccttt 240
aagaactgta acactcactg cgaggggtccg cgggttcatt cttgaagtga gtgagaccaa 300
gaaccacca gttctggaca caatttcaag tcctcaggtg ccatcaatat tctgaaaatg 360
gcagtgattt ttattcaacc tgtataaggc actttcacca tgtacctgga agcaacatct 420
acatcttttt cagaattgaa ccagagtcctg gtgaatatca gcactgggac cagttagcag 480
agggaaaagg aagaataaaa gcgaaaagaa tgaagagtca tatgattacc aacttttctc 540
ttttcatata aattgagtgat atattgggtct ggaacaacct gaatttccat caagtctctg 600
ctaacctcat tatgtcctat gaatattttt gactaatccc actttacatt aatctgtatt 660
gtgaatgtgg atattgtaatt atatttcttt gtaatcccat tatccaaaat ccagttcaga 720
gactattagt taccaatggt cactgtgaag gaaaaaaaaa aaaaaaaagc tcagaggata 780
aacatgtgat atggtttggc tgtgtcccca cccaaatctc atcttgattt gtagctccca 840
taattccac gtgttggggg agggagcccg tgggagataa ttgtatcatg ggggtggttc 900
ccccatacta ttctcatagt agtgaataag tctcacaaaa tctgatggtt ttatgagga 960
aaaccccttt cacctggttc tcattctctt ctctggtctg tcgtcatgta agacatgcct 1020
ttcaccttct ccaccatgac tgtgaggcct cccagccac gtggaactgt gagccatta 1080
aacctctttc acttataaat
```

```
SEQ ID NO: 23      moltype = RNA length = 1370
FEATURE           Location/Qualifiers
source            1..1370
                  mol_type = mRNA
                  organism = Homo sapiens
```

```
SEQUENCE: 23
gcttttatga gctgtaaacac tcaccgcgaa ggtccgcagc ttcactcctg aagccagcga 60
```

-continued

gaccacgagc	ctactgggag	gaacgaacaa	ctcccgaagc	gccgccttaa	gagctgtaac	120
actcaccgcg	aaggctctgca	gcttcactcc	tgagccagcg	agaccacgaa	cccaccagaa	180
ggaaaaaact	ccgaacacat	ctgaacatca	gaagcaacaa	actcgggaca	cgccgccttt	240
aagaactgta	acactcactg	cgaggggtccg	cggttctatt	cttgaagtga	gtgagaccaa	300
gaacccacca	gttctggaca	caatttcaag	tcttcaggtg	ccatcaatat	tctgaaaatg	360
gcagtgattt	ttattcaacc	tgtataaggc	actttcacca	tgtacctgga	agcaacatct	420
acatcttttt	cagtttcttc	tacgccaggt	gtgtgcttag	ctccatgaca	aaaggtgaca	480
gcttattctg	cagcacacac	acatcatcaa	agtgaggagt	ggtgagactg	gcacactgac	540
agtcgtctct	agcagatttc	agctcacact	gcaatctaga	tgctggggac	acaaggtcca	600
ccttccaggga	atatggccat	gacaccagaa	atcacaaaca	tgatgagaat	ggaatgactg	660
gggaagaagt	gccagatgct	tcacttgtaa	atgaagacct	agcctctggg	gatgcagata	720
ccacctccct	gaagaagctg	aatatctgca	gataagtgga	gttcaccaat	gatgaggagc	780
gggatggaga	aaggaggtag	ggagagtcac	ccaaggaaca	tgagcaacat	gttaaaagcc	840
aagtgggtta	tattctggag	atgggtgaacc	caagaggctc	tgctggggaga	caacaaaaat	900
aatgaagaat	tgaaccagag	tccggtgaat	atcagcactg	ggaccagtta	gcagaggaaa	960
aggaagaagt	aaaagcgaaa	agaatgaaga	gtcatatgat	taccaacttt	tcttttttca	1020
tataaattga	gtgtatatgg	gtctggaaca	acctgaattt	ccatcaagtc	ctggctaacc	1080
tcattatgtc	ctatgaatat	ttttgactaa	tcccacttta	cattaatctg	tatttgtaat	1140
gtggatatgg	aattatattt	ctttgtaact	ccattatcca	aaatccagtt	cagagactat	1200
tagttaccaa	tgttctactgt	gaaggaaaaa	aaaaaaaaaa	aagctcagag	gataaacatg	1260
tgatatggtt	tggctgtgtc	cccacccaaa	tatcatcttg	aattgtagct	cccataattc	1320
ccacgtgttg	tgggaggggac	cgggtgggag	ataattgtat	catgggggtg		1370

SEQ ID NO: 24 moltype = RNA length = 3812

FEATURE Location/Qualifiers

source 1..3812

mol_type = mRNA

organism = Homo sapiens

SEQUENCE: 24

gcggagagaa	ccctctgttt	tccccactc	tctctccacc	tctctctgcc	ttccccacc	60
cgagtgcgga	gccagagatc	aaaagatgaa	aaggcagtc	ggtcttcagt	agccaaaaaa	120
caaaaacaac	aaaaacaaaa	aagccgaaat	aaaagaaaaa	gataataact	cagttcttat	180
ttgcacctac	ttcagtgga	actgaatttg	gaaggtggag	gattttgttt	ttttctttta	240
agatctgggc	atcttttgaa	tctacccttc	aagtattaa	agacagactg	tgagcctagc	300
agggcagatc	ttgtccaccg	tgtgtcttct	tctgcacag	actttgaggc	tgtcagagcg	360
ctttttgctg	gggtgtctcc	gcaagtttcc	ttctctggag	cttcccgag	gtgggcagct	420
agctgcagcg	actaccgcat	catcacagcc	tgttgaactc	ttctgagcaa	gagaagggga	480
ggcggggtaa	gggaagtagg	tggaaagattc	agccaagctc	aaggatggaa	gtgcagtttag	540
ggctgggaag	ggtctaccct	cggccgcctg	ccaagaccta	ccgaggagct	ttccagaatc	600
tgttccagag	cgtgcgcgaa	gtgatccaga	ccccgggccc	caggcaccca	gaggcgcgca	660
gcgcagcacc	tcccggcgcc	agtttgctgc	tgtctcagca	gcagcagcag	cagcagcagc	720
agcagcagca	gcagcagcag	cagcagcagc	agcagcagca	gcaagagact	agccccaggc	780
agcagcagca	gcagcagggg	gagggatggt	ctccccaaag	ccatcgtaga	ggccccacag	840
gtacctcgtg	cctggatgag	gaacagcaac	cttcacagcc	cgactcgccc	ctggagtggc	900
accccagagag	aggttgctgc	ccagagcctg	gagccgcctg	ggccgcgcagc	aaggggtctgc	960
cgagcagctg	cccagcactc	ccggacgagg	atgactcagc	tgccccatcc	acgtttgtccc	1020
tgctggggccc	cactttcccc	ggcttaagca	gctgctccgc	tgaccttaaa	gacatcctga	1080
gcgagggccag	caccatgcaa	ctccttcagc	aacagcagca	ggaagcagta	tccgaaggca	1140
gcagcagcgg	gagagcgagg	gaggcctcgg	gggctccac	ttctccaag	gacaattact	1200
tagggggcac	ttcgaccatt	ttcgacaacg	ccaaggagtt	gtgtaaggca	gtgtcgggtg	1260
ccatgggcct	gggtgtggag	gcgttggagc	atctgagtc	aggggaacag	cttcgggggg	1320
attgcatgta	cgcgccactt	ttgggagttc	caccgcgtgt	gcgtccact	ccttgtgtccc	1380
cattggccga	atgcaaaagt	tctctgctag	acgacagcgc	aggcaagagc	actgaagata	1440
ctgctgagta	ttcccccttc	aaggagaggt	acaccaaagg	gctagaaggc	gagagcctag	1500
gctgctctgg	cagcgctgca	gcagggagct	ccgggacact	tgaactgccg	tctaccctgt	1560
ctctctacaa	gtccggagca	ctgggacgag	cgctgcgta	ccagagtcgc	gactactaca	1620
actttccact	ggctctggcc	ggaccgcgcg	ccctcccgcc	gcctcccat	ccccacgctc	1680
gcatcaagct	ggagaacccg	ctggactacg	gcagcgcctg	ggcggtgcg	ggggcgagct	1740
gccgctatgg	ggacctggcg	agcctgcacg	gcgcgggtgc	agcgggaccc	ggttctgggt	1800
cacctcagc	cgcgcgttcc	tcatctggc	acactctctt	cacagccgaa	gaaggccagt	1860
tgtatggacc	gtgtggtggt	ggtgggggtg	gtggcggcgg	cgccggcgcc	ggcgggcgcg	1920
gcggcgcgcg	cggcgcgcg	ggcgaggcgg	gagctgtagc	ccctacggc	tacactcgcc	1980
ccctcaggg	gctggcgggc	caggaaagcg	acttcaccgc	acctgatgtg	tggtaccctg	2040
gcggcatggt	gagcagagtg	ccctatccca	gtcccacttg	tgtcaaaagc	gaaatgggccc	2100
cctggatgga	tagctactcc	ggaccttacg	gggacatcg	tttgagagact	gccaggggacc	2160
atgttttgcc	catgactact	tactttccac	cccagaagac	ctgcctgac	tgtggagatg	2220
aagcttctgg	gtgactctat	gagctctca	catgtggaag	ctgcaaggtc	ttcttcaaaa	2280
gagccgctga	agggaaacag	aagtacctgt	gcgcagcag	aaatgattgc	actattgata	2340
aattccgaag	gaaaaattgt	ccatcttgct	gtcttcggaa	atgttatgaa	gcagggatga	2400
ctctgggaga	aaaattccgg	ggtggcaatt	gcaagcatct	caaaatgacc	agacctgaa	2460
gaaaggctga	cttgccctcat	tcaaaatgag	ggctctagag	ggctctagtg	gatagctctg	2520
agaaacctgg	cgtctgaggc	ttaggagctt	aggtttttgc	tcttcaaac	agactttgac	2580

-continued

```

gttgggggttg ggggctactc tcttgattgc tgactccctc cagcgggacc aatagtgttt 2640
tcctacctca cagggatgtt gtgaggacgg gctgtagaag taatagtggt taccattcat 2700
gtagtgttga gtatcatgat tattgtttcc tgaatgtgg cttggcattg gcaaagtgc 2760
ttttgattgt tcttgatcac atatgatggg ggccaggcac tgactcaggc ggatgcagtg 2820
aagctctggc tcagtcgctt gcttttcgtg gtgtgctgcc aggaagaaac ttgctgatg 2880
ggactcaagg tgtcaccttg gacaagaagc aactgtgtct gtctgaggtt cctgtggcca 2940
tctttatttg tgtattaggc aattcgtatt tcccccttag gttctagcct tctggatccc 3000
agccagtgc ctagatctta gctcaggcc ctgtcactga gctgaaggta gtatcgtatc 3060
cacagaagtt cagtaacaa ggaccagatt tctgcttctc caggagaaga agccagccaa 3120
ccccctctct caaacacact gagagactac agtccgactt tccctcttac atctagcctt 3180
actgtagcca cactccttga ttgctctctc acatcacatg cttctcttca tcagtgttaa 3240
gcctctcatt cttctcccaa gccagactca aatattgtat tgatgtcaaa gaagaatcac 3300
ttagagtttg gaatatcttg ttctctctct gctccatagc ttccatattg acaccagttt 3360
ctttctagtg gagaagtgga gtctgtgaag ccagggaac acacatgtga gagtcagaag 3420
gactctccct gacttgcttg gggcctgtct tccccacctt ctccagctcg tctaaacaca 3480
cacacacaca cacacacaca cacacacacg ctctctctct ctctccccc ccaacacaca 3540
cacactctct ctctctctca cacacacaca cacttcttct tctttccctt 3600
gactcagcaa cattctggag aaaagccaag gaaggacttc agggggggag tttccccctt 3660
ctcagggcag aattttaatc tccagacca caagaagttc ctaaatgttg attgaaaggc 3720
taatgaggtt tatttttaac tactttctat ttgtttgaat gttgcatatt tctactagt 3780
aaattttccc ttaataaagc cattaataca cc 3812

```

SEQ ID NO: 25 moltype = AA length = 644
FEATURE Location/Qualifiers
source 1..644
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 25

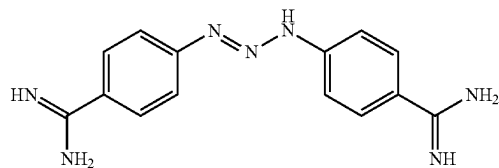
```

MEVQLGLGRV YRPPSPKTYR GAFQNLFSQSV REVIQNPGRP HPEAASAAPP GASLLLLQQQ 60
QQQQQQQQQQ QQQQQQQQQQ ETSPRQQQQQ QGEDGSPQAH RRGPTGYLVL DEEQQPSQPQ 120
SALECHPERG CVPEPGAAVA ASKGLPQQLP APPDEDDSA PSTLSLLGPT FPGLSSCSAD 180
LKDILSEAST MQLLQQQQQE AVSEGSSSGR AREASGAPTS SKDNYLGTS TISDNAKELC 240
KAVSVSMGLG VEALEHLSPG EQLRGDCMYA PLLGVPPAVR PTPCAPLAEC KGSLLDDSDAG 300
KSTEDTAEYS PFKGGYTKGL EGESLGCSGS AAAGSSGTLE LPSTLSLYKS GALDEAAAYQ 360
SRDYYNFPLA LAGPPPPPPP PHPHARIKLE NPLDYGSAWA AAAAQCRYGD LASLHGAGAA 420
GPGSGSPSAA ASSSWHTLFT AEEGQLYGPC GGGGGGGGGG GGGGGGGGGG GGGEAGAVAP 480
YGYTRPPQGL AGQESDFTAP DVWYPGGMVS RVPYPSPTCV KSEMGPWMDS YSGPYGDMRL 540
ETARDHVLPI DYYFPPQKTC LICGDEASGC HYGALTCGSC KVFFKRAEAG KQKYLCSARN 600
DCTIDKFRRK NCPSCRLRKC YEAGMTLGEK FRVGNCKHLK MTRP 644

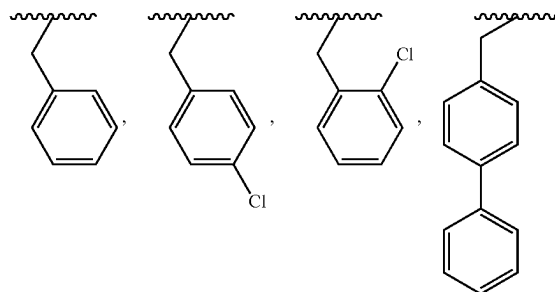
```

1.-2. (canceled)

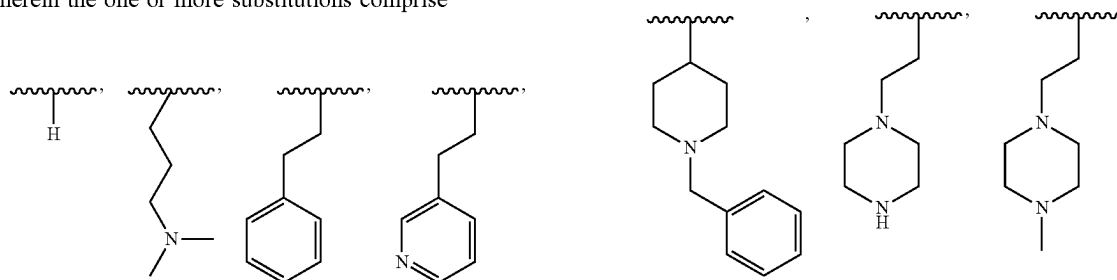
3. A compound of formula



-continued



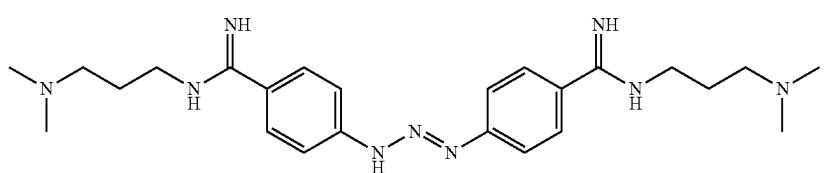
having one or more of ortho, para, and/or meta substitutions,
wherein the one or more substitutions comprise



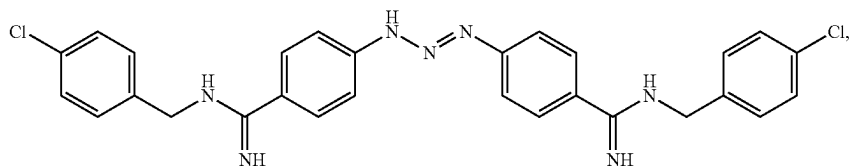
or any combination thereof.

4. (canceled)

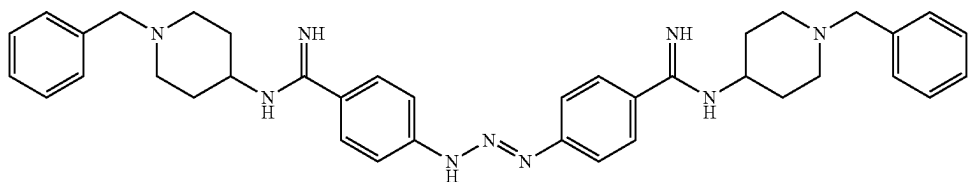
5. The compound of claim 3, wherein the compound is



(DMZ-P1)



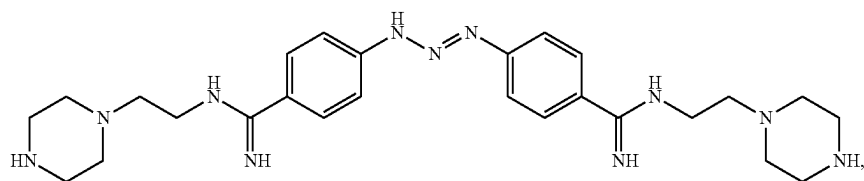
(DMZ-P5)



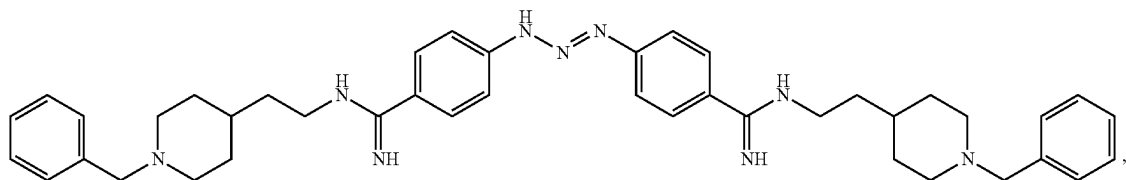
(DMZ-P8)

-continued

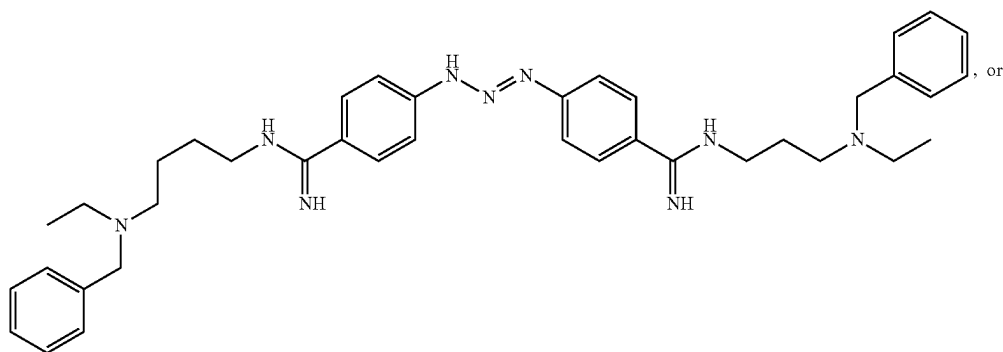
(DMZ-P9)



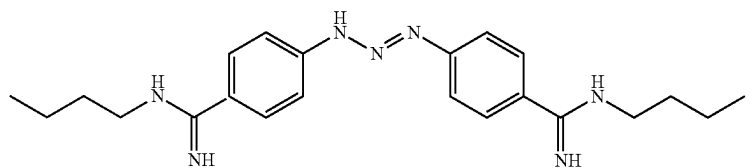
(DMZ-P13)



(DMZ-P14)



(DMZ-P17)

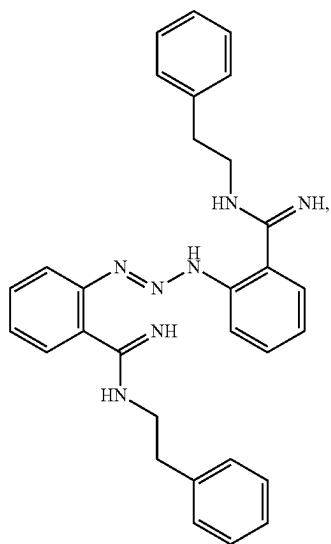
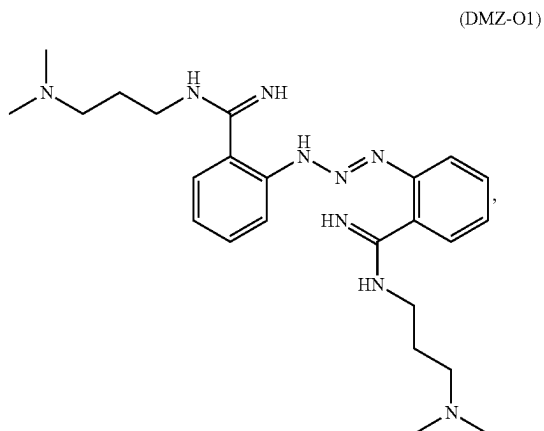


6.-11. (canceled)

12. The compound of claim 3, wherein the compound is

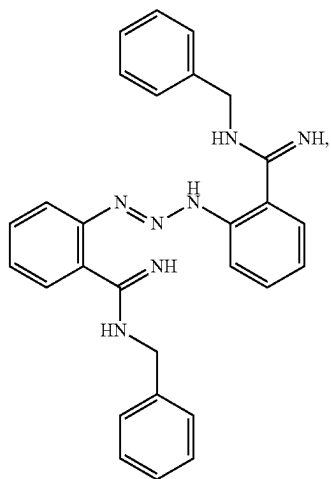
-continued

(DMZ-O2)



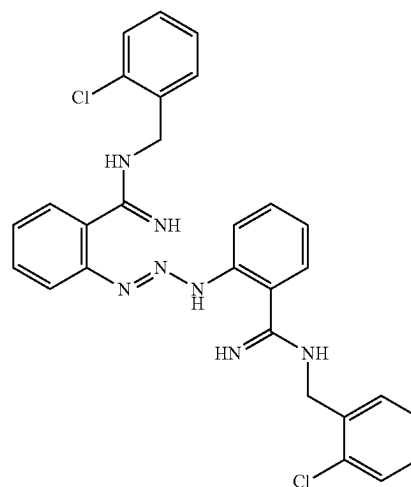
-continued

(DMZ-O4)

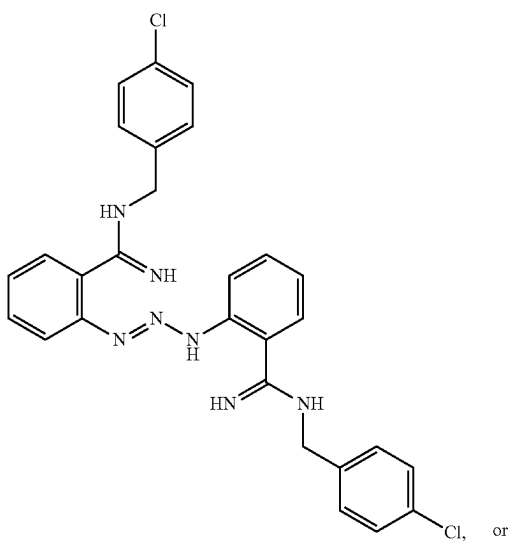


-continued

(DMZ-O6)



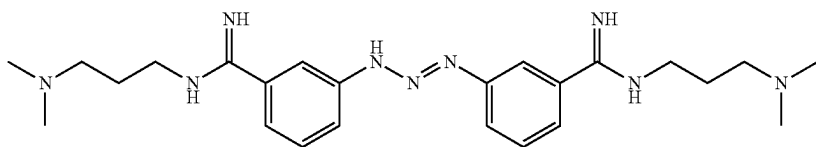
(DMZ-O5)



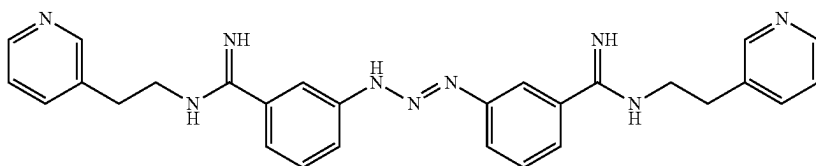
13.-16. (canceled)

17. The compound of claims 3

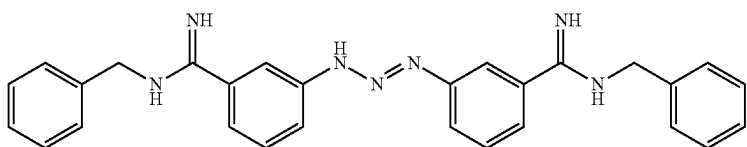
(DMZ-M1)



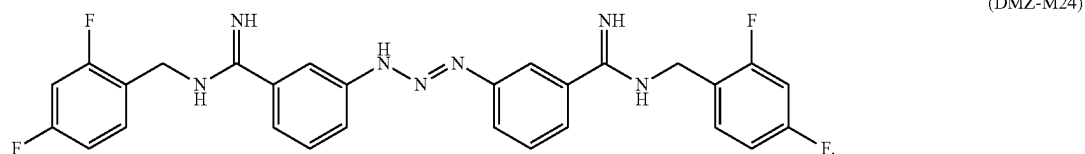
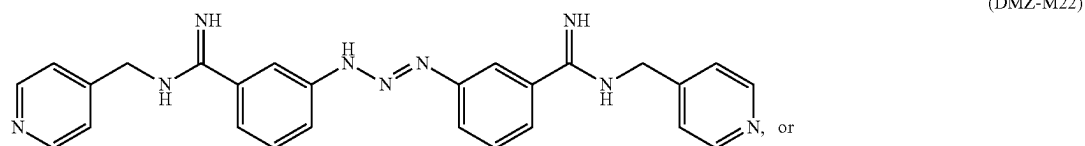
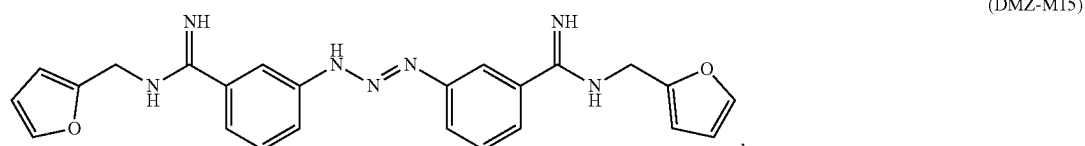
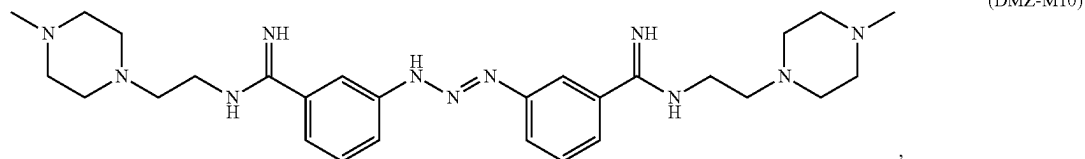
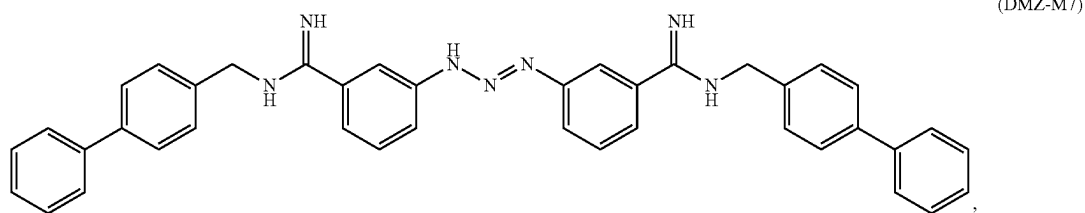
(DMZ-M3)



(DMZ-M4)

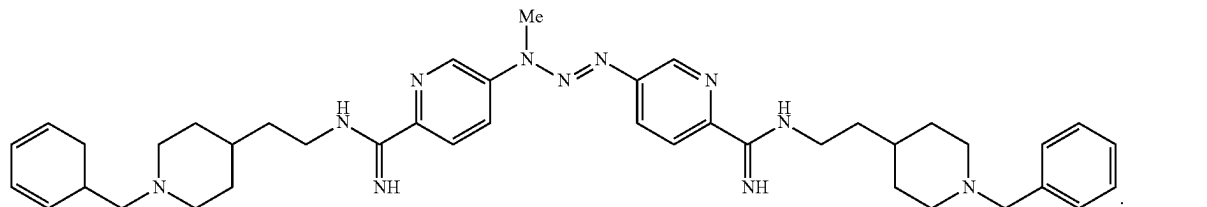
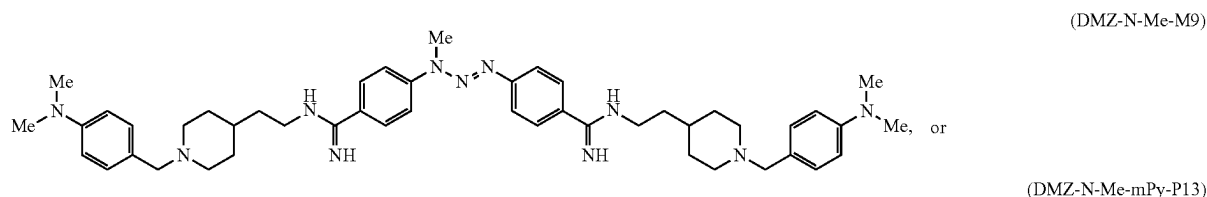


-continued



18.-24. (canceled)

25. The compound of claim 3, wherein the compound is



26. (canceled)

27. The compound of claim 3, wherein the compound modulates tertiary RNA structure.

28. The compound of claim 27, wherein the compound stabilizes the tertiary RNA structure.

29. The compound of claim 27, wherein the compound destabilizes the tertiary RNA structure.

30. The compound of claim 27, wherein the tertiary RNA structure comprises the MALAT1 triple helix.

31. A composition, comprising: a compound of claim **3**, and one or more pharmaceutically acceptable carriers and/or excipients.

32. A method of treating a subject in need thereof, the method comprising:

administering to a subject the composition of claim **31**, wherein one or more targeted RNA or targeted RNA structures are modulated.

33. The method of claim **32**, wherein the subject has cancer.

34. (canceled)

35. The method of claim **32**, wherein administering comprises delivery to one or more of the subject's body systems affected by the targeted RNA or targeted RNA structure, and wherein the one or more body systems affected by the targeted RNA or targeted RNA structure comprise the cardiovascular system, the digestive system, the endocrine system, the lymphatic system, the muscular system, the nervous system, the reproductive system, the respiratory system, the skeletal system, the urinary system, the integumentary system, or any combination thereof.

36. (canceled)

37. The method of claim **32**, wherein following the administering step, (i) the subject's symptoms are diminished and/or decreased, (ii) the subject's quality of life is improved and/or enhanced, and/or (iii) one or more of the subject's body systems experience and/or show signs of normal physiology and/or cellular homeostasis.

38.-39. (canceled)

40. The method of claim **32**, further comprising administering to the subject one or more therapeutic agents and/or active agents.

41. The method of claim **40**, wherein the one or more therapeutic agents and/or active agents comprise an anti-cancer agent.

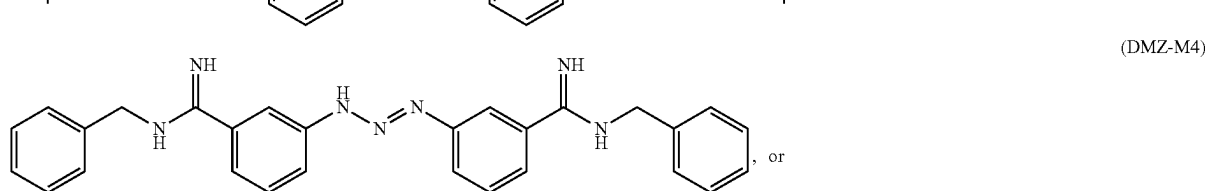
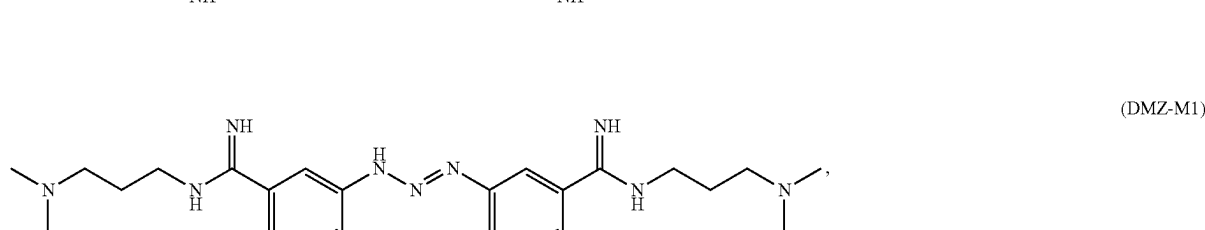
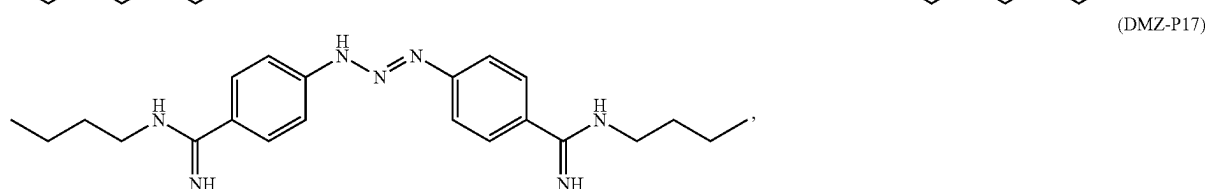
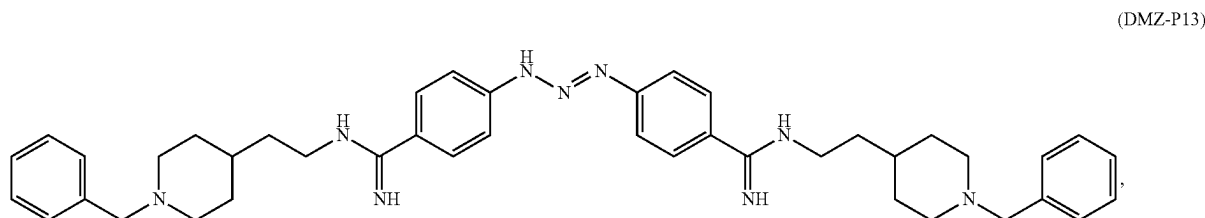
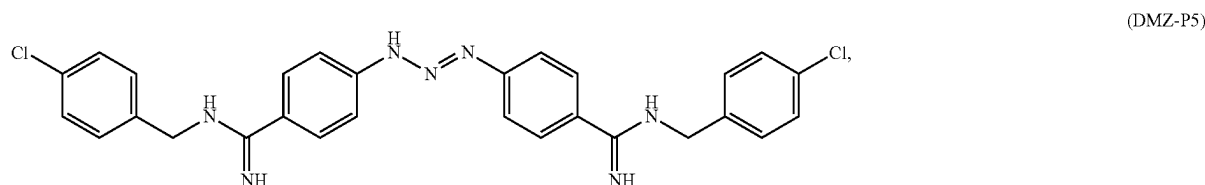
42. The method of claim **32**, further comprising monitoring the subject for adverse effects.

43. The method of claim **42**, wherein in the absence of adverse effects, the method further comprises continuing to treat the subject and/or continuing to monitor the subject, or wherein in the presence of adverse effects, the method further comprises modifying one or more steps of the method.

44. (canceled)

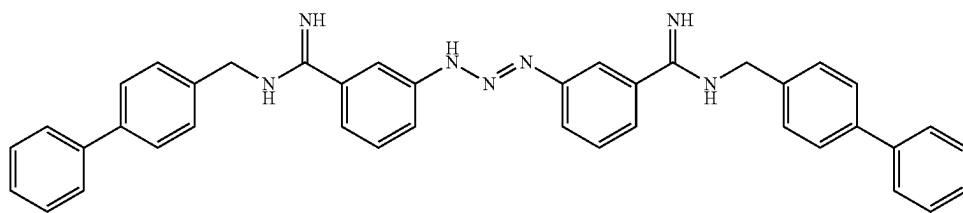
45. The method of claim **33**, wherein the cancer is prostate cancer.

46. The compound of claim **3**, wherein the compound is



-continued

(DMZ-M7)



* * * * *